Observer-Indexed Metatheory for Proof-Relevant Computation and Open-Ended Reflection

Dependent families, CBPV, coalgebraic worlds, and staged qualification

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Abstract

We develop a semantic envelope for a reflective language whose operational histories, dependent evidence, extensional readouts, effects, costs, and theory qualifications must coexist without being identified. The central discipline is that the proof-relevant dependent total space is primary, while every extensional or optimized face is obtained through an explicitly licensed observer. A call-by-push-value factorization is the intended basis for Prime's next evaluation design, making value/computation boundaries and qualified strategy changes explicit. The current nucleus supplies revision-keyed call-by-need; its migration and the supported combinations of eager evaluation, suspension and sharing remain proof obligations, not consequences of the CBPV name.

For open-ended reflection we study an observer-indexed family of set builders with opaque cross-observer references. We give exact hypotheses under which the proposed final-coalgebra construction is justified, explain why unrestricted powerset fixed points cannot be sets, and separate a well-founded Tarski–Grothendieck host from a non-well-founded object and its new membership relation. We also isolate a type-theoretic analogue of Grothendieck-universe selection: a family-enclosing Tarski-universe operator which places any named dependent family inside a universe closed under the standard dependent type formers. We separate its computational type-theoretic face from an extensional HOTG host, formalize the algebraic adequacy interface and its nonfaithful reverse boundary, and independently verify extensional family enclosure in HOTG. An open-ended PQI-inspired indexed decomposition relates physical descriptions, qualitative fibres, and indexical location without asserting an exhaustive basis or identifying philosophical and type-theoretic notions. An explicit tail reduced power supplies a choice-free relative-infinity baseline, with incomparability as its exact price. Principal ultrapowers are proved to collapse to their selected coordinate, whereas a free-ultrafilter model contains genuinely nonstandard points. A staged Grothendieck–Indra atlas remains a conditional organization of locally small views, not a stronger set theory or a constructed final coalgebra. Likewise, an alternating two-polarity qualification diagram can be linearized whenever it is countable and directed; its mathematical force comes only from the theories, interpretations, and models on its edges. A checked typed rho communication cell, a GSLT-generated graph-walk type theory, and a source-faithful formal-ethics development then serve as concrete consumers of the general boundary. We also separate a genuine multimodal dependent syntax from its semantic mode diagram, a family of native services over semantic targets from the modes themselves, and bootstrap diversity from object-theory strength. Local thinness of the selected operational–intensional–extensional comparison sector is recorded as a truncation certificate rather than as a global bound on cell dimension, and a typed, algorithm-neutral interchange core for higher-order problems separates parsing, elaboration, search algorithms, and proof authority. The result is a precise research programme rather than a declaration of a final native calculus. Proved results, conditional constructions, negative controls, and open design choices are kept distinct throughout. Compositional execution contracts place resource authority, actual read dependencies and committed observer

families over the same worlds and runs. Checked transport and adaptive-query laws connect these components without identifying permission, proof evidence, trusted assumptions or outstanding obligations; their complete native realization remains a separate boundary. Evidence is treated as a summary of observations whose operations are derived from stated observation contracts; absence is intuitionistic negation certified at declared frontiers; and nondeterministic choice yields probabilities only through a scheduler.

Contents

1 Scope and status discipline **4**
1.1 Higher-order MeTTa integration as the experimental objective 4

2 Digest: what is known about HO MeTTa Prime, and how firmly **6**

3 The semantic envelope **11**
3.1 The dependent total space is primary 11
3.2 Dependent and simple typing 12
3.3 Kernel selection and generation requirements 12
3.4 Quantitative observers without admission by similarity 17
3.5 Evidence: observation contracts, readouts and PLN 18
3.6 Infinite relational world models, Dobrushin shells, and coherent self-transcendence . 22
3.7 Draft-critical judgments and workload discriminators 24
3.8 Intermediate languages, open interfaces and local reasoning 34
3.9 Intensional, extensional and operational presentations 35
3.10 Biform theories and GSLTs as the semantic transformation waist 36
3.11 A theory graph rather than one maximal kernel 38
3.12 Observer refinement, dependent descent, and several identities 42
3.13 Compositional execution contracts over worlds and runs 43
3.14 An open-ended PQI-inspired indexed decomposition 47
3.15 A GSLT-shaped description of mathematics 48

4 Bubble type theory and hosted logics **49**
4.1 Bubbles: definitional equality declared where it is wanted 49
4.2 Why “weak”: the weakness order on bubbles 50
4.3 Least sufficient bubbles and the meet-closed (Horn) fiber 51
4.4 The empty top and the inconsistent-top method 52
4.5 Hosted logics and translations between them 52
4.6 HE’s types grounded in HOL 54
4.7 Demand, sharing and effects 55
4.8 Open decisions 57

5 Prime’s planned CBPV basis and qualified evaluation strategies **57**

6 Given-clause saturation as a CBPV consumer **68**
6.1 Strategy profiles and the guarantee lattice 69

7 A checked vertical specimen: dependent rho communication **70**

8 Observer-indexed non-well-founded worlds **71**
8.1 The visibility container . . . 71
8.2 Why the unrestricted set-level construction fails . . . 72
8.3 Route-rich and extensional coalgebras . . . 72
8.4 A constructed final coalgebra: circular propositions . . . 73
8.5 Truth over the circular universe is partial . . . 74

9 HOTG as host, not as hyperset membership **75**

10 Universe services for reflective reasoning **75**
10.1 A type-theoretic analogue of universe selection . . . 75
10.2 Shared material laws and explicit checking profiles . . . 94
10.3 Universe closure profiles and their different strength currencies . . . 94
10.4 Where the construction bill is paid . . . 96
10.5 A two-face adequacy target . . . 97
10.6 What a commuting bridge requires . . . 98
10.6.1 HOL predicates as dependent families . . . 99
10.6.2 Three different intensional–extensional bridges . . . 110
10.7 Why a reflective reasoner wants a universe tower . . . 111
10.8 Four stratifications that should stay apart . . . 112
10.9 Relative infinity without ultrafilter selection . . . 113
10.10 Principal collapse and free-ultrafilter divergence . . . 114
10.11 The independent-axis synthesis . . . 114

11 The Grothendieck–Indra atlas as a conditional design target **115**

12 Alternating qualification towers and the helix non-strength theorem **116**
12.1 Four claims that must not be conflated . . . 116
12.2 The internal claim realized in arithmetic: the consistency tower . . . 117
12.3 Conditional coupling of the atlas and qualification tower . . . 118

13 What is already clarified **118**

14 Formal ethics as a cognitive-architecture discriminator **121**
14.1 Exact equivalence and the cost of encoding . . . 121
14.2 Computability is a qualifier, not a counterexample . . . 122
14.3 Optimization, virtue learning, and uncertain concepts . . . 122
14.4 Native plurality for a multi-agent reasoner . . . 122

15 Modes, proof authorities, bootstrap, and the remaining braid **123**
15.1 A criterion for modes and their neighbours . . . 123
15.2 The mode-theory option ladder . . . 124
15.3 Local truncation certificates instead of a global cell ceiling . . . 126
15.4 Four concrete mode-theory candidates . . . 127
15.5 The checked modal core and the exact MTT gap . . . 129
15.6 Would Agda make the final MTT tranche easier? . . . 135
15.7 NIK is a family of native services, not another mode . . . 136
15.8 A typed interchange core for higher-order problems . . . 137
15.9 Bootstrap diversity and object-theory strength are independent . . . 138

15.10OSLF over equation classes . . . 139
15.11Vertical specimens that decide the architecture . . . 142
15.12The decision order . . . 145

16 Proof-theoretic and model-theoretic ledger **146**
16.1 Proof theory . . . 146
16.2 Model theory . . . 147

17 Status ledger: proved, refuted, conditional, open **148**
17.1 The typed diagram . . . 171
17.2 Proved by checked construction . . . 172
17.3 Refuted or proved impossible . . . 188
17.4 Conditional: proved under a stated hypothesis . . . 193
17.5 Open, with the smallest step that would decide each . . . 194

18 Open obligations and decision gates **198**
18.1 Modes and multimodal dependent syntax . . . 198
18.2 Native proof authority and NIK composition . . . 199
18.3 Bootstrap and independent qualification . . . 200
18.4 Vertical interconnections and decisive specimens . . . 201
18.5 Coalgebra and set theory . . . 201
18.6 Type theory and computation . . . 201
18.7 Ultrafilter and nonstandard semantics . . . 203
18.8 Formal ethics and multi-agent cognition . . . 203
18.9 World-model evidence and probability . . . 203
18.10Proof search and generated code . . . 204

19 Conclusion **205**

1 Scope and status discipline

This document organizes reusable mathematics. It does not define a particular product language, and names such as “native”, “extensional”, and “intensional” denote semantic roles rather than branded calculi. The eventual language design should be selected only after concrete calculi and runtimes have crossed the same experiments.

1.1 Higher-order MeTTa integration as the experimental objective

The immediate language-facing objective is an integrated environment for higher-order mathematics and reasoning: extensional HOL and HOTG theories, intensional dependent constructions, OSLF/GSLT-IL presentations and semantics, and NIK-qualified native computation should be usable together. It is not necessary to finish specifying every Prime binding convention, search presentation or runtime effect combination before experimenting with these connections. Conversely, merely placing several unrelated checkers behind one syntax does not establish their semantic integration.

The experimental unit is a declared mathematical task and its connected representations. Its premises identify the actual source calculi, assumptions, contexts and universe levels; the judgments and observations used by the task; and the interpretation and realization maps between them.

Preservation, substitution and no-invention claims concern these actual maps. A broader language profile may remain parameterized, provided each experiment fixes the operations it consumes and does not present unsupported cases as qualified. Such experimental choices do not ratify global defaults.

Four connected lines organize this objective:

1. Mathematical environments support definitions, quantification, induction, proofs and theorem application in their own right, not merely validation of an otherwise fixed runtime. Shared simply typed infrastructure does not reduce dependent mathematics to simple typing, and a finite collection implementation is not an abstract HOTG universe.

2. Intensional–extensional bridges explain how a predicate becomes a dependent family, a theorem supplies a dependent use, and code and model semantics correspond under substitution. Equality transport, scoped identity, retained presentations and universe enclosure belong to this line. A faithful comparison must state its image and assumptions; it need not be a global identification of HOL, DTT and HOTG.

3. GSLT/OSLF and NIK connect declared rules and mathematical judgments to justified realizations. The most appropriate qualified native operation may compute directly; trace replay is not the default meaning of integration. A fast route must preserve and reflect the selected observation, while profitability and runtime cost remain measured implementation claims.

4. Worked higher-order developments exercise these connections together: uniform induction, model discrimination, higher-order hypothesis formation, theorem reuse and specialized execution followed by justified reconstruction. The four-way comparison of a deep GSLT, HOL and DTT representations, and an explicit HOTG/model interpretation is a discriminating experiment, not a declaration that those representations are already equivalent.

The thirty-row candidate-specification inventory mixes these connections with source isolation, whole-language public policies and final handoff. Its row count is not a readiness measure or a prerequisite graph for higher-order experimentation. Its proved components remain usable at their exact scope. The concrete HOTG adequacy composition below, native dependent attachment of semantic bridges, and the relevant end-to-end runtime comparisons are genuine integration work. Unrelated syntax defaults and stronger universal models are not silently promoted to blockers. Further compatibility laws may constrain currently open policies; independence under a small ground model does not establish independence under the eventual combined higher-order requirements.

We use three statuses.

Proved. A theorem has a closed proof in the current formal development or follows directly from a cited theorem under hypotheses already instantiated.

Conditional. A theorem follows from stated categorical, smallness, or model-theoretic hypotheses, but those hypotheses have not yet all been instantiated for the intended construction.

Design target. A structure or comparison is mathematically specified but still requires construction, proof, or an empirical discriminator.

This distinction is especially important for final coalgebras. Writing νH is not a construction. One must name the ambient category, prove that H is an endofunctor of the required class, establish the relevant accessibility or small-based condition, and then instantiate an existence theorem.

2 Digest: what is known about HO MeTTa Prime, and how firmly

This section is a map of the current state, dated 19 October 2026. Each row names a part of the design, says what is known, and gives its status in the sense of the previous section, together with a PLN simple truth value STV$\langle s, c\rangle$ and credal indefinite truth value ITV$\langle [L, U], b\rangle$ (with prior weight $\kappa = 2$) recording how firmly the doctrinal decision and its Lean/C realization are established. *Decided* marks a design decision that is no longer under investigation. Section 4 explains the newest material, and Section 17 keeps the detailed ledger.

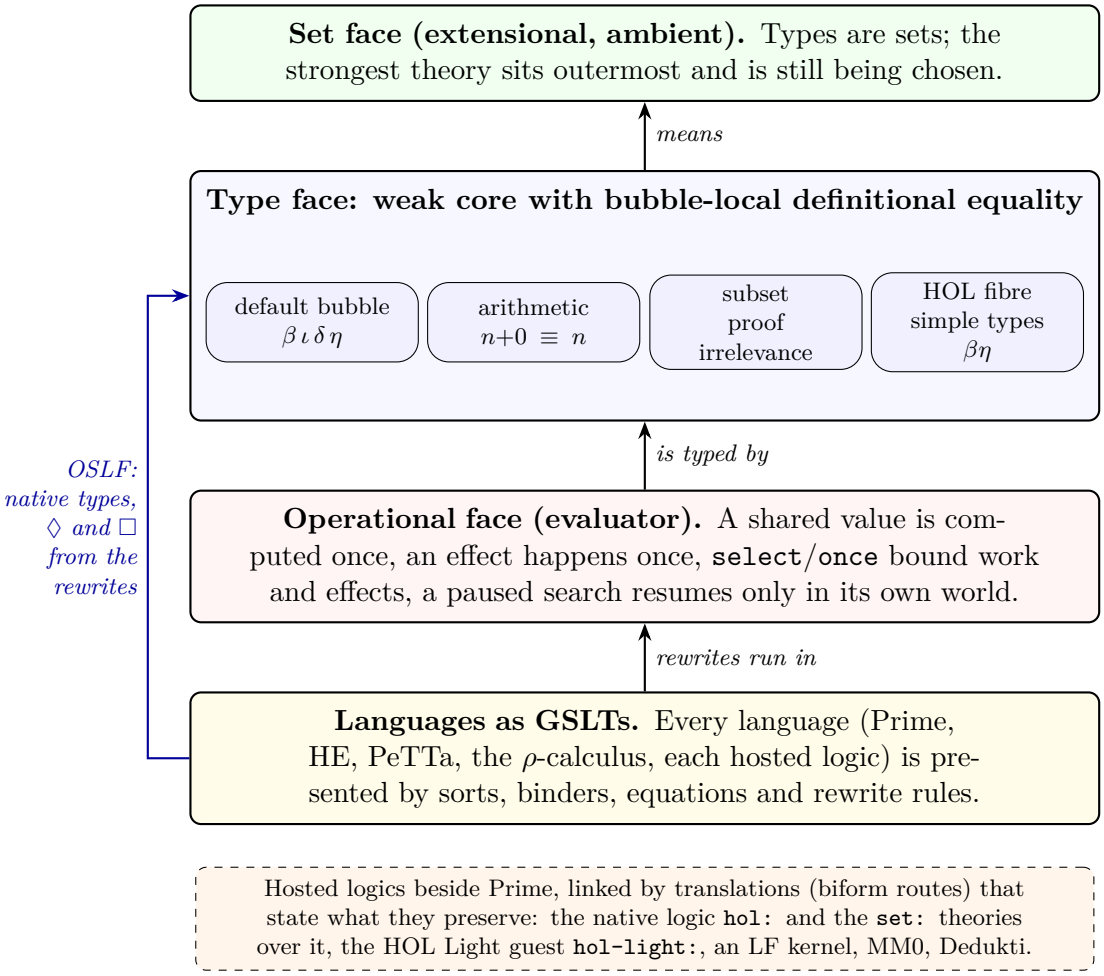


Figure 1: The three faces, the bubbles of the type face, and the hosted logics. Every language is presented as a GSLT: its rewrites run in the operational face, and from them OSLF derives native types and a modal logic in the type face. Every decidable or cheap fragment is a bubble carved inside the type face, and its good properties are theorems about that bubble.

Part	What is known	Status	PLN STV / ITV
Three faces	Every construct has a set face (meaning), a type face (checking) and an operational face (running), with links between them.	Decided	STV$\langle 1.00, 0.92\rangle$ $[0.92, 1.00]$
Sets ambient	A type is a set; the extensional side is the strongest. Membership is not yet connected to typing in the draft runtime.	Decided; realization open	Doc $\langle 0.95, 0.90\rangle$; Real $\langle 0.42, 0.86\rangle$ $[0.28, 0.64]$
Identity	J with ι in the core; no global K or UIP ("sans-K"); K and proof irrelevance in admitted per-space `(id:profile ...)` bubbles; UIP definable inside a non-UIP spine.	Decided	Doc $\langle 1.00, 0.92\rangle$; Real $\langle 0.89, 0.90\rangle$ $[0.80, 0.95]$
η	Kept, and type-directed; proof-producing $\beta\eta$ conversion recipes (`ConversionRecipes`) retain HOL equality witnesses.	Decided	Doc $\langle 1.00, 0.91\rangle$; Real $\langle 0.88, 0.89\rangle$ $[0.79, 0.95]$
HE's type engine	Prime never falls back to it; HE-style fragments are decided by the hosted HOL Light checker.	Decided	Doc $\langle 1.00, 0.94\rangle$; Real $\langle 0.95, 0.90\rangle$ $[0.86, 0.99]$
Candidate web	Prime is a web of explicit DTT candidates (`ParameterizedPiSigmaId`, `CumulativePiSigmaId`, `StagedScopedReflective`) and carved profile bubbles (Lean, Agda, Rocq), never one monolithic calculus.	Decided	Doc $\langle 1.00, 0.93\rangle$; Real $\langle 0.82, 0.89\rangle$ $[0.71, 0.91]$
Grounding & 3 kernels	Tier-0 grounding is correct-by-construction (Rung 1: minimal HOL proves the checker faithful once); certificates and per-proof exports to HOL Light, Megalodon and MM0 are interop/acceleration bridges, all checking the same fiber semantics.	Decided	Doc $\langle 1.00, 0.92\rangle$; Real $\langle 0.71, 0.88\rangle$ $[0.56, 0.84]$
Variables modulo E	Repeated pattern variables denote equality of model elements modulo the active equational theory E (presentation invariance; syntactic unification is the $E = \emptyset$ bubble).	Decided	Doc $\langle 1.00, 0.91\rangle$; Real $\langle 0.81, 0.89\rangle$ $[0.69, 0.90]$
Least bubbles & Horn fiber	Least sufficient bubbles do not exist in general (`no_least_sufficient_bubble`); a least bubble above every seed exists iff sufficiency is meet-closed (`tfae_mooreFamily`), for which Horn is the canonical finitary syntax (`models_hornTheoryOf`, `oneStep_rulesOf`).	Decided; proved	Doc $\langle 1.00, 0.92\rangle$; Real $\langle 0.95, 0.90\rangle$ $[0.86, 0.99]$

Part	What is known	Status	PLN STV / ITV
Inductive spine	Authored inductives join indexed polynomial containers $P = \Sigma_t \Pi_p \Delta_s$ (`IndexedPolynomial`) and representation-independent CeTTa backends (GMP exact integers) with proof-relevant ι policies (`ConversionRecipes.Policy`).	Decided; slice in progress	Doc $\langle 0.94, 0.89 \rangle$; Real $\langle 0.67, 0.88 \rangle$ $[0.52, 0.80]$
GSLT and OSLF	Every language is presented as a GSLT (sorts, binders, equations, rewrites); OSLF derives its native types, predicates on terms with $\Diamond$ and $\Box$ from the rewrite relation. Binding signatures are free monads of context-indexed polynomial functors; collections as quotients are proved for single languages, with the general case in progress.	Proved pieces; design	STV$\langle 0.80, 0.88 \rangle$ $[0.67, 0.90]$
Hosting	Three layers with their own crossing contracts: conversion bubbles, different logics, operational regions. A host preserves the counterexamples of its guest and declares what it adds.	Decided	Doc $\langle 1.00, 0.92 \rangle$; Real $\langle 0.82, 0.89 \rangle$ $[0.71, 0.91]$
Kernel shape	A weak core (syntax-directed dependent typing with identity evidence) and bubbles that declare their conversion, each step justified by a global proof. Soundness, widening, substitution and the weakness order are proved; the version for full models of type theory and the decidability of joins of bubbles are open.	Proved core; design recommendation	STV$\langle 0.84, 0.90 \rangle$ $[0.73, 0.93]$
Equality per bubble	Contextual by default: a paused read-only search that resumes only in its own world cannot tell contextually equal programs apart.	Proved for read-only searches; recommendation	STV$\langle 0.80, 0.88 \rangle$ $[0.67, 0.90]$
Propositions	In a set model an impredicative universe is proof-irrelevant, so impredicative propositions are truth values and proof-relevant propositions are predicative types; irrelevance is an opt-in bubble.	Proved obstruction; recommendation	STV$\langle 0.69, 0.87 \rangle$ $[0.53, 0.82]$
The top	The kernel's universe rules do not depend on the set profile; the top supplies universes as sets, at levels above V_ω exactly the V_κ for inaccessible κ. Which set theory is outermost (hypersets are the stated direction) remains open.	Proved layering, conditional on cofinal inaccessibles; top open	Layer $\langle 0.80, 0.88 \rangle$; Top $\langle 0.50, 0.86 \rangle$ $[0.30, 0.70]$

Part	What is known	Status	PLN STV / ITV
Operational contract	Twelve clauses: evaluation positions, binding, quotation, held data positions, shared and independent choice, ignored divergence, answer order, two kinds of effect, step/pause/resume, stale dependencies, `once` versus `select`. Function arguments are call-by-need by default while `let` is eager unless inside a `lazy` demand bubble.	Proved specification; runtime evidenced	Doc $\langle 1.00, 0.92\rangle$; Real $\langle 0.88, 0.89\rangle$ $[0.79, 0.95]$
Execution contracts	Resource authority is reified to machine states; complete occurrence-query certificates compose through adaptive reads and validated resumption; dependent observer families share the existing preservation and erasure boundary. Resource-safe interleaving does not establish observable commutation, and permission weakening does not discharge a duty.	Proved abstract laws and query/control profiles; complete native instances open	STV$\langle 0.82, 0.89\rangle$ $[0.71, 0.91]$
HE types	Each fragment on which HE's checker is one well-defined typing relation (static arrows, schemes, tuples, meta-types, `%Undefined%` as the gradual unknown) is exactly typing in HOL, with three-way kernel agreement across HOL Light, Megalodon and MM0 (`KernelRealizations`). The named binders of the `he-prime` profile are genuinely dependent ($\lambda\Pi$).	Proved fragments & 3-kernel agreement; dependent part open	STV$\langle 0.84, 0.90\rangle$ $[0.73, 0.93]$
Logic translations	The Grothendieck institution of a logic graph; borrowing a prover is sound exactly when the translation is conservative; proofs travel along routes.	Proved	STV$\langle 0.85, 0.89\rangle$ $[0.74, 0.93]$
HOL and sets	Classical HOL satisfies Farmer's seven virtues (`SevenVirtues`: Henkin completeness, Peano and COF categoricity, choice separation); intuitionistic HOL is complete for Heyting substitutional models (`HeytingGeneral`). Relativizing HOTG into its base logic is exact; HOL Light into HOTG and base into HOTG are not conservative.	Proved	STV$\langle 0.91, 0.91\rangle$ $[0.82, 0.97]$

Part	What is known	Status	PLN STV / ITV
Evidence logic	The four-valued reading of evidence (true, false, both, neither) is exactly double negation on the PLN evidence quantale. Counts summarize observations, and each operation is derived from a stated observation contract: additive revision for disjoint ledgers, an interval for overlap and for same-world conjunction, the paired product for pairs of sample bags, PLN's tensor for conditionally independent reports. Negation by evidence is monotone; PLN deduction is exact under screening off; conclusion confidence is derived through credal intervals.	Proved	STV$\langle 0.90, 0.90\rangle$ $[0.80, 0.96]$
Infinite MLNs & DLR shells	DLR Gibbs completions induce additive world models (`MarkovLogicInfiniteWorldModel`). Credal ITV width has two orthogonal sources (`PLNWorldModelPrimeBubbleBridge`): finite-sample/overlap dispersion (`idm_width_pos_of_finite_evidence`) and DLR phase-coexistence leaks (`queryObservation_leaks_iff_strictWidth`). Dobrushin interaction shells act as quantitative bubbles: static extensions preserve core WM truth values exactly (`TranscendencePath`), while distant rewrites decay as $2\|\Gamma\\|C^n$ (`DynamicTranscendenceStep`).	Proved	Doc $\langle 0.95, 0.91\rangle$; Real $\langle 0.88, 0.89\rangle$ $[0.79, 0.95]$
Weighted answers	One annotation semiring for spaces and answers; possibility composes exactly over positive semirings; commuting binds need commuting annotations; the answer readout is an observer of ordered runs.	Proved	STV$\langle 0.85, 0.89\rangle$ $[0.74, 0.93]$
Absence	Presence-only programs are coordination free; absence at a frontier is intuitionistic negation over reachable states, certified by the owner of the pattern's head and revoked only by it.	Proved; native source open	STV$\langle 0.80, 0.88\rangle$ $[0.67, 0.90]$
Choice, probability, strategy	Probability views a program with choice only through a scheduler, with envelopes at deterministic schedulers; consumption needs arbitration; bounded overtaking, realized by deadlines, certifies fair service.	Proved abstract laws	STV$\langle 0.80, 0.88\rangle$ $[0.67, 0.90]$

Part	What is known	Status	PLN STV / ITV
Empty and null	No universal null. Zero answers is not a value; an error is a value; an unknown call stays inert. Empty functions into different codomains are equal as sets and equivalent, not equal, as types.	Proved pieces; recommendation	STV$\langle 0.80, 0.88\rangle$ $[0.67, 0.90]$
Names	"Bubble type theory", glossed as a weak dependent type theory with bubble-local definitional equality ("fibered type theory" also fits). The native logic is `(hol:profile base)`; the HOTG set theory is the set profile `(set:profile megalodon-hotg)`, built over that logic.	Decided	Doc $\langle 1.00, 0.90\rangle$; Real $\langle 0.94, 0.89\rangle$ $[0.84, 0.99]$

The weakest parts are the choice of the outermost set/hyperset theory, the generic dependent eliminator derivation across the authored inductive spine, and the Kripke–Henkin scheduler extension for intuitionistic HOL. The strongest are the identity and η decisions, the meet-closed (Horn) characterization of least sufficient bubbles, Farmer's seven virtues for HOL, and the theorems that make bubbles and hosting precise.

3 The semantic envelope

3.1 The dependent total space is primary

Suppose an operation begins in state s, receives a response r, produces an outcome a, follows an occurrence-bearing route ρ, and incurs cost k. The lossless semantic object is not merely a but a dependent total space such as

$$\mathcal{E}(s) = \sum_{r:R(s)} \sum_{a:A(s,r)} \sum_{\rho:\mathsf{Route}(s,r,a)} \mathsf{Receipt}(s, r, a, \rho) \times K.$$

The response may select the result fibre. The route, receipt, revision, owner, and cost may be irrelevant to one observer and essential to another. Therefore none is definitionally reconstructed from the visible endpoint.

An observer is a map $q : \mathcal{E}(s) \to O$. A dependent family $B : \mathcal{E}(s) \to \mathcal{U}$ descends through q precisely when it is constant, up to the selected equality notion, on every fibre of q:

$$q(x) = q(y) \quad \Longrightarrow \quad B(x) \simeq B(y).$$

This is the master non-collapse law. It governs history erasure, extensional typing, quotation, memoization, cost erasure, and coalgebraic quotienting.

Proposition 1 (Observer licensing)**.** *An extensional readout may forget a coordinate of the dependent total space only when every consumer transported across that readout is invariant on the forgotten fibres. Endpoint equality alone is insufficient.*

The current formal theory contains positive factorization theorems for outcome-only continuations and negative canaries for history-sensitive families, independent costs, and erased routes. These results support an *intensional core plus licensed observers*; they do not yet select a concrete rule package.

3.2 Dependent and simple typing

The simple/extensional face should be a fragment or observation of the dependent face, not its replacement. Andrews-style simple type theory is the default mathematical and automated-reasoning waist when a notion is only a predicate $P : A \to \mathsf{Prop}$ and all conclusions are insensitive to the particular proof of $P(a)$ [4, 5]. Reifying the notion as the subtype $\{a : A \mid P(a)\}$ preserves existential and guarded-universal truth exactly; the same construction works for a predicate on a genuinely indexed carrier by using a dependent sum. This accounts for weak noun-like types in controlled mathematical language without promoting every guard to a new primitive type.

Constant families embed simple types into dependent types. Genuine dependency earns its cost precisely when a response, event, or proof selects the type of a later object, or when a consumer distinguishes inhabitants of the selected evidence fibre. Propositionally truncating a two-inhabitant evidence fibre to mere nonemptiness has no retraction, and a well-formed varying family exists outside the constant-family image. These are checked boundaries, not an appeal to expressiveness by slogan. Extensionality is orthogonal to simplicity: a dependent theory can admit extensional observations, and a simple theory need not identify all operational histories.

This boundary is internalized as a two-point capability order. Simple typing is the least sufficient profile for every literal constant family, while the dependent profile is the least sufficient one for a checked varying Boolean family. This semantic choice is separate from choosing the strongest licensed native implementation: first identify profiles that fully express the consumer's required semantics and prefer weaker sufficient assumptions; then choose a request-appropriate licensed native realization. A greatest capability profile need not exist, and greater capability is not a theorem of better runtime performance.

The current high-confidence intensional envelope contains contextual substitution, Π, Σ, identity with J, and cumulative predicative Tarski universes. The remaining choice is between concrete rule packages, conversion algorithms, universe mechanisms, and scoped equality principles. No generic envelope theorem can make that choice on their behalf. Following Martin-Löf's judgmental discipline, the shared identity capability must include formation, reflexivity, dependent elimination, its reflexivity computation rule, and stability of that computation under substitution [3]. A native tower specimen now validates this complete contract with proof-relevant computation receipts. Neither global uniqueness of identity proofs nor equality reflection follows from the contract; those remain separately licensed profile properties.

3.3 Kernel selection and generation requirements

The following are maintained design requirements, not additional axioms or a claim that a general kernel generator has been constructed. They distinguish the logical commitments of a development, the capabilities available in a native kernel, the trusted implementation, and the cost of using it.

KG1: Semantic coverage.
A request states its required judgments, computation and equality laws, dependent data, observations, and universe bounds independently of the selected implementation. A configuration is sufficient only through the corresponding interpretation and adequacy obligations. Syntax coverage, a capability label, or soundness alone does not establish full coverage. Required distinctions must be reflected or retained by a checked decoding; a quotient is permitted only when licensed by the request. No selection may silently weaken that request. Expressibility does not entail a total decision procedure for theoremhood or program behavior.

KG2: Minimal logical commitments.
Prefer the weakest sufficient object theory and equality/universe profile within the stated comparison class. Report the assumptions actually used by a development separately from those available in its ambient kernel. Where no least profile is established, retain incomparable minimal candidates rather than inventing a total strength ranking. Access to a stronger kernel does not by itself justify stronger object-theory assumptions. A local development can therefore work in ZF and explicitly acquire choice when needed, even if its host also offers stronger set or type theories. Returning to the weaker context requires accounting for the proof and the declarations it used, not merely recognizing its conclusion's vocabulary. Proof-local dependency reduction and weakening an object theory's model class are separate obligations: deleting an authored axiom does not remove the same law if the chosen model interface still requires it.

KG3: Appropriate native capabilities.
Among configurations meeting the semantic and trust requirements, select, host, compose, or specialize native decision, proof, operation, and external certificate services as appropriate. An efficiently implemented derived rule may increase native capability without increasing logical strength; its admissibility or conservative-extension obligation must be proved. Certificate replay is one service boundary, not the definition of NIK.

KG4: Checked kernel generation.
Generate feature-selected kernels and their implementations from admitted presentations and explicit feature-interaction laws. Their operational stages remain GSLTs; an implementation has an independently justified relation to that operational meaning. The generated artifact must be bound to the selected semantics and profile, with separate source-to-artifact and artifact-execution obligations. Generation does not certify itself, and correctness of individual features does not establish correctness of every combination. The chosen trust boundary and bootstrap remain explicit. Generation and specialization may be demand-driven: feature materialization can be deferred until needed. Admission may not be deferred past trusted use, nor may a newly supplied feature silently alter an earlier judgment's meaning. Cached results require the qualified dependencies of KG6. Formation, preservation of conversion classes, preservation of assigned types, and termination of an unfolding strategy are distinct obligations. A proof of one may not stand in for another; total conversion decision needs its own justified scope even for transparent definitions. Local computation admission must support a contextual preservation theorem, including transport of binder assumptions and dependent result types. Checking selected redex/reduct endpoints alone does not establish that law. A declaration/computation restriction that retains a proof and reflects admitted judgments is not thereby an independently qualified evaluator. Its retained rules must also retain the declaration and schema dependencies needed by their reducts, and satisfy the restricted package's preservation and conversion contracts. Proof replay and executing-kernel specialization are different services. Recovering a rule's typed parameters from an observed application is an explicit obligation. A named declaration can supply a principal spine; arbitrary unannotated terms need not have unique types. Constructor-conversion safety need not be proved by confluence of the authored runtime reduction. A proof-only completion is permitted when its conversion theory is proved identical and its joining argument retains the required constructors. Its extra proof steps are not executable rules or a conversion decision procedure without separate justification. Finite conversion-evidence checking is a separate service contract: accepted codes must imply the selected

conversion relation, and any claimed evidence completeness must cover that relation rather than only reflexivity. Termination on a supplied code does not imply a terminating search for one. Rejecting a particular code is not a proof that its endpoints differ.

KG5: End-to-end efficiency.
Compare generation, specialization, execution, checking, translation, memory, and retained evidence costs without trading away KG1. Native rules and compact proof terms can avoid costly expansion into a smaller kernel. Conversely a smaller trusted implementation may be preferred when the workload and trust policy justify it. Neither logical strength, feature count, nor trusted-code size is a substitute for a proved or measured cost comparison; no universally optimal generator is assumed. For the planned CBPV basis, lazy/eager rewrites must preserve the selected termination, effect, sharing, occurrence and result observations before their costs are compared. A strategy change is not an unconditional equation. Keeping code as a value does not by itself memoize its outcome. A rewrite may qualify each actually selected suspended body while allowing effects in its selection and consumer. Eliminating a forwarding cell requires an explicitly coarser observation or a proved resource/receipt correspondence; equality of returned dependent pairs alone does not justify that erasure. Selecting an effectful service provider once and selecting it at each call can preserve acceptance support while changing proof artifacts, correlated choices, occurrences and intents. Function-space interfaces do not license moving that selection across application.

KG6: Revisable qualification.
Bind each selection and generated artifact to a revision of its semantic requirements, feature assumptions, and dependencies. A revision needs justified transport or renewed admission; it cannot silently reinterpret an earlier artifact. Maintain positive and negative controls and distinguish proved, conditional, refuted, and open properties in the status ledger. For a fixed checker, a dependency manifest must include the lookup results actually protected by its contract, including absence, declaration transparency, and dependencies exposed by selected definitions or known propositions. An unchanged input name or type alone is not such a contract. A dependent cached reply must retain the full request that indexes its meaning, not only an equal result tag. A changed proof producer may leave an already checked proof usable at that unchanged request; a changed judgment or request needs independent transport or a new invocation. Rejecting a stale packet must not silently relabel it or refute the new mathematical proposition. A suspended producer's captured environment is distinct from its consumer continuation. Reusing a completed native value must neither rerun producer effects nor restore an earlier mutable-state snapshot. A retryable cache reset does not itself roll back already recorded effects; transactional behavior requires a separate contract.

KG7: Reusable kernel components.
Reuse binding, substitution, declaration lookup, rule instantiation and checking components through explicit contracts. An extension must state whether it preserves old judgments, conversion, proof identity and model meanings, and which interactions require new proofs. Adding a set-theoretic object theory need not add a kernel inference rule; adding dependency changes the formation and substitution contract and is not merely another axiom list. Count the transitive trusted dependencies of a selected service, including its primitive operations, rather than only the size of its outer dispatcher. Reified computation bodies must preserve scoped binding under substitution. Native-value binders and suspension-reference binders must retain their distinct scopes. A replacement that carries variables of several sorts must

be weakened along every binder sort it crosses. Scope and constructor conformance alone cannot rule out cross-sort capture: a captured reference can remain in bounds while naming a different cell. Returned values, computation functions, ordinary thunks and shared Need references must retain their distinct roles. Capturing a thunk stores lexical data, not a consumer world; caching its construction does not cache its executions. A local application or ordinary force must not invent an effect receipt or a sharing allocation to enter its captured code. Heap qualification relates each captured producer and cached answer to an independently declared cell type; lookup success alone is not mathematical admission. Whole-run admission must derive intermediate heap and continuation qualification from initial source typing and the primitive contracts, rather than assume those invariants anew at each observation. A commit retains its cell's declared type, while a consumer's result conversion retains its independent formation and conversion evidence. Fresh allocation must transport the captured environments inside cached functions as well as suspended producers. A returned executable closure remains coupled to its qualified final heap; an answer-only type projection is not a reusable closure environment. Allowed fault shapes must be distinguished from evidence about the event that produced a fault. Operation declaration formation, operational realization and preservation of the declared result family are distinct contracts. Agreement with an operation's own evaluator cannot replace the last contract. Source evaluation must also be stated independently of machine steps. Completed-outcome correspondence must retain the required heap, effects, choice occurrences and failure status, not just successful native values. Finite path existence does not establish result-list multiplicity, cost, termination or fairness; a bounded search with no answer is not a source refutation. Native admission remains conditional on the separately qualified source, environment, heap and primitive result contracts. A function consumer may execute a captured body and change the world. Its source interpretation must therefore retain an intermediate answer and world, not merely repackage a returned result. Final-heap qualification must concern that same world, not another execution with an equal answer. An execution equation must declare where substitution by that equation is licensed. Exact completed evaluation can be retained by an active consumer without being a congruence under quotation, source capture or allocation. Lexical body opening and source-term substitution are distinct operations when returned closures retain their source environments. An immediate-demand rewrite must respect the producer's computation polarity; its type qualification cannot be inferred from one successful run. A semantic value/computation adjunction must identify its induced sequencing with the existing computation operations. This alone does not supply all computation type formers or reified source substitution. Delayed native service data, its independent result-family qualification and any stored admission evidence remain separate; erasure compatibility does not recover proofs from unqualified raw execution. A raw mathematical-service reply must retain the exact request and article without assuming its own checker verdict. Data formation, structural decoding, selected-calculus checking and logical interpretation are separate obligations. A copied acceptance tag has no authority; scope labels require an independently fixed association with the selected calculus. Rechecking an untrusted reply and reusing authenticated evidence are different execution contracts, with different costs. A first-class checker retains that contract when passed through a caller; its function type alone cannot authenticate arbitrary checking code. Sharing a provider selection, sharing a returned reply and caching the provider's individual applications are different transformations. An executable proof materializer must reconstruct the submitted tree, interpret its actual rules and separately qualify the resulting native term; returning a well-formed data marker does none of these. A native extension must qualify the exact

combined rule package, not borrow a conversion boundary from a smaller one. Declaration formation, primitive preservation and conversion safety are separate obligations. A proof-only completion may establish the last while its executable checker continues to recognize only the original rules. Checking a reduction cannot replace independent admission of its source. A richer computation-object interface must preserve the existing free programs and supply its contextual universal properties. A terminal computation algebra is not an effectful program returning a singleton; interpreting in the former does not justify discarding the latter's work.

KG8: Integrated mathematical entry.
CeTTa is the metaprogramming home for mathematical construction, search, checking, and proof use. Its ordinary programs must be able to manipulate and invoke the same qualified mathematical services as an interactive client. Pure theorem development and verification must be able to use native NIK services without first acquiring a programming-language evaluation semantics. The direct ITP interface is a shared component boundary, not a requirement for a separate mathematical product. Proof search and elaboration remain distinct from their admitted judgments. A service's own operational implementation still requires its GSLT presentation and realization; this does not turn the mathematical object theory into the client's programming language.

KG9: Homoiconic integration with qualified elaboration.
Higher-order, dependent-type and set-theoretic constructions should remain ordinary inspectable program data, with scoped vocabulary such as `lang:`, `type:`, and `set:` rather than an undifferentiated top-level namespace. These names organize access; they do not identify three foundations or erase the distinction between a framework and an object theory. Elaboration may infer parameters, insert admitted translations, and obtain reusable checking evidence automatically. The elaborated terms, selected theory and revision, inserted maps, remaining obligations and trust dependencies must remain inspectable. Automatic handling cannot promote unresolved work to proof, discard required evidence, choose an ambiguous meaning silently, or identify distinct conversion laws. Each insertion requires the appropriate substitution, meaning-preservation and required-reflection laws; a common surface notation is not such a proof. A local abstraction scope also names its observation and equality discipline. Changing scopes may insert a checked transport or forget declared distinctions; it must not turn observational agreement or temporal lineage into intensional identity. Ordinary reasoning within a scope should not require manually repeating these qualifications at every use.

Current evidence covers parts of this contract: the two-profile `SimpleDependentCapabilitySelection` proves least sufficiency for constant and genuinely varying family specimens, not for arbitrary calculi; `CapabilitySelection` selects within a finite licensed family for a fixed semantic request; `SemanticService.Service` keeps all four native-service faces; and `MachineCompilationCorrectness` separates compilation provenance from execution refinement on its checker-facing boundary. These results do not yet synthesize arbitrary feature combinations, decide semantic sufficiency in general, or provide a universally weakest or fastest kernel. The equality specialization specimen supplies a further instance of KG3: a native generic definition and reflexivity proof are reused at every closed plain type, with source-level formation and checking proved independently. Its comparison with intrinsic extensional equality exposes an explicit model condition rather than importing set axioms or dependency. This is checked specialization of one native feature, not a feature-combining kernel generator. Checked definition expansion supplies another local instance:

partial constant-to-term substitution preserves successful results when its available replacement table is extended. Native unfolding agrees exactly with its typed interpretation at each fuel bound, and additional fuel preserves a successful result. Type preservation, semantic definition equations, and termination are separate obligations. These facts support demand-driven specialization; they do not establish incremental kernel generation or an end-to-end speedup. Exact environment-local reuse is now proved for the existing Lean implementation of native Megalodon checking, separately from its partial Henkin interpretation. A finite manifest containing the input dependencies, closed under selected definition bodies and known propositions, and preserving the selected lookup results gives exactly the same proof-checking verdict at every unchanged fuel bound. Executable manifest checks discharge these premises. This is a sufficient frame criterion, not a minimal-dependency theorem or logical conservativity of arbitrary added axioms. It applies to revisions of the environment of this fixed checker; changing its inference or conversion rules requires a separate extension proof.

The modular theory $\mathcal{U}$ of Blanqui et al. provides a particularly close precedent for KG2 and KG7 [93]. Its fragment theorem recovers a derivation in a subtheory when the context, proof and conclusion belong to that fragment. The fragment must be closed under declaration-type dependencies and all ambient rewrite rules whose left sides it contains; confluence and type preservation are substantive conditions. This is not the claim that every theorem stated in the smaller language has a proof there. Bauer and Petković Komel supply a complementary reusable equality algorithm for user-defined type theories, separating computation and extensionality rules [95]. Their soundness result does not supply termination or completeness for every feature selection. Together these works motivate dependency-sensitive proof reuse and separately qualified conversion components, not automatic compatibility of arbitrary kernel features.

The candidate `PrimeCandidates.EntryModeSelection.nativeProofOperation` supplies a claim/native-proof service interface independent of a programming-language run. Its comparison with replay uses two interfaces to the same guest kernel, not independent physical implementations. A direct interactive client remains to be implemented. Likewise, coGSLT-authored judgment and inference extensions provide additional mathematical input: `ConservativeExtension.extension_not_determined_by_core` refutes recovering all extension declarations from the core alone. A generator must consume such input and its laws; it cannot infer an arbitrary proof calculus from an operational language or a list of desired properties alone.

Small trace interfaces illustrate the same distinction. The MM2 execution description delegates matching and emission to sources and sinks, whose available operations include state changes and foreign computation [96, 97]. Moving execution into a sink moves its semantic and implementation obligations rather than eliminating them. The local MM21 sink-scheduling proposal is still an open specification question, not a proved two-rule verifier. The existing NIK finite-trace construction requires a sound local edge authority, including any premise discharger for queries, substitution, binders or capabilities. Finite edge checking does not establish termination, fairness or validity of infinite executions.

3.4 Quantitative observers without admission by similarity

The distinction calculus provides a quantitative companion to exact observation contracts [98, 99]. Its similarity kernel is not a proof kernel. The checked rational fragment in `Cybernetics.DistinctionCalculus` starts with a symmetric, reflexive matrix $\alpha : X^2 \to \mathbb{Q} \cap [0,1]$, without assuming transitivity. Its complementary distance is $d_\alpha = 1 - \alpha$. The paper's metric condition is a triangle law; separation is not assumed, so this is a pseudometric. Positive distance implies distinct states, but distinct states need not have positive distance under a lossy observer. Triangle coherence supplies cotransitivity of positive apartness and a setoid of zero-distance states. A goal descends

to the resulting quotient exactly when it is invariant on its zero-distance classes; the construction reuses `Cybernetics.MultiscaleGoal`.

Raw certificates derive upper distance bounds by authored edges, reflexivity, the unit bound, symmetry, capped triangle composition and weakening. The total checker agrees with a separately defined derivation relation. Its semantic theorem ranges over every metric similarity extending the seed, a nonempty model class. In particular,

$$\alpha \vdash_{\text{completion}} d(x,z) \le \min\{1, d_\alpha(x,y) + d_\alpha(y,z)\}$$

does not assert this bound in an incoherent seed. A finite proposed completion is admitted only after checking its observer laws, metric coherence, extension of the seed, and a path certificate for each proposed distance. `completionCheck_sound` proves that an accepted candidate is the *least* metric extension, not merely a feasible matrix. No particular shortest-path generator is trusted by this result. The graded control derives similarity 3/5 from adjacent similarities 4/5; an endpoint-collapse certificate is impossible. In the crisp three-node chain, completion really does identify previously distinguished endpoints, reducing uniform distinction from 2/9 to zero.

For a rational probability weight μ, the checked bracket is $g_\alpha = \sum_{x,y} \mu_x \mu_y \alpha(x,y)$, with distinction $h_\alpha = 1 - g_\alpha$. Migration distortion compares the declared source and transported target similarities. Zero distortion implies equality of those similarities on supported pairs; the global equivalence theorem requires full support. Route composition obeys a triangle bound under the same source pair law, and a balance theorem separates state change from observer change. These scores do not replace exact consumer compatibility.

`PolarizedNeedDistinctionObserver` instantiates this distinction on the executed Lean reference workload of a shared versus freshly selected inference provider. Every completed branch retains the same two native replies, but event frontiers distinguish the programs. Erasing those events has distortion 1/2 on the explicitly uniform two-program population, not a probability measure on runtime branches. A second result proves that actual article-admission verdicts cannot factor through the requested theorem alone: one article is accepted and another for the same request is rejected. Neither result authorizes replacing Need sharing by unshared evaluation or establishes a C implementation correspondence.

This supplies a precise application of observer-relative multiscale goals [100]. It does not identify biological agency with a quotient construction. In particular, observation coverage is not beneficence: a finite control gives equal distortion but opposite signed changes of a separately declared goal value. Structure-preserving repair likewise retains its value and misfit obligations; similarity alone cannot discharge them. Futamura specialization and Turchin supercompilation [101, 105] suggest producers for specialized checkers and transformations. Their semantic correspondence, termination, preservation of sharing and resource benefit remain separate requirements, not consequences of low distortion.

3.5 Evidence: observation contracts, readouts and PLN

A pair of counts (n^+, n^-) summarizes observations: how many samples supported a proposition and how many refuted it. Which operations on such pairs are sound depends on what was observed and how the observations relate. The checked development states the relation as an *observation contract* and derives each operation from it; no combination formula is postulated. Probability is read from evidence; the evidence is what a world model carries.

Identified observations. When every observation has an identity, a body of evidence is a ledger, a finite set of identities, each recording one world. Ledgers form a distributive lattice whose join

is union: merging two replicas is idempotent, commutative and associative, so ledgers replicate without coordination. The evidence of a ledger is a valuation on that lattice: the evidence of a union plus the evidence of the overlap is the sum of the two evidences (`EvidenceContracts.ledgerEvidence_union_add_inter`). Consequently additive revision is exact exactly for disjoint ledgers and otherwise counts the overlap twice (`additive_revision_exact_iff_disjoint`, `revision_counts_overlap_twice`); the disjointness check that non-axiomatic reasoning performs on evidential bases before revising [137] is this condition. Retraction of a ledger subtracts the evidence of its overlap by the monus of counts (`ledgerEvidence_sdiff`), and the abstract overlap world model of the tree is instantiated with the overlap read off the intersection (`ledgerOverlapModel`). When the overlap is unknown, the merged evidence lies between the coordinatewise maximum and the sum of the two evidences, and both ends occur for the same two evidences (`unknown_overlap_bounds`, `Controls.overlap_interval_ends`). The maximum is therefore the revision that is sound whatever the overlap, the counting analogue of covariance intersection [147]. The source-keyed revision of the layer atlas, which keeps each source's cumulative evidence and merges by maximum, computes the evidence of the merged ledger exactly when each source's observations on the two replicas are nested, as under cumulative publication, and under-counts otherwise (`ledgerEvidence_union_eq_sourceMerge`, `Controls.sourceMerge_undercounts`).

Pairs of samples and the weights of answers. Pairing every sample of one bag with every sample of another forces the paired product $(a^+b^+,\ a^+b^- + a^-b^+ + a^-b^-)$ (`EvidenceSpaces.conjunction_unique`); in the coordinates (positive, total) of non-axiomatic logic it is coordinatewise multiplication (`EvidenceSpaces.posHom`, `EvidenceSpaces.totalHom`), and its strength is the product of the strengths. With it the counts form a commutative semiring, so evidence can weight answers. Which laws of an annotation semiring a readout needs is now exact. Possibility composes for all answers if and only if the semiring is *positive*: sums vanish only when both summands do and products only when a factor does (`AnnotationCapabilities.support_bind_iff_positive`); this is equivalent to a zero-reflecting homomorphism into the two-element semiring (`positive_iff_exists_hom`), as in the semiring provenance of first-order logic [134]. Evidence counts with the paired product are positive, the coordinatewise product on pairs is not, and no nontrivial ring is (`positive_evidence`, `not_positive_pair`, `not_positive_of_ring`): signed weights retract anything exactly but lose compositional possibility, while the monus retracts only a known batch and keeps it. Two independent binds commute for all answers only if the annotations commute (`comm_of_bind_comm`).

The total of a paired product counts pairs, not independent samples of the conjunction. For independent bags of sizes n and m with success probabilities p and q, the variance of the number of positive pairs is

$$nm\,pq(1-pq)\ +\ nm\,pq\big((m-1)q(1-p)+(n-1)p(1-q)\big),$$

derived from the sampling model by finite sums (`PairedSampleDispersion.paired_variance`). The first term is the variance for nm independent joint samples; the excess vanishes for interior probabilities exactly when $n = m = 1$ (`paired_variance_eq_joint_iff`). At $p = q = 1/2$ the effective number of independent joint samples is $3n^2/(2n+1)$, below $3n/2$, not n^2 (`Controls.effective_size_half`). A confidence readout for a conjunction must therefore be derived from the factors' counts and strengths, not from the paired total.

Observations of one world. Two predicates evaluated on the same bag of worlds have marginal counts that do not determine their conjunction (`EvidenceSpaces.no_same_world_conjunction`

`_from_marginals`). The compatible conjunction counts are exactly the Fréchet interval $\max(0, a+b-n) \le c \le \min(a, b)$ [146]: every value in it is realized by a bag and none outside it (`EvidenceContracts.conjunction_counts_iff`). The product of the two frequencies lies inside (`independence_within_frechet`). For dependent observations the sound readout of a conjunction is this interval; independence is an assumption that selects one point.

Reports about one hypothesis. A report can instead be read as the pair of masses of a hypothesis and of its negation jointly with the report. For two reports conditionally independent given the hypothesis, under a uniform prior, the evidence of both reports is the coordinatewise product of the single-report evidences, up to the prior constant, and its strength is the posterior probability (`testimony_fuse_uniform`, `strength_fuse_uniform`). This coordinatewise product is PLN's binary-evidence tensor (`toPaired_tensor`). Its strength is the logarithmic opinion pool [145], or Dempster's rule on Bayesian mass functions over a two-element frame [140] (`strength_fuse`); its zero divisor is total conflict between two certain reports (`fuse_total_conflict`); and fusing two posteriors that each contain a non-uniform prior counts the prior twice (`testimony_fuse_counts_prior_twice`, `Controls.prior_counted_twice`). Likelihood ratios of independent tests multiply in the same way as betting scores do in game-theoretic statistics, where the product of independent e-values is an e-value [144, 143]. The tensor is thus a sound fusion operator for its own contract. It is not conjunction and not the direct path of PLN deduction: on two half-strength premises it has strength $1/2$, while the direct-path term $s_{AB}s_{BC}$ is $1/4$, the strength of the paired product (`Controls.fuse_not_direct_path`).

Readouts. Strength is undetermined with no samples (`EvidenceSpaces.readout_zero`), never strength zero. The interval of Walley's imprecise Dirichlet model [142] is $[0, 1]$ at zero counts (`idm_vacuous`); its width is one minus the PLN confidence $n/(n+s)$ (`idm_width`), and with samples it contains the observed frequency (`idm_contains_frequency`). Revision shrinks the width but moves the interval, so the intervals of successive evidence states are not nested (`idm_not_nested`). The vacuous interval coincides with the probabilities that schedulers assign to a pure nondeterministic choice between two outcomes (Section 4.7). Strength is not additive under revision (it is the mediant, `EvidenceSpaces.strength_add`), so the weights of alternatives must be carried as counts and read out once.

Negation by evidence, not by absence. Evidence against a proposition is recorded, not inferred from a missing record. Exchanging the two counts is an involution that is monotone in the information order and distributes over revision (`against_mono`, `against_add`); the truth-order meet and join of the evidence bilattice are monotone in the information order and exchanged by it (`tmeet_mono`, `tjoin_mono`, `against_tmeet`); and the four-valued reading as neither, true, false or both [148] is monotone (`belnap_mono`). This is the constructible falsity of Nelson [149], which the basic logic of evidence of Carnielli and Rodrigues reads as preservation of evidence [150]: every connective only grows with evidence, so a program built from them stays presence-only and coordination free. The intuitionistic negation $a \Rightarrow \bot$ of a nontrivial Heyting algebra is not monotone (`hnot_not_monotone`), in particular on PLN's evidence frame (`binaryEvidence_hnot_not_monotone`). Negation by absence therefore needs a frontier (Section 3.14); negation by evidence does not.

What PLN's rules become. Revision is derived for disjoint ledgers and refuted for overlapping ones. The strength of a conjunction is derived for independent sample bags and is only an interval

for observations of one world. Inversion is Bayes' rule in counts (`EvidenceSpaces.condition_pos_symm`). Deduction is exact under screening off: if, in the counts of a bag of worlds, C is independent of A given B and given $\neg B$, the frequency of C among the A-worlds is $s_{AB}s_{BC} + (1 - s_{AB})(p_C - p_B s_{BC})/(1 - p_B)$ (`EvidenceContracts.deduction_exact_of_screening`); without screening off the formula gives $1/4$ where the frequency is 0 (`Controls.deduction_fails_without_screening`).

Conclusion confidence through intervals. A conclusion's confidence is derived through credal intervals, in the sense of the confidence characterization of the tree: a truth value $\langle s, c\rangle$ is a readout of a credal envelope. The route has three steps. Premise evidence (n^+, n^-) becomes the imprecise Dirichlet interval with a declared prior strength s, an envelope theorem of the tree (`WalleyBinaryIDM.credal_envelope_matches_Walley_ITV_bounds`); its evidence weight reads back $n^+ + n^-$ (`EvidenceContracts.idm_intervalWeight`). The rule then maps premise intervals to a conclusion interval by the map its observation contract fixes: the product interval for independent pairs (`product_interval`), the Fréchet interval for conjunction on one world (`frechet_interval`), the corner interval of the bilinear deduction map under screening off (`deduction_interval`, `bilinear_between_corners`), and the vacuous interval $[0, 1]$ without screening off, where two bags agree on $s_{AB} = s_{BC} = 1/2$ and give C among the A-worlds the frequencies 0 and 1 (`Controls.unscreened_vacuous`). Finally a declared selector gives the strength and the width-complement chart the confidence, which is the PLN odds chart $w/(w + s)$ for the interval's evidence weight w (`widthConfidence_eq_odds`); the chart is forced by the Walley gate (`walley_width_complement_forces_plnOdds`). `deduction_confidence_route` assembles the steps for deduction.

Choice	Status	Fixed or justified by
Prior strength s (κ)	genuine degree of freedom	declared per context and source
Credal model of a premise	genuine backend choice	the imprecise Dirichlet model is the one whose width complement is the PLN chart
Range and normalization	convention	the declared range $[0, 1]$
Confidence chart	free until gated	forced to $w/(w + s)$ by the Walley gate
Propagation map	fixed	the observation contract of the rule
Strength selector	genuine decision knob	lower, midpoint, upper or point estimate; it does not affect confidence
Rule-level transport	computed in interval or weight space	raw-confidence transport double-damps: inputs $(0.5, 0.5)$ give 0.5 under minimum-support transport, not 0.333 (`truthInduction_conf_eq_min_capped`)

Against this route, the deduction evidence constructed in the tree, whose total is the sum of the premise totals (`EvidenceContracts.deductionEvidence_total`), is neither an upper nor a lower bound. On premises $(1, 1)$ and $(1, 1)$ with $s = 1$ it posits 4 (`Controls.deductionEvidence_posits_four`); the route derives the interval $[1/9, 4/9]$ and weight 2 at $p_B = p_C = 0$, and the interval $[4/9, 5/9]$ and weight 8 at $p_B = p_C = 1/2$; on premises $(0, 2)$ and $(1, 2)$ at $p_B = p_C = 0$ both give 5 (`Controls.deduction_posited_overstates`, `Controls.deduction_posited_understates`, `Controls.deduction_posited_agrees`). Likewise the weight-product conjunction of the tree, $w_{a\wedge b} = w_a w_b$, gives confidence $4/5$ on premises $(1, 1)$ and $(1, 1)$, where the route derives $2/3$ for independent conjunction and $1/3$ for conjunction on one world (`Controls.conjunction_through_intervals`, `Controls.frechet_through_intervals`); the weight product is the paired total, which the variance identity above shows overstates.

Neighbouring formalisms. The same counts underlie several formalisms that read them differently.

Formalism	Its reading of (n^+, n^-)	Relation to the contracts above
Non-axiomatic logic [137]	frequency w^+/w, confidence $w/(w+k)$	revision adds evidence after a disjointness check of evidential bases: the ledger contract
Subjective logic [138]	opinions in bijection with Beta evidence	cumulative fusion, for separate and independent evidence, adds the evidence parameters, and averaging fusion, for the same evidence, is idempotent [139]: the ledger contract at its two ends; the product of opinions coarsens the product frame and approximates the exact product [138]
Dempster–Shafer [141]	mass on hypotheses and on the frame	Dempster's rule on Bayesian masses is the likelihood contract, and total conflict is its zero divisor
Imprecise probability [142]	credal sets	the imprecise Dirichlet interval, vacuous at zero counts
Game-theoretic probability [143]	betting capital	independent betting scores multiply: the likelihood contract
Semiring provenance [133]	annotations	counts are a commutative positive semiring; exact overlap needs identities

Is evidential logic more fundamental than PLN? In a precise sense, yes. Its objects are bodies of observations with an information order in which both polarities only grow; its operations are derived from an observation contract (identities, one world, pairs, reports); its readouts are strength, confidence and credal intervals, each derived and each undetermined at no evidence. PLN's truth values are readouts of these objects, and its rules are derivations from them under stated assumptions: disjointness for revision, independence for conjunction, screening off for deduction, a uniform prior for fusion. What PLN adds is an inference discipline over a knowledge base: a set of rules applied to cached readouts of links, higher-order and quantified extensions, and the heuristic assumptions that let it chain without carrying the joint model. Those assumptions are where its results can fail, and each now has a checked counterexample.

3.6 Infinite relational world models, Dobrushin shells, and coherent self-transcendence

When a knowledge base ranges over a countably infinite open domain of atoms and clauses (`ClassicalInfiniteGroundMLNSpec`), its probabilistic semantics is governed by the Dobrushin–Lanford–Ruelle (DLR) specification of infinite-volume Gibbs measures (`MarkovLogicInfiniteExistence`, `MarkovLogicInfiniteUniqueness`). Every DLR probability measure μ on `InfiniteWorld` Atom induces a mass semantics with unit total mass whose singleton world-model query strength coincides with the Gibbs cylinder probability (`MarkovLogicInfiniteWorldModel.infiniteMLN_queryStrength_eq_gibbsProb`). When the uniform Dobrushin interdependence bound holds (`PaperUniformSmallTotalInfluence`, row-sum budget $C < 1$), the DLR completion is unique and the world-model query strength is determined by the specification alone

(infiniteMLN_queryStrength_unique_of_uniform), including across heterogeneous multi-agent trust networks (MarkovLogicInfiniteWorldOfViews.worldOfViews_wmBridge_unique).

Theorem 2 (Orthogonal sources of credal interval width; proved)**.** *In the world-model bridge (*PLNWorldModelPrimeBubbleBridge.pillar3_orthogonal_credal_width_sources*), credal interval width* $\mathrm{ITV}\langle[L,U],b\rangle$ *arises from two orthogonal mechanisms that must not be conflated:*

1. ***Evidential and overlap width*** *(*idm_width_pos_of_finite_evidence*,* unknown_overlap_bounds*,* conjunction_counts_iff*): For any finite nonnegative observation counts* (n^+,n^-) *and Walley IDM prior concentration* $s > 0$*, the credal interval has strictly positive width* $s/(n^+ + n^- + s) > 0$*. This finite-sample width remains strictly positive even when the underlying infinite relational specification lies in the Dobrushin uniqueness regime where the asymptotic DLR completion envelope collapses to a point (*zeroWeightGrid_originSpinUp_queryEnvelope_collapse*).*

2. ***Structural DLR phase-coexistence width*** *(*MarkovLogicInfiniteCredalBridge*,* GibbsCompletion.queryObservation_leaks_iff_strictWidth*): The family of DLR completions* DLRCompletion M *forms a projective credal set. When Dobrushin contraction fails (as on the geometrically reinforced half-line or low-temperature symmetric Ising grid), multiple Gibbs equilibria coexist and the asymptotic query envelope itself has strictly positive width (*reinforcedLineGeometric_originSpinUp_queryEnvelopeWidth_pos*). In the generative world-model semantics (*WorldModel.Generative*), strict DLR credal width is equivalent to a residual observation leak (*ObservationLeaks (completionSemantics M) () (queryObservation q) $\leftrightarrow$ dlrQueryHasStrictWidth M q*), whereas Dobrushin contraction implies observation stability (*queryObservation_stable_of_uniform*).*

Quantitative bubbles and open-ended self-transcendence. In a crisp world model, syntactic non-occurrence cannot isolate a local query from distant hidden axioms (CandidateDependentWorldModel.syntactic_nonoccurrence_counterexample). Combining BubbleWeakness, PLNWorldModelSupportForgetting, and the DLR interaction-shell hierarchy yields a three-tier account of local isolation and open-ended cognitive growth (PLNWorldModelPrimeBubbleBridge.pillar1_bubble_ledger_and_overlap_contract, pillar2_lawful_bubble_compaction_contract, pillar4_quantitative_wm_bubble_and_transcendence):

- **Tier 0 (Crisp conversion bubbles,** $C = 0$**):** A conversion bubble whose declared steps hold across a class of intended strict models leaves the propositional model class unchanged while strictly narrowing the strict model class (justified_bubble_propositionally_invisible), and induces identical ledgerEvidence on every identified ledger of intended models (justified_bubble_ledgerEvidence_eq). When compacting or forgetting a bubble scope, basic revision/extraction soundness leaves forgetting unconstrained (WMCalculusForgetBoundary.forgetting_differs), whereas a SupportTrackedForgettingLayer guarantees exact inverse retraction for deltas supported within the forgotten footprint and zero outside-scope leakage.

- **Tier 1 (Static interaction-closed WM bubbles,** $\varepsilon = 0$**):** Under Dobrushin budgets, expanding an individuated subsystem by adding new atoms and clauses outside an interaction-closed core Γ_{old} preserves BinaryWorldModel.queryStrength on Γ_{old} on the nose (MarkovLogicSelfTranscendence.TranscendenceStep.old_queries_preserved), and across any coherent multi-step growth path (TranscendencePath.original_queries_stable).

- **Tier 2 (Dynamic shell-protected WM bubbles, $\varepsilon = 2|\Gamma|C^n$):** When an agent rewrites distant clauses while preserving an n-step interaction shell `iterExpandRegion` M_1 Γ n around a core query region Γ, the change in world-model query probability on Γ decays geometrically as $2|\Gamma|C^n$ (`MarkovLogicDynamicTranscendence.DynamicTranscendenceStep.queryProb_approximately_preserved`). Along a coupled multi-agent development path, joint coordination score can strictly improve while cumulative drift on the original identity cores stays bounded by the shell-tail budget (`MarkovLogicSyntheticCognitiveDevelopment.SyntheticCognitiveDevelopmentPath.coordination_improves_while_joint_identity_drift_bounded`).

3.7 Draft-critical judgments and workload discriminators

The experimental target is higher-order reasoning and induction, mathematical metaprogramming, checked theorem use, and proofs about a specified language's behavior. Requirements KG1–KG9 constrain the admissible designs; they do not yet determine every formation rule, conversion algorithm or universe profile. A remaining choice should be resolved by a required consumer and a distinguishing proof or experiment, not by selecting a familiar kernel and calling its existing feature set optimal.

Workload	**Required meaning**	**Discriminator**
Higher-order chaining	Relations and functions are arguments; composition retains the requested intermediate witnesses.	Ordinary relational composition can be expressed in simple types. Keeping indexed proof objects requires native families or an evidence-preserving encoding.
Learned hypotheses	Sample agreement, structural admissibility, and independent rule soundness are different claims.	A checked extension receives meaning from each rule's interpretation, not from training fit or merely passing a structural checker.
Dependent guarded use	The admitted result depends on the value actually matched, including its bounds and selected evidence.	Matching a vector does not establish a valid index; guard support cannot reconstruct two distinct receipts.
Mathematical metaprograms	Quoted syntax, open candidates, admitted proofs and effectful execution coexist as inspectable data.	A quoted candidate can be manipulated without becoming a proof. Promotion must expose the formation, conversion and dependency obligations it discharged.
Proofs about execution	A theorem is interpreted in independently specified transition semantics with satisfied hypotheses.	Keeping proof syntax but altering the model interpretation can invalidate the proposed use.

Composition retains more than successful support. For a first evidence family $J(x, m)$ and a subsequent premise $K(x, m, e, y)$, the composite retains

$$(J\,;K)(x, y) = \sum_{m} \sum_{e:J(x,m)} K(x, m, e, y).$$

The checked `JudgmentComposition` module presents this by a GSLT with query, pending and answer states. Inversion of every complete finite route proves an equivalence with this total evidence space (`Sequential.evidenceRunEquiv`). Both directions keep the exact selected witnesses; a pending state is not an accepted answer. The two abstract service-invocation phases are not a

claim of two native instructions, a prescribed number of logical judgments, or a generic decision algorithm. The `bindExactEquivalence` law permits an exact replacement of the first service when the second premise is transported along its evidence map. A nonidentity Boolean-to-optional receipt example verifies that this is not merely identity replacement. This law has fixed query and answer types; cross-language changes of endpoints require their additional translation contracts. It is a condition for automatic service composition, not a claim that every available service supplies such an equivalence.

`MatchedIndexJudgment` instantiates the construction with the actual pattern matcher and a terminating vector-index checker. A match receipt retains its position in the matcher's answer list. The later evidence retains the recovered vector and an index in Fin(length vector). Every returned answer has that evidence, and every such complete run contributes an answer. The general `evaluate_count` theorem also counts repeated successful binding occurrences; a two-position bag control produces two equal answers. For first-order patterns, the existing binding-decision compiler gives the matching-language-run connection at its result-support boundary. Neither this connection nor the bounded-index example is a proof of arbitrary dependent typing or the physical C implementation.

Comprehension and guards have an exact evidence boundary. For any evidence family $E : X \to \mathrm{Type}$, projection

$$\sum_{x:X} E(x) \longrightarrow \{x : X \mid \mathrm{Nonempty}(E(x))\}$$

is surjective, and is injective exactly when every fibre is subsingleton (`GuardEvidenceComprehension.supportProjection_injective_iff`). It commutes with reindexing; reflection of injectivity additionally requires surjective coverage of the original context. Proposition-valued refinements give an equivalence, whereas two distinct receipts over one value give a checked failure of every witness-preserving reconstruction. This concerns forgetting evidence to predicate support, not an impossibility of representing dependent families by set-indexed total spaces. The actual engine guard connection additionally requires bound arguments, no builtin rows, exact echo-query behavior and an independently correct support recognizer. Arbitrary relation queries can introduce bindings and need not be filters.

Higher-order induction already supplies a reuse target. `NativeRelationLifting.ListComposition.chainEquiv` distributes proof-relevant relational composition through a list, preserving the whole intermediate list and both proof spines with inverse laws. The existing `MILCheckedChain` checks a real chain proof and rejects a changed middle occurrence. Its semantic-extension proof validates each authored rule against independent reachability semantics. The proof-relevant admission consumer retains the intermediate witness and premise derivations. These are reusable semantic and checking components, not an implemented general higher-order learner. The mother/father/chain calculus now has an executable exact-tree materializer, while this does not give materializers for arbitrary newly learned rule meanings. Finite-provider results do not supply finite enumeration of arbitrary infinitary relations. A zero empirical error asserts sample agreement only; the existing finite-observation indistinguishability theorem refutes a universally correct observation-only classifier for opposite unseen behavior.

The new `FormationSensitiveMIL` result admits every program of the existing primitive/chain syntax in the refined cumulative judgment. It proves actual closed formation of the hypothesis family and its two constructor types, then lifts the four independently checked quotation-leaf

obligations by structural induction over the existing program. No raw typing proof is silently promoted. The denotation and intermediate evidence remain unchanged: the concrete successor-chain program has a receipt from 0 to 2 and no receipt from 0 to 3; an undeclared primitive cannot be admitted. This port needs no normalization or object-level conversion premise. Constructor quotation does not use the dependent eliminator. The separate `FormationSensitiveMILElimination` now proves formation of its actual closed declaration, motive and branch types, and refined application at its parameters. `FormationSensitiveMILEliminationIota` gives formed canonical telescopes, independently typed primitive/chain redexes and reducts, and the corresponding typed substitution instances with the original Type-valued computation receipts. A dependent consumer identifies all endpoint codes while retaining two hypothesis receipts: both orders are admitted, their displayed motive types are syntactically different, and a crossed reduct has no receipt for the original redex. This is not inequality modulo conversion or confluence/normalization of beta plus iota. The source-derived inversion construction below now extends the typed schemas to arbitrary admitted primitive/chain redexes, under the package's explicit Pi-conversion boundary. Generic higher-order lifting and functional-map elimination still need their own formation and computation qualifications.

`FormationSensitiveNativeList` separately supplies the native list-relator connection. Six closed declaration-formation proofs cover the actual list, list constructors, relator and its two evidence constructors. Induction over the existing finite constructor spines admits their original quoted list and relation evidence, given independently checked leaf terms. The semantic result retains the polynomial evidence spine, and raw evidence equality is equivalent to equality of that semantic evidence at the fixed canonical endpoints. Two receipts with the same endpoints remain distinct; a missing tail receipt cannot be supplied by endpoint agreement. Constructor admission is distinct from the elimination and conversion qualifications below; it does not admit arbitrary raw terms.

`FormationSensitiveNativeListElimination` now independently forms the actual native List eliminator, its motive and both branches. The nil/cons endpoints are admitted in formed telescopes, and typed substitutions retain the original informative iota evidence and actual root steps. `FormationSensitiveNativeListEliminationPreservation` recovers arguments through the declaration telescope and preserves arbitrary displayed source types under the Pi-conversion boundary for the unchanged List/identity/mapRel package, now discharged below. A selected nil-tail instance has both endpoint judgments; redirecting only its recursive call fails the actual root relation, without claiming the altered term is untypable or nonconvertible.

`FormationSensitiveNativeIdentity` independently forms the actual identity eliminator and its endpoint-and-path-dependent motive. The relator elimination modules likewise form its nine-argument declaration and both canonical reductions. Declaration-spine recovery obtains both list indices, head and tail evidence, and the original motive from arbitrary admitted applications. Typed substitution constructs the selected reduct, then replays the source's displayed-type adjustments. This retains evidence dependency; agreement of the two lists alone cannot choose a different tail witness.

Proof use reaches an actual operational specimen. `TransitionInvariant.preservation_of_refinement` is an intrinsic HOL derivation: if relation R refines S and S preserves P, then R preserves P. Its finite-run theorem separates this proof from hypothesis satisfaction, admissible state interpretation and symbol meanings, allowing non-full Henkin models. Equation-interleaved runs additionally require equation invariance. `MainlineCallGuardHOLInvariant` supplies a concrete model of the existing compiler transition system; the independent step-denotation theorem establishes refinement. Using the HOL proof then establishes reference result preservation for

every finite run and exact results at a compiled halt. Changing only P to "the machine is currently running" refutes both hypothesis satisfaction and validity on a genuine running-to-halted step. This establishes intrinsic proof use, not a general external-source importer or correctness of the complete CeTTa runtime. The paired `HenkinTransitionInvariantProof` supplies an actual native proof of the same rule, with both open-sequent and closed theorem acceptance and independent proposition formation. Intrinsic type substitution and partial erasure identify its assumptions and conclusion exactly with the call-guard consumer's formulas. Removing the refinement premise rejects the native proof.

`HenkinProofSoundness` now gives the missing source-checker law for the supported monomorphic proof constructors: hypotheses, named facts, implication introduction and elimination, and term universal introduction and elimination. Successful inference yields a typed formula with the exact returned native syntax, valid in each admissible valuation satisfying the typed hypotheses. Named facts and definition equations have independently supplied model meanings. The proof retains every actual normalization site, including definition unfolding and beta/eta reduction; it does not require normalization to be a no-op or the Henkin function domains to be full. Prefix-polymorphic proof constructors and the remaining native proof constructors are outside this theorem. Its environment-wide definition and named-fact conditions are sufficient, not claimed to be minimal dependencies.

`HenkinTransitionInvariantImport` applies that law to the actual accepted native proof and the independent compiler model. The resulting finite-run theorem obtains the reference result at every reached compiled halt without using the separately authored intrinsic preservation derivation. Thus the native-proof service now has a general soundness law used by this consumer, not just a matching statement pair. The checked source is the Lean Mathdata checker; a parser, foreign binary, or equivalence with another checker implementation has not thereby been verified.

The normalization controls use the real declaration $q := id\ p$. Accepted proofs of the submitted $q \to p$ actually normalize it to $p \to p$. A model with q true and p false refutes the submitted formula and the definition equation, although the native proof still checks. Insufficient fuel separately rejects a formula valid in a satisfying model. Raw lookup can also accept an unscoped hypothesis that cannot satisfy the exact typed hypothesis interface. Model qualification, resource failure and formation failure are therefore not interchangeable verdicts.

The next choice is a formation and conversion contract. Three credible implementation organizations remain to be discriminated: a native formation-sensitive dependent value calculus with a shared simple fragment and set-theoretic library; a simple logical host with explicit set-family encodings; and multiple admitted calculi behind a common homoiconic elaboration interface. The second owes witness and computation recovery whenever native indexed evidence is requested; Boolean guards alone fail that test. The third owes coherence and visible qualification of inserted cross-calculus maps. The first most directly presents native dependent consumers. A candidate refinement now has regularity and substitution proofs as described below; its required declaration and conversion capabilities still need qualification. Existing tests exclude the permissive raw tower as an unrestricted trusted proof judgment; they do not prove a repaired calculus normalizing, mandate one universal conversion, or select an axiom package. Megalodon remains a tested source of reusable components, not an optimality theorem.

The available `RegularJudgment.subject_accessible` and `regularCheckBool_correct` theorems are useful formation and checking precedents, but concern the sealed regular two-sort fragment. Its `u0` is a ground type and `u1` an untyped formation marker, not two cumulative universes. Its syntax has neither type-level quantification nor the indexed declarations and dependent

eliminator of `IntrinsicMILHypothesis`; even full identity elimination is not part of that syntax. The existing cumulative hypothesis family does express those indexed consumers, but its typing results cannot inherit the regular fragment's normalization theorem without a proved connection. The draft-critical problem is thus to carry the formation and conversion discipline to the required universe/declaration fragment, not to rename the sealed kernel or re-prove its existing normalization result.

Formation is preserved; conversion remains a separate obligation. The `FormationSensitiveTyping` candidate uses the existing cumulative syntax but retains formation evidence at constant use, lambda introduction, pair introduction and conversion. Its erasure produces an actual raw typing derivation. Renaming, simultaneous substitution and binder instantiation preserve the refined derivation, including its intermediate formation premises. `FormationSensitive.Typing.regularity` proves that every result type is formed when the ambient telescope is formed and the universe rules satisfy their separate regularity interface. The existing cumulative universe rules satisfy that interface. No conversion injectivity, normalization or untyped top-sort exception is used in this proof.

A positive control admits the actual polymorphic term $\lambda A.\lambda x.x : \Pi(A : U_\ell).A \to A$ and checks its specialization and two beta steps. Another accepts an opaque constant with a formed declared type, while a negative control rejects a constant whose declared type names an absent declaration, even after conversion or cumulative adjustment. These tests concern the refined judgment itself, not only the formation of its final query.

Formation does not certify arbitrary equations. The `FormationSensitiveExamples.ConversionCollapse` control adds $G \equiv G \to G$. Both endpoints inhabit the same universe, and the added root rule is stable under renaming and substitution. Nevertheless the full refined judgment admits $\Omega = (\lambda x.xx)(\lambda x.xx) : G$, with a beta self-step and a proof that its reduction relation is not accessible at Ω. Thus formed endpoints and structural stability are insufficient for a normalization claim. The unsafe rule package is only a counterexample, not a Prime extension. The candidate likewise does not select Prime's final calculus or qualify arbitrary declared computation. Its beta relation describes proof computation; it does not select the runtime strategies of the planned CBPV foundation or replace the existing need protocol.

`FormationSensitive.Typing.betaPi` separately proves root beta preservation at the displayed result type. Its premises are formed context, universe regularity, and the explicit `PiConversionBoundary`: conversion retains Pi components and cannot identify a Pi with a head. Generation recovers the lambda's original formed domain, and substitution constructs the contractum; the original conversion/cumulativity tail is then replayed with its formation proofs. The theorem does not erase to raw typing. Dependent identity specialization has both a conditional instance of this general law and an independent direct derivation.

`PureConversion` discharges this boundary for the actual `Tower.rules` package. Complete development of the existing algebraic parallel relation proves Church–Rosser for the full cumulative syntax with no declared root equations, retaining semantic universe-head equality and opaque constants. It yields exact Pi-component conversion and Pi/head separation, so `FormationSensitive.Typing.betaPi_tower` requires neither an assumed conversion boundary nor an assumed universe regularity instance. A beta/universe-head peak joins in the actual reduction, and an outer Pi cannot become the opaque ground head even when its components are not normal. The generic parallel-substitution theorem also preserves repeated schema-variable coherence. Declaration-specific delta/iota extensions, including the separately treated native relator package below, require their own qualification. Church–Rosser does not imply strong normalization or a

total conversion decision procedure.

`DefinitionalExpansion` now supplies one such qualification for nonempty transparent definition packages. Closed constant expansion commutes with renaming, substitution and binder opening. Two primitive obligations suffice: each constant converts to its selected body in the source, and each source root equation becomes pure conversion after expansion. Induction then proves, for arbitrary open terms,

$$s =_R t \quad \Longleftrightarrow \quad \mathsf{expand}(s) =_{R_0} \mathsf{expand}(t),$$

where R_0 retains the same heads and head equality but has no declared root equations. This yields Pi-component injectivity and Pi/head separation for R, hence refined root beta preservation without assuming a Pi boundary about the extended package. The signature constructor checks transparent body agreement and additional declared equations separately. `FormationSensitiveDelta` proves the complementary type-preservation law: a closed body checked at its constant's actual declared type can replace that constant after any original cumulative/conversion tail, retaining the displayed type and all formation premises.

The actual signature specimen contains $A := G$, $B := A$, an opaque $v : B$, and a checked polymorphic identity definition. Specializing that identity at B and applying it to v follows one delta and two beta steps; every intermediate has refined typing at the displayed ground type. An alias of G cannot become a Pi type, and the earlier Pi/head collapse refutes every expansion qualification of this form. A different negative control is deliberately cyclic: $c : G := c$ passes conversion qualification and refined typing but has a delta self-loop and is not strongly normalizing. Thus these expansion laws alone certify neither a terminating unfolding strategy nor contextual type preservation. They do not provide an automatic acyclic-signature elaborator or a total dependent conversion decision procedure.

Contextual subject reduction from primitive obligations. `FormationSensitiveSubjectReduction` now proves preservation of the actual displayed type through every existing contextual step and finite run. Its premises concern only primitive head typing/equality, primitive declared root computation, universe regularity, and Pi/Sigma conversion boundaries. The theorem does not assume preservation for contextual steps. Induction over the refined derivation handles computation in binder domains, codomains, lambda bodies, dependent arguments, pair components, projections and identity endpoints. A checked identity substitution transports the body when its binder domain changes; conversion of instantiated families restores dependent result types. In particular, reducing a pair's first component also transports the type required of its second component.

`FormationSensitivePreservationInstances` discharges these obligations for the actual root-empty Tower. It also constructs the full preservation law for transparent signatures whose actual bodies are checked at their combined declaration types and whose expansion is qualified. Sigma component injectivity and Sigma/head separation follow from pure Church–Rosser and transfer through the same expansion interface used for Pi. `PreservationExamples.actual_runs_preserve` instantiates all premises for the nonempty alias/identity signature above. Its consumers reduce the type component of a dependent pair, retain the literal computational type of its second projection, and reduce a universe-valued binder domain with a genuinely dependent body. Their resulting judgments come from the general preservation theorem, not independently checked endpoints.

A separate negative signature declares $c : G := \lambda x.x$. The declaration's type is formed, and its expansion preserves conversion classes, but its actual delta reduct cannot have type G. Thus body checking is necessary in addition to conversion qualification. These results settle contextual preservation for the pure and qualified transparent packages, not for arbitrary additional iota rules.

Each required eliminator needs its own primitive preservation and conversion qualification, and termination remains a separate obligation.

Declared spines recover the dependent rule arguments. `FormationSensitiveDeclarationSpine` derives a principal type from an actual declaration lookup and its checked applications. Any observed typing of the same spine is reached by a directed conversion/cumulativity adjustment. Pi injectivity and Pi/head separation recover an application's argument at the declaration-derived domain, and the original formed result and adjustment tail can be replayed on a replacement term. `FormationSensitiveTelescopeSpine` lifts this construction to every finite declaration telescope: closing the existing context with Pi and applying its substitution in oldest-to-newest order recovers the complete formation-sensitive context morphism. No type-uniqueness hypothesis for raw terms is assumed.

The qualified alias/identity signature supplies a premise-free conversion instance for this recovery theorem. An application displayed at ground recovers the argument's original named type alias; applying the identity specialized at ground to a lambda has no typing at any displayed type. Conversely, the same raw identity lambda has both $G \to G$ and $(G \to G) \to (G \to G)$ as formed types, and these types are not convertible. The restriction to declaration-derived spines is therefore substantive.

`FormationSensitiveMILArgumentRecovery` uses this generic theorem for the actual hypothesis eliminator and both constructors. It recovers the motive and branch substitution, endpoint indices, primitive symbol or chain middle index, and both constituent programs from an arbitrary refined source derivation. `FormationSensitiveMILIotaPreservation` then instantiates the original independently checked rule reduct and replays the source's displayed-type adjustments. Every actual declared root preserves typing under Pi qualification; no quoted-input or primitive-preservation assumption remains. The full contextual finite-run corollary additionally uses Sigma qualification.

The native conversion boundaries are now discharged. `MILConversionCompletion` gives a proof-only presentation in which the four repeated metadata fields may differ syntactically when the authored conversion relation already relates them. Each completed contraction first converts the inner metadata to the outer representatives and then takes one actual native iota step. Conversely every authored root is included. Thus the completed and authored presentations have exactly the same conversion theory; their runtime-step relations are not identified.

`MILConversionParallel` retains those coherence derivations under parallel development, renaming and simultaneous substitution. Rigid partial eliminator and constructor spines recover the development of their actual arguments. Primitive and chain branches have cofinal developments; the chain keeps both hypotheses and both recursive eliminator calls. `MILConversionParallel.cofinal` then proves, by structural induction on every open raw term, a common parallel successor for all its parallel reducts. The resulting diamond joins every authored conversion in the auxiliary relation. Pi and Sigma retain their outer constructors and component paths there, yielding the actual native Pi/Sigma conversion boundaries without an assumed authored Church–Rosser property.

`FormationSensitiveMILQualification.steps_preserve` combines these proved boundaries with recovered declaration arguments and independently typed reducts. Arbitrary finite contextual runs of the unchanged native package preserve the original refined judgment, with no remaining conversion or primitive-preservation premise. A control reduces a native elimination inside an identity witness while retaining the identity's original endpoints in the displayed type. Another rejects incompatible Pi-versus-head metadata even in the completion; convertible beta metadata is

accepted there but is not an exact authored root. No raw execution confluence, normalization or total conversion decision theorem follows from this construction. Nor is the proof-only completion a runtime change.

`NativeRelatorConversionCompletion` separately qualifies the unchanged five-root List/identity/mapRel package. Its guards retain the authored conversion of every duplicated type, relation and list index. Renaming and simultaneous substitution preserve those guards; cofinal development handles all five branches and every open raw term. The proved `NativeRelatorConversionParallel.diamond` discharges this package's Pi/Sigma boundaries. `FormationSensitiveNativeRelatorQualification` then combines the independent declaration and reduct proofs with those boundaries: every admitted source retains its original displayed type through every finite contextual run. No conversion-boundary or primitive-preservation premise remains. This qualifies neither a union with MIL nor arbitrary future declarations, and does not assert original execution confluence or termination.

`FormationSensitiveSimpleFragment` connects the existing intrinsic atom/arrow calculus to this refined judgment. Its unchanged type, telescope, term and substitution translations now have formation-sensitive derivations. Substituting before translation or applying refined substitution afterwards gives exactly the same raw term, with a formed judgment along either route. The higher-order term $\lambda f.\lambda x.f(fx)$, specialized at an open function-valued lambda, checks capture avoidance beneath the new binder. Existing conversion preservation and reflection on translated simple terms therefore now have refined endpoints. This is not full HOL consequence or inhabitation reflection. In particular, two distinct intrinsic terms at the same displayed simple type still have one raw image because the existing translation forgets internal application-domain annotations. Refined target typing does not recover those annotations. Exact syntactic round trips must retain that information or explicitly restrict the requested observation.

`FormationSensitiveHOLInterface` separately represents the actual intrinsic HOL variables, constants, application, abstraction, implication, universal quantification and equality in the same presentation grammar. Independently formed closed symbol interpretations imply formation-sensitive admission; renaming and simultaneous substitution commute exactly with representation. The proposition carrier and logical operators are supplied by a signature: universal quantification is an operator applied to a lambda, not identified with dependent-product formation. Equality is an independently typed proposition-former, not native identity. Other HOL constructors, including conjunction and existential quantification, remain outside this partial interface. `FormationSensitiveHOLInvariant` supplies seven actual opaque declarations and independently forms the type-indexed universal and equality operators. It represents the existing call-guard conclusion exactly, including its two binder indices. Function-valued specialization beneath a quantifier and quantification over predicates give higher-order consumers. Henkin substitution and the call-guard invariant's validity remain independently proved source semantics; the altered-predicate model refutes validity without changing formation. These results are neither a chosen Prime proposition universe nor a translation of HOL derivations into dependent inhabitants.

Restriction follows derivations, not surface vocabulary. `FormationSensitiveDependencies` proves that every formed judgment has finite atomic support, including its entire telescope, recursive declaration formation and intermediate conversion derivations. Any rule package satisfying those primitive requirements reconstructs the judgment on the same syntax. `FormationSensitiveRestriction` constructs such a package: only requested constant declarations and the renaming/substitution closure of requested root equations are retained. The original judgment survives, and every judgment of the restriction reflects to the source. An actual unused polymorphic

declaration and its computation are removed in the example. Conversely, omitting a hidden type alias preserves the visible constant lookup and a raw proof but destroys every refined typing of that constant. This is declaration/computation restriction within the same dependent grammar: universe and head-equality policy are unchanged, and neither a smallest executable kernel nor STT inhabitation reflection has been established.

`FormationSensitiveRestrictionPreservation` makes the execution boundary concrete. A valid restriction of the already qualified transparent signature retains a type alias and its unfolding equation but drops the declaration of the alias produced by that equation. The original constant has a refined judgment, every restricted judgment reflects to the source, and conversion still has a qualified expansion. Nevertheless the retained unfolding produces a constant with no possible refined type. A formed conversion proof can even use this equation beneath an erasing lambda and return to its original type without requiring the intermediate declaration. Restoring the produced declaration repairs this example: primitive body typing and the restricted conversion boundaries yield preservation for arbitrary finite contextual runs, with their original displayed types. The repair still removes other declarations and even the restored alias's own unused unfolding. This is a checked failure and repair of execution qualification, not a failure of the earlier proof-restriction theorem or a general algorithm for finding the smallest executable package.

`PreambleDependencyRestriction` applies the native checker's dependency law to the existing powerset proof. Only `PowerI`, `In` and `Power` remain; the actual verdict agrees at every fuel. Removing or changing the used axiom rejects the proof. Native soundness gives its theorem in an independent interpretation satisfying the retained axiom, including the ZFC interpretation. A Boolean-carrier Henkin countermodel satisfies powerset introduction but refutes powerset elimination. The omitted axiom's own native proof also fails in the restricted environment. Thus preservation of this proof is not conservativity of the larger theory. The older set-operation model interface already requires the full powerset specification, so this strict weakening must be witnessed over the independent signature, not obtained by deleting names within that fixed model class.

Conversion evidence has an executable native boundary. `StructuralConversionCode` checks finite codes for the original structural rules, including capture-avoiding beta substitution, projections, semantic universe-head comparison and every congruence position. Symmetry and transitivity extend the codes to conversion. Transitivity checks the literal intermediate term: convertible but different representatives require an explicit bridge code. `MILRootConversionCode` and `NativeIndexedRootConversionCode` supply finite, proof-free root data and exact decoders for the existing MIL and List/identity packages, respectively. No proof-only completion rule participates in decoding. For each package and all raw open endpoints, conversion holds exactly when some finite code is accepted. The checker terminates on supplied evidence; the existence theorem does not find that evidence or decide nonconversion. `NativeRelatorRootConversionCode` also supplies an exact decoder for all five roots of the existing List/identity/mapRel package, retaining both witnesses and the recursive result. `NativeRelatorConversionChecking` extends the same finite-code laws to this unchanged package; none of these results qualifies a union with MIL.

`NativeConversionChecking` connects these results to admission. Its generic `Judgment.convertChecked` retains source admission and independent target formation. Its forward-step interface uses actual MIL contextual preservation, not symmetry. A dependent identity specimen checks both endpoint changes while separately forming the resulting identity type. Controls reject variable capture and a malformed path join; an explicit bridge repairs that join. Reflexive conversion of an undeclared constant checks, but no refined typing of that constant exists. Distinct retained codes can have identical endpoints, and the native Pi/head boundary rules out every purported

conversion code across that boundary. The relator qualification also transfers every accepted original forward step from an independently admitted source, including beneath an identity witness while its displayed endpoints remain unchanged. An actual accepted root code with an undeclared method has no possible refined source type. Thus correct reduction evidence is not itself mathematical admission.

`StructuralConversionCodeDependencies` extracts a finite list of root requests, retaining their binder scopes and repeated occurrences. Agreement on the complete decoding result of those requests preserves the exact step/conversion verdict, including rejection, with the head policy and structural algorithms fixed. The agreement test is executable. Removing the unused chain branch preserves the primitive identity proof's positive and negative verdicts, while changing a used primitive result fails the manifest check and invalidates the evidence. The removed chain branch is independently observable on its own code. This is sufficient proof-local support, not a minimal kernel or a replacement for formation and new-profile qualification.

The implementation gate should therefore settle the exact supported binder formation and conversion rules; the effect boundary for search and execution; the proof/declaration dependency closure of simple restriction; the required identity eliminations and universe bounds; and the elaboration law for each implicit lift. Only those commitments needed by the selected workload enter the initial profile. Full library import, universal kernel generation and unneeded stronger type formers are later work; soundness of any claimed initial feature is not. The surface qualification API and executable conformance workloads can be fixed before exhaustive lowering, but the trusted type checker cannot be specified by the raw permissive tower's name. This gate does not require strong normalization for every future language profile or completeness of proof search. The existing structural inference checker already separates finite derivation verification from finding a derivation. A supported conversion presentation can likewise validate finite positive evidence without claiming to decide every conversion question. The local rule semantics, formation and evidence-checking soundness still must be proved, and all required distinctions and expressive operations must remain represented. If a service instead promises a total conversion decision, its termination and completeness on that advertised domain become part of the gate. Replacing an unproved total decision claim by an explicitly partial service is legitimate only when the workload permits residual work; it cannot silently narrow the requested semantics.

Extension contract for experimental implementations. The `InferenceRuleSupport` theorem gives a concrete proof-local reuse law for the generic inference checker. Its finite manifest includes the rule identifiers of every child as well as the root. Exact agreement on their complete lookup results preserves both successful and unsuccessful verdicts, and transfers accepted derivations with the identical raw proof. A checked example removes an unused rule; another keeps the root rule but removes its premise rule, so the same proof fails. This is not a smallest kernel theorem or a proof that a simple conclusion needs only simple rules. The instantiation and side-condition algorithms are unchanged, the two definitions remain independently validated, and mathematical meanings remain separate. Declaration dependencies consulted by other native services require those services' own manifests.

At the shared client boundary a request identifies the judgment and theory revision, scoped input, declaration dependencies, admitted effects and required observation. A response distinguishes a successful result, an invalid candidate, and residual work. A proof of a negated proposition is itself a successful mathematical result, not the same thing as failure to check a candidate. Search may supply candidates, but cannot revise the selected logical rules merely by finding one.

The four NIK service faces remain available. A direct decision service need not materialize a

foreign certificate, and proof-irrelevant checking evidence may be erased where its contract permits. Proof-relevant output data and requested occurrence distinctions may not be erased on that basis. For each call-guard-style optimization an implementation must expose its source judgment, target run observation, primitive handlers, input admission, preservation and no-invention theorem, and dependency-qualified reuse law. Runtime ordering must not silently select a different mathematical meaning. The first end-to-end experiments should exercise higher-order composition with duplicate proofs, a guarded dependent result, transformed quoted syntax, and a theorem applied to an operational invariant; relevant and irrelevant environment changes supply paired reuse tests. These experiments discriminate interfaces and performance. They do not replace the general soundness and reflection obligations of the admitted profile.

The higher-order raw-checking experiment now has a concrete component path: `PolarizedNeedInferenceFunction.checkFunction` is passed through `PolarizedNeedFunctionPassing.passAndApply`; its actual run invokes the existing checker and exact-request qualifier. `PolarizedNeedHigherOrderInference.observed_materializes` then connects observed admission to `MILNativeMaterialization.materialize?`, a separate executable Lean function producing the submitted article's native hypothesis term. The latter is not yet a machine primitive or a C implementation. A future realization must preserve this connection and its independent quotation formation, not replace the native term by an acceptance marker. The current untrusted-data path may recheck an article; no checker-cost improvement or authenticated-cache optimization is claimed.

3.8 Intermediate languages, open interfaces and local reasoning

An intermediate language is not useful solely because many languages can encode into its syntax. Its value is the structure that translations expose and preserve. Ahrens characterizes sorted binding syntax and specified reduction by initiality, obtaining substitution-aware, reduction-respecting translations from the corresponding iteration principle [26]. This is a relevant foundation for language-definition generation, not a theorem that every encoding reflects every source distinction.

For separately compiled services, two compositions must be checked: consecutive translations through representations, and linking components that may interact. CompCertO explicitly treats language interfaces and calling conventions [27]. Direct open-module refinements further separate client-facing source/target semantics from intermediate compiler stages [28]. The three-dimensional refinement algebra adds composition across components of system state to module and abstraction-level composition [29]. These are precise models and proved constructions with their own scope conditions, not an already instantiated category of all Prime services.

This literature suggests an implementation contract: every service boundary names its input/output meanings and effects, each compilation pass has a refinement or exactness theorem appropriate to its observation, and separately optimized services must still link soundly. Linkage is not justified merely by transitivity of closed-run covers. Patterson and Ahmed's linking types make otherwise foreign behaviors explicit in the source-level reasoning interface [30]; semantic interoperability additionally qualifies the target glue that realizes conversions [31]. Automatic elaboration can hide routine plumbing while retaining those obligations as inspectable data.

Prime can consequently be a shared programming and intermediate-language workbench without requiring every object theory to become one logic. The proposed common simple-type component needs compatible scoped signatures, substitution and conversion routes into both the dependent calculus and the HOTG host. Conservativity of a dependent extension and the effect of adding HOTG axioms are separate questions: a shared term language does not prove that the combined

axiom theory conserves every old theorem. Likewise, interpreting dependent families in a set model does not reconstruct their intensional proof syntax. GSLT presentations describe operational interfaces; theory morphisms and model interpretations carry the mathematical meanings that must accompany them.

For the first experimental draft, the obligation is to qualify the required boundaries and demonstrate their composition, not to implement a universal compiler or a higher-dimensional framework before using the language. The planned CBPV basis must connect its explicit strategy choices to the current revision-keyed need layer. Shared syntax, proof conversion and a CBPV translation alone do not establish that operational correspondence.

3.9 Intensional, extensional and operational presentations

The three useful vertices are intensional mathematics, extensional mathematics, and a language/operation presentation in GSLT-IL. These are semantic roles, not three necessarily disjoint logics or three mandatory implementations. HOL has inspectable proof syntax; a dependent calculus can have extensional models. A single native dependent framework can expose a simple logical fragment, a dependent fragment and a set-theoretic theory through different interfaces, provided their interpretations and admitted extensions agree on the promised overlap. In particular, a proposed `hol:` interface need not imply a second proof kernel, and the `lang:` role need not be a fourth mathematical foundation.

The categorical requirement is comparison respecting contexts and substitution. In `ComputationalTrinity`, contextual presentations are presheaves and comparison maps are natural transformations forming a commuting triangle. Invertibility is a stronger property: its explicit information-loss control rules out a compatible inverse when a view forgets a distinction. This abstract framework is not yet an instantiation of the entire native DTT/HOTG/GSLT-IL triangle. The concrete family interpretation below fills part of an edge, rather than assuming the whole triangle in a record of required laws.

A simple fragment is a bridge, not automatic HOL adequacy. The formation-sensitive native embedding already preserves actual simple terms and substitutions and characterizes conversion on its image. Separately, the constant-family embedding is fully faithful in the selected set-family contextual models. Neither result says that every native inhabitant of a represented simple type comes from a source HOL proof. The existing native HOL interface represents formulas and their substitution; HOL equality is a declared proposition-former there, not native identity. Realizing a HOL proof consequently requires a proposition/proof family and the chosen logical rules, including the appropriate extensional principles.

There is a concrete interoperability hazard: HOL type variables range over nonempty types, whereas a dependent host can express empty types. A translation cannot silently extend that range to all host types. Assaf and Cauderlier address this mismatch with inhabited-type representations in their combined HOL/Coq development [103]. Quantifier domains, propositional truth versus proof inhabitation, universe levels, choice and the added set axioms must likewise be stated at the selected interface. These are focused mathematical obligations, not a demand to decide all future language conventions.

What must precede an accelerated realization. Three levels of evidence have different dependencies:

1. Mathematical coherence: actual interpretations of contexts, terms, proofs and substitutions, with the required equality and universe laws.

2. Operational adequacy: the GSLT-IL presentation and its logical observations implement the specified mathematical tasks on the stated input domain.

3. Realization and acceleration: the selected native implementation preserves those operations, and a comparison on the same task establishes any claimed performance benefit.

The first level does not depend on generated NIK code. The second needs an operational interpretation, but this may initially be a mathematical reference semantics. Executable NIK qualification belongs to the third. The useful reading of NIK as the language side hosting mathematics is therefore a meaning-preserving realization of operations, not an independent source of their truth. A fast implementation cannot establish the first two levels by its speed or its regression tests.

OpenTheory is a plausible bootstrap route: Assaf and Burel translated its standard library into a dependent logical framework while preserving binding, typing and reduction [102]. Their result is a precedent, not an adequacy theorem for this native tower. Conservativity also has its own theorems and hypotheses, as the functional-pure-type-system embeddings studied by Assaf illustrate [104]; it does not follow from merely sharing arrow syntax. HOTG supplies another coexistence and theorem-transfer precedent [58], not permission to identify all the extensional and intensional judgments globally.

3.10 Biform theories and GSLTs as the semantic transformation waist

Farmer's biform theory separates a formal language and axioms from executable transformers, and attaches a meaning formula to each transformer [7, 8]. His little-theories discipline additionally organizes mathematics as small theories connected by inclusion and interpretation rather than one undifferentiated foundation [9]. The useful correspondence here is not to introduce another intermediate representation. A semantic stage is a GSLT, an implementation is a host machine equipped with a faithful encoding, and a transformer is represented by retained operational evidence. Its *meaning formula* is a sentence of the native mathematical theory relating the denotations of its inputs and output; it must not be identified with an operational simulation condition. The formal biform seam therefore pairs a closed theory in a π-institution with a proof-relevant GSLT, assigns a native sentence to every retained primitive event, and requires each such sentence to be a theorem of the closed theory. A biform route carries a logical theory morphism, a proof-relevant operational cover, and a naturality equation saying that translation of the source meaning sentence is the meaning of the translated event. Hence event meanings transport compositionally while remaining independently justified by the native consequence relation.

This contract has a nontrivial Henkin simple-type-theory discriminator. A Boolean machine retains one event, while its source meaning states equality of two typed constants. A non-injective signature map merges those constants; the event and its meaning transport to the reflexive target equation, which is valid in the target's empty theory. If the source axiom is removed, the machine still executes, but a separating Henkin model proves that the same proposed meaning assignment is not sound. Thus executable behavior cannot certify its own mathematical interpretation.

A closed theory in a π-institution also induces a one-fibre NIK theory family without selecting a checker: claims are native sentences, and scope and meaning are the independently supplied closed theorem set. Any proof calculus with the exact theorem-image property supplies the corresponding native evidence discipline, while a little-theory morphism supplies a heterogeneous NIK semantic route. The carrier of signatures in this one-fibre object is the singleton selected signature, rather than the entire ambient signature category. This restriction is categorically necessary: identity and composition of little-theory morphisms then map to identity and composition of NIK routes on the whole object, not merely on its distinguished fibre. The construction is consequently a genuine

functor from closed theories to semantic NIK theory objects. Naturality of proof projection proves that mapped proof objects establish the translated sentence.

This bridge deliberately stops before Boolean replay: a checker and an exact certificate boundary remain additional data. For a biform theory, every retained event has an inhabited native proof fibre for its meaning sentence. Attaching primary NIK authority requires a claim-indexed equivalence between the checker's accepted-certificate fibre and that native proof fibre; ordinary scope authority is derived only after this stronger boundary is supplied. A paired discriminator equips the same valid Henkin equation first with the canonical thin proof calculus and then with a Boolean-tagged calculus. The first fixed proof fibre is subsingleton; the second contains two provably distinct witnesses with the same projected theorem. No checker whose raw certificate type is the singleton can be fibrewise equivalent to the latter. Hence extensional theoremhood and certificate existence do not determine intensional proof identity.

Proof authority therefore qualifies the biform graph rather than being reconstructed from it. A certified biform theory pairs a biform theory with an exact executable certificate boundary for a selected native proof calculus. Its arrows carry a biform route and a certificate map, constrained by a replay square that uses the sentence translation of that same biform route. These objects and arrows form a category with functors both to the biform category and to the heterogeneous NIK authority category. The image of an arrow under the joint projection is exactly the replay-compatible locus. A Boolean-tagged discriminator supplies two qualified endomorphisms with the same biform identity: one retains the proof tag and one flips it. Both replay the same native conclusion, so the biform projection identifies them, while the authority projection separates them. Thus native certificate transport is additional checked data even after operational meaning and theoremhood are fixed.

The proof-relevant translations themselves form a category without first quotienting their executable choices. Equality of arrows includes the term map, forward evidence map, and local evidence-lifting algorithm; identity and composition preserve all three. Biform theories consequently form a category with separate functors to the little-theory category, the proof-relevant GSLT category, and—by composing the first functor with the bridge above—the semantic NIK category. These projections have deliberately different faithfulness. In a checked discriminator, one endomorphism retains each of two Boolean occurrence tags and another flips them. Their native logical map and every event meaning coincide, while the operational projection proves the arrows distinct. Thus strict categorical composition does not require collapsing algorithm identity, and any later observational quotient must name the consumer whose observations license that collapse.

Retained primitive events themselves form a covariant functor from the proof-relevant GSLT category to types. The corresponding biform event functor is its composite with the operational projection. Consequently the observer-relative constructions below depend only on this event action and on their declared value map; they do not depend on the logical projection except when a biform meaning theorem is also requested.

The joint logical–operational projection is sharper than either forgetful view. On every hom-set, its image is exactly the locus of pairs consisting of a little-theory map and a proof-relevant operational map for which the event meaning square commutes. The pair determines the biform route uniquely, but an arbitrary pair need not be a route: a checked tag-sensitive example has well-typed logical and operational components whose meaning square fails. Conversely, a second Henkin discriminator keeps the operational map fixed while changing the translation of typed constants, so the operational projection is not faithful either. Thus the biform category is not a product of independent IR components; it is the compatibility locus inside that product. This gives the categorical statement of why a useful intermediate language retains both an executable transformation and its mathematical meaning without identifying them.

Observer-relative control factors through the proof-relevant event action at an even weaker layer. A map of events pulls target occurrence indices, observers, client contracts, and scheduler readouts back to source histories, while concrete changes and their complete removal ledgers map forward. These actions respect route composition. If the target occurrence index is exact, its pullback is exact precisely when the event map is injective. A checked Henkin biform route exhibits the boundary: it preserves the native theorem assigned to every event and preserves a length observer and an accounted reordering, while merging two Boolean occurrence witnesses into one target event. Exact occurrence identity therefore fails to pull back. Mathematical meaning, client observation, and proof-occurrence identity are three distinct contracts, even when they travel along the same route.

Separately, an exact transformer between semantic stages may carry the following two operational directions:

$$s \longrightarrow_S s' \implies f(s) \longrightarrow_T f(s'), \qquad f(s) \longrightarrow_T t \implies \exists s'.\, s \longrightarrow_S s' \wedge f(s') =_E t.$$

The second clause is intentionally modulo the target equations. Together the clauses define an equation-class cover; such covers compose and induce exact transport of the generated modal observations. When proof occurrences are public, a stronger arrow carries an equivalence of the complete evidence fibre at each pair of endpoints. More general compiler transformations may instead map one source step to a finite target path; fusion, lowering, and exact covered translation are deliberately different grades rather than one overloaded notion of morphism.

An equation-class cover is not yet a literal covered map of representatives. Quotient completion sends every represented GSLT to its term classes and every equation-class cover to a covered translation whose lifted endpoint is now literally equal. Consequently any functor from a pass index into represented languages induces the existing covered GSLT diagram after quotient completion. Formula transport requires the additional observation and label maps, with their satisfaction and action naturality laws; it does not follow from a bare term translation.

This gives parser compilation the same shape as symbolic mathematics. The scannerless source judgment is one proof-relevant GSLT, the compiled ParserPack judgment is another, and compiler correctness is an exact GSLT arrow between them. Tables, charts, forests, and host-language plans implement the target step relation but are not semantic pipeline nodes. OSLF and the native type theory are generated from the two GSLT endpoints. Successful compilation is sealed as one package containing only the target proof-relevant GSLT and the exact arrow from the source; the concrete plan is hidden behind that package. A real RFC 8259 instance preserves and reflects the entire root-derivation fibre, including production occurrences, spans, zero-width steps, and administrative end-of-input evidence; a malformed target profile cannot acquire an exact arrow.

The same criterion disciplines a Felix- or ForTheL-like mathematical surface [12]: parsing, notion elaboration, logical elaboration, optimization, and runtime realization are composable GSLT arrows. Their host data structures remain private realizations. A weak notion normally lands in the simple higher-order predicate fragment; an evidence-indexed notion lands in the dependent fragment. Quotation and evaluation are admitted only through a declared syntax boundary and a meaning theorem, rather than by silently identifying syntax with denotation. The explicit parse-tree data advocated by that work is retained as a term and evidence object of the parsing GSLT; it does not become an unverified semantic intermediate language.

3.11 A theory graph rather than one maximal kernel

The preceding operational waist is only one face of the desired system. A GSLT records terms, an equational congruence, and dynamics; it does not by itself provide the models, satisfaction

relation, dependent judgments, or foundational strength of every logic that may use those dynamics. Conversely, a mathematical theory with no useful operational rules need not be forced into a vacuous GSLT merely to enter the library.

Institution theory supplies the outer, logic-independent interface [13]. It separates a category of signatures, sentence translation, contravariant model translation, and satisfaction, with the satisfaction condition requiring truth to be invariant under change of notation. The little-theories graph can therefore contain classical simple type theories, dependent type theories, set theories, modal logics, and proof systems without identifying their native judgments. The formal development now contains this model-valued structure directly. Every institution induces its semantic-consequence π-institution, and every model-valued comorphism projects to a consequence-preserving comorphism; the earlier consequence graph is therefore a forgetful view rather than a competing formalism. This projection is a functor between the corresponding heterogeneous categories: it preserves identity and composition, but is not faithful because distinct native model actions can induce the same consequence route. Predicate models generated from an arbitrary semantic carrier functor give a reusable test family: sentence translation follows the carrier naturally, models reduce by inverse image, and the satisfaction square commutes. A direct-image falsifier on Booleans proves that the contravariant model direction is semantic content rather than bookkeeping.

The model-valued institutions are also bundled with their native signature categories into one heterogeneous category whose arrows are institution comorphisms. This supplies composition across genuinely different logical languages without first translating them into a common syntax. The constant-family simple-to-dependent comorphism is an actual arrow of this category and is provably noninvertible: a two-sided inverse would put the varying Boolean family in the essential image of the constant-family functor. Thus the theory graph supports fluid movement between simple and dependent bubbles while retaining a checked reason that the latter cannot be globally collapsed to the former.

Conservative transport admits an isomorphism-safe refinement. Literal model coverage is sufficient for reflection of semantic consequence, but is stronger than the mathematical requirement when model presentations differ. Coverage up to model isomorphism composes and also reflects consequence provided the source satisfaction relation is invariant under model isomorphism. The latter condition is not derivable from the bare institution laws, which govern change of signature rather than change of model representative. A checked negative institution contains two isomorphic model objects that disagree on its unique sentence: a comorphism covers both up to isomorphism, but consequence reflection fails. Ordinary predicate institutions satisfy the invariance premise. Hence theory-graph routes may use representation-independent coverage without silently assuming that every native notion of observation is extensional in model identity.

The first concrete logical node is now present for a precisely qualified Church-style simple type theory: the alphabet of base types is fixed, typed constant signatures and closed formulas vary functorially, and extensional Henkin models reduce contravariantly. The satisfaction condition is proved by commutation of term denotation with constant translation. The model class includes the function-extensionality property required by the extensional derivation calculus, so finite derivability is proved to imply institutional semantic consequence. A non-injective signature map that merges two constants preserves the satisfaction square but provably fails to reflect validity; this separates an arbitrary institution morphism from a conservative embedding.

The corresponding property-explicit theory-graph node adds excluded middle, schematic Hilbert choice, and Dedekind infinity at one selected base type. It enters as a logical-only node: no transition system is fabricated for it. A later machine may provide an operational face only by mapping each equation class of states to a qualified Henkin model. This construction is not yet a full category of HOL signatures varying base-type operators, and it does not make choice, infinity, or

extensionality implicit in every simple type theory. Concrete institutions for the selected dependent and set-theoretic proof systems remain obligations; the generic institution interface itself does not supply their soundness or completeness.

An inter-bubble route can consequently carry two independent contracts:

1. an operational GSLT translation, graded as forward, covered, or proof-fibre exact; and
2. an institutional translation or comorphism preserving the declared consequence and satisfaction judgments.

Only a commuting pair earns a claim about both execution and mathematical meaning. This is the appropriate place for linking-type annotations: a component reasons in its own language, while the boundary explicitly records which foreign behavior is admitted and which equivalence theorem survives linking [15]. NIK supplies admission and replay authority for such routes; it is not a substitute for either semantic law. The formal theory graph realizes this separation by giving every node a native closed theory and only an optional operational face. Nodes with an operational face and commuting routes form a category with separate forgetful functors to the logical theory category and the equation-class-covered GSLT category. Thus a pure mathematical theory needs no dummy transition system, while a compiler route cannot acquire a mathematical meaning theorem merely from step preservation. Composing the logical projection with the closed-theory NIK functor gives the semantic authority view of every equipped node. This view is intentionally lossy: a checked hidden-state refinement has the same NIK object as its visible source, yet contains a concrete private operational state outside the source representation's image. Therefore the operational, logical, and authority projections coexist in one commuting architecture without any one of them being advertised as recoverable from another.

The model and operational readings now meet constructionally. A model-qualified operational face maps each equation class of machine states to a model of its native signature. The interpretation of a sentence is simply satisfaction in that state-indexed model, so equation invariance and semantic consequence soundness are derived rather than separately postulated. Forgetting the models only after this construction recovers the consequence-level operational face.

There is also a first nontrivial simple-to-dependent institution route. Over a fixed context, the simple side uses one value type at every context point and the dependent side admits an indexed family; total-space predicates supply the models on both sides. Constant-family inclusion induces an institution comorphism whose satisfaction square is exact. Over the two-point context its signature functor is faithful but not full, and the family with a singleton false fibre and a Boolean true fibre is provably outside its essential image. The route also covers models: every simple predicate model is literally the reduct of the same predicate on its constant-family carrier. A generic theorem therefore proves preservation and reflection of semantic consequence on the translated sentence image. This hypothesis has content beyond satisfaction preservation: a checked comorphism collapsing two Boolean sentences to one still satisfies the institution square but cannot lift the source model $\{\mathsf{false}\}$, so it does not reflect consequence by this criterion. This is the precise scoped answer to whether dependent types are justified: not for objects already in the constant-family image, but necessarily for a consumer that must preserve genuinely varying fibres. It remains a semantic specimen, not yet a complete institution for either a concrete Church-style simple type theory or a selected dependent proof calculus.

The selected-theory layer now varies the consequence institution as well as the signature. Its objects are closed theories in their native π-institutions. An arrow consists of an institution comorphism, a map from the translated source signature into the selected target signature, and a proof that every selected source theorem lands in the target theory. These arrows compose, and the

ordinary category of closed theories over any fixed institution embeds faithfully as a vertical fibre. Thus the heterogeneous theory graph replaces neither the existing little-theory categories nor their native sentences; it organizes them in one total category.

The biform layer has the corresponding heterogeneous total category. Its arrows pair a cross-institution closed-theory map with a proof-relevant GSLT translation, subject exactly to the commuting event-meaning square. The image of the projection to the independent logical and operational arrows is proved to be the compatible locus, and every fixed-institution biform category embeds faithfully as a vertical fibre. The simple-to-dependent comorphism supplies a nontrivial positive specimen: two distinct retained operational occurrences cross the logical boundary while their translated meaning theorem is preserved. A negative specimen changes the target meaning sentence while retaining the same logical and operational arrows; the compatibility square fails and no biform arrow exists. This is the checked categorical form of a multi-level intermediate representation: native logics remain native, GSLT events remain proof-relevant, and only their justified commuting pairs can be composed as meaning-preserving transformations.

Executable proof authority now lifts over this heterogeneous biform category as well. Each object selects its own native proof calculus and an exact certificate boundary; each arrow adds a certificate map whose Boolean replay commutes through the very same cross-institution sentence translation used by the biform arrow. The fixed-institution authority categories embed faithfully as vertical fibres, and the joint image is exactly the replay-compatible locus. In the simple-to-dependent specimen, native thin proof objects cross the logical boundary exactly. Keeping the same biform arrow while sending its source certificate to a different target theorem is a checked negative: theorem preservation and event-meaning compatibility do not manufacture replay authority.

This also sharpens the role of GSLT-IL. It is an operational language for sitting at a represented stage and moving along typed routes between stages, not a universal proof foundation and not a lowest-common-denominator term format. A reflective surface language may orchestrate that graph, but no surface becomes the "ultimate IR" merely by being able to quote all the others. Stronger bubbles stay available without imposing their conversion, identity, or universe rules on weaker consumers.

The categorical accounts of operational semantics explain why this separation is productive rather than bureaucratic. Turi and Plotkin combine an algebra of syntax with a coalgebra of behavior through a distributive law, obtaining a compositional operational model and a canonical denotational model under explicit hypotheses [17]. Recent categorical semantics for SSA likewise gives an IR its own equational theory and proves soundness and completeness against categorical axioms [18]. MLIR supplies the engineering analogue: several dialects retain the invariants needed by their own transformations and lower progressively rather than reconstructing high-level facts from one premature low-level form [19]. An equality-saturation dialect sharpens the same lesson: retaining the e-graph in the intermediate representation lets ordinary passes coexist with nondestructive equational rewriting instead of extracting one representative after each phase [20]. In the present setting this is one possible operational GSLT dialect, not a global identity principle: its equations, extraction observer, and cost model remain explicit. The semantic content of the multi-level practice is a diagram of GSLTs and independently justified models, with the strength of each arrow stated rather than inferred from the word "compilation".

Collapsing the internal execution of one representation into a coarser step does not erase that representation's equality. Given a run protocol, a raw run is a finite path from the protocol's entry state to a designated exit state, while its semantic run view is

$$E \,;\, \mathsf{Run} \,;\, E.$$

This view inherits the represented language's generated setoid, and a cover by runs lifts target

executions only up to that same setoid. Syntactic equality is recovered by theorem when the presentation generates no equations; it is not a parallel equation-free interface. A four-constant discriminator makes both endpoint changes essential: with $A =_E B$, $C =_E D$, and a protocol recognizing only $B \mapsto C$, the raw relation does not connect A to D but the semantic run view does. Thus a compiler or parser pipeline may use language definitions and GSLT morphisms at every semantic node while retaining private host data merely as an implementation of an arrow. No result here depends on rho-calculus minimal contexts: those are needed only for the rho-specific contextual labels in its modal adequacy theorem [24].

3.12 Observer refinement, dependent descent, and several identities

There is no single useful chain called equality. At least the following relations have different elimination principles and different consumers:

1. judgmental conversion, used silently by a checker;
2. the authored and derived GSLT congruence E, used to choose semantic operational states;
3. intensional identity or path evidence, used by dependent elimination;
4. behavioral equivalence relative to a chosen family of observations and labeled interactions;
5. extensional equality in a selected mathematical model; and
6. diachronic lineage, represented by retained routes between temporally distinct states rather than by endpoint equality.

These relations may coincide in a scoped profile, but the coincidence is a theorem, not conversion. In particular, quotienting by the symmetric closure of operational steps computes connected components. A two-state system with one transition has equal component classes although its source and target are not bisimilar. This checked counterexample prevents the operational–intensional–extensional diagram from silently reclassifying reachability as identity.

Observer refinement gives the precise refinement law. Suppose every coarse atom and label has an exact representative for a finer observer, while the fine observer may ask additional questions. Every coarse modal formula then has the same truth value after translation, and fine behavioral identity implies coarse behavioral identity. Hence there is a canonical contravariant map

$$\mathsf{Class}_{\mathrm{fine}} \longrightarrow \mathsf{Class}_{\mathrm{coarse}}.$$

If the atom and label maps are also surjective, this map is injective and the two observers induce the same identity. Without that hypothesis it can merge classes: a checked inert Boolean system is indistinguishable to an observer with only the always-true question and distinguishable to one that may inspect the Boolean state.

The dependent consequence is exact. All equality families anchored at fine behavioral classes descend through the coarse class map if and only if that map is injective. Individual families can still descend through a lossy refinement when they are constant, up to the selected equivalence, on every merged fibre. This is the formal version of fluid abstraction: ordinary reasoning inside a coarse bubble remains legitimate, while refining back into a distinction that was forgotten requires retained evidence or a reconstruction theorem.

It also gives a non-rhetorical answer to whether dependent type theory is justified. Simple type theory is the preferred mathematical interchange when types and conclusions are constant on the

relevant fibres. Dependency is earned when an earlier response, proof, revision, observation, or route selects the type of later data. For example, $n - 1 + 1 = n$ is false for natural numbers at $n = 0$; the missing lower-bound evidence is real information, not a defect of dependent syntax. One may instead work in integers, retain the guard as a dependent premise, or cross a boundary that states exactly what is forgotten.

Observational type theory and two-level type theory are therefore valuable candidate bubbles rather than mandates for the common core. The former keeps decidable definitional equality while supporting an explicit substitutive extensional equality [21]; the latter separates strict and weak equality without requiring global equality reflection [22]. The shared contract remains dependent identity elimination with its computation and substitution laws. Primitive J, cubical transport, observational equality, and scoped UIP are alternative realizations whose use must be selected and qualified locally.

Hofmann's translation programme explains why this should be a bridge rather than a global switch: extensional equality reflection makes typing undecidable, while partial-equivalence-relation and family models recover extensional quotients inside an intensional host; the simple PER model does not support genuinely varying dependent families, and the richer family model pays for that power with more complicated coherence and some conversion laws only up to propositional equality [6]. Accordingly, an extensional bubble is a certified quotient/model readout of an intensional one when such a model theorem is available. It is not permission to add equality reflection to every checker.

3.13 Compositional execution contracts over worlds and runs

An execution contract selects a program semantics, the resources and services it may use, its observable outcomes and commitments, and the applicable progress and cost comparisons. These are obligations over a common execution object, not interchangeable meanings of one numerical weight. Epistemic authority qualifies a judgment and its evidence; resource authority describes held rights. A service assumption can justify conditional reasoning about a run without becoming checked evidence or an execution grant.

Resource authority and reification. Let A be a separation algebra, with compatibility $p \# r$ and composition $p + r$ defined lawfully on compatible resources. A reification $V : A \to \mathcal{P}(S)$ interprets authority as sets of machine states. For an action relation $a : S \to S \to \mathsf{Prop}$, the frame judgment is

$$\mathsf{AP}(V, a, p, q) \quad \Longleftrightarrow \quad \forall r,\ p \# r \Rightarrow \forall s \in V(p + r),\ \forall s',\ a(s, s') \Rightarrow q \# r \ \wedge\ s' \in V(q + r).$$

This is the existing `Contracts.ActionPreserves`, following the Views interpretation of concurrent reasoning [1]. Rely–guarantee claims instantiate the same separation algebra: assertions are stable under their rely, and compatibility requires each guarantee to lie within the other's rely [2]. These laws prove safety of completed interleavings. They do not equate their visible traces or establish that a particular waiting task receives service. Enabledness is separate: an action that never runs from the admitted view can satisfy this safety judgment vacuously.

Relational transport with a frame back law. Suppose $f : A \to B$ is a resource homomorphism and $R : S \to T \to \mathsf{Type}$ is a proof-relevant relation. The reification square is

$$t \in V_T(f(p)) \quad \Longleftrightarrow \quad \exists s \in V_S(p),\ \mathsf{Nonempty}(R(s, t)).$$

Target steps must reflect along the selected relation witness: from $R(s,t)$ and $b(t,t')$, obtain an s' with $a(s,s')$ and $R(s',t')$. The resource back law lifts every compatible target frame:

$$f(p) \# u \quad \Longrightarrow \quad \exists r,\ p \# r \ \wedge\ f(r) = u.$$

Proposition 3 (Transport and composition of the frame judgment)**.** *Under these three conditions,* $\mathsf{AP}(V_S, a, p, q)$ *implies* $\mathsf{AP}(V_T, b, f(p), f(q))$*. Two such adapters compose by resource homomorphism composition and by the existing horizontal relation*

$$(R;Q)(s,u) = \sum_{t:T} R(s,t) \times Q(t,u).$$

Proof. Lift an arbitrary target frame to a source frame, recover a related source state through the reification square, reflect the target step, and apply the source frame judgment. Reify its successor and map its compatible frame back to the target. For composition, retain the actual middle state and both relation witnesses; frame lifting and step reflection compose through that same middle representation. The checked endpoints in the `ActionPreserves` namespace of `GSLT.Contracts.ReifiedActions` are `transport_relation`, `ReifiesThrough.comp`, `ReflectsThrough.comp`, and `transport_relation_comp`. □

The controls are substantive. Exchanging two owned cell locations transports the write judgment. Adding an effect log gives a relational representation whose frame-safe actions can emit $[\mathsf{false}, \mathsf{true}]$ or $[\mathsf{true}, \mathsf{false}]$ while leaving identical owned cells. Conversely, forgetting one cell's authority admits a reification square and step reflection for a target action that overwrites both cells; it breaks the forgotten cell's independently held frame. Thus the first two conditions alone do not justify the general transport rule, and a frame-safe adapter still owes its selected observation comparison.

World-indexed resource transport has the covariant opfibration constructed in `Contracts.IndexedAuthority`. Its base arrows are already selected changes of world; the cocartesian lift does not authorize an actor to take an arbitrary arrow. Cartesian lifts require the stated right-adjoint law. The constructed all-access revocation example fails that law; this is not a claim that every information-losing transport lacks a right adjoint. Contravariant predicate reindexing remains a different, compatible role.

Actual adaptive reads determine transaction dependencies. For a finite ordered occurrence store s, let $\mathsf{rows}(q, s)$ be its full list of matching identifier/payload pairs. The certificate of a read records both the complete ordered identifier list and the presence of each corresponding payload. Under the explicit integrity condition that one identifier has one payload in the current store, validation of this certificate is equivalent to equality of the full row list. The equality retains multiplicity, including distinct occurrences with equal printed values.

`Contracts.Dependencies.valid_readDependencies_iff` connects those concrete certificates to the existing `Machines.ReadOnlyQuery` interpreter's actual read agreement. `validated_adaptive_query` consequently preserves its entire observation, including the answer-selected query path. `readDependencies_bind` appends the actual continuation's dependencies; `validated_adaptive_resume` validates the performed prefix before querying a fresh suffix. A coexisting environment run establishes that same validation premise. Unrelated insertions preserve the constructed adaptive query; inserted child matches, changed payloads and ambiguous identifiers distinguish invalid certificates. An empty child answer still contributes a complete-query dependency.

These are laws for the pure occurrence-query profile with fixed program and query meanings. They do not supply native higher-order matching, rule-version agreement, cross-space dependency

completeness or stateful own-write continuation. Snapshot continuation and validated-fresh resumption retain their separately stated meanings.

Committed observers and nondiscardable obligations. For a dependent observer family $O(s) : \prod_{i:I} D_i$, administrative preservation holds exactly when it holds componentwise. Erasure likewise preserves a committed family O against retained family H exactly when each O_i agrees with H_i at every erased administrative dead end. The checked endpoints are `OrderedSystem.administrationPreserves_family_iff` and `OrderedSystem.erasurePreserves_family_iff`. The family may therefore include separate permission, duty, disclosure or history readouts without coercing their carriers into one algebra.

Frame-preserving updates permit relinquishing authority. They do not thereby settle an obligation already incurred. The constructed `Contracts.Joins.duty_family_requires_retention` control permits erasure at an empty permission observer and refuses it at an outstanding-duty observer unless the duty is retained outside the rejected leaf. Authorized settlement, transfer, cancellation, crash recovery and compensation still require actual service transitions and disposition evidence. A check after publication can expose an obligation without providing a compensator; irreversible and uncertain outcomes cannot be reclassified as rollback.

Presence, consumption and fair service. Presence-only programs, whose rules are monotone and inflationary, reach the same quiescent store under every schedule and every network delivery order, and an inflationary program that computes a function without coordination computes a monotone one: the two halves of the CALM theorem [135, 136] (`Contracts.Coordination.coordinationFree_of_presenceOnly`, `monotone_of_coordinationFree`). Absence breaks confluence, and firing absence rules only at a frontier restores it (`stratified_determinate`); frontiers are certified as in Section 3.14. Consumption is not confluent either: two consumers of one token reach two stores (`Controls.consumption_not_confluent`). What it needs is arbitration, a single decision of which consumer takes the token. The commit order is one arbitration: in every schedule the first consumer to fire takes the token and later ones change nothing (`Contracts.Arbitration.first_committer_wins`); optimistic validation refuses a later commit whose dependency on the taken occurrence fails (`Contracts.Dependencies.printed_value_does_not_rescue`). Exclusive authority is another: if each consumer may fire only while it holds the consume right and the two rights can be held side by side, at most one holds it, the gated program is confluent, and its outcome is the holder's whatever the schedule (`gated_confluent`, `gated_winner`). Neither removes the choice; each moves it from the schedule to an order or an assignment fixed once, which is the controlled nondeterminism that coordination supplies.

Fair service is a separate obligation. A least-first scheduler can starve an item behind regenerated work (`PriorityFairness.Controls.priority_not_weakly_fair`). If at most B items arrive ahead of x from time t on, x fires within $|\text{ahead}| + B$ steps (`PriorityFairness.Run.fires_within_overtaking`). The overtaking bound is the per-item certificate that the recurrence of stable states (`Contracts.Joins.ranked_stable_within_rank`) does not give. Deadlines realize it: if every arrival carries a deadline later than its arrival time, for instance arrival time plus a slack chosen by priority, and at most k items arrive per step, an item with deadline d fires within $|\text{ahead}| + kd$ steps (`deadline_fires_within`). Urgent work is served first among items due together, and no stream of urgent work starves an older item; epochs are deadlines rounded to a period (`PriorityFairness.epoch_fires_within`).

Conserved service credit. Service scheduling and financial permission to continue are distinct. The existing `ParallelFuelLease.Ledger` threads one authoritative account of balances b_i and incurred work s with invariant $\sum_i b_i + s = T$. Concurrent siblings do not spend copied pre-debit snapshots. A reservation transfers existing balance to a child; spending decreases that child's balance and increases s; cancellation returns only the unused balance. Copying a reference does not create another balance. An unmetered provider is selected separately: neither finite zero nor a largest machine integer denotes it.

The maintained cursor now derives its charge word from the source provider's actual replies and continuations. `Cursor.Lease.advance_exact` and `resume_exact` identify the funded run with the same source outcome and its exact spending; `advance_erases` forgets only the financial coordinate. The source world, full residual and occurrence history remain. A local bound B per request supplies a pre-execution reservation of qB for q requests (`reserved_advance`), without inspecting future answers. An unaffordable algebraic debit is not an admitted financial execution. The shared-producer instance connects these charges to the independent `runActions` and existing `RunAccount`: shared production is paid once, and distinct deliveries retain their separate declared prices. Its concrete later-tranche control spends five units, refuses an unfunded positive suffix, transfers the remaining two units into the same child and resumes to total spending seven without replaying the prefix (`LeaseExecution.Controls.conserved_tranche_resumes_the_same_execution`). A genuine failed primitive also retains its changed world and paid work (`Cursor.Lease.Controls.a_faulted_request_is_not_free`).

The native owned-execution adapter selects a funded provider explicitly and debits the existing controller-step counter, including work without answers. Its lease survives movement, pending-loan refusal and shared production; a later transfer changes funding without restarting the retained computation. The ordinary provider body contains no credit calls. These are serialized, process-local accounts, not authenticated grants or durable credit issuance. Their protocol and controller prices are not CPU time or complete service-slot prices. Preparation, copying, callbacks, validation, cleanup and external delivery retain their separate costs and progress obligations; conservation alone does not discharge those bounds or prove the compiled implementation.

Trust, observation and the remaining realization boundary. Entrusted service assertions retain their identity, scope, meaning, epoch where relevant, and assumption dependencies through composition. Reputation can inform a scoped reliance or delegation policy; it cannot inhabit a checker's evidence-soundness field merely by adding a basis tag. Established proofs remain proofs after reputational revocation, while new reliance or effect publication must satisfy the current admission contract. Distributed profiles additionally state tolerated fault families, authenticated local evidence and authority for membership or rule changes. Resource overlap alone proves neither honest quorum intersection nor a distributed protocol.

Observation refinement, run accounts, oracle meanings and cost/progress simulation carry the complementary comparisons. Resource transport does not discharge any of those independent comparisons. Work, parallel span, retained storage and recording expense use their stated composition laws; listing them as fields does not make them all additive. An optional history or reflective graph view is a selected observer with its own retention and expense obligations. Execution still retains the state needed for sharing, authority and exact resumption when recording is disabled; that does not require eagerly materializing a history graph.

Joint coefficient, demand and ownership contracts. An ordered coefficient ledger with a common inherited prefix is a restricted realization, not a general parallel composition law.

Independent same-world work may share a producer after its exclusive extensions; alternative worlds instead retain their distinct accounts. A factor view or its compatible subsequences do not determine the source's admissible global order, and noncommutative coefficients can distinguish proposed orders. The existing scoped ledger now also validates occurrence captures: `SharedCoefficientLedger.Scoped.extendsOccurrenceCapture` checks unique productions, retained factor identity, dependency and coefficient, captured order, and supported inherited claims. Selection keeps fresh unclaimed identities in the established world order, rather than reusing an integer offset after other factors have been inserted. Its claim and transport laws preserve the account; complete captured-world admission, pending producer authority and concurrent claim arbitration remain separate execution obligations. A native producer reply must retain the chosen world, bindings, factors, claims, rights and unfinished work alongside its value. Weight-only qualification therefore does not justify a production port before these ownership, resumption and parallel joins are established.

Cardinality as an execution consumer. The existing `AnswerCardinality` intervals apply to physical answer occurrences before coefficient aggregation. Weighted sequencing sums the counts of its reachable continuations; multiplying two counts needs a uniform continuation bound or an independent product construction. Unit regrading counts occurrences while preserving their order, but an arbitrary coefficient total does not determine that count. These laws now connect directly through `WeightedResumption.sequence_cardinality` and `WeightedResumption.total_countingContributions`. The checked consumer `WeightedBranchingCardinality.singleResult?` accepts only a complete cut with at most one returned occurrence, preserving its entire packet and coefficient. Its reconstruction and resumption laws compare the direct frontier with the free handler without forcing a second answer. Refused conversion returns the complete cut; continuing that retained representation agrees with the independently computed combined-budget run. A singleton prefix cannot establish closure; duplicate values remain distinct occurrences, and zero coefficients do not delete them. The actual Need instance retains its world and transition account. A halted fault is a reply, but not a successful value, so the value observer counts it separately. These are completed-cut representation laws, not an inferred determinism contract or a native fast-path qualification. Early specialization still requires independent mode-sensitive evidence, validated source-world dependencies, and preservation of the selected effects, faults and residuals. Termination, choicepoint elimination, cost composition and parallel admission retain their separate proof obligations.

The native frame and contextual-variable readbacks in `CeTTaNeedHistory`, together with `CeTTaFrameWriteback`, provide a concrete component of retained-world realization: selected frame readback and write receipts preserve the shared logical image. Complete payload graphs, producer worlds, force rights, Need identities and native memory/commitment correspondence remain further joins. A model-level `OracleRealization.Meets` theorem also retains a separate dependency on the actual compiled service. These constructions support compositional contract instances; they do not establish a minimal final architecture.

3.14 An open-ended PQI-inspired indexed decomposition

Chalmers's scrutability programme distinguishes physical, qualitative, and indexical information as part of a larger proposed basis [59]. We retain only the mnemonic $P/Q/I$ separation. We do not include its exhaustiveness component: the theories considered here are open-ended, and no stage asserts that its vocabulary captures every possible future distinction. The following is therefore a structural non-collapse device, not a formalization or truncation of Chalmers's thesis.

Let I be a family of indexical positions, let $P(i)$ be the extensional or publicly describable state at position i, and let $Q(i,p)$ be the further intensional or qualitative fibre over $p : P(i)$. The lossless object is

$$\mathcal{E}_{PQI} = \sum_{i:I} \sum_{p:P(i)} Q(i,p), \qquad \pi_{PI} : \mathcal{E}_{PQI} \longrightarrow \sum_{i:I} P(i).$$

The projection forgets the selected Q-inhabitant while preserving the indexical coordinate exactly. If every $Q(i,p)$ is inhabited, the projection is surjective. If any one fibre contains two distinct inhabitants, the projection is not injective. Thus an extensional account may cover every visible state without recovering all intensional distinctions. Later theories may enlarge I, refine P, or add new Q-fibres together with explicit transition maps. Such enlargement is not an exhaustiveness assertion at the earlier or later stage; it is an open sequence of explicitly related theories.

Absence without a totality clause. Refusing an exhaustiveness clause has a cost: a negative answer cannot be read off the absence of a record, since a later stage may add it. The checked replacement is a closure that is declared and certified for one question, not asserted for the whole theory. Read the states a program can reach as the worlds of a Kripke model ordered by reachability. Reads of what is present are persistent when the program only adds, and the positive connectives keep persistence (`FrontierNegation.presence_persistent`, `Persistent.and`, `Persistent.or`, `Persistent.exists`): presence-only programs read the positive fragment of intuitionistic logic. Intuitionistic negation, failure in every reachable state, is also persistent, but it quantifies over the future; its finite evidence is an inductive invariant that excludes the pattern (`knot_iff_invariant`). For the over-approximation of firing in a self-modifying space, a pattern is absent in every reachable state exactly when it is absent now and no pending executable may eventually produce it (`FrontierNegation.knot_iff_absent_frontier`). So frontier absence is intuitionistic negation, and a snapshot absence read, the closed-world reading of negation as failure, is never contradicted later exactly at a frontier. A frontier is sound for every refinement of the over-approximation (`refinement_sound`) and not complete for refinements: when a producer can never fire, the pattern is absent in every actually reachable state though no frontier exists (`Controls.refinement_incomplete`).

A frontier is declared by naming a pattern and certified by ownership. If the space obeys the namespace discipline of the closed-resolver results (an executable outputs only atoms whose heads its own module owns or nobody owns) and the module that owns the pattern's head is frozen at a set of its atoms closed under its own outputs and free of matches, the absence is final (`ownership_certifies_frontier`). The check reads only the owning module's atoms. Atoms of other modules never revoke the certificate (`foreign_extension_certified`); the owner revokes it by installing a producer, after which a previously final absence becomes false in a reachable state (`Controls.owner_revokes`). A verdict read under a certificate holds for that certificate's rule set, and a change of the rule set begins a new epoch of the module. This is a local, checked "that is all" for one question and one owner, revocable by that owner alone. Evidence against a proposition needs no such closure, since it is recorded rather than inferred (Section 3.5).

3.15 A GSLT-shaped description of mathematics

Each subject-matter object should be described through separate layers:

Data	signatures, sorts, constructors, contexts, occurrences, and indices
Equations	congruence, substitution, structural laws, and quotient candidates
Dynamics	oriented steps, schedules, interactions, and generated computations
Semantics	models, denotations, satisfaction, and invariant properties
Observers	declared readouts and exact descent/non-collapse boundaries
Certificates	finite witnesses, replay, authority, and provenance
Valuation	value, rank, evidence, cost, budget, ownership, and revision
Qualification	soundness, completeness, consistency, and strength assumptions

This prevents an equation list, a rewrite strategy, a checker, and a semantic theory from being mistaken for one another. It also prevents a large case table from masquerading as a language semantics.

4 Bubble type theory and hosted logics

This section gathers the results that make two slogans of the semantic envelope precise. The first: definitional equality is local, so that a theory with very little global computation can still offer convenient computation where a program needs it. The second: other logics are hosted beside Prime and connected by translations that state what they preserve, rather than absorbed into one shared logic. Both rest on the GSLT presentation of languages (Section 3.10): a bubble's declared steps form a small rewrite system read as definitional equality, OSLF derives the native types of each presented language from its rewrites, and a translation between logics is a biform route carrying a theory morphism, an operational cover and the transport of meanings. The Lean names cite the formal development; status labels follow Section 2.

4.1 Bubbles: definitional equality declared where it is wanted

Definition 4 (Bubble)**.** A *bubble* over the core judgments declares a set of directed steps between expressions. Its conversion $\equiv_b$ is the congruence generated by those steps. Typing in the bubble is global typing followed by one conversion:

$$\Gamma \vdash_b t : B \iff \exists A.\, \Gamma \vdash t : A \ \wedge\ A \equiv_b B.$$

A step $l \mapsto r$ carries a *justification*: a proof of the global identity $\mathrm{Id}\, l\, r$.

The global layer, the *weak core*, is the bubble with no steps: typing is syntax-directed, J and its computation rule provide identity evidence, and nothing computes inside a type unless a proof says so. Today's conversion (β, ι, δ and type-directed η) becomes the *default bubble*, which is the one most programs use.

Proposition 5 (Coherence of bubbles; proved)**.**

1. *If every declared step of a bubble is justified, every conversion of the bubble is a global identity (*`Bubbles.sound`*). A bubble with an unjustified step admits no such translation (*`UnjustifiedControl.unjustified_step_not_sound`*).*
2. *A map of declared steps maps conversions, functorially (*`Bubbles.widen`*,* `widen_refl`*,* `widen_comp`*), and conversions are stable under substitution whenever the steps are (*`Bubbles.substitute`*).*

3. *Typing in a bubble is global typing followed by one conversion, and the empty bubble is the global judgment (*`hasTypeIn_iff`*,* `hasTypeIn_none_iff`*).*

Figure 2 shows what entering and leaving a bubble means for a vector of length $n + 0$. Inside an arithmetic bubble whose steps are the recursion equations of addition, $\mathsf{Vec}\,(n+0)$ and $\mathsf{Vec}\,n$ are the same type. Outside it, the program carries one transport along a proof of $n + 0 = n$.

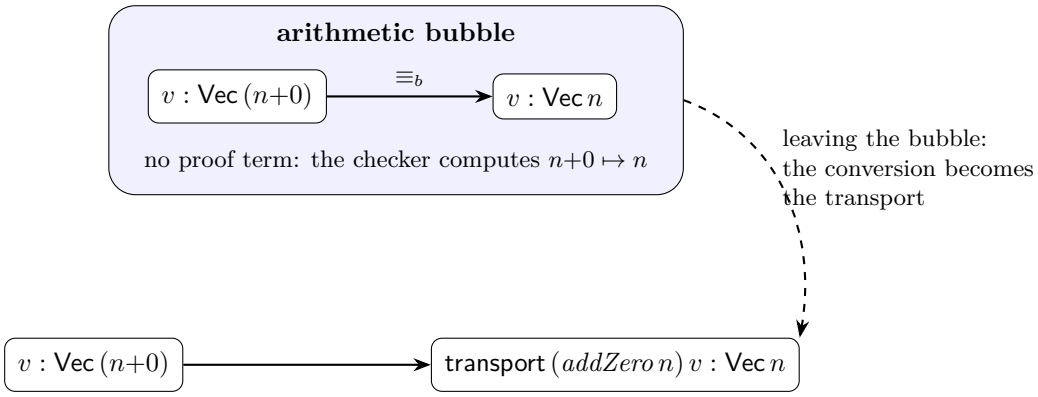


Figure 2: Entering and leaving an arithmetic bubble. Inside, the conversion is free; outside, the same step is an explicit transport whose proof is the step's justification.

The arithmetic bubbles are fully characterized, positive and negative cases included (Table 2). `arith` contains the two recursion equations of addition; `arithPlus` adds the left unit $0 + t \mapsto t$, justified by induction.

Judgment	global (weak core)	`arith`	`arithPlus`
$v : \mathsf{Vec}\,(n+0)$ given $v : \mathsf{Vec}\,n$	no	yes	yes
$v : \mathsf{Vec}\,(0+n)$ given $v : \mathsf{Vec}\,n$	no	no	yes
$v : \mathsf{Vec}\,(n+1)$ given $v : \mathsf{Vec}\,n$	no	no	no

Table 2: Which vector judgments each bubble accepts (`vec_addZero_inside`, `vec_addZero_not_global`, `vec_zeroAdd_plus`, `vec_zeroAdd_not_arith`, `vec_succ_not_inside`). Each bubble's conversion is decided by a normal form (`Arith.conv_iff_norm`).

The draft runtime realizes two bubbles. An arithmetic bubble arises when addition is admitted with its equations in a theory space. A subset bubble arises from `(hol:irrelevant F J)`, which makes all proofs of $F\,x$ definitionally equal. It is admitted only when the justification J proves $\Pi x\,p\,q.\ \mathrm{Id}\,(F\,x)\,p\,q$; an unjustified declaration is not admitted, and its verdict is undetermined, with the reason stated. A non-degenerate control remains: the pair $(0, p)$ is still distinct from $(2, q)$.

4.2 Why "weak": the weakness order on bubbles

Read a bubble's steps as equations, and call a congruence of the core's term syntax in which they hold on the nose a *strict model* of the bubble. Bennett's weakness of a theory is the size of its class of models [106]; Goertzel's quantale weakness generalizes the count.

Theorem 6 (Weakness of bubbles; proved)**.**

1. *A bubble's conversion is exactly what its steps entail over strict models (*`convertible_iff_entails`*).*

2. *One bubble has fewer strict models than another exactly when it converts more (*`models_equations_subset_iff`*); every monotone measure of model classes, Bennett's count included, falls as conversion grows (*`weaknessOf_le_of_convertible`*).*

3. *The weak core is the weakest bubble (*`global_weakest`*), strictly weaker than any bubble converting two distinct expressions (*`models_ssubset_noSteps`*).*

4. *The weakest bubble sufficient for a program declares exactly the program's conversion obligations (*`isWeakestAdmissible_obligations`*,* `obligationBubble_types`*,* `obligationBubble_weakest`*).*

5. *Over a finite catalogue of interpretations, among bubbles correct on what has been observed, the chance of remaining compatible with an unknown exchangeable future is monotone in weakness (*`bubble_survival_mono`*,* `survival_antitone_in_conversion`*).*

So "use the weakest bubble that suffices" is the choice of the weakest sufficient hypothesis. Three limits keep the slogan honest. A justified bubble adds nothing to the propositional theory (`justified_bubble_propositionally_invisible`): weakness lives in the judgmental layer only. The order is stated model by model, and may vanish up to weak equivalence of models [68, 107]. The obligation bubble is not stable under substitution (`obligation_not_substitution_stable`).

Remark 7 (Names). "Weak Type Theory" already names an unrelated system for mathematical vernacular [108], and theories with no judgmental computation are called objective or axiomatic type theories. The settled name here is *bubble type theory* (or *fibered type theory*), with *weak core* for the global layer, which is exactly the layer the theorem above shows to be weakest.

Remark 8 (Prior art). Varying judgmental equality by modal classification goes back to Pfenning's intensional, extensional and irrelevant modes [109]. Multimodal type theory varies context structure per mode, but every mode keeps the same conversion [40]; the extension needed here makes β and η mode-indexed and adds an identity former as the global target. The closest theorem for the crossing discipline translates rewrite rules into axioms and conversion into transport, conservatively [110]. Two-level type theory and observational equality are candidate bubbles of the same kind [22, 21].

4.3 Least sufficient bubbles and the meet-closed (Horn) fiber

When does a request have a single canonical *least* sufficient bubble above every seed? In general it does not (`SufficientBubbles.no_least_sufficient_bubble`): conversion-preserving and evaluation-based typing requests can admit incomparable minimal bubbles (`ComputationBubbles.syntactic_not_convex`, `ComputationBubbles.evaluation_no_least`), and minimal inclusion can disagree with minimal checking cost (`CertificateBubbles.cost_inclusion_disagree`).

Theorem 9 (Meet-closed sufficiency and the Horn fiber; proved)**.** *Let L be a complete lattice of candidate bubbles (such as $\mathcal{P}(G)$ over a finite generator set G) and let $F \subseteq L$ be the bubbles sufficient for a request.*

1. *Seven characterizations of F coincide (*`ClosureSystems.tfae_mooreFamily`*): (a) F is closed under arbitrary meets; (b) F is the set of closed elements of a closure operator; (c) F is the set of prefixed points of a monotone operator; (d) F is the set of elements respecting a family of implications $a \le x \to b \le x$; (e) the inclusion of F has a left adjoint; (f) F is the category of*

Eilenberg–Moore algebras of an idempotent monad on L; and (g) every seed has a least member of F above it.

2. *Over $L = \mathcal{P}(G)$, single-conclusion implications are definite Horn clauses: Horn is the natural, canonical finitary syntax of meet-closure (*`models_hornTheoryOf`*, Aczel's rule presentation* `oneStep_rulesOf`*,* `leastAbove_iff_hornDefinable`*).*

3. *Within any meet-closed fiber, chaotic fair iteration reaches the least sufficient bubble independently of rule order or search strategy (*`ChaoticIteration.schedule_independent`*,* `effectiveSteps_le_card`*).*

4. *When meet-closure fails, the canonical object is the Smyth upper set $\uparrow F$ of sufficient bubbles ordered by reverse inclusion (*`DisjunctiveBubbles`*), which is principal if and only if a least sufficient bubble exists.*

Thus a meet-closed (Horn) fiber is welcomed in Prime wherever obligations are implicational—such as declaration closure and syntax-directed typing (`ComputationBubbles.syntactic_isLeast`)—without making Horn or depth-first clause order the global default where conversion or disjunction breaks meet-closure.

4.4 The empty top and the inconsistent-top method

In the theory–model Galois connection, the theory of all sentences is the top. Under classical satisfaction it has no model. If the candidate structures include a degenerate one, a structure where everything holds, then that structure is the top's only model, and no axiom can carve it away: excluding it means choosing a different universe of structures. The model that is actually useful sits between the two: a *permissive* model in which every type is inhabited while conversion stays non-degenerate.

The same picture gives a method for metatheory. Syntactic properties such as Π-injectivity and subject reduction do not need consistency, so they can be proved for a maximally permissive, even inconsistent, theory with a universe $U : U$ through a domain-theoretic model [111]. The method has been ported to Lean's own theory [112]. For Prime its contract is: separations transfer to every carved fragment; closure results such as subject reduction transfer only along a conservative embedding; consistency and normalization never transfer, and each fragment needs its own model with an empty type.

4.5 Hosted logics and translations between them

Layer	What varies	What a crossing carries
Conversion bubbles	which computations count as definitional equality, within shared judgments	a justification for each step; substitution and transport laws
Different logics	judgments, equality, rules, axioms and models	an explicit interpretation that states what it preserves
Operational regions	machines, representations, scheduling, observation	preservation of the declared observable behaviour

Table 3: Three layers, three crossing contracts.

A host must preserve the distinctions and counterexamples of its guest, not only its truths, and must advertise any commitment it adds. The requirement is not cosmetic: a host can preserve every truth of a guest and still create new ones when its interpretation represents too few guest models (`InstitutionBridge`). The institution machinery makes the requirement checkable [13].

Theorem 10 (Hosting; proved)**.**

1. *A logic graph flattens into its Grothendieck institution, with a canonical comorphism from each node that preserves and reflects entailment (*`GrothendieckInstitution`*) [113].*
2. *For a sound and complete target, borrowing its prover through a comorphism is sound exactly when the comorphism is conservative (*`borrow_sound_iff_conservative`*) [114]; borrowing along a collapsing translation derives a true conclusion from false premises (*`Control.collapse_borrows_false_conclusion`*).*
3. *Proof-carrying routes compose, and certificates return only along conservative routes (*`Theory GraphProofTransport`*).*
4. *Route choice prefers a conservative route, then the route adding the fewest commitments, then the weakest; two optimal routes can be incomparable (*`Control.incomparable_optima`*).*

The native higher-order logic has its own namespace. A theory space selects the logic alone with `(hol:profile base)` and is judged with `hol:`. Its metatheory is anchored in two proved packages:

- **Classical HOL and Farmer's seven virtues** (`SevenVirtues`): Church-style extensional HOL with a single excluded-middle axiom $\forall p.\, p \vee \neg p$ is sound and Henkin-complete for countable signatures with closed sort inhabitants (`SevenVirtues.Completeness.provable_iff_valid`, `consistent_iff_satisfiable`), categorical for second-order Peano arithmetic (`Peano.categorical`) and complete ordered fields (`OrderedField.categorical`), and strictly separated from Hilbert's choice operator ε by a Fraenkel–Mostowski permutation model (`Choice.choice_separates`).
- **Intuitionistic HOL** (`HeytingGeneral`): parameter-free theories are sound and complete for Heyting-valued substitutional models (`provable_iff_heytingConsequence_param_free`).

Tier-0 grounding follows *correct-by-construction* (Rung 1): minimal HOL proves the candidate or native-inference checker faithful once, while per-run certificates and translations to HOL Light, Megalodon and MM0 (`hol.mm0`) serve as interop and acceleration bridges checking the same fiber semantics. Every set theory is a set profile over that base logic and is judged with `set:`: the shared material fragment by default, and the set theory HOTG, HOL with set axioms, selected with `(set:profile megalodon-hotg)`, never by default [58]. A set profile's signature includes the base logic's declarations and rules, and `(set:signature base-logic)` answers `(hol:profile base)`. The hosted HOL Light kernel keeps its own operations under `hol-light:`. Figure 3 shows the routes between them that run today.

The same judgments are importable from the other dialects: `import hol` and `import set` give Prime, HE and PeTTa programs the judgments of the logic and of its set theories, each over an explicit theory space and with answers returned as data. The HOL Light guest is not offered in HE until a translation into HE is qualified.

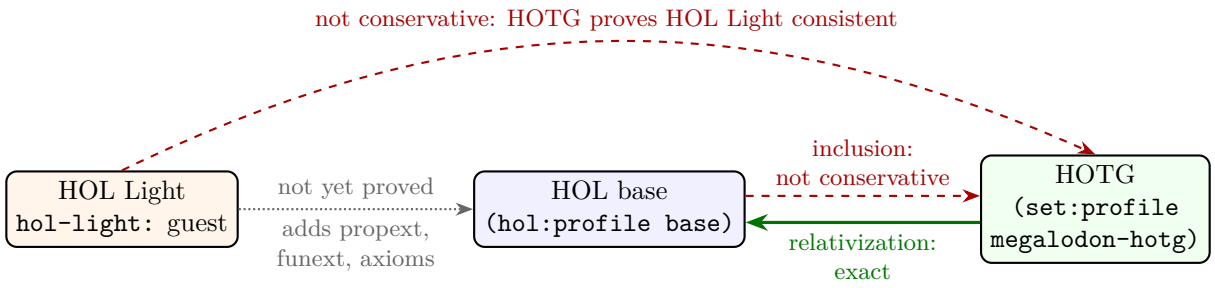


Figure 3: Routes between HOL and set theory that run in the draft runtime. Solid: conservative (proved). Dashed: not conservative, with the added commitments named. Dotted: conservativity not yet proved.

4.6 HE's types grounded in HOL

HE's types can mention symbols that also name terms: its own tests type length-indexed vectors with `(: Cons (-> $t (Vec $t $x) (Vec $t (S $x))))`. The indices are matched by syntactic unification. HE has no binder that names an argument, so `(: remember (-> Fact (WitnessOf $p)))` gives `(remember fact)` the type `(WitnessOf $p)` for every `$p`; and it has no conversion, so `(Vec Nat (plusZ Z))` does not match `(Vec Nat Z)` even when `(plusZ $n)` reduces to `$n`. Read as propositions, HE's types are all inhabited: `if` has the type `(-> Bool Atom Atom $t)`, so `(if True Z Z)` is accepted wherever any type is expected, including a declared type with no constructors. This is the situation of a programming language with general recursion, not a paradox: `Type` itself has the unknown type `%Undefined%`, not `Type`. Statically accepted terms are checked again at run time; `(needEmpty (if True Z Z))`, with `needEmpty` expecting the constructor-free type `Empty`, is accepted statically and evaluates to `(Error Z (BadType Empty Nat))`, as gradual typing prescribes.

On a fragment, HE's checker is one well-defined typing relation, and each such fragment is exactly HOL typing, where every type denotes a non-empty set and the truth of a proposition is a theorem of type `bool`, not the inhabitation of a type: saturated applications of declared static arrows (`hol_iff_exact`), type variables as polymorphic constants with HE's answer as the principal scheme (`Schematic.schematic_hol_iff_exact`), tuples as products (`Tuples.reference_iff_tupleHOLTyped`), meta-types as a classification theory (`MetaTypes.metaAccepts_iff_entailed`), and `%Undefined%` as the unknown type of abstract gradual typing (`Gradual.corePlusR2Consistent_iff_top`) [115]. Furthermore, the first-order HOL typing contract is proved to agree across all three native kernel realizations—HOL Light, Megalodon (`MathdataKernel.inferTerm`) and MM0 (`Preterm.infer`) (`KernelRealizations`). At run time Prime answers HE-style declarations inside these fragments from the hosted HOL Light checker, never HE's type engine; a Prime candidate may implement that contract only as a replaceable backend with proved correspondence. A failed check inside the fragments is a genuine refutation; outside them the answer is undetermined and never a reason to prune an alternative.

The `he-prime` profile extends HE with named binders, as in `(-> (: $p Fact) (WitnessOf $p))`, for which upstream HE gives no type. These are genuinely dependent and lie outside every simple fragment. The proposed treatment is a stable $\lambda\Pi$ contract, of which the HOL contract is the simple slice, checked by a hosted logical-framework kernel. Until then such declarations are left undetermined.

4.7 Demand, sharing and effects

The operational contract requires that a shared value be computed once, that an effect happen once, and that a bounded observation bound both work and effects. Figure 4 contrasts an eager and a bounded run of `(select 2 (gen))`, where each of four alternatives writes an atom before answering.

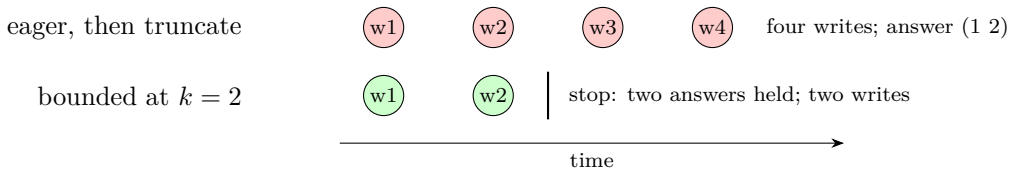


Figure 4: Bounded selection in the illustrated sequential policy. The run stops at its k-th successful answer and avoids later effects. An infinite producer must reach those successes in finite work for this observation to terminate.

Three runtime defects of this kind were found and removed in the draft runtime. Each now has a route-agreement fixture, so the default route, the reference route and the fuel-bounded evaluator must agree, and each has a mutation that reintroduces it and is killed:

1. selection of $k \geq 2$ answers ran every alternative's effects before truncating, and did not terminate on infinite producers (`bounded_selection_stop`, `bounded_selection_effects`);
2. a shared pending value reaching two strict operands of a primitive was computed twice (`strict_operand_sharing`);
3. a binding whose closed source produced no answer ran that source twice (`empty_source_once`).

Retained search must distinguish a snapshot continuation from validated-fresh resumption. The latter checks the dependencies of already performed reads; an unrelated write need not invalidate them, while a changed complete result or absence must not reuse the old answer. Stateful own writes require their own continuation law. Section 3.13 establishes the pure adaptive-query join; the complete effectful retained-world and native resumption comparison remains an implementation obligation.

Answers, choice and probability. The speculative layer is read through weighted answers, $\mathsf{Spec}\, K\, C\, W\, X = W \to \mathsf{Ans}_K(X \times W \times C)$, which aggregate equal outcomes, forget their order and drop branches without answers together with their cost. This readout is an observer of ordered runs, not a replacement for them: an ordered run returns its branch outcomes in order, and a failed branch keeps its world and the cost it paid. Observing a sequenced run is sequencing the observations, so the observer is a monad morphism (`SpecRunObserver.observe_pure`, `observe_bind`); it forgets order and the cost of failed branches (`Controls.observe_forgets_order`, `Controls.observe_forgets_failed_cost`). Laws proved for Spec are laws of what ordered runs show through this observer.

Nondeterministic choice and probability are distinct effects. No distributive law of finite distributions over finite sets exists [151, 152] (`NondeterminismProbability.no_distributiveLaw`); in one weighted monad, additive choice and convex probabilistic choice are both algebraic, and the support readout identifies them (`convex_bind`, `support_convex`). Probability becomes a view of a program with choice only once a scheduler resolves each choice. Without choice every

scheduler gives the same probabilities (`SchedulerResolution.prob_choiceFree`); with choice, the envelope of probabilities over all randomizing schedulers is attained by deterministic ones (`exists_deterministic_le`, `exists_deterministic_ge`), so the honest view is the convex set of distributions between those extreme points, the convex powerset of mixed powerdomains [153]. A pure choice between two outcomes has every probability in $[0, 1]$ (`choice_probabilities`), the interval readout of zero evidence. Counting paths is one scheduler among many: normalizing the additive readout of a three-way choice built from two binary choices gives $1/3$, fair coins give $1/2$, and the scheduler weighting each choice by its number of paths gives $1/3$ (`Controls.counting_not_fair`). Evidence weights differ from both: they count observations, probabilities sum to one, and a nondeterministic choice carries no weight at all.

Where each element lives. The theorems above fix where each element belongs in a theoretical Prime, without selecting a candidate calculus. A *primitive* cannot be recovered from the others; a *library* is defined over primitives with the laws it needs; a *profile* is a declared choice of strategy or semantics that consumers can rely on.

Element	Place	Typing and authority	Justifying theorem
Annotated space and weighted answers	primitive	one semiring per space, with declared capabilities (merge; meet, monus and join where they exist)	`Ans.bind_assoc`, `support_bind_iff_positive`, `no_space_sub_undoes_merge`
Evidence algebra	library	counts as a positive annotation; ledgers carry identities; revision of overlapping ledgers is typed by a disjointness proof or falls back to the maximum; readouts return "undetermined"	`additive_revision_exact_iff_disjoint`, `unknown_overlap_bounds`, `conjunction_counts_iff`
Absence at frontiers	primitive guard	an absence read carries its certificate, the owner of the pattern's head and the frozen rule set; only the owner revokes	`knot_iff_absent_frontier`, `ownership_certifies_frontier`, `foreign_extension_certified`
Call	primitive step	a synchronous call is a step of the kernel, not sugar over rewriting	`call_not_locally_eliminable`
Seals and owned namespaces	primitive	sealing and the right to extend a head are capabilities; kernels resolve helpers in a closed environment	`isEmpty_sealElimination`, `late_bound_forgery`, `owned_sound`
Consumption	primitive step, arbitration by profile	the consume right is exclusive; commit order, optimistic validation or exclusive authority arbitrates	`consumption_not_confluent`, `first_committer_wins`, `gated_confluent`
Strategy	profile	priority keys are semantic; a fairness certificate is an overtaking bound, realized by deadlines or epochs	`priority_not_weakly_fair`, `fires_within_overtaking`, `deadline_fires_within`
Probability and choice	library and readout	probability is an annotation; credal sets are the readout of choice under all schedulers	`no_distributiveLaw`, `exists_deterministic_le`, `choice_probabilities`
Knowledge-based programs	library	three-valued tests answered by grounded inquiry; ranked programs are decided	`stage_sound`, `least_decides`

Element	Place	Typing and authority	Justifying theorem
Observer-indexed truth	library	knowledge is the interior of an observer's partition	`knows_le`, `knows_knows`
Speculative readout	observer	`Spec` observes ordered runs; order and paid work stay with the runs	`observe_bind`, `Controls.observe_forgets_failed_cost`

4.8 Open decisions

1. The stable $\lambda\Pi$ contract for dependent HE declarations, and the hosted kernel that checks it.

2. Pairing conversion bubbles (`BubbleWeakness`) with axiomatic little-theory nodes (`HenkinInstitution`), cross-base feature composition, and the weakness order for full models of type theory rather than term congruences.

3. Generic derivation of the complete dependent eliminator from an authored inductive signature across the indexed-polynomial container spine (`IndexedPolynomial`) and proof-relevant ι conversion policies (`ConversionRecipes.Policy`).

4. Full Kripke–Henkin completeness for intuitionistic HOL via a scheduler provider extending the proved Heyting substitutional completeness (`HeytingGeneral`).

5. Bounded `select` and `once` in HE's extended profiles, and eliminating any fixed implementation bound on contextual frame identities in the runtime.

5 Prime's planned CBPV basis and qualified evaluation strategies

The intended next foundation for Prime is CBPV-based, with a theoretically qualified mixture of eager evaluation, suspension and sharing. This is a design direction, not a completed migration or a choice of every surface default. The current Prime nucleus includes a call-by-need layer: computation is demanded when required, and subsequent uses of the same qualified need cell reuse its cached outcome. Ownership, revision and occurrence identity determine which uses may share that cell.

Call-by-push-value (CBPV) distinguishes value types from computation types and relates them through the familiar $F \dashv U$ organization. Call-by-name (CBN) and call-by-value (CBV) arise as translations into this common value/computation language [34]. Plain thunk/force does not itself memoize a result. McDermott and Mycroft extend CBPV with a need binder and effect-sensitive equations for evaluation-order reasoning [37]. This is a candidate construction to compare with Prime's existing need protocol, not an automatically adopted calculus. A strict primitive, guest calculus or optimized fragment must preserve its declared observation and surrounding demand/sharing contract; the existence of a CBV or CBN translation is not permission to change it.

For example, eagerly evaluating an unused diverging argument changes termination; rerunning a shared nondeterministic choice can change a correlated pair into two independent results; moving a read across a write can change its value. The permitted rewrite therefore depends on demand, effects and the observer. Effect/coeffect tracking offers a way to certify both effects and variable-use requirements [38]; it is not a proof that any one strategy minimizes total cost. A proposed optimization must separately establish semantic preservation and its cost advantage on the advertised workload.

`ContextualStrategyQualification` supplies a first general law for the existing contextual-effect syntax, without introducing another evaluator. A structurally read-only program—no choice, write or intent—can be evaluated once and its value reused in place of two consecutive evaluations before any effectful continuation. It can also be discarded when unused. The proof preserves the isolated-world evaluator's complete ordered result: branch paths, answers, final states and intent lists. Its observation does not count reads, so this is not a work or full-trace equality theorem. The controls distinguish independent Boolean choices from a correlated reused choice, equal answer support with four versus two result occurrences, repeated state increments, and repeated deferred intents. Even equal answer support does not permit changing multiplicity. These are sufficient qualification laws at demanded sites, not a general need implementation, a necessary criterion, or a result about divergent computations.

The value/computation boundary now has a concrete semantic instance in `ContextualKleisliAdjunction`. For the existing contextual program monad T, the existing Kleisli category supplies

$$F : \mathbf{Type} \rightleftarrows \mathbf{Kl}(T) : U, \qquad F \dashv U, \qquad U(B) = \mathsf{Program}(S, |B|, I).$$

Here F sends a value map to pure return; U acts on an arrow by sequencing with its program. In particular, U does not simply extract an answer type. The natural hom equivalence and both triangle laws are checked, and the induced monad's multiplication is the existing `bind`, not a new evaluator. A real choice is outside the pure unit's image. This is a Kleisli resolution of the monad, not a complete CBPV model: computation products and value-indexed computation function objects are not supplied by the adjunction alone [35]. There is now a checked obstruction, not merely missing infrastructure: `ContextualKleisliProductBoundary` proves that the singleton-state, singleton-intent instance has no terminal computation object. Every program has a positive finite answer-occurrence count, and duplicating it by choice doubles that actual handler observation. Any supposed terminal target would have to identify an arrow into it with its duplicated-choice arrow. Ordinary value types still have their terminal singleton; applying F does not make it terminal on the computation side. The free-returner category therefore lacks even the empty computation product. A full CBPV model needs a richer computation category or an explicitly justified change of observation; neither is supplied by relabeling the existing adjunction. This does not refute every binary computation product or require erasing occurrences.

An algebra-based candidate now resolves this structural obstruction without changing the existing programs. `ContextualComputationAlgebra` uses Mathlib's Eilenberg–Moore algebras for the same T, its free/forgetful adjunction and the comparison functor

$$\mathbf{Kl}(T) \hookrightarrow \mathbf{Alg}(T).$$

The comparison is fully faithful: evaluating a free-algebra homomorphism on pure inputs recovers its original Kleisli arrow, and the homomorphism law proves the converse. Both encodings retain literal choice/effect programs; the new adjunction induces the same sequencing. `ContextualComputationAlgebraProducts` constructs varying-family products by Mathlib's creation of limits. Their carriers are dependent functions, and their actions and projections are pointwise. The empty product is terminal, but the free returner $F1$ is not: appending an intent and doing nothing are distinct algebra endomorphisms. This does not quotient programs or authorize effect erasure.

`ContextualAlgebraSequencing` supplies the contextual interfaces: computations are functions from value contexts into algebra carriers; stacks are context-indexed algebra homomorphisms. Sequencing into any target algebra satisfies unit, associativity and stack compatibility, while a free target recovers actual `Program.bind`. Contextual continuations transpose to stacks from free returners. Products, value-indexed computation functions and value coproducts have their computation/stack universal properties, with context reindexing and resume compatibility. These are the

module-relative semantic interfaces of adjunction semantics [35]; they do not themselves reify a dependent CBPV source language. In particular,

$$F(A \to B) \quad \text{and} \quad A \to FB$$

are different interfaces: an argument-dependent intent lies outside the canonical map from computations returning pure functions.

The distinction reaches a real mathematical consumer in `NativeConversionAlgebraService`. Its computation-function object is $\prod_q F(\mathsf{Reply}(q))$, using the existing native conversion request, code, checker and reply family. Projection invokes the exact request; accepted outputs license native use only with independent source admission and target formation. Selecting one provider before two calls produces two correlated pairs of actual accepted proof codes; per-call selection produces four pairs, including mixed codes, and repeats the selection intent. Both accept every returned candidate, so acceptance support hides this difference. Revision and endpoint mismatch still reject packet reuse. This is semantic higher-order service composition, not computation-lambda source syntax, Need memoization or adoption of a complete Prime profile.

`ContextualThunkStrategy` uses the same programs as first-class answer values. An effectful selection may return a packet $(a, M_a) : \sum_{a:A} T(B(a))$ without running M_a. Forcing executes its body in the selected world's current state and returns $(a, b) : \sum_{a:A} B(a)$. Delay followed by force agrees with the existing dependent bind. The read-only reuse/discard laws now lift through arbitrary outer selection and indexed effectful consumers; only each actually selected body needs that sufficient qualification. A stored choice forced twice yields four independent Boolean pairs and repeated intents; reusing its result yields two correlated pairs and one body intent. Eagerly forcing an unused body likewise changes intents and occurrence multiplicity. Thus a computation-valued thunk is not itself a memoized need cell.

`ContextualDependentSequencing.runWorldsAt_bind` now gives the general sequencing law for this same finite effect syntax. Each continuation runs in the selected world's state and branch history, and the earlier deferred intents precede the later ones. The equality is of ordered result lists, so occurrence multiplicity is retained. Its dependent instance keeps the selected index and its fibre value. Reindexing to a different continuation requires pointwise compatibility; a collapsed index does not supply it. `FormationSensitiveDependentComputation` then uses the actual native Sigma introduction rule to construct $(a, b) : \Sigma(x : A).B(x)$ from the effectfully selected admitted $a : A$ and admitted $b : B(a)$, with independent Sigma formation. Native value renaming and typed substitution commute with this construction. A native identity-family control rejects the other branch's reflexivity value even through conversion and cumulativity tails; resetting a continuation to the initial state separately changes its intents. `FormationSensitiveThunkComputation` extends this interface to raw delayed-service packets containing only a native index and a program body, with no admission fields. A separate semantic qualification requires typing of that index and of the body's actual results in its indexed fibre. The native Sigma rule then admits every forced pair. Forcing is the sequencing induced by the checked adjunction; erasing supplied native evidence before forcing agrees exactly as a raw program with erasing the admitted answer afterwards. Native answer substitution commutes with forcing and transports the qualified result family. This is semantic answer reindexing, not substitution through a newly reified thunk grammar or an automatic checker for the qualification contract. The delayed identity service records only its selection intents until forced. Writing the consumer state before force changes the body's read without changing the retained index. A body returning the other index's reflexivity term executes but fails qualification. Logical dependency therefore does not imply capture of a mutable-state snapshot, and raw execution does not restore erased admission evidence. That construction uses

admitted native values as its answers. A separate reified connection now removes this restriction: `ScopedComputation` has raw native terms, operation names and syntactic bodies under real binders. Capture-avoiding renaming, substitution, composition and binder opening reuse the native structural algebra. `ScopedComputationTyping` independently types return, choice, operation calls and both sequencing forms. Ordinary sequencing requires a result type independent of the selected value; the dependent form retains that value in a native Sigma result. Operation declarations separately form their closed input and argument-indexed result family.

`ScopedComputationSemantics` gives direct ordered-world semantics and an interpretation into the existing contextual handler. Primitive realization implies exact list agreement, and source substitution commutes with the actual interpreted program. The handler has no abort constructor: every execution returns a nonempty finite list, whereas an empty primitive world list has no realization in it. This is not native-term normalization or progress for a full CBPV language. `ScopedComputationPreservation` proves native result typing for every independently typed source computation under every refined environment, given the primitive result contract. The contract is discharged for actual state-marking and reflexivity operations, so it covers arbitrary admitted compositions over them, not only one example. The example selects x, updates its private state and returns $(x, \mathsf{refl}(x))$ with ordered intents. Capture, repeated choice, state reset and a misindexed primitive are separating controls. An altered implementation can realize its altered operational specification exactly while still failing the declared dependent family. This supplies the scoped sequencing fragment, not full value/computation types or a need implementation by itself.

`ScopedNeedComputation` extends that raw syntax with a separately scoped, nonrecursive need binder, force and effect emission. Native substitution lifts under value binders; handle renaming lifts under need binders. These operations commute, and the old sequencing syntax embeds injectively. Independent computation typing weakens existing handle-result types under a native binder and extends only the handle context under a need binder. Incorrect handle redirection is rejected even through the source conversion rules. `ScopedNeedMachine` interprets the captured code through the unchanged owned Need machine. A producer captures native substitutions and cell references, never its caller's continuation. Need binding allocates without forcing; ordinary sequence allocates and immediately demands; Sigma sequence assembles the native pair before the enclosing commit. General exact transition laws retain the arbitrary outer stack, the two choice occurrences and cached-force independence from the primitive producer.

The raw machine values remain native terms. A separate cell-type assignment relates source typing, captured environments, cell identities and completed cache entries. Typed native-body opening, handle-body opening, Sigma continuation completion, successful allocation and qualified cache replacement preserve their respective invariants. These local laws do not by themselves establish all-control-stack preservation. Cache replacement's type law also does not license an overwrite or establish equality with the producer's computed result; those require the operational ownership and execution laws. `ScopedNeedMachineAdmission` combines captured source typing and the independent heap invariant to qualify actual cached-force successors and native continuation resumption, retaining source conversion tails. A genuine shared source returns $(x, \mathsf{refl}(x))$ in its declared identity family. An injected cache can still be returned by the raw machine while assigning it the other input's identity fibre refutes both heap typing and native admission. The separate `ScopedNeedMachineControlTyping` and `ScopedNeedMachinePreservation` now close the whole-control typing obligation for this scoped source fragment. Demand and allocation resumptions have distinct typing relations. Conversion-aware native continuations retain the source's independently formed conversion targets without adding runtime conversion instructions. Every actual branch, action, resumption and commit preserves the declared final type; fresh allocation extends the cell-type assignment without changing older declarations. The argument covers ownership failures and

retry as non-value outcomes, not as proofs.

Independent source typing and a typed captured environment establish the initial empty-heap invariant. Primitive qualification then implies preservation along every finite execution and bounded frontier, including unfinished states. In a formed ambient native context, every actually returned value has its native judgment. No intermediate-heap assumption, normalization theorem or termination premise is inserted. The primitive contract is discharged for argument retention and argument-indexed reflexivity in every native context, so arbitrary independently admitted compositions over those operations inherit the result. A shared effectful choice returns $(x, \mathsf{refl}(x))$ with one producer effect per branch; its two native judgments follow through the whole-machine theorem. An altered reflexivity primitive still returns a term operationally but fails the declared identity fibre and the primitive contract. `NeedAllocationBound` separately proves that every occupied slot stays below the next allocation counter under every actual machine step, for arbitrary language specifications. Consequently the actual allocator succeeds in every world reached from an empty heap, including resampling generations. Combining this result with the local logical heap invariant gives a successfully allocated, typed extended heap. Malformed supplied heaps still fail closed. This natural-number reference result does not qualify a finite C counter or exclude a primitive explicitly returning a collision tag. Actual bounded executions also distinguish one shared choice, which returns two correlated pairs with one producer effect receipt per branch, from two independently allocated choices, which return four pairs with two receipts. An unused suspension emits nothing; stable faults are cached, retryable faults restore suspension without erasing previous receipts, and insufficient fuel retains an unfinished frontier. These examples project outcome status and chronological effect receipts, not full machine paths: the latter include administrative singleton forks. Effects here are receipt events, not an implicit mutable-state handler. An independent sharing-aware natural semantics now supplies the finite source–machine correspondence. This follows the methodological separation between a heap-based natural semantics and a machine implementation [32, 33]; it is not a formalization of those papers' complete calculi. In `ScopedNeedNaturalSemantics`, mutually inductive source evaluation and force rules inspect authored constructors and heap entries, without invoking machine transitions, continuations or bounded frontiers. A forced producer retains its selected occurrence, evaluates that source, and commits or retries against the current owned cell. Retries preserve earlier effects and allocations; lost ownership cannot overwrite another result.

`ScopedNeedNaturalAdequacy` proves that a natural derivation exists if and only if the actual machine completes from the corresponding empty consumer stack with exactly the same outcome and world. The forward argument is structural on source derivations, under arbitrary native consumers and outer stacks. The reverse argument interprets pending continuations by source computations and reconstructs this interpretation backwards through every actual step. It does not assume a derivation of the initial source. Heap contents, branch paths, allocation/evaluator counters and receipt histories are retained; runtime work counters and path lengths are existential. Every answer exposed by a bounded frontier therefore has a source derivation. A three-step unfinished frontier has no answer even though a complete natural derivation exists, refuting the inference from budget exhaustion to source failure. The result is not a bijection of derivations or an ordered-list, cost, fairness or termination theorem.

Whole-control preservation then gives native admission of naturally evaluated values from independent source/environment/heap typing and primitive qualification. The same source can evaluate under an altered reflexivity primitive while its returned term fails the declared dependent type. Semantic execution evidence alone is consequently not mathematical admission. This native-returning construction does not itself supply first-class CBPV value/computation types, recursive suspension, full mutable-effect semantics or adequacy for all requested observations of the eventual language.

An exact demand-point law now connects two existing source forms: `ScopedNeedImmediateDemand` transforms natural derivations in both directions between `letNeed P (force 0)` and `sequence P (return 0)`. Both allocate the same producer and demand it at the same point. The law retains every completed outcome and full world, including choice, effects, native/stable/retryable faults and allocation collision; actual-machine correspondence follows from natural adequacy. Independent typing of the sequencing form qualifies the lazy form's same native result. This is not prewarming, frontier equality or a cost theorem. Erasing a pure suspension still removes its cell and four protocol receipts. The dependent control `letNeed P (sequenceSigma (force 0) K)` and `sequenceSigma P K` returns the same admitted native pair, but the former allocates an additional forwarding cell: the observed cell/receipt counts are $(2, 8)$ and $(1, 4)$. Any optimization identifying these worlds must state its permitted observation loss or correspondence explicitly.

Two source-level contracts constrain the first-class interface. `SortedABTRenaming` and `SortedABTSubstitution` reuse the existing sorted abstract binding trees. Simultaneous substitution lifts a replacement along the complete field binder list, with identity, composition, renaming interchange, opening coherence, scope preservation and exact signature preservation. A value replacement can contain an outer Need reference. Inserting it beneath a new Need binder must shift that reference from zero to one. The selected-axis operation instead retains zero; both results can pass the same scope and signature checks. An exact field observer then forces the selected cell on the existing Need machine and distinguishes cached answers ten and twenty. This is a binding counterexample with an actual machine observation, not an evaluator for a newly adopted ABT language or a proof that the supplied cached heap arose from admitted source.

Separately, `ScopedNeedComputationFunction` represents a native-domain function by its raw domain, dependent result family and existing scoped computation body under one more native variable. Application uses the actual source substitution. Independent formation and body typing imply the opened result judgment; native substitution and handle renaming commute with application. Materializing the machine's captured value-body environment agrees with source substitution followed by application. Qualified natural evaluations therefore yield native results in the selected argument's fibre, using the independently typed heap and primitive contracts. The dependent identity example forces its suspended proof twice, returns reflexivity at the selected argument, and emits only one effect. Reflexivity at the other argument is rejected. This function-body interface does not add ordinary thunks to the native term grammar, computation-valued runtime results, or a first-class computation-lambda evaluator. The generic ABT laws and this native function interface do not alone supply their combined source-language refinement.

A separate native-indexed polarized candidate now makes the first-class source connection explicit. `PolarizedNeedSyntax` has mutually defined values and computations with native, value and Need-reference scopes. Ordinary thunks, computation abstraction/application and native-index/value packets are source trees. The native term grammar is unchanged. Its type codes distinguish FA, UB, value-domain computation functions and native-dependent computation functions. A packet can retain an effect-selected native index together with a closure whose type depends on that index. It does not let native types depend on arbitrary effectful closures. `PolarizedNeedSubstitution` lifts value replacements along all three crossed scopes. Identity, composition, renaming interchange and native, value, Need and simultaneous native/value opening laws are proved. The complete old source embeds injectively, with exact substitution and renaming compatibility.

`PolarizedNeedTyping` supplies independent value/computation formation and typing, preserving every existing scoped Need derivation. Need slots declare arbitrary computation types, including computation functions. Regularity is qualified by the native universe-regularity contract and formed contexts; native payload and native-return inversion recover the existing formation-sensitive typing judgment. `PolarizedNeedTypedSubstitution` proves simultaneous typed substitution from a

native context morphism, independently admitted value replacements and exact Need-type reindexing. Native application, value application, packet opening and ordinary thunk force preserve their displayed source type, including final native conversion. These are principal source openings, not an evaluation strategy or whole-machine preservation. `PolarizedNeedTypingExamples` retains dependent identity evidence through thunking, higher-order passage and application. Its general packet consumer works for every independently admitted native producer: after unpacking and forcing, the selected index is returned with its proof. Discarding that index is not a derivation of an escaping dependent result type. Wrong-index evidence and value/computation-polarity confusions are excluded by the independent judgment.

`PolarizedNeedMachine` executes these constructors using lexical environments. Its answer carrier distinguishes returned values from native-argument and value-argument computation functions; Need can cache any of those answer forms. A cached function resumes in the current consumer's continuation, while an ordinary thunk stores neither that continuation nor a world. `NeedLocalSteps` provides silent local transitions without modifying the reference machine: each changes control and charges one transition, leaving the complete world and outer stack unchanged. Disabling interception preserves every reference specification's ordered bounded frontiers exactly. `NeedRepresentation` separately proves exact step and frontier commutation under explicit representation maps and specification compatibility. Its occurrence law retains list position and duplicate alternatives, with finite-path lifting in both directions from represented states. `PolarizedNeedEmbedding` instantiates this contract for every old scoped Need machine. No new local transition occurs on that image. Cells, ownership, branch paths, cached values/faults, full receipts and work are retained, not merely native answers.

Actual first-class programs distinguish ordinary execution twice, with four choice pairs and two effects per branch, from shared execution, with two correlated pairs and one effect. Sharing the thunk's construction retains ordinary reexecution. A shared computation function emits its prefix once and its body once per application. A returned higher-order closure retains an outer native index and Need reference beneath later binders; a packet's retained-index consumer returns its selected index alongside the obtained evidence. A separate raw projection intentionally forgets that index and claims no admitted outer dependent result. Wrong raw polarities produce explicit machine faults, not false mathematical propositions.

`PolarizedNeedDependentExecution` connects first-class execution to a genuinely dependent native judgment. An independently typed one-native-variable source template passes a thunked identity computation function through a value-level consumer, forces it and applies it. For every native argument in an arbitrary target context, seven checked machine transitions return its reflexivity term, preserving the empty world and recording exactly seven local/protocol transitions with no allocations or receipts. Independent argument admission supplies the typed loading environment and qualifies any returned native term at $\mathsf{Id}(\mathsf{ground}, t, t)$. Wrong-index reflexivity is excluded. This is a symbolic-argument program theorem, not general runtime preservation.

General preservation is now supplied by `PolarizedNeedRuntimeTyping`, `PolarizedNeedMachineLocalTyping`, `PolarizedNeedMachineHeapTyping` and `PolarizedNeedMachinePreservation`. Runtime values, closures and function bodies retain independent source derivations and typed lexical environments. Their qualification transports through a growing cell assignment, while old cells retain their exact computation types. An older cached function may consequently receive a new argument capturing a newly allocated cell without discarding its own environment. Body opening uses the actual captured code; it does not assume the result of applying the function to every future argument.

The local proof distinguishes intercepted application/force/packet opening from the fallback reference action. Typed continuations and complete commit stacks then qualify every actual suc-

cessor, including allocation, caching, effects and retry reset. Every bounded frontier retains the invariant from independent initial source/environment typing and the primitive contract, even when unfinished. Every observed native result is admitted by the formation-sensitive judgment when the target context is independently formed. Returned functions retain a qualified final machine as a witness; their captured shared cells cannot be recovered from the answer alone. Native-tagged failures and protocol-shaped retries remain possible, but administrative polarity faults are excluded for every bound, not just tested executions. Fault qualification does not certify event provenance: a native primitive may itself produce a protocol-shaped retry reason. Heap typing likewise does not identify a cached answer's denotation with its origin.

`PolarizedNeedPreservationExamples` exercises the combined contract on a shared dependent computation function. Its construction emits one effect, and each of two applications emits a separate effect and returns identity evidence at its supplied native argument. Independent source typing qualifies the general two-argument program. The concrete run returns the two reflexivity proofs, retains the full function body and lexical environment in its actual heap, and records the effects $[3, 4, 4]$. Native admission and exclusion of administrative faults for every bound follow from general preservation, rather than from reducing that single run. A state-retaining consumer extracts the cached body's typing from the preserved final heap, not from the cache's answer tag.

Independent rich source adequacy is now supplied by `PolarizedNeedNaturalSemantics` through `PolarizedNeedNaturalAdequacy`. The proof-relevant source evaluation and forcing judgments inspect source constructors, lexical environments and heap data. They contain no machine-control, path or bounded-execution premise. They cover computation functions, ordinary thunks, native index packets, shared cells, allocation failures, cache ownership and raw polarity faults. An arbitrary primitive interpretation suffices for this correspondence; primitive soundness is needed separately for logical admission.

The consumer interpretation is world-indexed: applying a function can run its saved body after the function itself has been evaluated or retrieved. The forward proof measures finite source and consumer evidence together, not arbitrary program execution. The reverse proof reconstructs source meaning through each actual local or protocol step and begins at a halted endpoint, without assuming a derivation of the initial source. Consequently, source evaluation to an outcome and final world is equivalent to a completed finite machine run with that same outcome and world. Existential fuel gives the corresponding answer-membership statement. The complete receipts, branch path and heap are retained; this does not assert a bijection counting duplicated derivations, a runtime-cost equality or termination.

The admission transfer applies whole-machine preservation along the constructed path. Its heap and answer qualification concern the source derivation's exact final world, so executable cached answers remain coupled to their supporting heap. A native judgment still requires independent source/environment qualification, a sound primitive contract and a formed target context. Source evaluation alone does not manufacture proof evidence. `PolarizedNeedNaturalExamples` reconstructs a source derivation from the shared-function example's actual completed frontier. It retains the cached function and effect projection $[3, 4, 4]$, not $[3, 4]$: shared construction does not memoize later applications. A separate source-derived captured-thunk execution retains an older lexical value rather than the caller's argument; an empty one-step answer list does not refute that finite source derivation.

`PolarizedNeedNaturalEquations` now proves the principal active evaluation equations against that independent source judgment. Ordinary *force of thunk* enters the enclosed body, native- and value-argument lambda application open the corresponding lexical environment, and a native packet opens its index and payload binders. Each is an equivalence for every initial world, raw outcome and exact final world, under arbitrary primitive interpretations. The application right sides are

captured bodies with extended environments, not syntactically substituted source terms.

`PolarizedNeedEvaluationEquivalence` lifts these equations through the same active first-class consumer. Its intermediate answer and world are identical on both sides, so the consumer may apply a retrieved computation function and perform further effects. Independent consumer meaning is equivalent to an actual completed machine run, extending the previous done-consumer adequacy. An effectful function control returns its supplied native argument after recording the specified effect. If one complete source consumer is independently typed and the primitive contract is sound, any completed replacement run retains that exact final heap's qualification and native result admission. This does not infer formation of a replacement source solely from its evaluation behavior.

`PolarizedNeedImmediateDemand` compares shared binding followed by immediate force with the existing allocated returned-value sequencing. Both allocate and demand the same producer, retaining the same final world. The raw result relation must nevertheless retain a projection: a computation function is a valid Need answer but is rejected by the returned-value consumer. Independent source typing at a returner type excludes that mismatch and yields literal outcome equality. Native effectful results and returned thunks provide positive controls; a native computation function distinguishes the two unqualified interfaces. This does not license allocating less or forcing an unused suspension early.

`PolarizedNeedEquationBoundaries` makes the observer boundary explicit. Two actively equivalent producers, one wrapped in ordinary thunk/force, leave provably different retained heap origins when stored in an unused Need cell. Exact source inversion excludes alternative runs with the other final world; exhibiting different possible runs alone would not refute a nondeterministic equivalence. Native and value beta examples also return different literal thunk records after lexical opening versus actual syntactic substitution. Both forms have independent source typing, and forcing each returned thunk produces the same native term in the same current world. That positive probe is not a contextual-equivalence theorem. These are representation-sensitive distinctions, not refutations of beta under a separately justified coarser behavioral observation.

Proposed replacement	Established observation and boundary
Force an ordinary thunk / enter its body	Same completed answer and full world, also under the same active consumer.
Apply a lambda / open its body	Same answer and world with extended lexical environments; not unrestricted syntactic beta substitution.
Unpack a native packet / open its body	Same retained native index, payload, completed answer and world.
Immediate Need demand / allocated value bind	Same world; equal outcomes under independent returner typing, otherwise an explicit polarity projection.
Store equivalent computations as unused producers	Not equality of retained source origins or full heaps.
Allocate a pure return / return directly	Native-answer agreement does not erase allocation, receipt or work differences.

`NativeWireData` and `RawInferenceService` connect ordinary data to selected mathematical checking. The existing CeTTa wire carrier is encoded by native constants and applications under an opaque data signature, with exact decoding and substitution laws. Data formation does not imply that decoding succeeds, much less that a represented proof is valid. A request retains authority/revision labels and its complete goal; a reply retains the candidate article and an untrusted verdict. The service uses the independently selected validated calculus, never a rule table

supplied by the candidate. Exact-request validation replays its real checker and reconstructs a derivation whose erasure is the submitted decoded article. Completeness retains the runtime's explicit closed-constructor payload condition. Independent rule meanings supply the derivation's interpretation.

`PolarizedNeedInferenceService` authors the integration as a source program: check a candidate, bind the actual reply through the owned Need protocol, and submit that bound data to exact-request admission. This uses the existing source semantics and machine, not a second evaluator. The primitive's native result typing asserts only Data preservation; the mathematical conclusion comes separately from the actual checker. Rechecking is deliberate at this untrusted-data boundary and is not a claim of once-only checker execution. A bare admitted-result marker remains forgeable data; only a qualified computation can license its logical use. For every completed execution of this source from a slot-bounded heap, independent source inversion reconstructs the exact verdict and cached-reply world. Thus observed admission implies actual validation of the bound candidate, not merely the existence of an unrelated successful checker run. The exact completed world remains slot-bounded, so another service invocation can reuse it without resetting the heap or history. This bound supplies fresh allocation, not logical admission of arbitrary old cache contents.

`RawInferenceMILWorkload` reconstructs arbitrary accepted relational articles as the existing proof-relevant native hypothesis programs. Refined native admission uses independently formed quotation leaves and context, discharged for the concrete two-premise chain. A bad chain is rejected while a different proof of that exact goal is accepted; a forged acceptance tag cannot bypass replay. The original existential construction remains available, but `MILNativeMaterialization` now supplies an executable alternative for this same learned relation calculus. Generic reconstruction recursively retains every submitted rule instance and ordered child; erasure uniquely determines the reconstructed derivation. Executable rule lookup and interpretation produce the native constructor quotation, independently proved equal to the existing exact-erasure artifact. Success is exactly acceptance by the selected raw checker, and independent quotation formation recovers the refined native judgment. This does not assign semantic meanings to arbitrary new learned rules or combine native rule packages.

`PolarizedNeedInferenceWorkload` connects those general laws to actual source executions. The two-premise MIL chain is admitted and supplies its exact raw native program and relational evidence. The wrong-middle chain is never admitted at any fuel, although the same goal has another accepted proof. Wrong expected requests and stale scopes are rejected at every bound; zero fuel remains an unfinished computation, not a negative judgment.

`PolarizedNeedInferenceFunction` reifies that checker/qualifier source as a computation function reading its supplied native argument. Its independent Data typing and exact completed semantics hold over every supplied lexical environment and slot-bounded world, including malformed wire inputs. The allocated checker origin retains its argument binder and is provably not the earlier closed checker closure. `PolarizedNeedFunctionPassing` proves that passing a function through a value-level caller using an ordinary captured thunk preserves completed outcomes and full worlds, also under a later active consumer. The general law retains faults and possibly dependent result typing; it is not source substitution or memoization of the function's applications. `PolarizedNeedHigherOrderInference` combines these contracts: observed admission after actual higher-order passage yields the executable materializer's native term for the exact submitted article, with separate formation and proof-relevant relational meaning. A two-premise chain has a constructed completed run. Rejected articles, wrong requests and stale scopes cannot be admitted at any fuel; malformed input completes to a distinct malformed-admission datum. Native materialization remains a separate executable Lean operation, not an installed source primitive.

`PolarizedNeedProviderWorkload` exercises a different sharing boundary on actual source functions. Two tagged providers invoke the same fixed checker on their supplied arguments; they differ in their retained per-call events. Sharing the selection through Need yields two correlated provider histories, whereas independent selections yield four, including the mixed histories. Both executions retain the literal pair of raw replies: an accepted mother article and a rejected wrong-middle article for the same goal. All listed frontiers complete. Generic source typing and the existing preservation theorem separately exclude administrative domain faults at every fuel for arbitrary supplied candidates. Thus the shared function is not a constant-reply stub, and sharing its selection does not memoize its two applications. The events count explicit source actions, not checker instructions or verified C costs. The two programs are not interchangeable at this effect observation.

The inherited allocating sequencing still has a fine-observation boundary: sequencing a pure native return through its identity consumer agrees with direct return on the native answer, but performs one allocation instead of zero. Thus this execution is not a proof of the full CBPV equational theory at the retained receipt/work observation. Runtime preservation and exact-world finite source–machine adequacy are distinct from normalization and the remaining value/computation equations. The active equations above do not equate the allocating source bind with the abstract computation monad's unallocated sequencing. Native payloads remain represented terms, not normalized CBPV values. The candidate does not change a default evaluation strategy or adopt a full dependent effect model.

The same audit exposes a limit of the current Lean Mathdata adapter. `MathdataMixedSubstitutionBoundary` checks a term replacement with a free type variable inserted beneath a type abstraction. The raw native substitution captures that type variable, while the simultaneous sorted-ABT operation preserves the outer reference. Both results pass raw type inference and scope checking, and globally closing the complete source under another type binder does not fix the local operation. Exact parity between two implementations of the selected-axis operation therefore does not prove cross-sort binding correctness. The existing Henkin interpretation rejects this type-abstraction syntax; a checked no-erasure result excludes the input from its normalization and proof-soundness premises. Those results remain qualified, not refuted. The external OCaml term substitution has a different traversal boundary at type syntax, so this is neither a counterexample to its admitted documents nor a proof of unrestricted source-port equivalence.

For dependent effects, the distinction is sharper. The result of a computation may determine the type of what follows. Vákár's dependent CBPV framework separates a base fragment from stronger dependent Kleisli extension; the latter is useful but not automatically compatible with every effect [39]. Thus a language should claim dependent sequencing only for the effects for which the required extension and subject-reduction laws are proved.

Definition 11 (Proof-relevant dependent bind)**.** A default dependent bind retains the selected value and its computational receipt:

$$x \leftarrow M;\, N(x) \quad : \quad \sum_{x:A} \mathsf{Receipt}_M(x) \times B(x).$$

An observer may later erase the receipt if B and all downstream operations descend along that erasure.

For a pure fragment and a result observer insensitive to repeated evaluation, call-by-name can serve as a reference for a call-by-need correctness theorem:

$$\mathsf{Obs}(\mathsf{need}(M)) = \mathsf{Obs}(\mathsf{name}(M)),$$

This is a conditional comparison, not a universal identification. Work counts can distinguish recomputation from reuse. Effects and nondeterministic choice can distinguish their results as well: sharing one choice correlates its uses, whereas reevaluating each use independently need not do so [36]. Such fragments require their own sharing-aware reference semantics, ownership, revision and no-double-commit laws. The runtime may omit latent histories only after a proved realization/erasure square says the selected observation cannot distinguish them.

The existing Prime nucleus explicitly includes a revision-keyed need layer. The generic need protocol proves `Trace.from_cachedValue_no_evaluation`, and `GradualExecutionPlan` proves that the first demanded check claims one evaluation while a cached observation claims none. These checked demand contracts do not by themselves prove a complete runtime/CBPV correspondence. An actual-machine connection is now supplied by `NeedDependentService`. A total producer returns a packet in the family indexed by its complete immutable request. The existing owned need machine evaluates and commits it; the resumed consumer checks the packet index before returning it at the requested fibre. Exact frontier equalities give seven protocol transitions for a fresh consumer and five for a cached consumer, preserving the full worlds and receipts. These counts exclude native producer and request-comparison cost. The cached route is independent of a replacement producer, and the existing answer-erasure theorem transfers its exact answer list to the execution machine. A wrong request yields an operational mismatch while the old immutable cache remains unchanged; no automatic revalidation or recomputation is asserted.

`NativeConversionNeedConsumer` instantiates this connection with the actual MIL finite conversion checker. A request fixes the scoped endpoints and revision; the service fixes the authored logical package. The packet retains its proposed code and verdict. Accepted conversion enters native typing only beside an admitted source and independently formed target. An admitted packet can then feed the existing effectful dependent-pair consumer, whose raw results retain the specified native judgment. The controls retain an identity-type conversion across first and cached demands, reject a changed revision or target, and preserve the old checked proof when a new producer proposes a failing candidate at the same request. Candidate rejection is not conversion refutation. Arbitrary externally constructed heaps are not proved origin-correct: first-force construction establishes matching origins, and the consumer independently validates every packet's requested index.

The draft-critical gate is to select the workload-required computation formers and equation observer beyond the proved polarized typing, substitution, preservation and active-opening laws, qualify any further sharing/strategy rewrites, and connect the selected semantic formers to their reified operational consumers. Surface defaults and implicit elaboration must then agree with those rules. This does not require general optimal evaluation or lowering every future effect before the first usable draft.

Warning 12. The planned CBPV direction is not itself a correctness theorem. Runtime migration must be qualified on the supported dependent operational fragment; the current need laws and required observations cannot be silently discarded.

6 Given-clause saturation as a CBPV consumer

The given-clause loop is a useful discriminator because it consumes nearly every coordinate of the envelope. Its values include clauses, occurrence identities, frontier snapshots, inference cursors, and replay receipts. Its computations select a given clause, enumerate partners lazily, simplify, insert, schedule, and emit costs. Nondeterminism, mutable state, fairness, provenance, and budgets are effects rather than fields of one overloaded score.

CBPV role	Given-clause instance
Value	immutable clause occurrence and frontier cursor
Computation	one resumable inference/simplification action
Choice	selection policy among eligible occurrences
State	passive/active stores and persistent inference duties
Receipt	source occurrences, rule, substitution, and target occurrence
Cost	independently composed work valuation
Observer	clause consequences, proof replay, or performance trace

The central implementation rule is to store a lazy frontier of inference duties, not a materialized Cartesian table. Fairness then concerns eventual service of persistent duties. The verified Isabelle given-clause development proves dynamic refutational completeness from static calculus completeness and fair queues for several standard loop architectures [51]. A corresponding implementation should therefore prove frontier-store refinement, unique inference ownership, fairness, schedule replay, and preservation of occurrence identity. Precomputed tables may remain a measured specialization for a finite hot fragment, never the default semantics or compiler protocol.

6.1 Strategy profiles and the guarantee lattice

Search strategies are not one knob. In the saturation framework, inference soundness is a property of the calculus and is invariant under every selection order, simplification policy, indexing scheme, memoization bound, and resource budget; dynamic refutational completeness is instead conditional on static completeness of the calculus together with fairness of the loop [80, 51]. Termination, resource bounds, and fairness itself are therefore properties of a *profile*, and a profile may deliberately sacrifice them—an incomplete but fast heuristic, a bounded-depth probe, a restart policy—without touching soundness. The guarantees form a lattice: soundness at the bottom, unaffected by any profile; completeness above it, conditional on fairness; termination and fairness as profile-selected layers. Every strategy knob should carry the class of guarantee it can affect, so that a consumer of the resulting derivations knows which claims survive the choice.

The checked finite-evidence interface makes the profile-relative negative precise. Finite positive evidence is a monotone certificate: acceptance at a stage is sound and persists under further work, whereas eventual completeness is a separate property that never converts finite non-acceptance into refutation. A profile that abandons fairness may lose proofs; it cannot manufacture one, and its "no proof found" is an exhaustion report that must carry the profile under which it was obtained. This is what licenses aggressive profiles: derivations produced under any profile are valid everywhere, while exhaustion claims are valid only relative to their profile.

Memoization is the same law seen from the store. A memo key is an observer of inputs, and a table keyed by it is correct for a consumer exactly when that consumer descends along the key: equal keys must force equal observations. Under that condition every table reachable by storing is coherent, and restricting a coherent table to any subtable changes no answer, so eviction and memo caps are resource knobs. The checked canary is a compact-syntax key serving a cost-bearing consumer: two inputs with equal erased syntax and different elaboration colour have different cost, the syntax-keyed table returns the wrong cost on a two-input sequence, and the same key is sound for a consumer that reads only the syntax. Soundness therefore belongs to the pair of key and consumer, never to the key alone.

Profiles attach naturally to the execution spaces that own a search rather than to the language. A stable default profile may govern a long-lived knowledge space, while a probe runs in a leased

subspace or a separate process under its own, possibly per-run, profile. Results return to the stable space only as replayable certificates carrying the profile receipt. For a query spanning several spaces the intended composition law is the meet in the guarantee lattice, the weakest participating guarantee: soundness composes unconditionally, whereas joint completeness requires completeness of each part together with fairness of the combined schedule. The meet law and the combined-schedule fairness theorem are design targets; they belong to the comparison-profile algebra listed among the vertical obligations below.

7 A checked vertical specimen: dependent rho communication

The observer law is not supported only by finite Boolean examples. A checked vertical specimen uses an exact typed communication occurrence from the rho calculus. Its retained data include the source and target processes, a strict communication receipt, source typing, target typing up to the established subject equivalence, and a semantic revision. The same object enters the typed communication protocol, the protocol modulo equations, and the established rho `LanguageDef` GSLT. Static process equations, operational occurrence evidence, and typing remain separate components.

For an exact communication e, define the continuation family

$$K(e) = \mathrm{Fin}(\mathsf{revision}(e) + 1).$$

The finite fibres are a cardinality canary; they do not assert that operational history is finite. Revision-zero and revision-one occurrences can have the same visible source and target while their fibres are respectively Fin(1) and Fin(2).

Theorem 13 (Dependent rho observer boundary)**.** *For the checked communication pair:*

(i) the endpoint-indexed strict-communication receipt family factors through endpoint observation;

(ii) the revision-indexed family K does not factor through that observation;

(iii) K is not equivalent to any constant family, hence does not lie in the selected simple-type image;

(iv) dependent sequencing retains each selected communication together with its continuation token in a dependent sum; and

(v) sequencing is natural under the ordered-list-to-occurrence-bag observer, although that observer is not claimed faithful.

Moreover, endpoint observation identifies two distinct retained dependent outcomes and is therefore noninjective.

The same closed occurrence now also crosses the authority and cost faces.

Theorem 14 (Typed communication authority–cost braid)**.** *For the revision-zero communication:*

(i) an exact NIK certificate retaining the strict communication occurrence accepts its literal endpoints and rejects the same certificate paired with a forged target;

(ii) a funded operational Cost1 occurrence forms a genuine one-wave schedule with complete singleton chronology, work one, and span one, and its canonical rho erasure reaches the exact endpoint equivalence classes;

(iii) independently, the declaration-generated Cost1 source and exhausted target are both well sorted, and the authored whole-redex rule reduces the former to exactly the latter under ordinary kernel evaluation;

(iv) erasing the two generated static colours from their computational cores recovers the literal typed communication endpoints; and

(v) the complete generated term inhabits the selected Cost2 displayed fibre, where its provenance key is exact although the global compact Cost2 key is not exact.

The runtime-resource Cost1 algebra and the declaration-generated Cost1 language are not thereby identified. The theorem gives explicit comparison data for this closed channel and proves that both executions meet at the same rho endpoints. A general two-sided comparison remains an obligation.

This theorem narrows what may be erased, not what may be constructed. A set or HOTG model can encode the whole dependent sum when its size hypotheses hold. What an endpoint-only extensional quotient cannot do is recover a family that varies over a revision it has discarded. Dependent type theory makes the indexing and transport computationally native; an extensional host makes the total space and its models mathematically available. The two sides therefore have complementary strengths rather than an expressibility winner.

The specimen is a discriminator, not yet an end-to-end language theorem. The remaining braid must give this same occurrence an explicit dependent-CBPV polarization, a concrete universe-enclosure/HOTG interpretation, admission by the plural NIK dispatcher, and a two-sided CeTTa realization. The constant-family control and the revision-dependent negative result must survive every candidate DTT profile.

8 Observer-indexed non-well-founded worlds

8.1 The visibility container

Let C be a type of observers and let $V(c,d)$ be the type of visible routes from c to d. For a C-indexed family X, consider

$$(H_V X)_c = \mathcal{P}_{\mathrm{s}}(X_c) + \sum_{d:C} V(c,d) \times X_d. \tag{1}$$

The first summand contains small same-observer collections. The second contains *opaque references*: c retains the identity of an object at d and the route by which it is visible, but local membership does not recurse through that reference. Using a route type rather than a mere proposition preserves multiplicity and provenance.

Equation (1) is meaningful only relative to an ambient category and a notion of small map. Van den Berg and De Marchi prove an indexed final-coalgebra theorem for small-based functors in a Heyting pretopos with a suitable class of small maps, and instantiate it for the small-powerclass functor [52]. Polynomial M-types provide a related final coalgebra theory for non-well-founded trees [53].

Obligation 15 (Existence of the observer-indexed coalgebra)**.** To infer νH_V from the indexed final-coalgebra theorem, prove:

(i) the observer object and every route fibre used by a stage are small;

(ii) sums, reindexing, and $\mathcal{P}_{\mathrm{s}}$ form an endofunctor on the selected indexed category;

(iii) H_V is small-based and has the required weak-pullback behaviour; and

(iv) the ambient small-map axioms support the chosen anti-foundation profile.

Until these are instantiated, νH_V is conditional notation.

8.2 Why the unrestricted set-level construction fails

If $\mathcal{P}_{\mathrm{s}}$ in (1) is the full powerset on ordinary sets and a final coalgebra has set carrier E, Lambek's lemma would make its structure map an isomorphism, forcing $E \cong \mathcal{P}(E) + \cdots$. Cantor's theorem forbids this. The repair must be explicit: use a bounded/finite powerset, move to a category of classes with small maps, or place the carrier in a strictly larger universe. An opaque reference prevents foreign membership exposure; it does not repeal Cantor's theorem for the local powerset summand.

Finite immediate support also does not support literal all-to-all reflection over infinitely many observers. If one observer has a finite list covering every other observer, adjoining the observer itself gives a finite surjection onto C, so C is finite. This obstruction is now machine checked. It does *not* make infinite reflection finite: a system with one successor reference per observer has finite local branching and realizes every finite reflection depth. Infinite depth and infinite immediate width are different commitments.

For finite powerset, the final sequence has its own transfinite subtlety; coherent finite views need not be exhausted at the first ω stages. Worrell's final sequence analysis and Abramsky's limit-completion account of finitary non-well-founded sets are therefore relevant acceptance references [56, 55].

8.3 Route-rich and extensional coalgebras

Let F_V be a route-rich functor retaining occurrence, provenance, revision, or cost labels, and let

$$\varepsilon : F_V \Longrightarrow H_V$$

erase selected labels. If both final coalgebras exist, then $R = \nu F_V$ acquires an H_V-coalgebra structure by composing its destructor with ε_R. Finality of $E = \nu H_V$ yields a unique coalgebra morphism

$$q : R \longrightarrow E. \tag{2}$$

Thus the extensional readout is derived, not postulated. However, surjectivity of q and the non-collapse canary

$$\exists x \neq y, \quad q(x) = q(y) \tag{3}$$

do not follow from finality alone. They require explicit realization and tagged witnesses. Different observers may use natural transformations $\varepsilon^o : F_V \Rightarrow H_V^o$, producing a family of readouts q_o rather than one universal equality.

Proposition 16 (Route-rich utility test)**.** *The route-rich carrier R earns semantic status only if there is a declared consumer D and $x, y \in R$ such that $q(x) = q(y)$ but $D(x) \neq D(y)$, or an equivalent dependent-family obstruction to descent. If q is an isomorphism, or every admitted consumer factors through q, then R is observationally redundant and supplies no additional semantics or foundational strength.*

A top-level hosting discipline consequently retains a compatible tuple of profile, presentation, residual computation, observed class and material reading. Each coordinate is linked by an actual representation relation; an arbitrary product of independently chosen coordinates supplies no such coherence. Hosting constructive and classical profiles requires separate logical commitments and certified interpretations. It does not combine incompatible object axioms into one untyped theory or identify the operational, intensional and extensional carriers. A finite checker can check a bounded artifact while its declared denotation is infinite or cyclic; the checker's finite run does not set the denotation's cardinality.

The intended consumers are operational rather than consistency-theoretic: provenance-sensitive replay, route-dependent continuation types, independently valued costs, revision control, and explanations that distinguish duplicated occurrences. The R-construction does not by itself strengthen the object logic, prove a consistency statement, or justify a larger universe.

8.4 A constructed final coalgebra: circular propositions

Equation (1) remains conditional notation; the following final coalgebra is constructed. Fix a type B of base facts and let Σ_B be the polynomial functor with one node for each base fact, a truth node Tr, negation, conjunction and disjunction, with arities 0, 1, 1, 2 and 2. Its final coalgebra $\mathsf{Prop}_B = \nu\Sigma_B$ is built as an M-type; no well-foundedness condition is imposed, so a circular proposition may contain itself. $\mathsf{Tr}\,p$ reads "p is true".

A *system* of self-referential judgments assigns to each judgment j a strong Kleene formula $\mathsf{spec}(j)$ whose atoms are base facts and truth claims $\mathsf{Tr}\,j'$ about judgments. Finality gives each system exactly one unfolding map into Prop_B that commutes with one step of unfolding (`System.unfold_unique`). This is the solution lemma of anti-founded set theory [54] for this signature.

Prop_B is itself the judgment set of one *universal* system: the specification of a proposition is its root connective applied to the truth of its parts. Grounded truth is a property of propositions, not of presentations. A formula holds or fails in the least fixed point of a system exactly when its unfolding does in the universal system (`System.unfold_mem_least`), and morphisms of systems preserve and reflect grounded verdicts (`System.Hom.mem_least_iff`). The universe is therefore terminal for the semantics as well as for the structure: every presentation maps into it by one map, and that map neither creates nor loses a grounded verdict.

The collapse is a collapse of meaning, not of objects. Two liars, each calling the other false, unfold to the same proposition as one liar calling itself false (`Controls.mutualLiars_unfold`). The unfolding map is the readout q of (2), and it has the collapse of (3). Literal self-reference exists and is unique: the self-negating proposition is the unique p with $\neg p = p$, and the falsity liar is the unique p with $\neg\mathsf{Tr}\,p = p$ (`selfNegation_unique`, `falsityLiar_unique`).

Over encodable base facts the universe reads faithfully into hypersets (`material_injective`). Grounded truth and grounded falsity are then hypersets of the same universe, the truth schema and the negation schema hold, and no proposition lies in both (`hypersetUniverse`). Groundedness is a property of evidence, not of shape. The proposition "the fact, or this proposition" is circular and grounded-true (`orSelf_held`), so the truth hyperset is not well-founded (`Controls.truthSet_not_wf`).

These are three faces of one meaning. Stages of the jump are the operational face, well-founded evidence is the intensional face, and the fixed point is the extensional face. All three agree on every verdict (`System.mem_least_iff_exists_stage`, `System.mem_least_iff_evidence`), and evidence of height n is exactly a verdict reached at stage $n+1$ (`System.mem_stage_succ_iff_evidence`). The comparison maps are not injective: two pieces of evidence for one verdict differ (`Controls.twoEvidence_ne`), and different stages yield the same verdict.

Status. Proved. Choice enters through the M-type and the classical dual; the checked results use at most `propext`, `Classical.choice` and `Quot.sound`, and several use none. Nothing here constructs the indexed coalgebra νH_V of the obligation above. The universe is propositional: it has no quantifiers and no provability predicate.

8.5 Truth over the circular universe is partial

Grounded truth is Kripke's least fixed point in the information order under strong Kleene evaluation [116, 117]. It is consistent (`System.least_consistent`), and it leaves both liars undetermined (`selfNegation_undetermined`). Its status map sends negation to Kleene negation and the self-negating proposition to the Kleene fixed point, *undetermined* (`groundedStatus_neg`, `groundedStatus_selfNegation`). The greatest fixed point is circular justification: with bivalent base facts, it holds a judgment exactly when some sound valuation does (`System.mem_greatest_iff_sound`). There the liar is self-justifying in both polarities (`System.liar_selfJustifying`). For finite systems the least fixed point is reached within $2|J|$ jumps and is decided by the kernel (`System.stage_eq_least`, `System.groundedDecide_spec`).

The universe cannot carry a stronger truth notion. Each of the following is proved.

- *No two-valued meaning.* No Boolean valuation of circular propositions commutes with negation (`no_boolean_valuation`). This is the negative reading of Lawvere's diagonal argument [118], with the fixed point supplied by finality instead of by a code map.
- *No classical deduction.* Every Hilbert system over circular propositions that has modus ponens and the axioms K and S for the material conditional proves every proposition (`hilbert_trivial`). The proof is Curry's paradox [119], obtained as Löb's theorem for the identity provability operator (`ProvabilityStage.curry`). Grounded truth escapes only because it is not classical: axiom K is not grounded at the self-negating proposition (`least_not_axiomK`).
- *The gap is invisible from inside.* No strong Kleene formula holds exactly at the valuations that leave a given judgment undetermined (`System.undetermined_not_definable`). The least fixed point holds neither the liar nor the true sentence "the liar is not true" (`System.liar_revenge`). That sentence is stated one level up. This is the revenge phenomenon recorded at the end of Kripke's paper.

Consistent self-reference in one universe is therefore partial. Coded self-reference behaves differently. In arithmetic the diagonal reaches only the provability predicate, classical deduction stays consistent, and the price is incompleteness (Section 12.2). Both routes feed the same abstract theorem (`ProvabilityStage.loeb_of_hasLoebSentences`). Final coalgebras answer which cyclic objects exist. They do not answer which propositions about those objects are true: a cycle can exist as data while truth about it stays partial.

Knowledge-based programs. A one-round knowledge-based program translates to such a system of judgments, "agent i does a in local state ℓ", with knowledge evaluated in strong Kleene logic over the worlds an agent cannot distinguish. Implementations are its two-valued fixed points, every finite stage of the grounded inquiry is below every implementation, and a decided inquiry has exactly one implementation (`KnowledgeBasedPrograms.implementation_fixed`, `stage_sound`, `implementation_eq_of_decided`); this is the one-round case of the constructive interpretation of Knapp, Mühlberger and Reus [154], who ask for criteria that ensure a decided least fixed point. One criterion is a rank that strictly decreases along every reference to the truth of a judgment.

With bivalent base facts, a ranked system decides every judgment of rank below n at stage n, so its least fixed point decides every judgment and a ranked program has exactly one implementation (`RankedDecision.stage_decides`, `least_decides`, `ranked_implementation_unique`). "Act if I know I will act" refers to itself, is not ranked (`RankedDecision.Controls.teller_not_ranked`) and is undetermined. The multi-round case over reachable states is not covered.

9 HOTG as host, not as hyperset membership

An arithmetic library illustrates how a rich ambient construction can preserve earlier identities. Brown et al. first construct finite ordinals and ω, then choose a surreal-number representation in which ordinals remain those same sets. The resulting number domains satisfy literal inclusions $\omega \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$; all surreals form a proper class, not a further set. Agreement of the arithmetic operations is proved separately [94].

The transferable design requirement is compatible extension of representations and operations, not that each small calculation must invoke the whole ambient theory. Even shared elements do not identify every operation: natural-number truncated subtraction sends $(0, 1)$ to 0, whereas field subtraction sends it to -1. A common domain therefore needs explicit operation restrictions and their agreement laws. Similarly, smooth STT–DTT transport must distinguish literal reuse of a term from preservation of typing, judgmental conversion, propositional identity, proof fibres and computational cost. An internal model of a calculus is not, without an adequacy and recovery theorem, an inverse translation of that calculus.

Higher-order Tarski–Grothendieck set theory supplies a powerful well-founded ambient universe and a precise Grothendieck-universe operator. Brown and Pąk formally relate higher- and first-order formulations of Tarski–Grothendieck set theory and construct the corresponding universe principles [57]. This makes HOTG a serious host for the coalgebraic programme.

It is nevertheless essential to distinguish two membership relations:

$$\in_{\text{host}} \qquad \text{and} \qquad \in_{\nu} .$$

The first is the well-founded membership of HOTG. The second is the destructor induced relation of an internally constructed final coalgebra. A quine-like element $\Omega \in_{\nu} \Omega$ does not violate host Foundation because it is not the assertion $\Omega \in_{\text{host}} \Omega$. HOTG and anti-foundation can therefore coexist as host and interpreted object; identifying their memberships would be inconsistent with the intended host.

Universe shifts must also be priced. If the stage-n observer family and container are small in a Grothendieck universe U_{n+1}, construction of their final coalgebra may require the category of U_{n+1}-small objects or even a further host universe, depending on the existence theorem. "Every set lies in some Grothendieck universe" is a completed-totality principle. Open-ended staged ascent is a different principle: no largest stage is asserted, and each finite obligation is discharged later. Neither should be silently substituted for the other.

10 Universe services for reflective reasoning

10.1 A type-theoretic analogue of universe selection

A literal Grothendieck universe is an extensional set-theoretic object. It is therefore a category mistake to ask an intensional dependent type theory to contain that very object without first supplying an internal set interpretation. The useful design question is instead whether dependent

type theory can recover the same *one-motion ergonomics*: start with the family currently under discussion, move to a universe containing it, and continue using the ordinary closed constructions there.

The natural generator in dependent type theory is not a single untyped set but a dependent family

$$A : \mathcal{U}_i, \qquad B : A \to \mathcal{U}_i.$$

A *family-enclosing Tarski universe* over (A, B) consists of codes U, a decoder $T : U \to \mathcal{U}_i$, codes for A and every $B(a)$, and code constructors closed under a declared collection of type formers. The useful profile here contains 0, 1, $\mathbb{N}$, binary sums, dependent products, dependent sums, identity types, and W-types. Decoding laws are equivalences, not definitional equalities; this avoids silently installing equality reflection or a particular conversion algorithm.

A *family-enclosing universe operator* is then a uniform operation

$$\mathsf{Enclose}(A, B) = (U_{A,B}, T_{A,B}). \tag{4}$$

Palmgren introduced precisely this universe-above-a-family idea and distinguished it from a superuniverse which is itself closed under the universe operator [72]. Equation (4) is the clean type-theoretic counterpart of "choose a Grothendieck universe large enough for the present data." It is not an assertion that types are sets, nor that there is one completed universe of all types.

The current formal development establishes the following capability-level result.

Theorem 17 (Predicative family enclosure)**.** *At every ambient type level, every dependent family has a family-enclosing Tarski universe with the closure profile above. Its* Π- *and* Σ*-closure instantiate the common universe-capability interface. A nonconstant family with fibres* 1 *and* 2 *is enclosed as a positive dependent control. The ambient code carrier cannot code itself at the same level.*

That capability-level witness uses the next metatheoretic level as its code carrier. A separate generated construction strengthens it. For a family (A, B) at level i, define a strictly positive, type-indexed grammar

$$\mathsf{Gen}_{A,B}(T) : \mathcal{U}_{i+1}, \qquad U_{A,B} = \sum_{T : \mathcal{U}_i} \mathsf{Gen}_{A,B}(T), \qquad \mathsf{El}(T, g) = T.$$

Its constructors retain the base and fibre generators and the complete formation derivations for $0, 1, \mathbb{N}, +, \Pi, \Sigma, \mathsf{Id}, W$. Dependent constructor premises range over their decoded domains; the grammar therefore admits arbitrary nonconstant families of previously generated codes. The code carrier consists of generated derivations, rather than the entire ambient type universe with its identity decoder.

Theorem 18 (Generated enclosure and material decoding)**.** *The generated grammar encloses every generating family. Substitution of generators preserves its decoding and satisfies identity and composition laws. Successor embeddings have coherent decoding comparisons for* $\Pi, \Sigma, \mathsf{Id}, W$*, including their application, projection and tree operations.*

For arbitrary material sets $X : \mathsf{HSet}_i$ *and material families* $B : \mathsf{El}(X) \to \mathsf{HSet}_i$*, the dependent core grammar generated by their actual member types has a constructed material interpretation. Every derivation built using* $0, 1, \Pi, \Sigma, \mathsf{Id}, W$ *receives a graph at bound* $i + 1$*, an explicit inverse member decoder, and uniform term graphs. These constructions include arbitrary nested nonconstant families and W-types with arbitrary dependent position carriers. Their complete interpretation commutes with substitution. The core code carrier lies at type level* $i + 2$*; its interpreted type values form a material enclosure at graph bound* $i+2$*. Raising the original material family preserves interpreted values through proved decoder equivalences and embeds the lifted enclosure into the enclosure generated over the raised family.*

The constructions are checked in the general type-theory and hyperset library. They require no selected Prime or Megalodon theory, no chosen representatives of quotient classes, and no independently supplied section carrier. The material core theorem does not assign host-valued decoders to the broader grammar's natural-number and binary-sum conveniences. Graph labels that distinguish natural numbers are a different capability from a decoder that recovers a host natural number from an arbitrary material member.

The material interpretation uses discrete identity fibres and their dependent eliminator. This is a stated model profile, not a global rule imposing UIP or equality reflection on native identity. The separate presentation-groupoid comparison retains automorphisms and proves dependent elimination for natural motives; its occurrence-sensitive controls distinguish identity witnesses that material equality forgets. An actual functor from presentation isomorphisms to discrete material values preserves dependent elimination for coherent observed motives. The material member family is nonconstant, but every loop acts trivially after discrete descent. The occurrence family has a nontrivial loop action and therefore cannot descend, even through a natural equivalence of fibres. This is a proved obstruction to erasing identity evidence, not a global axiom of equality reflection.

The discrete model's successor-site comparison also retains the complete two-endpoint and identity-witness context. Dependent J, its computation rule, naturality and commuting substitution are checked for arbitrary natural motives at the raised site. Actual lifted lower material motive dictionaries have corresponding J-value and inverse-member decoder laws. The latter comparison concerns those translated dictionaries; it does not assert descent of every arbitrary upper material motive.

The contextual construction retains the complete displayed family and its material dictionaries. Its dependent products observe every future context and actual arrow; its inductive trees satisfy hereditary naturality across those arrows. Substitution of full families, and W-type comparison across actual base-presheaf substitutions, are checked with decoder, restriction, constructor and fold comparisons. Equal present carrier sets cannot be used as complete universe codes: the growing family and the unit family have the same carrier now and different carriers along a real future arrow. This obstruction survives universe lifting. The full-family code quotient instead has proved substitution equations and a natural family decoder. A constructor-closed quotient now descends $\Pi, \Sigma, \mathsf{Id}, W$ through proved constructor congruence and dependent quotient elimination. Its decoder retains the complete family rather than only present carriers. The equations preserve formation recipes: two codes can decode to identical complete families and still remain distinct. Fresh formation after a substitution is compared through actual natural member equivalences and material-value laws; semantic agreement does not by itself identify codes. In particular, a noninjective context substitution can change the labels of freshly constructed W-leaves. Their values need not equal those of trees transported from the original construction.

The successor construction now closes a new code grammar using previously constructed lower codes as its cumulative seeds. Independent upper $\Pi, \Sigma, \mathsf{Id}, W$ formation remains available, with constructed dependent comprehension comparisons to lifted lower formation. A seed-sensitive trace proves that the cumulative code embedding is injective, including for distinct codes with identical complete decodings. Two actual successor sites and code grammars are constructed by re-instantiating the seed, model and arrow data; their members, natural sections, restrictions, decoders and substitutions commute with the two successive lifts. Fresh upper recipes need not be cumulative seeds. Recursive lifting of a lower generation derivation separately rebuilds every constructor from primitive seeds and transports dependent bodies through the constructed comprehension equivalences. These are actual constructions at each displayed successor level, not iteration of an assumed enclosure operation. They do not provide a term-indexed sequence of variable host universe levels or internal transfinite closure.

Bare material carriers with authored member restriction maps also receive a constructed contextual interpretation. Uniform membership graphs present every lifted carrier, and an explicit inverse member equivalence builds its decoder without selecting an original graph representative. The site and graph bound rise by one level. Arbitrary bare dependent bodies over the actual raised comprehension support generated $\Pi, \Sigma, \mathsf{Id}, W$; they need not be constant along the projection. Complete family substitution and inverse natural-section comparisons are checked. Original-level smallness of each individual fibre must still be distinguished from original-level smallness of the entire family-parameter space.

The general authored grammar also has a constructed mixed-bound material enclosure. With context bound u and external parameter bound $\max(u, v)$, its actual generated carriers are collected at graph bound $\max(u + 1, v + 1)$. The inverse decoder recovers a native term from an enlarged member of a retained code, and the whole-section decoder retains every restriction map. Its material comprehension is naturally isomorphic to the pullback of the constructed universal small family. Classification, decoding and parameter substitution commute as whole natural maps. Further maximum-bound embeddings preserve and reflect membership and agree with direct embeddings. An infinite observed-family control has equal present carriers and different future support, even after enlargement. These enclosures therefore retain their codes and dictionaries; a bare present material carrier is not a decoder for the full family or its formation recipe.

The contextual power construction retains a stable predicate on every future context, actual arrow and argument. Its existential direct image and inverse image satisfy an order adjunction; an explicitly constructed argument-family pullback satisfies the complete-future Beck–Chevalley equation. Singleton and union are natural in both context and argument, with both unit laws and associativity. For arbitrary small parameter families B and argument families A, natural maps $B \to \mathcal{P}(A)$ correspond, with constructed inverses, to stable predicates on the actual product $B \times A$. This correspondence commutes with parameter substitution. Faithful material labels for all future arguments supply the full member decoder. Whole material values are preserved when those labels are transported; independently fresh labels preserve decoded truth through an equivalence. Actual dependent substitutions construct inverse future-argument comparisons and inverse natural-section comparisons, including at the raised site.

Infinite labelled-history controls separate the full contextual power from its present subset, even for natural classifiers with an actual parameter family. Two predicates can have the same empty present part and different future material values. A context functor that omits a future can destroy injectivity; arbitrary context functors therefore require their stated coverage and lifting hypotheses. These power constructions concern interpreted small argument objects. A small powerclass of arbitrary large objects, its quotient and collection properties, and indexed finality remain distinct construction obligations.

For argument objects at an independently larger host level, a further construction retains stable predicates with small receipt covers at every complete future index. The existence of a uniform receipt family is part of the admitted predicate, not a choice of its enumeration. Restriction and direct image construct the new covers; identity, composition and the natural singleton map are checked. On originally small argument families, the actual truth subtypes construct an inverse comparison with the full contextual power. The output bound includes both the context level and the argument level, while its receipt bound stays fixed.

This larger construction has a discriminating size obstruction. Any surjective small cover of all hypersets would construct a small membership graph whose decoration and range form a universal hyperset, contradicting separation. Every individual singleton is covered, but the full ambient predicate is not. Its occurrence as the inverse image of a covered terminal singleton proves that arbitrary inverse image need not preserve these small covers. An independently raised infinite

growing family has infinitely many distinct covered future predicates with identical empty present truth, so the larger construction does not collapse to its present argument carrier.

The larger covered power also classifies relations with independently larger parameter objects. The stable relation is a predicate on the actual parameter–argument product; for each retained parameter it has a small enumeration of all related future arguments. Constructed inverse maps identify these relations with natural maps into the covered power. Both full inverse laws and arbitrary parameter substitution are checked. The ambient membership relation is classified by the identity. No enumeration is selected from its propositional existence. These laws classify this precise future-enumerated relation class, rather than every map with merely pointwise small fibres.

Arbitrary natural observations between larger argument families have constructed pointwise kernel quotients, image factorizations and natural universal descent. Effective kernel-pair descent persists after arbitrary argument pullback; dependent quotient elimination builds the resulting factor without choosing a representative. Images have an actual pullback comparison with inverse maps. Supplied small fibre enumerations transport through pullback, and through surjective commuting triangles, by explicit maps of their original receipts. The latter law requires surjectivity: an authored map omitting a tag also omits the corresponding fibre. Duplicate receipt codes remain permitted in a cover and do not provide a decoder.

The covered power sends an argument pullback to a weak pullback. For two covered predicates with equal full future images, filtering the product of their small receipts constructs the stable relation of all matching pairs. Its projected images are the original predicates. This complete matching relation supplies a natural right inverse to the canonical power comparison, without selecting an individual match. It does not make that comparison injective. In the infinite growing tagged family, the diagonal and opposite-tag relations are distinct, while both of their projected images are the full source. The complete matching relation retains both kinds of pair, and new pairs become available at every stage.

Actual small generators and reachable subcoalgebras are also constructed from a contextual coalgebra with authored uniform future-enumeration data. A generator stores finite paths of context moves and admitted future branches, together with its terminal context arrow. Its endpoint map covers precisely the reachable values. Filtering those small generated receipts constructs every successor cover of the literal reachable subfamily. Both the surjective covering map and the injective inclusion satisfy the complete future-coalgebra equations. Observed material-member controls admit a Quine-atom child from an initially empty value, while different context and branch histories can have the same endpoint. This constructs genuine reachable subcoalgebras; it does not supply a selected inverse to their covers or uniform enumeration data for every merely covered coalgebra. Arbitrary small-index union, universal small-map representability, Collection and indexed finality remain further construction obligations.

The indexed covered power is constructed in each argument slice. Its support law retains the base parameter at every complete future index. An argument pullback has an explicit comparison with the pullback of the indexed power projection: direct image supplies one direction, and the reverse predicate retains the saved parameter equality. Both whole natural inverse laws are checked. Identity and successive base changes agree through the constructed argument comparisons, including for noninjective parameter maps. Present-only support does not imply this future support law.

For any covered contextual coalgebra, the union of all stable future-matching bisimulations is itself a bisimulation. Its actual quotient has a natural coalgebra, an exact bisimulation kernel, and no remaining bisimilar distinct values. Complete matching also constructs a coalgebra on the argument pullback of two maps to a common target; both projections satisfy the whole future-image equations. The matching coalgebra need not be a unique choice on that pullback.

A faithful material encoding has the same contextual-bisimulation kernel. It constructs a small labelled graph with distinct context-transport and child-transition tags, retaining the actual source, target and arrow. Deterministic identity-context edges force related flat states to share their context, and the other context edges recover the stability law. Authored world and arrow graph dictionaries construct every material label; no dictionary for argument values is required. Material rows recover the context action and future child values, and full coalgebra maps preserve these readings across different small source systems. The infinite labelled-path site supplies actual faithful dictionaries and distinguishes parallel histories and arbitrarily many result labels. The collected material range has coverage and an injective quotient reading; existential coverage alone is not claimed to select an inverse into the raw quotient type. Readings distinguish contexts, so this encoding is not a natural map into constant ambient membership.

Declared atomic readings also have actual material rows, distinct from context actions and execution children. This applies even to states with no execution children. The resulting equality kernel is precisely a single stable complete-future bisimulation preserving every declared reading. Ordinary future bisimilarity together with agreement of only present readings is insufficient: an infinite contextual control agrees now but exposes a different result later. Whole coalgebra maps with exact atomic observation laws preserve both the material value and every declared modal formula; mere forward simulation fails when the target introduces an unmatched child. Quotient elimination constructs the observed quotient coalgebra and its atom valuation, with exact compatible operation factorization and uniqueness.

Native predicate classification now uses that actual observed quotient, rather than an assumed observer package. Its full sieve classifier factors through the quotient if and only if the predicate is invariant under the same stable, atom-preserving future bisimulation. The factor is unique as a whole natural map. Pulling an actual displayed family back from the quotient gives precisely the same native support image. This support statement does not select a dependent section. Infinitely many declared result predicates give distinct classifiers, while a stable predicate reading an erased receipt tag has no factorization. Ordinary future bisimulation alone loses the result predicates on terminal states.

An authored two-sided history category now supplies an infinite concrete contextual instance. Its arrows retain ordered event scripts, injective renamings and parallel frames; composition renames the earlier script before adjoining the later one. The observed quotient retains declared results, faults, costs, pending direction and occurrence receipts. For its exact profile, the readout kernel is equality of fresh configurations; coarser profiles have their own proved kernels. Successor diamond descent is characterized by outgoing matching modulo that kernel, and predecessor box descent by incoming matching. Context insertion has both forward lifting laws, but source-back lifting requires an empty inserted frame, and target-back lifting additionally requires a surjective renaming. Thus adding a parallel frame does not automatically preserve modal observations. A clamped state cost fails context congruence; retaining the unclamped potential or restricting to neutral frames gives distinct proved repairs. The unbounded authored dynamics never reach a finite stabilization stage. Bounded certificates constrain the dynamics, rather than merely capping observation depth.

For an actual small observed source, its authored execution graphs construct a material member family. Stability of the observed kernel proves that transporting all representatives produces a singleton before union computes the value. The collecting graph supplies an actual original-bound class face, inverse member decoding and inverse maps on compatible whole sections. Matching future children constructs its covered coalgebra at that same branch bound, with the full observation square. An infinite result-indexed control has genuinely growing small dependent fibres over these material members: each actual fibre is compared with its independently formed finite interval, and

its full future code decodes the whole family. A newly admitted member is never an image of an old member. Erased provenance has no factor through the material observation. This construction uses the actual source bound; it does not turn existential coverage of arbitrary wider sources into a chosen occurrence inverse.

Over this actual material-indexed family, two constructed dependent products agree on every current application but differ when a later argument becomes available. Their distinction is proved by transporting the whole products and evaluating the newly admitted argument, using the actual adjunction computation rule. The class-level atom valuation is also constructed: its observed material readout equals the member decoder and is injective. The stronger observed separation belongs to that valuation together with the future coalgebra, not to the child transition structure alone.

At a fixed small context bound, the coproduct of all actual small argument functors and their covered coalgebras lives one host universe higher. Its bisimulation quotient receives exactly one coalgebra map from every originally small source. Common-target pullbacks prove that two small presentations with the same execution endpoint have the same canonical reading. Consequently, a larger source with authored uniform branch enumerations has a constructed natural readout: read each value through its generated small rooted coalgebra. Its full coalgebra equation and exact contextual-bisimulation kernel are checked, and changing those enumerations does not change the readout. This does not select uniform enumerations from their pointwise existence or prove finality in every slice. Infinite result labels remain distinct despite empty present child predicates, while a one-loop and a two-cycle agree without recovering their distinct raw receipts.

The parameter-relative quotient retains strictly more than the ordinary behavioral quotient. For an actual coalgebra over a parameter map, its support square forces every future child to lie over the parameter transported along that same arrow. The conjunction of contextual bisimilarity and equal parameters is itself a stable bisimulation. Pairing the parameter with the ordinary quotient gives exactly this kernel; quotient elimination constructs both its future coalgebra and its parameter map. On an infinite growing source, ordinary bisimilarity can identify every natural-number parameter while this quotient retains all of them.

An actual small-source recipient is constructed in each independently larger parameter slice by pairing the retained parameter with the universal small-coalgebra reading. Its children pair an actual behavioral child with the uniquely transported parameter, retaining the original branch bound. Every originally small indexed source, and every wider source with authored uniform branch data, has a unique whole coalgebra map over the parameter. Its exact kernel is contextual bisimilarity together with equal parameters. For arbitrary parameter substitution, explicit inverse natural maps identify this recipient with its literal pullback. The pulled coalgebra is built independently using indexed-power base change; both comparisons satisfy the complete future and indexed coalgebra equations. Whole sections, identities and successive substitutions agree. A noninjective substitution can merge parameter tags in its behavioral map while the dependent pullback retains both tags and their future support.

The same universal small-source behavior also has a constructed material carrier. Each small code supplies its own faithful labelled graph at the original bound. Their collecting graph is one host level larger, with actual material members one type level above that graph. Cross-code equality has exactly the coproduct's contextual-bisimulation kernel. Transport ranges over all representatives and proves its material row to be a singleton before union computes the result. Full occurrence-fibre predicates give an actual smaller class face with inverse material-member decoding. Its representative-independent future-child predicate has a cover at the original branch bound, and both faces satisfy the whole observation equation and behavioral separation. Small-source and authored-enumeration wider-source maps, exact kernels and enumeration independence

are constructed. These are contextual material children with their actual future index, rather than an identification with constant ambient membership. No inverse into the raw quotient or uniform branch enumeration is selected from existential coverage.

Under an explicit external host-Choice profile, uniform branch enumerations are selected and the generated readout receives every covered source. The ordinary recipient is a genuine final coalgebra of the covered-future-power endofunctor in the ambient fibre level $\max(u+1, v)$; its structure map has the constructed Lambek inverse. The material class recipient is independently final, and the two universal maps give an actual isomorphism. The indexed covered power is also an actual endofunctor on the ambient over-category, with independently constructed raw and material terminal coalgebras over every parameter. Their exact kernels retain both contextual bisimilarity and equal parameters. The receipt bound stays u, while collecting graphs, material members and class codes retain their distinct levels. The dependency ledger exposes both the direct branch-enumeration chooser and the external Choice dependency of the common over-category interface. These are optional host-model results; they neither select a native foundation nor establish a full internal set theory from tagged graphs.

The optional model now also has constructed internal membership and set operations on that same varying final carrier. Extensionality quantifies over every actual future; equality of present member sets is insufficient. Empty set, singleton, pairing, union, arbitrary stable proposition-valued separation and infinity have natural constructions and exact future-member laws. Every stable graph with receipt-small edge projections has a unique natural decoration. This is a stated anti-foundation interpretation of that graph class, not an identification with well-founded host membership.

Powersets retain stable duplicate-saturated predicates on the parent's actual future receipts. Their original-bound code carrier has an inverse comparison with all future subsets. The assembled power set contains exactly those subsets at every future. A singleton already has infinitely many distinct internal subsets whose current members are all absent; present-only subset tests therefore do not characterize this power set. For Strong Collection, actual member projections and the constructed covering diagram supply small witness rows. Their values assemble an actual collecting set satisfying both clauses at every future. No natural witness function on the original parameter family is assumed. A relation total only now can fail to admit any all-future collector. These internal axioms are proved individually; they do not replace the separate generic small-map powerset axiom.

Generic powerclass smallness is proved separately for arbitrary wider source and target objects. An inverse-image receipt retains both an original-bound target receipt and an original-bound map-fibre receipt. Stable, identity-saturated predicates on these receipts classify every full future subpower; filtering their codes by the exact image equation constructs the actual fibres of the power map at the original bound. The route from authored uniform enumeration data uses no selected inverse. The optional external profile obtains these data from pointwise smallness using explicitly recorded host Choice. An infinite tagged source has distinguishable future subobjects with the same current members and the same image under the power map; its small power-map fibres do not shrink the wide source itself.

The interpretation also validates a fully specified intuitionistic first-order deduction system over the same actual set family. Its implication and universal quantifier inspect every future arrow; its existential quantifier supplies a witness in the current fibre. Variable substitution, binder lifting and transport of assumptions are proved for arbitrary formulas and proof trees. Arbitrary interpreted formulas induce stable predicates, so the independently constructed separation and Strong Collection operations validate the corresponding formula schemata. The interpreted extensionality formula uses full future agreement. An independently stated object theory adopts empty set, pairing, union, powerset, infinity and extensionality, together with arbitrary formula Separation

and both clauses of Strong Collection. Its adopted instances are closed under variable substitution. The same changing set model validates every sentence and schema instance at every world and assignment; induction on actual deduction trees then proves soundness for all derivations from those instances. The object-theory adoption relation is an inductive syntax declaration, not an unproved host axiom. Foundation, Choice and excluded middle are absent from this theory; anti-foundation has the separate graph-decoration interpretation above.

The actual set-axiom model also refutes excluded middle and double-negation elimination. One of its infinite threshold subsets contains the empty value only at later stages. That membership atom is false now, its negation is false now, and its double negation holds at every stage. Consequently neither classical principle is derivable from any adopted instances of this set theory. A late-only child and the empty value can agree on current members while failing the quantified extensionality premise. These controls distinguish external host Choice used to construct the model from classical rules inside its interpreted logic.

The fixed-successor presheaf ambient has finite limits and colimits, effective equivalence relations, extensivity, regularity, actual Heyting quantifiers and an indexed natural-number object. Its small-map representation and Collection are constructed from the small site, with external host Choice explicitly retained. Every monomorphism is receipt-small by an independently proved subsingleton-fibre construction. The actual membership projection is classified by the same small-family universe, and its decoder is naturally equivalent to literal material members. Whole sections, member-dependent sums, full-future products and extensional identity elimination use this decoder over the same site. An infinite member family grows from the empty set's sole child to admit a cyclic child; identity and constant-empty functions agree on all its present arguments and differ on the later cyclic argument. This is a model of changing material members and compatible dependent use, rather than a constant-family instance. Its externally classical construction does not assert internal excluded middle or internal Choice.

Arbitrary argument-dependent material families now use this same membership decoder. A natural set-valued body on the actual member comprehension supplies a new member fibre for each typed argument. Constructed sum decoders retain both the argument and its literal body member. Constructed product decoders retain literal body members at every future typed argument, with inverse comparisons for inner naturality and outer contextual restriction. Their whole compatible sections agree with actual mixed-universe dependent sections. Independently formed parameter changes preserve these comparisons; identity and W formation use the same genuine comprehension coordinates. An actual singleton body gives later cyclic arguments a cyclic answer, whereas a filtered body can have every present answer and no full future section. A signature with a position in every future has no W tree; introducing an actual empty-position constructor supplies a tree.

The resulting generated code grammar retains whole contextual family functors and authored formation recipes. Its seeds are actual material member maps, and its constructors form dependent sums, complete-future products, discrete identities, contextual W types and reindexed families. There is no supplied family-closure operator. Each generated family has an actual small-family classifier, inverse universal-projection pullback decoder and whole-section comparison. For wider external parameter bound $L = \max(u + 1, v)$, recipe codes live at $L + 1$ and decoded fibres remain at u. Identity reindexing may preserve the full decoded family while changing the retained recipe.

The observed source also has an actual interpretation in this same contextual set model. Its independently constructed observed class is retained beside its unlabelled final set value. The source-to-class and class-to-set coalgebra maps satisfy a proved whole natural square. The structured readout identifies exactly the stable bisimulation that preserves every declared atom. Forgetting its observed-class coordinate is sound and complete for that equality precisely when ordinary contextual bisimulation preserves those atoms. Natural material membership tests encode exactly the

transport-stable atoms; arbitrary changing observations remain in the structured reading.

Observed child classes construct an actual original-bound dependent continuation family. Its natural map onto literal set members has the unlabelled behavioral kernel, and can merge children with different terminal results. No inverse into those observed children is selected. The child reading constructs genuine argument-dependent member bodies, with complete-future products, actual native Hom comparisons, sum and W formation, identity elimination and whole substitution/section laws. An infinite authored instance acquires terminal and cyclic members; its selected body is empty at the terminal argument and inhabited at the cyclic argument. A filtered body has every present answer and no full future product. Actual diagonal/off-diagonal identity fibres and a later empty-position W leaf discriminate the dependent constructions. Neither set-value equality nor membership alone recovers raw edge occurrences or proof provenance.

A separate untyped graph-presentation interpretation keeps logical truth as constructed data. Equality supplies a coherent matching strategy, membership supplies a literal child together with matching evidence, existence supplies a witness, and universal quantification supplies a function over the full graph carrier. Empty set, pairing, union, infinity and extensionality have computed realizers. Bounded formula Separation, arbitrary formula Strong Collection and the stated small indexed Subset Collection instance have constructed witnesses; the adopted intuitionistic deduction trees compute realizers while preserving hypothesis occurrences. These statements describe a proof-relevant static set model. Its literal receipt families separately have actual changing-world dependent decoders. A positive erasure into constant material membership alone cannot validate an implication-bearing Collection schema in a changing-world model.

A separately constructed varying graph universe now closes that particular logical join. On a category whose objects and arrows lie in Type_u, each authored node functor and its literal child fibres remain at u; the untyped collection of rooted diagrams lies at $u+1$. Equality is a constructed coherent matching limit that answers at every future and actual arrow. Membership retains a literal child and that child's future matching evidence. Arbitrary first-order realizers lie at $u+1$, while guarded bounded realizers stay at u. Implication and universal realization inspect the complete future; existential realization supplies a current value. Persistence and equality substitution are constructed for the whole formula grammar, and actual deduction trees compute these realizers while retaining premise positions.

The same universe validates the independently stated first-order empty, pairing, union, infinity, extensionality, bounded Separation, arbitrary Strong Collection and Subset Collection sentences and schemata. Collection's actual graph indexes witness origins by a future context, an arrival arrow and a literal source child. A node also retains its terminal arrow, with the factorization of the original arrival kept in each root edge. Formula realizers and arbitrary material values are not used as an unbounded node index. Both Collection clauses hold at every future arrow for arbitrary bodies, including implication and unbounded quantifiers. Bounded Separation adds an original-small bounded realizer occurrence to this construction. Its conversions with guarded first-order realization are maps in both directions; no inverse equivalence of all proof-relevant realizer actions is asserted. The union construction similarly retains future double-member origins, and its infinity object contains mutually distinguishable finite ordinals of arbitrary bounds.

Subset Collection constructs a full-future catalogue of literal source-to-target receipt functions before the arbitrary relation parameter is supplied. Its image graphs retain all past origins and both collecting clauses at every composite arrival. The catalogue uses original-small worlds, arrows and receipt functions, not arbitrary formula realizers. An identity image agrees materially with its varying source, including members newly available at every natural stage. A present-only function need not extend to a complete future function; an actual changing labelled graph separates these two carriers.

Every original-small authored graph diagram has an actual natural decoration in this universe. Coiteration constructs comparison with any other decoration satisfying its two-sided member equations. Uniqueness is full future matching, rather than native identity or equality of all matching strategies. This states the realized anti-foundation theorem at its graph bound. Powerset, unrestricted Separation and a full internal powerclass fixed point do not follow from these constructions; stronger profile interpretations retain their own construction obligations.

The actual observed-class coalgebra also determines a diagram in this universe. Its literal value reading has exactly the declared observed bisimilarity kernel. Its independently formed successor-receipt family is naturally equivalent to the existing authored continuation family, with inverse comparisons of whole compatible sections. An infinite growing instance and its terminal-versus-cyclic dependent body exercise the same comparison. Material matching of the edge-only diagram remains a distinct observation: different terminal result classes may have equal material values. The retained observed coordinate or a declared labelled encoding is therefore still needed for complete result and fault readings. Raw successor occurrences project naturally and surjectively, as a proposition, to the observed receipts without choosing a representative. Two distinct whole compatible source sections over all positive natural stages can have the same observed section. A native family with an inhabited fibre at one retained tag and an empty fibre at the other cannot descend through that projection.

Primitive matching and membership have sound current material pictures at every declared future. Even equality of all those separate pictures does not supply full future matching. Two diagrams over every natural stage witness this: one initially has two childless occurrences, the other one; later the latter adds the missing occurrence while one child acquires a member. Each current picture agrees, but the initially required response cannot follow that child's future. Compatible matching evidence is stronger than a succession of extensional snapshot agreements.

A separate proof-relevant comparison retains coherent graph-matching witnesses for equality and actual child receipts for membership. Two matching witnesses on the same cyclic graph induce distinct actions on literal membership receipts while erasing to the same material equality proof. Consequently those actions cannot be recovered from that erased proof. Matching equality is not thereby adopted as native identity with dependent J.

Context restriction now acts coherently on the retained matching strategies themselves. Restricting by an identity gives the original strategy after endpoint transport; successive restrictions agree with restriction by the composite. An independently authored restriction has the same complete forward and backward replies as the existing coiteration. Indexed coalgebra finality proves their equality, rather than pointwise erasure of their evidence.

The same varying graph universe represents actual native families by literal member receipts. On an original-small parameter family, one constructed global graph decodes complete native sections. On a wider parameter family, an explicit successor site and graph bound provide the global representation. The resulting enclosure includes every lower future-family classifier code, including codes available only at later contexts. Dependent sums and complete future products, discrete identity elimination and hereditary W initiality compare through the literal decoder. These receipt comparisons do not assert that an unlabelled material graph faithfully encodes every native type.

Material dependent sums now retain actual Kuratowski pairs of the declared argument and result readings. Their matching kernel is exactly matching of both components. Material dependent products retain the complete graph of actual applications at every future world and arrival arrow. Such a graph need not be a function on material argument values: two native arguments may have matching readings while their results do not. An original-small subfamily classifies precisely those native functions carrying a uniform full-future compatibility strategy. Given that strategy, material application is single-valued on the argument kernel. Classification by propositional nonemptiness

does not select a strategy for evidence-sensitive consumers. Compatible whole sections retain abstraction, application, β/η and substitution through inverse natural receipt comparisons.

Successor lifting raises the site, parameter family, native fibres and their actual material bodies together. Independently formed upper sums and complete future products have inverse natural comparisons with the raised lower families. The product comparison retains every future arrow and actual argument, and its evaluation square matches the actual result readings. Uniform functional compatibility strategies transport in both directions. The material carrier comparisons quantify over arbitrary upper elements, not only elements already supplied as lower images. In the growing observed model, two distinct compatible native functions still have matching material graphs after lifting, while the incompatible function remains excluded.

Actual parameter substitution commutes with this raising. The inverse comparisons act on whole native sections and on the dependent body sections, not only on individual fibres. Independently formed upper abstraction agrees with raised lower abstraction by evaluating both at every future argument. The literal receipt decoders and substituted material carriers satisfy the same squares. A growing identity abstraction qualifies at every stage, whereas an occurrence-sensitive abstraction remains outside the material functional subfamily after raising.

Generated contextual W families also have a constructed material body, rather than a supplied tree readout. Native labels and complete future branches generate the labelled graph nodes and edges at the original bound. The exact matching kernel is their full-future unfolding observation. Inverse literal decoders and whole sections connect this graph with the actual generated `Code .w` family and its successor-enclosure decoder. Native fold computation is retained; descent of an arbitrary fold still requires its compatibility with the declared observation. Actual generated continuation trees appear only at later contexts, and matching label readings can conceal native distinctions inspected by a fold.

A second construction attaches a declared natural material body to each literal receipt, retaining all of that body's nodes and edges. Its membership criterion is a native receipt together with full-future matching to its body; its readout kernel is precisely the declared body's kernel. Independently rebuilt substitutions have inverse natural receipt comparisons and matching material carriers. The successor graph raise preserves and reflects matching and membership on the lower image. In particular, attaching the actual member family of any lower graph reconstructs its raised material carrier, with both membership directions covering arbitrary upper elements. This last result is stronger than checking only upper elements already given as lower images.

Attached discrete identity contexts also have inverse natural native decoders. Whole dependent motives retain their natural body readings under J, with computation and substitution laws. An infinite growing endpoint motive distinguishes a cyclic zero result from terminal positive results. Distinct positive receipts can nevertheless have matching bodies and no native identity witness. Thus material matching is insufficient to authorize dependent J; the construction neither adopts global equality reflection nor supplies a higher identity profile.

The independently formed upper discrete identity context has inverse natural maps to the raised lower context, retaining both endpoints and their witness. Its material identity carrier agrees by full-future matching in both directions for arbitrary upper members. An arbitrary upper motive pulls through those context maps without being assumed to have a lower presentation. The two independently formed dependent eliminators agree as whole sections, with computation and parameter-substitution squares and preservation of any declared natural motive reading. A motive that grows at every natural stage exercises this comparison. The interpreted identity profile is discrete; this does not select global UIP or equality reflection for other native identity profiles.

Independently rebuilt material W readings also commute with actual parameter substitution. The native destructor comparison constructs both branch replies, matching their declared argu-

ment bodies and recursively reached children at every future. Coiteration then proves equality of the complete material readings and carrier sets. Inverse natural receipt maps compare whole sections. A noninjective parameter map can identify the readings of two distinct retained parameter occurrences; the same substituted family is initially empty and later inhabited. This comparison therefore retains the dependence and changing worlds of the original generated W construction.

The upper material W family is also formed independently over raised parameters, labels and dependent positions. Native hereditary future-tree transport preserves root constructors and constructs both directions of the branch comparison. The resulting graph matching compares the lifted lower reading with the freshly formed upper reading. Its carrier comparison covers arbitrary upper members, and inverse whole-section maps retain the literal tree decoder. The raised label and position families agree with those used by the common material-family raising, so this W comparison shares the same universe transition as sums, products and identity. For every generated `Code.w`, the successor-enclosure receipt and independently formed upper receipt decode to the same native tree, and their declared material body readings match. Neither the receipt enclosure nor native tree equivalence by itself licenses an arbitrary fold to descend.

The positive first-order fragment of this realization has a constructed interpretation into the constant material membership model over an arbitrary context category: equality, membership, conjunction, disjunction and both unbounded quantifiers are preserved. Universal soundness uses quotient induction into propositional truth, rather than selection of graph representatives. Variable substitution commutes with this interpretation. Implication and negation are outside this comparison: their soundness would additionally require turning arbitrary material premises back into matching and membership evidence. Positive soundness alone is not a full interpretation of the two logics.

The observed interpretation now carries the actual reduction span as well. Its event carrier is retained, with both endpoints mapped through the structured readout. Outgoing endpoint lifting follows from the labelled, atom-preserving behavioural kernel. Incoming endpoint lifting is equivalent to past matching for that kernel: this is exactly the extra condition for the predecessor box to commute with observation. The existing native GSLT diamond and predecessor box agree with these event-span modalities. Stronger occurrence lifting has a separate criterion requiring recovery of the literal endpoint of the same event; ordinary endpoint matching does not supply that recovery.

An actual contextual event family supplies a small dependent receipt family over both observed endpoints. Its comprehension has constructed inverse natural maps to the whole original event family, retaining the receipt even when its endpoints coincide with those of another event. The classifier decodes the complete receipt family, and its contextual products compare with actual dependent native sections for independently wider consumers. Parameter substitution preserves the comprehension, classifier and whole-section comparisons. Native predicate classifiers factor through the structured readout exactly when their predicates are invariant under its declared kernel; comprehension support descends without selecting a term or a source representative.

The authored history instance retains written and actual entries, residual scripts, configurations and copy identifiers. Its event restriction transports both endpoints and commutes with context identity and composition. The commuting predecessor comparison requires its stated incoming lifts; a future-only, atom-forgetting instance distinguishes those requirements. A separate constructive observed interpretation retains its observed class beside the material value and constructs actual continuation sums, complete-future products and W families without external Choice. Its controls distinguish a function available on present arguments from a compatible whole-future section, and refute recovery of an erased occurrence by a dependent consumer.

A fixed-site cumulative successor retains each exact lower recipe as an upper seed. It raises

external parameters from L to $L+1$ and codes from $L+1$ to $L+2$, while decoded fibres remain at u. Retained lower payloads prove the code embedding injective independently of any material observer. The lower family and its upper decoding agree through the actual inverse parameter wrapper, including restrictions, whole sections and substitution. Independently formed upper sums, identities, W types and complete-future products have whole-family comparisons with lifted lower formations. Product evaluation agrees at every future argument. A lifted lower recipe can differ from the freshly formed upper recipe even when their decoded families agree. This cumulativity does not identify codes with internal material sets, enlarge the material carrier or context site, or provide transfinite closure.

In the explicit host-Choice material-member profile, a separate cumulative comparison enlarges the actual material model and its context site. Original worlds and arrows have explicit raised presentations. The transported lower coalgebra keeps its original values and every actual future, with receipts raised to the successor branch bound. Its unique upper final readout is injective and covers exactly all upper members of an embedded parent, at each current and future arrow. Literal member decoders, whole sections and arbitrary parameter substitution have natural inverse comparisons. The source set carrier at $u+1$ becomes small on the raised site, while upper set values live at $u+2$.

This embedding is genuinely proper at every world. A constructed full-future Russell predicate on the lower carrier has a truth-subtype cover at the successor bound; its embedded image assembles into a natural upper set outside every lower value's image. Naturality propagates an alleged lower reading through all futures, where the constructive diagonal argument refutes it. No cardinality assumption or global inverse is used. Raising only the context presentation preserves every first-order formula on the old carrier. The material embedding additionally preserves and reflects the bounded fragment, whose quantifiers range over members of named parents. This comparison includes full-future implication, variable substitution and composite membership embeddings. It does not assert unrestricted elementary reflection into the larger carrier or identify this model lift with the fixed-site recipe successor. Independently formed upper dependent operations now have actual comparisons over the enlarged set model's member and member-dependent body families. The sum comparison keeps both literal coordinates; the complete-future product comparison commutes with abstraction, application and the dependent adjunction. Discrete identity contexts have inverse natural maps, and the two independently formed dependent J operations agree for arbitrary larger natural motives, including computation and parameter substitution. Natural W trees retain every future branch. Their constructed comparisons preserve whole cone algebras, arbitrary larger consumers, initiality and the resulting full-section folds. These results do not identify all larger carriers or assert an additional isomorphism of entire unrestricted polynomial endofunctors.

The upper controls use genuinely changing members and dependent bodies: every present argument can have an answer while the complete upper product is empty. A body giving each future argument its own singleton has a constructed complete product. An everywhere-inhabited position family has no W tree; an actual empty-position constructor supplies a leaf. Infinitely many whole future trees remain distinct even when all present branches agree. The independent upper material-member comparisons retain their exact external Choice dependencies. The generic coordinate, future-cone, W and substitution comparisons are checked separately without that dependency.

The optional native over-category comparison is also constructed against its actual selected pullbacks and right adjoints. A literal pullback has an independently verified limit and both projection squares. The original-bound product represents the full Hom functor of every native slice consumer. Its inverse isomorphism to the selected dependent product is fixed by both counits and the transpose equation; an explicit fibre decoder commutes with restriction. The native dependent sum agrees through an actual coordinate isomorphism and its full Hom law. These comparison

interfaces retain their exact external Choice dependencies, separately from the pure coordinate constructions.

Actual observer comparisons now use the final contextual set readout. Natural behavioural consumers factor uniquely through its constructed quotient, and agreement of all such consumers is exactly its bisimulation kernel. Presentation occurrences map naturally onto literal interpreted members, but a symmetry moving an occurrence prevents a natural inverse from the discrete material family. The presentation identity eliminator can observe that symmetry. For any original-small graph, its actual contextual decoration agrees with the embedded ordinary hyperset reading, with precise preservation and reflection conditions for a proposed regular identification. Rooted unfolding isomorphism supplies a bisimulation comparison; it is not claimed to construct full Scott, Finsler or Boffa anti-foundation universes.

Membership accessibility in the actual model exposes a further substitution boundary. A value with no current members can later acquire a cyclic member, so present well-foundedness is not a stable family predicate. Requiring accessibility at every future gives a constructed stable predicate. On embedded ordinary hypersets, this persistent predicate is equivalent to ordinary membership accessibility. The actual full-future Quine section is unique, while equality or bisimilarity of present member graphs alone need not identify contextual set values.

Constant ambient hyperset membership supplies a separated covered coalgebra, with equality precisely its contextual bisimilarity. It is not the general contextual final model: an actual contextual coalgebra with a late-only child has no coalgebra map into this constant membership coalgebra. Thus neither separation nor the ambient anti-foundation theorem justifies replacing genuinely contextual membership by constant material membership.

A coherent small-family universe is also constructed. A code at a context is an actual small-valued functor on its complete future category; restriction precomposes with the actual arrow-prefix functor, and the displayed decoder evaluates at the identity future. Codes live one host level above the decoded fibres. Every authored small displayed family over an independently larger parameter functor has a natural classifier. Decoding recovers the whole displayed functor, and its total family has constructed inverse natural maps with the literal universal-projection pullback. Whole sections and arbitrary parameter substitution satisfy the same comparisons. Equal present decoded types do not identify different future actions.

Small dependent sums and products are constructed over independently wider parameter families. A product fibre consists of a compatible section on the complete small future cone, including the future arrow and dependent argument; it is not just a function on present arguments. The whole contextual adjunctions, abstraction/application, both inverse laws, evaluation at arbitrary futures and independently formed substitution comparisons are checked. Nested comprehension has actual flattening and inverse maps. Two products can agree at every present application and differ on a later admitted argument. A present inhabited fibre can also have an empty full future product. Identity formation uses the actual small equality-witness carrier; its dependent eliminator accepts arbitrary wider natural motives, with computation and substitution laws. This is the discrete set-family model's identity interpretation, not global UIP or equality reflection for the native language.

The dependent comparison extends to actual codomain slices with independently wider sources and consumer fibres. Literal pullbacks retain the original source, parameter and argument coordinates. Explicit inverse natural maps compare their fibre families with nested comprehension. Codomain composition realizes the dependent-sum adjunction; a full-future original-bound function family realizes the dependent-product adjunction. The resulting Hom equivalences hold for every such wider consumer, with constructed evaluation, the actual counit equation, both computation laws and source substitution. Their argument and function fibres remain at the original

bound even when the source admits no cover at that bound. The optional comparison with selected over-category pullbacks and right adjoints has its own external dependency account.

The original-bound family universe also has genuine W-type closure over independently wider parameters. Complete future trees carry the natural constructor and destructor with inverse laws. For arbitrary natural polynomial algebras, the constructed fold satisfies its computation equation and is the unique whole algebra map. The consumer can have an independently larger fibre universe: mixed-universe polynomial actions, constructor equations and fold uniqueness keep the tree carrier at its original bound. Algebra morphisms satisfy whole fold fusion even between differently sized consumers. Independently formed substitution preserves constructors and folds through explicit inverse comparisons, including noninjective parameter maps. The comparison with native dependent sections retains every small future arrow and branch. An actual wider consumer pairs arbitrary material values with varying trees; the fold recovers both coordinates, although this consumer has no original-bound cover. Infinitely many full trees in a single fibre can agree on every present branch and differ later; a signature with a constructor requiring an everywhere unary branch has no well-founded trees. Neither control supplies an internally self-containing universe.

For an arbitrary original-bound displayed family over a wider parameter object, stable predicates on its complete future cone form an actual original-bound power family. Whole compatible sections correspond with inverse laws to stable subfamilies. More generally, maps over an arbitrary parameter change correspond to stable subfamilies of the independently reindexed displayed family; the member square has a unique whole classifying map. These comparisons commute with actual substitutions. Infinite future-threshold subfamilies have equal current members and distinct full power codes. This construction uses full proposition-valued predicates and exposes that strength; it is distinct from proving the generic small-map powerclass axiom for arbitrary wider argument objects.

The graph receipt interpretation is now joined to that generated universe through one natural diagram. Literal graph-receipt comprehension, native comprehension, enlarged material-member comprehension and the universal-family pullback have computed inverse maps. Complete sections, cumulative composition and arbitrary parameter substitution commute with the diagram. The material coordinate agrees with the independently constructed graph reading, while the generated recipe retains its native future action.

This diagram extends to the actual nested discrete identity contexts. The diagonal and left-endpoint projection commute with the decoder; arbitrary natural motives and complete reflexive methods have an independently formed literal-receipt J operation. Its whole-section comparison with native and material decoding preserves computation, motive naturality and parameter substitution. The identity-witness family is itself a generated recipe with an enclosed graph. In the infinite growing observed model, a genuinely varying endpoint motive computes the selected cyclic receipt through both decoders; diagonal receipts exist and distinct terminal/cyclic endpoints have none. These discrete identity witnesses cannot faithfully encode the distinct matching strategies that act differently on literal members while having the same proof-irrelevant material equality reading.

Coherent small-map data have constructed fibre inverses, composition and independent pullback. Bounded Collection filters an actual witness relation and computes the exact parameter-cover criterion. A more general choice-free Collection construction retains future generators, arrival arrows and every small witness receipt; it constructs the whole quasi-pullback diagram rather than a selected natural witness section. Generated material dictionaries instantiate these constructions. Small families descend along an actual material cover by saturated receipt predicates and singleton-union decoding, without a natural splitting of that cover. Under external host Choice, every pointwise small map and cover has the separate full Collection construction at the fixed

successor ambient level. No unrestricted constructive Collection principle or complete internal set-axiom interpretation is inferred from a bounded dictionary instance alone.

For a material fibre with a small receipt enumeration, saturated receipt-fibre predicates form an actual small code type. Bounded separation inside its given material carrier proves that the matching row is a singleton; union decodes it without choosing a receipt or graph representative. Encoding and decoding are proved inverses. Authored material restrictions therefore construct a coherent small displayed fibre family. Above the full-future enumeration-data parameter cover, this family supplies two actual pullback diagrams, with unique natural lifts: one recovers the original material map, and the other classifies the small family through the constructed universal decoder. These are representability diagrams for that precise covered material class. They use full proposition-valued receipt subsets and do not establish unrestricted Collection or representability of every merely pointwise small map. Infinite varying cyclic fibres, new members at every stage, and duplicated receipts discriminate these boundaries.

For authored material families, full-future small-map classification needs no extra enumeration-data parameter cover. The actual dictionaries and their inverse member decoders construct fibre receipts for every future of each material map. The decoder makes the original map itself an explicit pullback of the universal small-family projection, with inverse natural maps, whole compatible-section comparisons and unique lifts from arbitrary wider sources. The generated material dictionaries instantiate this construction for the existing dependent type formers. Actual receipt factories for these maps do not constitute Collection for every wide cover.

Within any such covered bound, empty predicates, binary union and intersection have constructed covers, and arbitrary stable predicates support bounded proposition-valued separation. Future implication quantifies over every actual later argument and morphism; its bounded version satisfies the relative Heyting adjunction. Restriction preserves this implication through constructed lifts of its complete future arguments. In the raised infinite growing family, a predicate that first becomes true after stage zero has empty relative negation and inhabited double negation now, while being false now. Double-negation elimination therefore fails in this actual bounded contextual logic. Original small covers are retained by each of those logical constructions.

The separation between constructed codes and classified small maps is also present in the reference model theory. Hofmann–Streicher universes retain complete slice presheaves and evaluate their decoders at the identity arrow[1]. Representability and Collection require separate arguments even when finite limits, images and dependent products are inherited by a presheaf model[2]. The Hofmann–Streicher presheaf construction admits a constructive internal presentation, including its stated restricted realignment properties. General realignment for the alternative Grothendieck-topos construction remains a separate constructive problem. Classification of every merely small family and these realignment laws are stronger than decoding the constructed generated codes; a classical construction does not discharge their constructive versions[3].

At the original graph bound, a common material bound on all fibre members permits a different construction of arbitrary bare dependent products. Kuratowski pairs lie in the double powerset of the union of the domain and codomain bounds. Separation constructs the dependent-pair set and each actual function graph; a further separated powerset collects every total single-valued graph. Bounded row separation and singleton union supply evaluation and its inverse graph decoder, with

[1] Steve Awodey, *On Hofmann–Streicher universes*, sections 1 and 4, https://awodey.github.io/papers/universes.pdf.

[2] Benno van den Berg and Ieke Moerdijk, *Aspects of Predicative Algebraic Set Theory III: Sheaves*, sections 3.1–3.2, https://arxiv.org/pdf/0912.1242.

[3] Daniel Gratzer, Michael Shulman and Jonathan Sterling, *Strict universes for Grothendieck topoi*, sections 2, 4 and 7.1, https://www.danielgratzer.com/papers/strict-universes-for-grothendieck-topoi.pdf.

β and η laws. Graph reindexing and the dependent-product Beck–Chevalley comparison retain the changed argument coordinates rather than asserting literal equality of differently labelled graphs. The infinite material natural-number domain has a genuinely varying bounded family with distinct complete sections. This construction needs no selected presentation or small section carrier, but uses full powerset and proposition-valued separation. A collected fibre range supplies the common bound by union; the construction does not itself prove unrestricted Collection for an arbitrary unbounded bare family.

The observed model constructs these families and all compatible natural sections from actual material dictionaries. Its infinite labelled-history control has genuinely argument-dependent identity fibres and dependent products: every current result fibre can be inhabited while a full future product is empty. Parallel histories can act identically on operational states while retaining distinct future-function labels. Provenance-sensitive sections therefore descend exactly when their proved fibre compatibility allows it, not merely because their positive supports descend.

Restriction-stable predicates construct contextual separated families at the original graph bound, with their complete member decoders. Dependent sums construct every witness of a typed stable relation, and functional relations yield unique natural sections by bounded material computation. Collection projections can cover every argument without admitting a natural section: an infinite-loop model advances members of the actual material natural-number set and has no fixed compatible choice. This distinguishes collection from coherent dependent choice in the interpreted family model.

Over an actual small contextual category, a further constructed witness cover has fibre $\sum_g (g \to x)$ at x and transports receipts by postcomposition. Authored witnesses in a possibly wider family determine an actual natural map from this small cover; evaluating such a map at the identity receipts recovers the witnesses, with both inverse laws. The displayed version covers every member of a small displayed family and retains its transported witness. On the infinite advancing natural-number model, receipts reach every material ordinal, but there is no compatible section or natural splitting. The wider host sort of the natural-member presentation is not a proof of mathematical non-smallness: that family already has a constructed small member model. A separate comparison chooses witness data from pointwise nonemptiness using explicitly recorded host choice. These small external-site constructions do not establish the category-wide Collection axiom or an internal material CZF/AFA model.

Whole natural sections also have an exact refinement theorem. Completing a finer observation fibre to its entire coarser fibre constructs a natural class map, and the same authored material source action descends through both observers. The independently decoded families have equal material values along this map. Coarse sections embed into fine sections precisely as those whose pulled selected values agree on the coarse kernel; their existing contextual naturality is retained. The section maps satisfy identity and three-observer composition. In the infinite growing control, an occurrence tag fails the middle observer, while a position-dependent section descends through the middle observer and fails the terminal one. Family invariance alone therefore licenses neither coarsening.

The same infinite observed model has a concrete non-Boolean internal control. A separated unit family expresses positive history length; its initial fibre is empty and its fibre after an extension is the singleton containing the empty material set. Negation is interpreted by the actual dependent product into the empty family. Its exact support criterion requires absence of a witness at every future context and arrow, rather than absence only now. Every point has an extension with a witness, so the negation family is empty everywhere. Double negation nevertheless has a constructed natural section and singleton material fibres. There is no natural map from this double negation family to the original proposition. These are checked internal logic distinctions in an actual infinite

model, not uses of host excluded middle or double-negation elimination.

These semantic codes still live one metatheoretic level above their decodings. A native internally small universe calculus owes its own code/decoder rules, judgmental computation, conversion and normalization or an adequate reduction-free replacement. The checked semantic substitution laws are not a proof of those native metatheorems. In the older erased material model, typed collection inside generated decoded families and collection of arbitrary material witnesses at a raised bound do not establish unrestricted Collection at the original material bound. The varying proof-relevant graph model above instead constructs an original-small future-occurrence witness cover and validates Strong Collection for arbitrary first-order bodies in its realized semantics. This does not extract witnesses from arbitrary propositional existence in the erased model; the two logical interpretations remain distinct. Inductive–recursive definitions are the standard mechanism for internal universe syntax; stronger Mahlo-style universes can internalize closure under prescribed family transformers, but at substantial and explicit proof-theoretic cost [74, 77]. External successor embeddings alone do not supply internal closure or transfinite growth.

An explicitly classical comparison constructs unrestricted Strong Collection at the original graph bound: choose one relational witness and one graph for each actual root occurrence, and take their material range. Replacement then follows from uniqueness. This supplies an actual instance of the displayed set-forming hyperset ledger, including replacement and a Quine atom; its dependency manifest records host choice. It is neither a constructive proof of original-bound Collection nor a necessity theorem saying that Collection requires choice. Selecting enough witnesses and collecting every possible witness are different operations: the everywhere-true relation needs only a singleton collecting set, while its complete raised witness collection is the entire smaller material carrier and cannot descend to an original-bound set.

The constructive model also contains an actual infinite material set at every graph bound. Decreasing natural-number graphs supply its finite ordinals; their material range is closed under zero and successor, satisfies induction for every material predicate, and is the least inductive material set. It and its members are well-founded, while the ambient hyperset carrier still contains a Quine atom. Moreover, every relation on the members of any material set has a unique decoration at the original graph bound. The set-indexed theorem constructs its small carrier using full nonempty observation fibres inside its existence proof. Neither statement assumes a uniform presentation selector, nor supplies a host-natural decoder from propositional surjectivity.

Logical commitments also arise from readout conveniences. Full separation constructs the actual truth set

$$V_P = \{z \in \{\varnothing\} \mid P\}.$$

The checked laws give

$$\varnothing \in V_P \iff P, \qquad V_P \neq \varnothing \iff \neg\neg P.$$

Consequently, a procedure deciding equality of all material hypersets entails excluded middle. The assertion that every subset of a singleton is either empty or the whole singleton is equivalent to excluded middle; converting every truth set's inequality with the empty set into positive membership is equivalent to double-negation elimination. These strength comparisons are proved without adopting those principles. A native service that offers one of these capabilities must expose its logical commitment alongside its programming contract.

Thus dependent type theory is not intrinsically condemned to less elegant universe use than Tarski–Grothendieck set theory. The clean interface exists. What remains unresolved is which native rule package realizes it with the best combination of ergonomics, computation, predicativity, and metatheoretic cost.

10.2 Shared material laws and explicit checking profiles

A shared material calculus can be stated before selecting an ambient native foundation. Its independently authored first-order sentences express empty-set existence, pairing, union, infinity, extensionality and bounded Separation. Strong Collection and Subset Collection are separate adopted extensions; their validation does not make them consequences of the weaker calculus. Assumption occurrences remain indexed in derivations, and substitution transports both the formula and its adopted assumptions.

Theorem 19 (Interpretations of the shared material calculus)**.** *Every derivation of the shared calculus, or of its two Collection extensions, has an interpretation in the varying constructed-witness graph model. The independently stated logical rules are sound there. Formula substitution commutes with reindexing assignments, and substitution of deduction trees retains their indexed assumptions. The same sentences have ordinary first-order interpretations in the hyperset carrier and in the well-founded set carrier. The latter two Strong Collection interpretations retain their named host-choice dependency. Neither first-order Foundation nor excluded middle is derivable in the constructed-witness interpretation, even after adjoining both Collection extensions.*

The checked construction is packaged in `CommonSetProfiles`. The positive interpretations apply to arbitrary well-scoped bodies, rather than a finite list of schemas. Their negative controls use the literal cyclic singleton and the genuinely varying future-context excluded-middle countermodel. The constructive interpretation supplies witnesses as data; it does not extract them from arbitrary propositional existence.

The common native profile adopts the validated shared laws and both Collection extensions. Three explicit fragment profiles add first-order Foundation, excluded middle, or both. Anti-foundation, choice and universe closure remain separate commitments. This is a selected common theory with checked interpretations, not an algorithm computing the complete intersection of all candidate theories, and none of these fragments is presented as a complete CZF, ZF or AFA implementation.

Logical compilation has two guarantees. A general theorem identifies the first-order semantics with evaluation of compiled logical terms; a separately constructed lexical decoder reconstructs its closed encoding. The native schema adapter is compared with freshly generated outputs of this checked compiler, including bound variables and the distinct Separation and Collection grammars. Those executions are runtime evidence, not a general soundness proof for the larger higher-order C checker. The Church encoding of existence has propositional semantics and supplies no witness-extraction theorem.

Native receipts retain the selected checking profile, actual type declarations, admitted computation rules and construction origins. Removing a used dependency invalidates the receipt; a local binder does not acquire an unrelated global dependency merely by sharing its spelling. The foundation manifest exposes these context commitments separately from the profile-source digest. Agreement of a formula across profiles therefore does not silently transfer its proof receipt, and executable cyclic material values remain a separate implementation obligation.

10.3 Universe closure profiles and their different strength currencies

The universe-above-a-family operator, a superuniverse, and a Mahlo universe are not three names for the same convenience. Their structural contracts form an increasing sequence inside a suitable dependent type theory.

Family enclosure.
Given one family (A, B), construct a closed universe containing its base and fibres.

Superuniverse. Fix a next-universe operator. A superuniverse internally represents the universe generated by that operator for every family already represented inside it. Iterated operators yield super^n-universes.

Mahlo reflection.
For every internal operator f on represented families, the Mahlo universe contains an ordinary subuniverse U_f closed under the restriction of f.

The last clause is strictly more general in form: the next-universe operator is one selected f, whereas Mahlo reflection ranges over arbitrary internal family operators. This is the type-theoretic analogue of reflecting closure points, not merely adding another externally numbered level [72, 74, 77].

Theorem 20 (Checked universe-closure boundary)**.** *Internal family embeddings compose. Closure under one monotone family operator contains every finite iterate of that operator. Mahlo reflection supplies an operator-closed subuniverse for each selected operator. A one-level-up ambient type universe presents the output of any already supplied small family-universe operator, but its code carrier cannot be represented at the same decoded level. Finally, the finite Tarski universe* $n \mapsto \mathrm{Fin}(n)$ *is closed under the identity operator yet admits no subfamily closed under strict successor growth; ordinary universe data therefore does not imply Mahlo reflection.*

This theorem formalizes the implication shapes and a strict negative control; it does not construct a native superuniverse or Mahlo kernel. Such a construction still owes its inductive–recursive rules and metatheory.

There are at least four inequivalent meanings of "stronger" here.

Ordering	Evidence required
Constructor capability	One profile derives or internally presents every operation of another.
Interpretability	A translation preserves derivability, and reflects it when equivalence rather than an upper bound is claimed.
Proof-theoretic strength	Ordinal analysis or another calibrated reduction over pinned rule packages.
Classical model strength	Lower bounds or equiconsistency over a common set theory, not merely a large-cardinal assumption sufficient for one model.

For Palmgren's specifically pinned systems without elimination rules, Rathjen's analysis gives

$$|\mathsf{MLU}| = \Gamma_0, \qquad |\mathsf{MLS}\uparrow| = \Phi_{\varepsilon_0}(0), \qquad |\mathsf{MLS}| = \Phi_{\Gamma_0}(0),$$

where MLU has the universe operator and MLS adds the superuniverse [72]. Setzer's pinned theory with one Mahlo universe has the exact ordinal $\psi_{\Omega_1}(\Omega_{\mathsf{M}+\omega})$, slightly beyond the recursively Mahlo theory KPM [74, 75]. Dybjer and Setzer show that ordinary and super^n universes are definable within their theory of induction–recursion, while the internal Mahlo universe goes beyond that

theory; their newer work gives Martin-Löf-style meaning explanations and set-theoretic models for the Mahlo rules [77, 78].

HOTG is calibrated differently. Brown, Kaliszyk, and Pąk construct a standard HOTG model from a 2-inaccessible cardinal—their term for a regular limit of inaccessible cardinals—and note that this is the same sufficient assumption used for their model of plain TG [58]. One set-theoretic model used in the modern Mahlo analysis assumes a Mahlo cardinal and an inaccessible above it, which is a stronger *sufficient ambient hypothesis*. This does not prove that the Mahlo type theory is stronger than HOTG: a cross-foundation conclusion would need lower bounds, mutual interpretations, or an equiconsistency result over a common base. Conversely, HOTG's extensional universe selector does not by itself reproduce Mahlo's internal computation and reflection over family operators.

The relevant "reverse mathematics" programme is therefore mostly an interpretability and proof-theoretic ablation programme rather than classical second-order reverse mathematics. Start from a real consumer theorem, remove Mahlo reflection, then superuniverse closure, then inspectable enclosure, and record the first theorem, program, or categorical construction that fails. Going beyond the apparently necessary profile is useful when it discovers such intermediate conservation boundaries; installing the strongest rule globally before that experiment would erase the information.

The known consumers are already nontrivial. A superuniverse can internalize transfinite and iterated universe formation. Mahlo reflection supports generic family transformers and higher-order universe operators. Takahashi uses the Mahlo reflection property together with accessibility to interpret constructive set theory with inaccessible sets of all transfinite orders, and mechanizes the main construction in Agda; with extensional identity, the accessibility predicate can be defined using W-types [73, 79]. These are practical mathematical gains, not merely larger ordinal labels.

10.4 Where the construction bill is paid

The contrast between HOTG and dependent type theory is not that one may postulate universes and the other may not. Either foundation can expose a primitive universe-selection operation. The difference is what that operation promises. An extensional set-theoretic selector may be specified by membership and closure properties; it need not compute to a canonical representation of the selected universe. This makes ordinary set and category arguments concise, but packages the existence, choice, and large-universe strength into the host theory.

In a computational dependent type theory, a universe may occur in programs, proofs, conversion, and normalization. If the selector exposes first-class codes, its formation, elimination, substitution, and computation behaviour must therefore be specified. It is possible to imitate the extensional interface by making the selector opaque and giving only propositional decoding equivalences. That is a legitimate profile, but it is an axiom-like service rather than a computational construction. Asking for canonical generated codes or judgmental decoding is a stronger and more involved profile.

This yields six distinct implementation profiles.

Profile	Contract and price
Level-polymorphic enclosure	An elaborator chooses a sufficiently high existing level. No new kernel rule; the enclosure is usually not inspectable data.
Opaque family selector	A primitive $\mathsf{Enclose}(A, B)$ with propositional closure laws. One-motion ergonomics, but little computational content.
Inductive–recursive universe	Generated codes and a decoder with computation rules. Supports reflection over codes, but owes substitution, conversion, canonicity, and a semantic model.
Superuniverse	Internal coding of universes produced by one selected universe operator, including finite iteration. Stronger than bare enclosure in both constructor capability and the pinned Palmgren–Rathjen analyses.
Mahlo reflection	An operator-closed subuniverse for every internal family transformer. Supports generic reflection but requires a substantially stronger rule package and proof-theoretic justification.
HOTG universe selector	Extensional set-level enclosure and categorical smallness. Concise for model mathematics; existence and choice are paid in the set-theoretic host rather than by code reduction.

Inventing a kernel rule is therefore not intrinsically objectionable. It is justified when a named consumer needs the stronger profile and the rule passes the following admission gate.

Obligation 21 (Universe-rule admission)**.** A primitive family-enclosure rule must have precise formation, introduction, elimination, equality, and computation rules; stability under weakening and substitution; a decidable or explicitly scoped conversion account; a model with the level shift visible; a canonicity, normalization, or adequate reduction-free decision theorem; an explicit strength and conservativity profile; and a stated erasure or runtime interpretation. Same-level self-coding must either be refuted or its consistency cost stated. Without these items the rule is a design target, not a foundation.

This gate is particularly important for Mahlo rules. Palmgren showed that a natural-looking elimination rule for the original Mahlo universe is inconsistent; later extended-predicative and meaning-explanation programmes carefully restrict and reconstruct the acceptable rules [72, 76, 78]. Thus "invent a kernel rule" is a legitimate research move, but the exact elimination and computation behaviour is part of the mathematical claim, not an implementation detail.

The default should consequently be the weakest profile that serves the actual consumer. First-class inductive–recursive codes are warranted when the language must inspect, transform, serialize, or generate type descriptions. Closure under the universe operator is warranted only when an internal theorem or program genuinely quantifies over such generation. Neither should be added merely to mimic the visual simplicity of a set-theoretic axiom.

10.5 A two-face adequacy target

The extensional and computational strengths can be combined without identifying them. Let $[\![-]\!]$ interpret a native dependent calculus in a pinned HOTG model. For a family (A, B), collect its interpreted data in

$$X_{A,B} = \{[\![A]\!]\} \cup \{[\![B(a)]\!] \mid a \in [\![A]\!]\}, \qquad G_{A,B} = \mathsf{UnivOf}(X_{A,B}). \tag{5}$$

The type-theoretic face returns inspectable codes when required; the HOTG face supplies the extensional universe in which their denotations and categories of models live.

Obligation 22 (Family-enclosure adequacy)**.** For every native code $c : U_{A,B}$, prove $[\![T_{A,B}(c)]\!] \in G_{A,B}$; prove that the interpretations of 0, 1, $\mathbb{N}$, sums, Π, Σ, identity, and W-code constructors agree with their set-theoretic operations; prove naturality under substitution; and state every universe enlargement used by the interpretation. Reflection from arbitrary host sets back to native codes is not assumed. If claimed, it requires a separate definability or reification theorem.

Two relevant endpoints are now machine checked, with different scopes. On the type-theoretic side, an abstract extensional universe algebra exposes host objects, their elements, and the operations interpreting 0, 1, $\mathbb{N}$, sums, Π, Σ, identity, and W-types. A family-universe adequacy interface requires a total code interpretation, element equivalences, and constructor-commuting laws; consequences of those supplied fields are proved. This interface is not an independently constructed HOTG interpretation. On the HOTG side, the selected host

$$\mathsf{UnivOf}(\{[\![A]\!], \{[\![B(a)]\!] \mid a \in [\![A]\!]\}\})$$

is proved to contain the base and every fibre, to be transitive and ZF-closed, and to be minimal among hosts with those properties. It is also proved not to be a member of itself.

The remaining design target is the *composition*: instantiate the abstract host algebra by the actual HOTG sets, interpret the syntax and contexts of a chosen native calculus, and prove substitution naturality. The current results therefore establish both endpoints and the shape of their joint, not yet the end-to-end model of a concrete dependent syntax in HOTG.

10.6 What a commuting bridge requires

At the object-and-element layer the forward square has the form

$$\begin{array}{ccc} T(\mathsf{op}(\vec{c})) & \xrightarrow{\delta_{\mathsf{op}(\vec{c})}} & \mathsf{Elem}([\![\mathsf{op}(\vec{c})]\!]) \\ \downarrow \epsilon^{\mathrm{native}}_{\mathsf{op}} & & \downarrow \epsilon^{\mathrm{host}}_{\mathsf{op}} \\ \mathsf{op}_T(T(\vec{c})) & \xrightarrow{\mathsf{op}(\delta_{\vec{c}})} & \mathsf{op}_{\mathsf{Elem}}(\mathsf{Elem}([\![\vec{c}]\!])), \end{array}$$

where the vertical maps are the native decoding and host element equivalences for the selected constructor. The substantive requirements are:

$$[\![c[\sigma]]\!] \cong [\![c]\!][[\![\sigma]\!]], \qquad \delta_{c[\sigma]} \text{ is natural in } \sigma, \qquad [\![\mathsf{op}(\vec{c})]\!] \cong \mathsf{op}_G([\![\vec{c}]\!]). \tag{6}$$

Contexts and substitutions should be organized as a category with families or a related comprehension structure, so that (6) is one structure-preserving interpretation rather than a list of unrelated lemmas [60]. Explicit universe polymorphism supplies a precise syntax of levels and initial-model account [61, 62]; strict universe constructions in Grothendieck topoi show how semantic universes can be made coherent enough for intensional type theory [63]. Type-theoretic interpretations of constructive set theory provide further precedent for moving between intensional and extensional foundations [64, 66]. These ingredients make the programme credible, but none by itself is the desired HOTG adequacy theorem.

10.6.1 HOL predicates as dependent families

An actual higher-order predicate gives a more informative bridge than sending each simple type to a constant family. Fix a Henkin model M, an admissible parameter valuation ρ for Γ, and an actual HOL formula φ in context $A :: \Gamma$. Write $\mathrm{Adm}_M(A)$ for the admitted domain, rather than silently replacing it with the full ambient function space. Define

$$F_{\varphi,\rho}(a) = \mathrm{Lift}(M \models \varphi[\rho, a]), \qquad R_{\varphi,\rho} = \sum_{a:\mathrm{Adm}_M(A)} F_{\varphi,\rho}(a).$$

The lift presents a truth-valued proposition as a type. In `HenkinPredicateFamilyInterpretation`, the refinement is equivalent to the corresponding subtype, and the actual syntactic substitution law is proved:

$$F_{\varphi[\sigma^+],\rho}(a) = F_{\varphi,[\![\sigma]\!](\rho)}(a).$$

The resulting comprehension equivalence keeps the underlying admitted value. Moreover,

$$M \models (\forall a.\varphi)[\rho] \iff \mathrm{Nonempty}\left(\prod_a F_{\varphi,\rho}(a)\right), \qquad M \models (\exists a.\varphi)[\rho] \iff \mathrm{Nonempty}(R_{\varphi,\rho}).$$

An actual HOL proof supplies a section on valuations satisfying its hypotheses, under the model's explicit extensional-congruence condition. A proof of $\forall a.(\varphi(a) \Rightarrow \psi(a))$ therefore transports a witness of $R_{\varphi,\rho}$ to $R_{\psi,\rho}$ without changing its underlying value. The positive control uses an actual proof quantified over two predicate parameters, $\forall P, Q.\forall a.((P(a) \wedge Q(a)) \Rightarrow P(a))$. A Boolean interpretation has an inhabited true fibre and an empty false fibre, proving that this family is not constant. With an independently supplied family envelope, the universal also has a dependent-product code whose decoding is equivalent to the section type.

These statements neither extract computational witnesses from arbitrary existential proofs nor identify truth fibres with native dependent proof syntax. Two distinct HOL proof trees are proved to yield the same truth section: inspection must retain the supplied proof separately. Attaching these families and sections to actual native types and terms, and composing that attachment with the intended HOTG interpretation, remain distinct construction tasks.

Proof actions and their substitution square. `HenkinPredicateFamilyCoherence` constructs actual HOL proof trees for the universally quantified identity implication and for composition of two universally quantified implications. If $\mathcal{R}(p)_\rho$ is the map of refinement witnesses induced by such a proof, the interpretation satisfies

$$\mathcal{R}(\mathrm{id}_\varphi)_\rho = \mathrm{id}, \qquad \mathcal{R}(q \circ p)_\rho = \mathcal{R}(q)_\rho \circ \mathcal{R}(p)_\rho.$$

Simultaneous substitution is a traversal of all constructors of the retained `ProofSyntax`, including quantifiers, equality and extensionality; it is not a reconstruction from the proposition that a derivation exists. A valuation satisfying the substituted hypotheses pulls back to one satisfying the original hypotheses. Writing $c_{\varphi,\sigma,\rho}$ for the existing comprehension-substitution equivalence, the proved square is

$$c_{\psi,\sigma,\rho} \circ \mathcal{R}(p[\sigma])_\rho = \mathcal{R}(p)_{[\![\sigma]\!](\rho)} \circ c_{\varphi,\sigma,\rho}.$$

The section supplied by an arbitrary proof, and the dependent section supplied by a universal proof, obey the corresponding substitution laws. These equalities compare the actions on truth-valued witnesses, not the identity of the retained proof trees. A non-ground control substitutes $P := \lambda x.(x = y)$ and $Q := \lambda x.\top$ into the higher-order intersection proof: the free y survives both binders, the four rule nodes remain, and changing y changes membership.

Induction is carried through the same interpretation. `UniformListPredicateFamily` interprets the actual object-HOL map-fusion induction proof as a section over every admitted list. Applying object-HOL equality congruence to an arbitrary predicate P constructs proofs in both directions of

$$P(\mathrm{map}\ f\,(\mathrm{map}\ g\,xs)) \quad \Longleftrightarrow \quad P(\mathrm{map}\,(f \circ g)\,xs).$$

Their interpreted refinement equivalence preserves the input list. The ordinary-list model exercises arbitrary Boolean endofunctions, lists and predicates, without requiring the predicate to be decidable. The displayed fusion equation is obtained from the object proof through soundness, not by substituting an independent metatheoretic map-fusion theorem.

The incorrect composition order has an inhabited empty-list fibre but an empty singleton fibre. It passes all empty-list and equal-function samples, but supplies no uniform section. Separately, the equation-only junk model has neither an induction witness nor a universal map-length witness. That second control concerns induction and map length, not a claim that this particular junk model refutes map fusion. These are already connected HO mathematical examples. The native dependent proof-term realization and operational fusion refinement are constructed below; neither follows from this semantic paragraph alone.

Retained proofs compiled to native dependent terms. `FormationSensitiveHOLProofFamily` extends the uniform HOL declaration interface with a proposition-to-type decoder and two equations:

$$\mathsf{Prf}(P \Rightarrow Q) \longrightarrow \prod_{p:\mathsf{Prf}(P)} \mathsf{Prf}(Q), \qquad \mathsf{Prf}(\forall x : A.\, P(x)) \longrightarrow \prod_{x:A} \mathsf{Prf}(P(x)).$$

These are explicit equations of a candidate presentation, not consequences of naming a constant Prf, and not a global language selection. Their formation and capture-safe structural laws are proved. `HOLNaturalDeductionNativeTranslation` then traverses the actual retained HOL proof syntax. Hypotheses, implication introduction/elimination and universal introduction/elimination produce native variables, lambdas and applications. Unsupported rules produce no term. A successful traversal preserves the complete native judgment in a formed telescope whose object and proof environments realize the source assumptions. The emitted term commutes with native substitution; supplying a proof argument gives exactly the substitution computed by native beta reduction.

The closed quantified transitivity proof yields $\lambda P, Q, R, f, g, x.\, g(f(x))$, with its actual dependent typing judgment. Applying the existing higher-order list induction sentence yields the native application of its retained principle to its predicate, base and step. That is an application of an assumption, not a derivation of a native recursor from induction-shaped syntax. The full equality-based fusion proof is not covered by this initial compiler; a separate equality-aware compiler is treated below. Nor does the typing-preservation theorem establish consistency, conservativity, or a semantic model for the extended decoder calculus. In particular, the universal decoder accepts arbitrary native-small types, a larger domain than the represented HOL simple fragment.

`FormationSensitiveHOLProofConversion` supplies additional noncollapse results without assuming the missing injectivity laws. A concrete untyped expansion respects decoder conversion and separates Pi/Sigma from universe heads; universe conversion equates exactly levels with the same interpretation. In particular, U_0 and U_1 remain distinct. This one-way expansion forgets information and is not a typed semantic model. Typed substitution proves the beta laws for translated implication and universal introduction/elimination. Composing the implication law with the actual compiler proves that supplying an argument produces the same substituted proof term with the same complete native judgment before and after reduction. `HOLNativeProofSource`

`Substitution` additionally traverses substitution on the original retained HOL proof, rather than merely substituting its emitted native term. Its compatibility theorem relates represented source inputs to an independently formed native context morphism. Compilation commutes with this source traversal, including under mixed object and proof binders. Universal specialization and the native beta step consequently retain the same complete judgment. Substitution does not enlarge the compiler's supported proof fragment.

`FormationSensitiveHOLProofConfluence` establishes stronger properties for the exact decoder presentation itself. A parallel complete development proves Church–Rosser for the two decoder roots together with the underlying beta and projection reductions; no additional completion rules are introduced. Pi and Sigma conversion then recover component conversion, and their heads remain distinct. Inverting the actual declared application spines proves preservation for each decoder root; the structural preservation theorem extends this to arbitrary contextual steps and finite paths in a formed telescope. These are subject-reduction theorems for this presentation, not normalization or consistency, and do not automatically extend to a union with native List or identity-elimination rules.

Retained equality and native predicate consumption. `HOLLeibnizProofComparison` constructs retained proofs in both directions between primitive HOL equality and $\forall P : A \to \mathsf{prop}.\ P(x) \Rightarrow P(y)$. The reverse direction specializes the predicate to equality with x; neither direction adds a logical axiom or identifies this relation with native identity. Predicate reflexivity, transitivity and consumption use only rules already translated by the native compiler. Their emitted terms include $\lambda x, P, h.\, h$ and $\lambda x, y, z, h_{xy}, h_{yz}, P, h_x.\, h_{yz}\, P\, (h_{xy}\, P\, h_x)$, with complete native typing judgments. The primitive comparison proofs themselves still use unsupported native rules. The comparison commutes with source renaming and substitution; a retained proof's round trip preserves admission but need not reproduce its original proof tree.

`HOLLeibnizZFSetInterpretation` connects these retained equality proofs to the same set model described below. Predicate transport induces both an actual inclusion between separated sets and a map between dependent refinements, and the two maps agree on witnesses. The equality-induced maps commute with source proof substitution. A beta-equality example keeps the source proof's nontrivial inference tree; changing its parameter to the powerset of the empty set invalidates the required fibre and admits no corresponding equality proof in this model. This semantic square does not identify primitive HOL equality with native identity or supply a model of the full native decoder calculus.

`FormationSensitiveHOLLeibnizInterface` constructs a separate logical signature whose equality operator is the native closed expression $\lambda x, y. \forall P.\, P(x) \Rightarrow P(y)$. Its typing is proved before constructing that signature, using the unchanged quantifier and implication declarations. Equality application exposes the predicate formula by two ordinary beta steps. The existing two proof-decoder rules then expose the required dependent function type, without a new equality root. `FormationSensitiveHOLLeibnizRules` types introduction, specialization, consumption and reflexivity; reflexive consumption returns its original input proof by two beta steps, including at function carriers. The original primitive-equality signature remains available and unchanged.

`NativeHOLLeibnizDenotation` interprets both the actual equality application and its beta contractum using the existing constructorwise native denotation judgment. On admissible valuations, when the Henkin functions respect its equality relation, their predicate meaning is equivalent to the original HOL equality. This uses the retained source comparison and does not assume full predicate domains. The same square commutes with native substitution whose components have independently constructed interpretations. Unequal Boolean operands give a concrete negative

control; coherence excludes any alternative true denotation of that same expression. These are expression-interpretation theorems, not a model of arbitrary native proof inhabitants.

`FormationSensitiveHOLLeibnizDerived` constructs native symmetry, transitivity, application congruence and propositional transport from these predicate operations. `HOLLeibnizNativeProofTranslation` traverses the original retained proof syntax with those actual terms. Every successful translation preserves its complete formation-sensitive judgment and commutes with native substitution. Unsupported function extensionality, lambda equality and propositional extensionality rules explicitly return no translation; they are not needed by the retained fusion proof.

In `HOLLeibnizMapFusionNative`, the original `UniformListMapFusion.mapFusionProof` now successfully compiles, including its induction application. Abstracting the original five ordered theory assumptions gives a closed native proof of the corresponding conditional theorem. None of those assumptions is added as a native proof constant. The body is the original proof tree, not a separately written fusion argument. This advances beyond the earlier implication/universal-only compiler, which remains unchanged as a separate instance.

`HOLLeibnizRepresentationSemantics` expands equality throughout actual HOL terms. The old representation of an expanded term is exactly the alternative representation of the original. Independent Henkin semantics proves extensional agreement on admissible valuations, including equality nested beneath functions and quantifiers. Native denotation coherence therefore fixes the original truth value of every represented formula. `HOLLeibnizNativeProofAdequacy` joins this result with actual compilation: its native typing judgment, interpreted conclusion and source-justified truth refer to the same emitted proposition. The original fusion proof is an instance. A represented false arithmetic equation has neither a true native denotation nor a closed source proof in the model. This establishes source adequacy of successful translation, not soundness for arbitrary native inhabitants. Interpreting the source sequence carrier by native List and supplying the native induction/equation witnesses remain separate construction tasks.

Actual set operations and dependent refinements. `ZFSetHenkinInterpretation` supplies a full-domain HOL model on extensional `ZFSet`, with actual membership, union, powerset, separation and higher-order replacement. Its seven closed set-theory sentences, including higher-order set induction, are validated independently of proof acceptance. HOL domains are inhabited and Hilbert choice is supplied by the ambient classical model. For a set bound B and a HOL predicate φ, the bounded refinement is equivalent to the elements of $\{x \in B \mid M \models \varphi(x)\}$. This equivalence retains x, and separation commutes with actual HOL substitution. A retained universal implication proof induces inclusion of the separated sets; the resulting inclusion and dependent refinement map commute on witnesses. Two such implications imply extensional equality of the sets, not identity of their proof trees. The same model has both inhabited and empty refinement fibres and refutes an internal universal-set sentence. This construction is not by itself the HOTG universe operator or an interpretation of the native decoder.

Least internal universes under a stated model assumption. `ZFSetUniverseClosure` constructs the additional set operation. An inaccessible-rank segment V_κ is transitive and closed under union, powerset and higher-order replacement. Replacement closure follows from the cardinal bound on a member and regularity of κ, rather than an assumed universe-adequacy field. Inside any such enclosure of N, separation constructs the intersection of all closed universes containing N. Closure, membership of N, leastness, and independence of the initial enclosure are proved. Cofinally many inaccessible cardinals at the fixed set level suffice to make this operation UnivOf total. This is an explicit relative assumption, not a theorem supplying that cofinal family. Separately, a lower

Lean universe supplies one concrete closed rank segment at a higher set level and a least enclosure of the empty set; that example does not establish the global cofinal assumption.

`ZFSetUniverseInterpretation` extends the actual HOL set model by this operation. The four universe sentences state membership, transitivity, closure and leastness, matching the retained HOTG universe conditions. Their validity combines with the seven set sentences into a satisfied eleven-formula theory. Its retained proofs supply dependent truth sections in the same model. The signature reduct is the earlier set model; old bounded predicates therefore retain their separated sets. New predicates may themselves mention UnivOf, and their separated sets lie in the universe enclosing their bound. Membership and foundation imply $\mathsf{UnivOf}(\mathsf{UnivOf}(N)) \neq \mathsf{UnivOf}(N)$: this is not an idempotent subset-closure operation. Exact translation of the full imported HOTG preamble, and its common interpretation with the native dependent decoder, are not consequences of these fragment results.

`ZFSetDependentProducts` adds actual dependent set codes to these universes. A dependent sum is a set of Kuratowski pairs; a dependent product is a set of total single-valued graphs with the stipulated fibres. Their closure follows from replacement, powerset and separation. Graph lookup is uniquely determined, and encoding followed by decoding, and decoding followed by encoding, are proved inverse. The pair encoding is bijective. A nonconstant family is inhabited, while another example has an empty product because one fibre is empty although its dependent sum remains inhabited.

`ZFSetUniverseLevelInterpretation` uses the same relative cardinal hypothesis and a supplied seed set to construct $U_0 = \mathsf{UnivOf}(\mathit{seed})$ and $U_{n+1} = \mathsf{UnivOf}(U_n)$. Codes at level n are members of U_n; decoding takes their elements. Cumulative lifts preserve the underlying set, and the next level contains a code for the preceding code carrier. Dependent products and sums lie at the maximum of the input levels and decode equivalently to dependent functions and pairs. The existing native zero, successor, maximum and level-substitution semantics commute with this tower. The seed is a parameter, not an implicit choice of all bottom-level mathematics. These constructions supply universe and type-former model content; an interpretation of all native syntax, identity elimination and inductive interpretation require additional work.

`ZFSetContextualInterpretation` constructs contexts, substitution, comprehension and terms for these actual set codes. Lambda encodes a total graph, application performs its unique lookup, and pairing uses Kuratowski pairs. Their beta, eta and substitution laws follow from the independent encoding inverses. A nonconstant family and a noninjective context map exercise dependent computation and reindexing; a wrong constant result is excluded. `ZFSetContextualUniverseInterpretation` then instantiates the contextual Tarski hierarchy with these same operations. Its strict product, sum, lifting and substitution equations compare actual set codes; they do not identify graph carriers definitionally with Lean function types. The earlier HOL carrier consisting of the entire external ZF set type is not itself an internal set at that same ambient level. Connecting that carrier to these internal codes requires an explicit universe shift or another justified large-type interpretation.

`ZFSetUniverseLift` supplies that shift. Recursive pre-set lifting descends to an injective map of ZF sets, preserving and reflecting membership. Every larger-level subset of a lifted set is proved to be the lift of a lower-level separation; this establishes preservation of the actual target powerset, not just its lifted members. Empty set, union, pairs, separation and replacement also commute. The range of the lift is a larger set whose elements decode bijectively to the entire original carrier. It is transitive and closed under the actual universe operations. Conversely, no same-level set can have a decoding bijective with that entire carrier. These results need no inaccessible-cardinal hypothesis. `ZFSetLiftedCarrierInterpretation` places this code in the fixed unbounded contextual tower and supplies actual decoded terms and operation equations. That placement

explicitly assumes cofinal inaccessibles at the larger ambient level; it does not silently promote the lower-level assumption.

`ZFSetLiftedUniverseClosure` also constructs the least closed enclosure of a lifted input inside this larger carrier. Closure is preserved and reflected by the lift. Classification of bounded subsets turns that larger enclosure back into a lower closed set; the two minimality properties then show that it is exactly the lift of the original UnivOf. Consequently the actual carrier operation, not an operation defined by transporting the old answer, commutes with decoding under the original lower-ambient hypothesis. Comparing it with a globally available larger-ambient universe operator states the additional hypothesis separately. Containment and minimality are proved, and membership irreflexivity excludes idempotence of this universe operation.

`ZFSetHOLTypeInterpretation` codes all types of that same HOL model recursively: propositions by two truth values, the set sort by the shifted carrier, and arrows by actual total graphs. `ZFSetHOLTermInterpretation` independently traverses every source term constructor using these graph and set operations. Induction on source syntax proves agreement with the original Henkin denotation, including quantifiers, extensional equality, the universe operation and substitution. `ZFSetHOLContextualInterpretation` then identifies valuations with contexts of the existing set-coded CwF. Its explicit comprehension equivalence, graph lambda/application comparison and substitution squares connect the same interpreted terms to the dependent contextual structure. A variable-identifying substitution cannot reproduce an off-diagonal truth valuation; these lawful context maps are not assumed invertible.

`ZFSetHOLProofInterpretation` interprets retained source proof trees into actual truth fibres and maps between separated refinement sets. These maps agree with the earlier Henkin dependent-family construction, including simultaneous substitution. Bounded refinement of the shifted set carrier is exactly the lift of the original separated set. A retained function-eta proof supplies a higher-order positive example, while a changed predicate rejects the original point and the false fibre has no inhabitant. This extensional interpretation does not recover proof syntax from truth.

There is a strictness obstruction to extending this particular assembly directly to the native proof decoder. In `ZFSetHOLProofDecoderBoundary`, the truth fibre of $\top \Rightarrow \top$ is $\{\varnothing\}$, whereas the actual function-graph code from $\{\varnothing\}$ to itself does not contain $\varnothing$: its unique element is a nonempty graph. These sets are unequal even though their element carriers have an explicit equivalence. Thus this particular graph-code interpretation does not validate the native Prf-implication conversion. The following constructions separate a weak graph comparison from a strict trace-code repair; neither imposes a global proof-erasure or identity policy.

`ZFSetHOLWeakProofDecoding` constructs the two weak comparisons for arbitrary implication and universal quantification over the original HOL carriers. The comparison maps commute with source substitution; introduction produces actual graphs and application satisfies beta–eta. A retained higher-order function-eta proof is interpreted over function graphs, while a nonconstant predicate with an empty fibre has an empty universal product. These comparisons do not turn an equivalence into equality of universe codes. Separately, `ZFSetHOLContextualProofConsumption` makes each retained universal-implication proof's refinement map a first-class term of the same set-coded CwF. Its graph application agrees with the original Henkin map, preserves the represented value and commutes with contextual reindexing.

The function representation itself supplies a stricter alternative. Aczel's trace encoding replaces the graph of a set-valued function by $\{(x, z) \mid x \in A,\ z \in f(x)\}$, with application recovering the fibre. Lee and Werner use this uniform encoding to handle the interaction between proposition proofs and dependent function types [65]. `ZFSetTraceProducts` constructs this encoding on actual sets, its typed application, inverse comparison with function graphs and beta–eta laws. The encoding is faithful at a fixed typed domain, but an untyped trace need not recover its domain:

different domains can give the same empty trace.

`ZFSetTraceContextual` equips the same set-coded CwF with these dependent products. Lambda, application and graph comparison commute with arbitrary contextual substitution. This is a uniform construction for arbitrary set-valued fibres, not a special case for proof types. For $\mathsf{Tr}(P) = \{\varnothing \mid P\}$, `ZFSetTraceProofDecoding` proves the actual set equality

$$\Pi^{\mathrm{trace}}_{x \in A} \mathsf{Tr}(P(x)) = \mathsf{Tr}(\forall x \in A.\, P(x)).$$

The implication case follows by taking the domain to be a truth fibre. These equalities are stable under reindexing and may be used beneath arbitrary families depending on the actual code; the dependent consumer keeps its original set element. In particular the true-implies-true case that fails for graph products succeeds here, while true-implies-false still has empty code.

`ZFSetTraceUniverseInterpretation` uses the existing actual cumulative hierarchy, not a new assumed hierarchy. Trace product codes lie at the maximum input level and their decoding and cumulative lifts commute. The displayed truth-code equality also proves that a proof product over any set domain in the same ambient set universe lies in the bottom internal universe. This establishes these model components and decoder equations, not yet soundness for every native identity, universe and inductive rule together. Retained proof presentations remain separate from this extensional meaning.

`ZFSetHOLTraceTypeInterpretation` uses these trace products at every simple arrow type, recursively including higher-order domains. The set carrier and truth-value code are unchanged. An independently recursive interpreter in `ZFSetHOLTraceTermInterpretation` handles every original HOL term constructor, with the same actual set operations and universe interpretation. Structural comparison with the graph interpreter proves agreement with the original Henkin model and simultaneous substitution. Graph and trace arrows are explicitly proved unequal as set codes; the comparison does not silently identify them.

`ZFSetHOLTraceProofDecoding` attaches the generic strict laws to these actual source expressions. Implication and universal proof fibres are literally contextual trace products, including lifted source substitution. Retained source proof trees cross the existing proof-family interpretation, and universal elimination uses actual trace application. A retained higher-order eta proof is consumed at the trace-coded powerset function. The false implication has empty fibre and no source proof under the stated model assumptions. This is a source-to-model result, not yet the full semantics of its separately compiled native proof term.

Actual lists and their universe boundary. `ZFSetList` represents nil by the empty set and cons by an ordered pair of the head and tail code. The constructors are distinct and injective; the actual list set has a dependent eliminator with both computation equations and a uniqueness theorem. `ZFSetListContextual` proves contextual substitution laws for these constructors, maps and eliminators on the same CwF. `ZFSetTraceListContextual` independently builds the same contextual list operations using trace lambda and application. Constructor computation and substitution hold, and its actual eliminator is compared with the graph eliminator through the typed trace map. A varying motive has different nil and cons results, rejecting a constant-result replacement.

Closure under the earlier finite set operations and replacement alone does not imply that the entire list set is an element of the universe. `ZFSetIndexedClosure` constructs the counterexample V_ω, and `ZFSetListClosure` proves that the list set on its singleton empty-set carrier is not an element of it, although every individual encoded list belongs to it. A separate natural-index seed supplies the missing closure: replacement builds the sequence of constructor stages and their union is proved equal to the actual list set. Starting the existing least-universe tower from the parameter

and this index seed consequently supports list codes at every level, with literal compatibility under cumulative lifting. List smallness is thus derived from an independent seed, not assumed as a premise of itself.

`ZFSetUniformListModel` interprets the exact original `mapFusionProof` over these actual set-coded lists. All five original assumptions are independently validated; the induction assumption uses the constructed dependent list eliminator. This does not replace the source proof by a separate semantic proof of map composition. `ZFSetUniformListProofConsumption` interprets its existing predicate-consumption proof as maps between actual separated list sets. Graph and trace applications agree with the same retained-proof interpreter and preserve the input list code. The map codes belong to the same universe under the explicit natural-index premise. A two-element carrier validates the five assumptions while refuting the reversed-composition HOL sentence. Empty element carriers are permitted by this choice-free fragment; full HOL choice over such carriers is not asserted.

`ZFSetUniformListTraceTypeInterpretation` now gives this exact uniform-list theory a trace-coded simple-type interpretation: the three ground carriers are unchanged and every arrow is an Aczel trace product. `ZFSetUniformListTraceTermInterpretation` traverses every original HOL term and proves, structurally, that decoding its trace value gives the existing set-model denotation. Thus this is a change of function representation within one validated list model, rather than a second model chosen to agree only on the final theorem.

`ZFSetUniformListTraceProofBridge` applies the retained `mapFusionProof` three times through the literal trace products. At each functions-and-list instance, the resulting source proof fibre is the same set code as contextual dependent identity between the two map terms. The actual contextual (J) operation can therefore consume that source proof under an arbitrary set-valued motive. Reversed composition has no inhabitant in the same trace interpretation, by the concrete two-element countermodel rather than by failure of proof search.

On the native side, `NativeTraceLambdaSemantics` gives actual native variables, lambdas and applications a typed trace denotation. `NativeHOLNaturalDeductionTraceAttachment` independently compiles the corresponding natural-deduction fragment and proves that erasure is exactly the output of the existing compiler; implication identity supplies a semantic positive case. Constants are not hidden behind an opaque term. Instead, `NativeHOLUniformListTraceRepresentation` interprets the exact native representation of every supported constant-bearing HOL term constructor-by-constructor and proves agreement with the trace term interpreter. `NativeHOLTraceMixedTermSemantics` combines object values and retained proof sections in telescopes where their binders may interleave.

`NativeHOLTraceLeibnizProofSemantics` then interprets the exact native proof combinators used by the equality-aware compiler. Native lambda and application realize implication and universal introduction and elimination through the literal trace-product decoder. Leibniz equality is interpreted as transport through all trace predicates; symmetry, transitivity, congruence and propositional transport are consequently derived from the emitted terms rather than added as semantic operations. `NativeHOLTraceLeibnizCompilerSemantics` proves by induction over the complete retained source proof that every successful compilation in this supported fragment denotes a section of the original conclusion's truth family. Source objects and hypotheses are independently interpreted, and weakening, instantiation and mixed object/proof binders commute with the construction. Unsupported proof rules remain unsuccessful compiler cases.

That partiality now has an exact executable boundary rather than an informal constructor list. `HOLLeibnizNativeProofCoverage` defines a structural classifier over retained HOL proof trees and proves, for every native object and hypothesis environment, that the real compiler returns a term if and only if the classifier accepts. Compiler support is therefore environment-invariant even though

emitted terms depend on the environment. Positive implication proofs are accepted, while valid instances of function extensionality, equality under lambda, and propositional extensionality are explicitly classified as unsupported. Consequently failure to compile one of these proofs is a coverage result, not a refutation; extending the boundary requires a separately justified extensionality realization.

Finally, `NativeHOLCompiledMapFusionTrace` applies this general theorem to the compiler's exact open output for the original `mapFusionProof`. Its five ordered source assumptions form an explicit semantic telescope and receive witnesses from the independently validated uniform-list model. Evaluating the retained section at that environment and applying three trace-coded universal eliminations yields the map-fusion proof value. Its truth fibre is literally the contextual identity family between the two map programs, so the existing dependent eliminator J consumes that same compiled value under an arbitrary set-valued motive. This is one proof crossing source syntax, compiler output, set-valued trace meaning and dependent consumption; it is not a second handwritten fusion proof. It still does not establish conservativity for arbitrary native inhabitants or the full common HOTG/native universe interpretation.

`NativeHOLCompiledMapFusionUniverse` composes this crossing with the existing seeded internal universe tower under the explicit cofinal-inaccessibles assumption. Every simple type of the uniform-list theory, every proof-hypothesis family in the mixed telescope, the exact open compiler section, the specialized fusion proof value, its contextual identity family and the concrete endpoint motive consumed by J are codes or values of the same bottom universe. This is a theorem about the actual sets used by the construction, not an interface postulating smallness. The negative control remains essential: elementary set closure and membership of every individual encoded list do not imply membership of the collecting list type. Thus the natural-index seed is a mathematical premise of this common universe construction, not bookkeeping that may be erased.

The uniform-list element carrier can now be chosen to be the actual shifted code of the entire lower set universe. In `NativeHOLCompiledMapFusionHOTG`, recursively translating each set-theory simple type into the uniform-list signature gives literal equality of its trace code with the full set/HOL trace interpretation. In particular, the set and set-to-set function codes agree definitionally at the bridge. The powerset constant and the independently constructed least-universe operation may therefore be supplied directly as the two functions of the exact compiled fusion proof. Their applications decode respectively to $\mathcal{P}(x)$ and $\mathsf{UnivOf}(x)$, and the specialized proof is consumed by the same dependent J operation. No user-facing coercion or new fusion axiom is introduced.

This literal agreement requires a real universe shift. There is no set at the lower ambient level whose elements are equivalent to all sets at that same level, since such a code would yield an internal universal set. The least-universe operation uses the stated lower-ambient inaccessible assumption; placing the resulting list/proof construction in an independently selected unbounded tower one level higher uses a separately stated higher-ambient assumption. Neither is inferred from the other.

`NativeTraceContextMorphisms` isolates the categorical structure needed for that remaining theorem. Its context morphisms are oriented like substitutions, compose strictly, lift through comprehension while retaining the newest component, and reindex trace products, abstractions and applications by the existing Aczel-trace substitution laws. A non-surjective control records why reindexing by itself cannot establish reflection or conservativity. The displayed syntax attachment remains a separate obligation: dependent context records are not identified merely to avoid their fibre transports.

The rule data actually execute. `DeclarationAwareSubstitutionExecution` orients all thirty original binding schemas into recursive operations of the generic `LanguageDef` interpreter.

Unsupported schema forms are refused; none of the thirty authored cases is lost. Structural proofs compare execution with independently defined weakening and substitution, including explicit sufficient derivation-depth bounds. `DeclarationAwareSubstitutionExecutionBeta` identifies the result with native scoped instantiation. A binder-crossing example returns the correctly shifted free variable and never returns its captured variant. No native substitution callback or supplied derivation is an input.

`HOLNativeMixedRuleData` puts these helpers, ten computation roots and fifteen native context positions into one flat fifty-five-rule language. These contexts do not descend into encoded names, indices or universe-expression metadata. The generic closed-subsystem theorem proves that helper queries have exactly the same output lists after extension, including order and multiplicity. `HOLNativeMixedRuleDataCompleteness` proves that every computational step and finite derivation of the unchanged native profile is realized by this actual interpreter. `HOLNativeMixedRuleDataSoundness` proves the converse on canonical encodings: every matcher output reconstructs an independently defined native computational step. The equivalence lifts both to the OSLF contextual-step interface and to finite derivations. The proof exhausts all fifty-five rules: thirty binding rules are separated by their operation heads, while every root and contextual computation rule is reconstructed. It does not claim soundness for malformed encodings or add determinism, confluence or normalization.

The full profile additionally compares semantically equal universe-level spellings. `HOLNativeMixedOperationalDecomposition` gives an exact step and finite-derivation decomposition, with an independently proved exact normal-form query for head equality. The expression $\max(u, u)$ can step to u in the full relation but not in its purely computational component. A finite computational rule inventory is therefore not claimed to enumerate the full equality relation.

Native execution consumes the HOL-derived property. `IntrinsicNativeListMapComputation` proves directed beta/iota execution of the existing native List eliminator on arbitrary finite constructor spines with open element terms. Both pure map programs reach the same native constructor output; function composition is an ordinary lambda, not an added fusion rule. For the GSLT built from this actual reduction relation, the generated finite-execution OSLF observation satisfies

$$\Diamond_{\mathrm{fin}} O(t) \quad \Longleftrightarrow \quad \exists u.\ t \longrightarrow^{*} u \ \wedge \ O(u).$$

`NativeListHOLPredicateObservation` observes native spines using the independently defined native denotation of their element terms. The original object-HOL induction/congruence proof transports an arbitrary property of the unfused value, which this observation consumes after either native program. Code, paths and observations respect interpreted context substitution. With f constant true, g Boolean negation and input [false], correct composition observes [true]; the reversed-composition constructor output cannot satisfy that observation.

The result concerns existential finite execution, not every execution, equal traces, effects or costs. The reduction-based GSLT is not the separate textual `LanguageDef` interpreter. Native List spines have a concrete observation here, but the abstract HOL sequence carrier is not thereby identified with native List, and the full common typed execution and native dependent proof-consumption judgments are distinct obligations. `FormationSensitiveNativeHOLMapExecution` now joins the actual native List/relator declarations and retained HOL declarations at a common level, with explicit rule morphisms. It independently formation-types the existing eliminator-based map, its generic applications and encoded outputs. Both fusion programs and their common result have complete judgments in one formed telescope, and both full reduction paths belong to those rules. The HOL-derived predicate observation composes with that admission. A reversed-composition result is well typed but rejected by the observation, separating typing from correctness

of the requested equation. Endpoint typing and path inclusion do not establish unrestricted subject reduction of the mixed profile, full native equality-proof attachment, or the common HOTG interpretation.

`FormationSensitiveHOLProofListIntegration` further combines the proof decoder with the native List/relator presentation. Explicit morphisms lift the actual proof compiler's complete judgments and the native map execution paths into one extension. In a common formed telescope, the emitted predicate consumer $h_{xy}\,P\,h_x$ has its dependent proof type while a nonempty singleton map computes by the existing eliminator rules. The HOL sequence and native List variables remain distinct: sharing a telescope is not an interpretation equating their carriers. The conditional fusion proof translation is supplied by the alternative compiler described above. Actual native implementations of all its source theory inputs remain a further obligation.

`FormationSensitiveConstantInterpretation` supplies a generic typed substitution of closed bodies for declared constants. Preservation of contexts, substitutions, conversion and complete judgments follows from independently typed declaration images and root conversions. `HOLNativeListDeclarationInterpretation` instantiates it with actual native List, constructors, map and an eliminator-defined length. `HOLNativeListInduction` constructs the original represented predicate-quantified induction premise using the native dependent eliminator. `HOLNativeListEquationInputs` proves all four original map and length equations by native beta/iota computation. Finally, `HOLNativeListProofInputs.native_map_fusion` applies these five witnesses to the original retained compiler output and obtains a complete empty-context native judgment of the original uniformly quantified formula. No source premise is silently dropped, and no equation or induction witness is an opaque added declaration. A concrete instance uses the existing small count carrier, without asserting a natural-number interpretation for it. This discharges the native-input obligation just described; it does not yet give an independent semantic model of all native inhabitants or full HOL conservativity.

`HOLNativeListProofConsumption` specializes that very same compiled proof at arbitrary typed native functions and lists in a formed caller context. For $P : \mathsf{List}(E) \to \mathsf{prop}$, it checks

$$((\mathit{compiledFusion}\ f\ g\ xs)\ P)\ h : \mathsf{Prf}\big(P(\mathsf{map}(f \circ g)\, xs)\big)$$

from $h : \mathsf{Prf}(P(\mathsf{map}\, f\, (\mathsf{map}\, g\, xs)))$. The caller's functions, list and predicate need not be source-HOL expressions. The small element type is closed, and the predicate ranges over HOL propositions; this is not transport along native identity into an arbitrary universe-valued family.

`FormationSensitiveInterpretationComposition` proves that the declaration interpretations used here compose on actual bodies, terms, contexts and typed substitutions. Successive translation agrees with the single composed translation. Associativity holds, and identity is qualified by actual declaration formation. An explicit constant-expansion example shows that changing interpretation order can change the resulting program.

`FormationSensitiveMixedListElimination` checks the generic dependent eliminator on arbitrary mixed-typed motives and branches. `FormationSensitiveMixedHOLProofRules` similarly admits proof introduction and elimination on predicates over these native List types, reusing the independently constructed decoder equations.

`FormationSensitiveHOLNativeMixedPreservation` proves full contextual subject reduction for the unchanged joint profile, including all five native and both proof-decoder roots. Pi/Sigma conversion boundaries are obtained through a separately proved conservative auxiliary development; each auxiliary contraction is convertible by the authored rules, but is not installed as an executable rule. Thus every finite operational endpoint of an admitted term retains its displayed type, including endpoints selected by the generated OSLF observation. A decoder step in one duplicated metadata

argument temporarily blocks an exact native iota root; an explicit authored path reconciles both copies and joins the peak. A separate formed example computes a nonempty map beneath a proof-family binder. The development proves these conversion boundaries and preservation, not raw execution confluence, normalization or textual `LanguageDef` adequacy.

10.6.2 Three different intensional–extensional bridges

Even at a fixed context, three mathematically different constructions must not be called the same "commutator." Let $\mathsf{Fam}(\Gamma)$ be the category whose objects are families $A : \Gamma \to \mathbf{Type}$ and whose morphisms are fibrewise functions. There is a checked adjoint triple

$$\Sigma_\Gamma \dashv \Delta_\Gamma \dashv \Pi_\Gamma, \qquad \Sigma_\Gamma A = \sum_{\gamma:\Gamma} A(\gamma), \quad \Delta_\Gamma X = (\lambda\gamma.X), \quad \Pi_\Gamma A = \prod_{\gamma:\Gamma} A(\gamma). \tag{B}$$

The two hom-set equivalences are dependent-sum and dependent-product currying. The family category maps fully faithfully to the fixed-Γ fibre of the set-families CwF, so (B) is not an artifact of a weakened semantics.

However, $\Delta_\Gamma : \mathbf{Type} \to \mathsf{Fam}(\Gamma)$ is generally not full. For $\Gamma = \mathbb{B}$, the fibrewise map from the constant unit family to the constant Boolean family that returns the current index cannot be the image of one uniform function $1 \to \mathbb{B}$. It is faithful whenever Γ is inhabited. By contrast, the live simple-type fibre over Γ already admits display maps that depend on $\gamma : \Gamma$; its constant-family inclusion into the dependent fibre is fully faithful. The two embeddings agree on objects but have different source hom-sets. Thus the ordinary constant-diagram functor must not be declared fully faithful merely because the contextual simple-to-dependent inclusion is.

The third construction is quotient completion. The checked equivalence $C/\!\sim_d \simeq \mathrm{EssIm}(d)$ below is the exact quotient of one denotation map in sets. It is not yet an elementary quotient completion of a dependent doctrine, nor a coherence or conservativity theorem connecting a chosen intensional calculus with an extensional one. Those stronger bridges require stable quotients, substitution/comprehension coherence, and a theorem such as conservativity or Morita equivalence [67, 68, 69].

The backward direction is necessarily more selective. A host object can be reified only when it carries a native code witness, for example

$$\mathsf{Coded}(X) = \sum_{c:U} (\llbracket c \rrbracket = X).$$

For such objects, interpretation after reification recovers X. There is in general no total inverse from arbitrary host sets to codes. A checked tagged-code model makes the obstruction exact: two distinct native codes can have the same extensional denotation, so a left inverse would falsely prove those codes equal. A genuine equivalence is available only after restricting to the coded essential image and either retaining code witnesses or quotienting codes by the chosen semantic equality. "Back and forth" is thus a span with a total forward interpretation and a code-indexed reverse, not an unqualified bijection between syntax and sets.

The reversible core is now an exact checked theorem. For any denotation map $d : C \to H$, let $c \sim_d c'$ mean $d(c) = d(c')$, and let $\mathrm{EssIm}(d)$ be the subtype of host objects having a code. Then

$$C \longrightarrow C/\!\sim_d \;\simeq\; \mathrm{EssIm}(d) \hookrightarrow H. \tag{C}$$

The factored denotation $C/\!\sim_d \to H$ is injective, and it is surjective exactly when the original d is. Hence quotienting repairs nonfaithful intensional multiplicity but cannot manufacture codes for

extensional-only objects. The forward factorization and injectivity are choice-free in the formalization; packaging the essential-image bijection as an inverse-bearing equivalence uses Lean's standard classical choice.

This yields three honest host zones rather than a forced syntax/set identity: objects with multiple intensional codes; the semantic quotient or code-witness-retaining essential image; and extensional-only objects with no native code. A Boolean-tagged code model realizes the first zone and becomes equivalent to its host after quotienting. An injective two-code map into $\mathbb{N}$ realizes the third zone: the host object 2 remains unrepresentable even after quotienting. HOTG's freedom to contain sets lacking canonical computational codes is therefore useful mathematical flexibility, but it is not a reverse compiler.

This also explains the relationship between Lean and Megalodon. Lean can state and verify the native adequacy interface, and it can formalize a syntax and semantics for HOTG. It does not thereby identify its ambient hierarchy of `Type` universes with Megalodon's object theory. Megalodon independently checks the extensional HOTG theorems; a cross-system certificate or a common foundational replay layer would add implementation diversity, while the mathematical bridge still consists of the interpretation and naturality proofs above. HOTG is a pinned extensional theory rather than a magical external universe imported definitionally into dependent type theory [58, 57].

10.7 Why a reflective reasoner wants a universe tower

The practical value of the tower is controlled reflection rather than largeness for its own sake. A sufficiently capable reflective reasoner would use it for the following tasks.

1. **Scoped totalities.** A task may quantify over all objects in one declared universe without postulating a universe of everything. The next universe is available when the scope must itself become data.

2. **Internal metatheory.** The syntax, derivations, checker, and models of a lower theory can be represented and compared at a higher level. This is the size discipline needed by proof-producing self-improvement; it is not a same-level self-soundness principle.

3. **Categorical closure.** Categories of small structures, functor categories, presheaves, sheaves, natural transformations, and categories of models typically live one size above their components. Explicit enlargement makes these constructions reusable rather than bespoke.

4. **Plural perspectives.** Several observer worlds or candidate models can remain locally small while a later universe contains their comparison diagram. Enlargement enables comparison without collapsing the observers.

5. **Trust and provenance.** A theorem can record the universe and assumption profile used by its model, checker, and consistency qualification. Stronger principles are then invoked locally rather than inherited silently by every computation.

6. **Graceful extension.** New mathematics, sensors, agents, and theory fragments can be admitted by moving upward while preserving earlier judgments. Conservativity and interpretation theorems say precisely what is retained.

7. **Efficient implementation.** When programs do not inspect universe codes, level constraints and enclosure witnesses may be elaborated and erased. Reflective programs retain only the codes and provenance they actually use.

These benefits suggest a consumer-driven architecture: a small native universe service, a proof-relevant interpretation into an extensional host, and optional strong profiles attached to named tasks. They do not justify installing the strongest known universe principle in every kernel.

Theorem 23 (Finite support and open ascent)**.** *For every finite list S of natural-numbered universe requirements, there is a strict host stage*

$$\mathsf{next}(S) = \max\{s + 1 \mid s \in S\}$$

(with a fixed base value for the empty list) such that every $s \in S$ lies below $\mathsf{next}(S)$. There is no final natural-numbered stage.

This theorem is machine checked and captures the size-theoretic part of the promise that a reflective reasoner can keep moving upward: every finite metareasoning task can be hosted later, and no finite stage is declared final. It does *not* by itself prove that every future theory extension is sound. Safe indefinite ascent additionally requires coherent embeddings, preservation or explicit revision of old judgments, a named model or qualification for each new rule package, finite support for certificates, and resource policies which eventually service each accepted obligation. Under those contracts the reasoner has no fixed universe ceiling; it does not acquire a same-level proof of its own global correctness. This is the precise, non-paradoxical content of "reason up the stack indefinitely." For category-theoretic consumers, universe-change invariance must likewise be proved for the constructions being transported [70, 71].

10.8 Four stratifications that should stay apart

Open design note. The checked results of Sections 8.4, 8.5 and 12.2 each contain a "level", but the levels are of four different kinds. A reflective language should keep them as independent indices rather than compress them into one integer.

Stratification	What the index measures	What it prevents
Reflective level ℓ	Which interpreter's code and state a computation may inspect; successor with no top	Uncontrolled same-level access to one's own evaluator
Computational delay $\triangleright$	Available approximation steps of a recursive definition	Unguarded recursive feedback
Truth definedness	Information about truth: true, false or undetermined, with grounding stages	Totalizing self-applied truth (the liar)
Logical reflection strength	Which theory proves a statement, and which consistency or reflection principles it has	Treating "this theory proves" as harmless evaluator reflection

The identifications that suggest themselves fail for stated reasons.

- *Truth stage is not delay.* Kripke's construction and the later modality both replace a self-referential equation by approximations [116, 123], but along different orders. For arithmetic the least truth fixed point closes only at the first nonrecursive ordinal [116], whereas the standard model of guarded recursion is indexed by ω [124].

- *A guard does not make truth total.* A guarded Löb rule is valid because its antecedent refers to a strictly earlier approximation. Same-level reflection of truth or provability has a different strength: Curry and Löb collapse a classical system over literal self-reference (`hilbert_trivial`), and a sound arithmetic stage cannot prove its own consistency (`SoundStage.not_proves_own_consistency`).
- *Existence is not truth.* A final coalgebra supplies cyclic objects, but not a total truth predicate on them (`no_boolean_valuation`). Cyclic live state and circular propositions may therefore need different coalgebras, with different carriers and notions of observation.
- *Reflective level is not universe level, and not theory strength.* A small term can sit at a high reflective level, and a large type need not be reflective. An interpreter at level $\ell + 1$ need not embody a sound proof theory for level ℓ. The consistency tower ascends in theories, not in interpreters.

The literature on reflective towers supports keeping the reflective level unbounded while charging nothing at run time for unused levels. The interpreter tower of 3-Lisp [126] has an ordinary non-reflective denotational account [127], its levels can be given an explicit semantics [128], and multi-stage specialization can collapse a tower to residual code [129]. The open questions lie at the interfaces. Brown and Palsberg built a self-interpreter for a strongly normalizing calculus by indexing code by type [130]. Bauer's note shows that their stated criterion also admits a value-carrying "interpreter" for System T, and that the type of codes cannot be of lower complexity than the type of the program. It asks for a definition by a structure-preserving homomorphism between calculi, since commuting with substitution alone is not general enough [131]. With untyped code and internal type checking, the normalization barrier returns [132]. For Nakano's original system with subtyping, the known inference algorithm is sound but not complete, and decidable inference is proved for a variant without subtyping [125]. None of these choices is made here, and no candidate calculus is selected by this note.

10.9 Relative infinity without ultrafilter selection

Constructive analysis suggests a useful intermediate profile. Quotient natural number sequences by explicit tail equality,

$$f \sim_{\text{tail}} g \quad \Longleftrightarrow \quad \exists N\, \forall n \geq N,\ f(n) = g(n).$$

The class $\omega_{\text{tail}} = [n \mapsto n]$ is strictly above every constant class and is not in the diagonal image of $\mathbb{N}$. This produces a relative infinity without selecting a free ultrafilter.

Theorem 24 (Choice-free relative-infinity baseline)**.** *In the explicit tail quotient,*

$$\forall k : \mathbb{N}, \quad [k] < \omega_{\text{tail}}, \qquad \omega_{\text{tail}} \notin \operatorname{im}(\Delta).$$

The formal proof uses quotient soundness and propositional extensionality but no choice or ultrafilter-selection principle. The even-high and odd-high oscillating classes are incomparable in the eventual order.

The incomparability is the exact price. This reduced power is not a linearly ordered non-Archimedean field and it has no full Loś transfer theorem. Constructive real analysis can enrich representatives with Cauchy moduli, locatedness, or apartness, while filters and nets can index observer-relative scales. Those additions preserve explicit computational evidence but do not magically supply total order or first-order transfer. A free ultrafilter is a separate classical profile when those stronger properties are actually needed.

10.10 Principal collapse and free-ultrafilter divergence

Let M be a model, I an index type, and $\mathcal{F}$ an ultrafilter on I. The ultrapower $M^I/\mathcal{F}$ quotients functions by equality on an $\mathcal{F}$-large set. The principal and free cases have opposite meanings.

Theorem 25 (Principal collapse)**.** *If $\mathcal{F}$ is principal at $i \in I$, evaluation at i and the diagonal map are inverse model homomorphisms:*

$$M^I/\mathcal{F}_i \;\cong\; M.$$

Consequently the diagonal is bijective and the principal ultrapower adds no new points or model-theoretic strength.

This is the indispensable negative control. A principal ultrafilter may encode a selected observer or coordinate, but it cannot supply the desired ultrainfinite divergence.

For a free ultrafilter on $\mathbb{N}$, the diagonal need not be surjective. The current formal model proves that the growing sequence represents a point outside the diagonal copy of $\mathbb{N}$. In the corresponding hyperreal construction, the canonical element ω is larger than every embedded real, so the real embedding is not surjective. Both the principal and free carriers can be enclosed by the family-enclosing universe operator; enclosure organizes their dependent types but does not create their nonstandard points.

This yields a rigorous *magnitude-separating* model, not yet a new foundation. The free-ultrafilter construction uses classical choice and belongs to ordinary nonstandard model theory. It contributes a genuine divergence witness—unlike a principal ultrapower—but it does not by itself provide non-well-founded depth, open-ended reflection, or stronger proof-theoretic principles.

10.11 The independent-axis synthesis

The clean construction keeps five mechanisms distinct.

Mechanism	What it contributes
Family-enclosing universe	Native-looking dependent organization and closure around any selected family; no nonstandard points by itself.
Tail reduced power	Choice-free relative infinity with explicit finite-tail evidence; partial order and no full Loś transfer.
Principal ultrapower	A coordinate-selection semantics and collapse canary; canonically equivalent to the base model.
Free ultrapower	Nonstandard magnitude and transfer-relative perspectives; classical, and not automatically foundational.
Indexed final coalgebra	Potentially non-well-founded depth and infinite observation; existence and non-collapse require separate proofs.

The strongest coherent target is consequently a composite model, not an identification: an intensional dependent construction language equipped with a family-enclosing universe operator; a named set-theoretic or categorical host which prices smallness; tail-reduced perspectives when explicit relative scale is enough; free-ultrafilter perspectives when total nonstandard magnitude and transfer are wanted; and an indexed final coalgebra when non-well-founded observation is wanted. Observer maps then state exactly which distinctions survive. No one component is allowed to borrow the advertised effect of another.

11 The Grothendieck–Indra atlas as a conditional design target

A disciplined candidate for mutual reflection is not one set containing a transparent copy of every object in a proper class. It is a staged atlas of locally small, globally open-ended perspectives. The word "Indra" names the intended pattern of mutual visibility; it does not add a set-theoretic axiom or an existence theorem.

Definition 26 (Grothendieck–Indra atlas)**.** A Grothendieck–Indra atlas consists of:

(a) a strictly open sequence of host universes or stages $U_0, U_1, \ldots$;

(b) a stagewise small observer object C_n and small route family V_n;

(c) an observer-indexed small-based functor H_n and, when constructed, its final coalgebra E_n in a declared later host;

(d) veil-preserving transition maps from earlier to later stages;

(e) coherence of those maps with destructors and observer readouts;

(f) *eventual total visibility*: for any two earlier observers or finite diagram of views, some later stage contains a perspective referencing all of them; and

(g) *finite factorization*: every finite proof, certificate, and bounded observation of the atlas factors through one stage.

Warning 27 (No strength from the atlas shape). The atlas data alone supplies neither a larger universe nor a final coalgebra. Its foundational strength is exactly that of the host theories and construction principles used to instantiate its stages. Before those stages and coherence maps are built, the atlas is an indexing contract. A finite colored graph is a negative boundary specimen for local visibility, not a small model of the claimed infinite object.

Every jewel is locally incomplete: its foreign images are opaque and its support is small. Yet the atlas has no final jewel, and every finite constellation is reflected later. This strengthens a two-color knot in three ways: observers form an extensible category rather than a fixed pair; routes retain identity and provenance; and the global object is a coherent class diagram rather than a fictitious set of all perspectives.

If the obligations are instantiated, this yields "infinite reflection" in a mathematically disciplined sense. Unbounded depth and stage growth need not be finite merely because each local support is small. No local carrier enumerates a proper class. Opaque cross-world reference remains distinct from local containment.

Experiment 28 (Atlas falsifiers)**.** Reject or revise the atlas if any of the following occurs:

(i) an intended stage functor is not small-based;

(ii) coherence forces transparent foreign membership;

(iii) every route-rich distinction collapses before the declared observer;

(iv) the global atlas can be replaced by one ordinary set without changing any claimed theorem; or

(v) a finite proof uses unboundedly many stages.

12 Alternating qualification towers and the helix non-strength theorem

Let polarities be exchanged by a fixed-point-free involution $p \mapsto \mathsf{op}p$. For each stage n, let $T_{p,n}$ be a finitary rule system and let $M_{p,n}$ be a semantic model. The basic alternating edge is

$$M_{\mathsf{op}p,n+1} \models T_{p,n} \quad \text{and} \quad M_{\mathsf{op}p,n+1} \not\models \bot. \tag{7}$$

Rule soundness in (7) implies $T_{p,n} \nvdash \bot$. The current formalization proves this for arbitrary finitary rules and arbitrary world-indexed models. A negative theorem shows that a direct-bottom rule admits no such qualification.

If, for fixed p, the rules grow monotonically with n, every derivation in the union

$$T_{p,\infty} = \bigcup_n T_{p,n}$$

uses finitely many rule nodes and therefore lies in one common stage. Hence stagewise no-bottom implies $T_{p,\infty} \nvdash \bot$. Directedness is essential: there exist two individually bottom-free rule sets whose undirected union derives bottom by taking a premise from one and an elimination rule from the other. Both the positive theorem and this negative canary are machine checked.

Proposition 29 (Alternating-diagram linearization test)**.** *Suppose the two-polarity stages form a countable directed system and every edge is merely a model qualification or rule inclusion. Any enumeration followed by successive common upper bounds yields an ordinary ascending tower containing the same finite derivations and model-qualification obligations. Consequently, the alternating or helical shape alone has no additional proof-theoretic strength.*

Proof. Enumerate the stages as $a_0, a_1, \ldots$. Put $b_0 = a_0$ and, recursively, choose b_{n+1} above both b_n and a_{n+1} by directedness. Then (b_n) is a cofinal ascending chain. Every finite set of rule uses and qualification edges occurs below one b_n, so replacing the original index diagram by this chain does not change its finite derivations or external no-bottom consequences. □

Polarity can still have architectural value: it can enforce representational or implementation diversity, prevent accidental same-level circularity, and keep two incomparable semantic traditions visible. It acquires mathematical content only when the strands carry genuinely different theories or model constructions and the braid proves translations, conservativity, or comparison theorems not already present in either strand. Any additional strength then comes from those theories and reflection principles, not from drawing the index category as a helix.

12.1 Four claims that must not be conflated

1. **External qualification:** a metatheory constructs $M_{\mathsf{op}p,n+1}$ and proves (7).
2. **Checked model certificate:** a later checker verifies a finite certificate describing the relevant model and rule-soundness facts.
3. **Internal earlier-consistency theorem:** $T_{\mathsf{op}p,n+1}$ represents the syntax of $T_{p,n}$ and proves its encoded consistency statement.
4. **Global tower theorem:** one ambient theory proves that all alternating edges exist and cohere.

Only the first is supplied by a semantic qualification record. The third needs arithmetization, a theory translation, and enough induction. The fourth reveals the actual strength of the formal host: if Lean or HOTG proves the entire tower, then Lean or HOTG remains part of the metatheoretic account. Strict ascent avoids same-level self-certification; it does not make Gödel's and Tarski's constraints disappear.

12.2 The internal claim realized in arithmetic: the consistency tower

The third claim is now realized for arithmetic. A *sound stage* is a theory of arithmetic with a Δ_1 presentation of its axioms that extends $\mathsf{I}\Sigma_1$ and is true in the standard model $\mathbb{N}$. *Zoom-out* adds the stage's own consistency statement, $S \mapsto S + \mathrm{Con}(S)$, and the result is again a sound stage.

The diagonal step is shared with the rest of the library. Every Δ_1 extension of $\mathsf{I}\Sigma_1$ is a provability stage whose code map applies a formula to the numeral of a Gödel number. Gödel's diagonal lemma is exactly its relative Lawvere step (`Arithmetic.stage_diagonalizable`). Löb's theorem and the second incompleteness theorem then follow from the abstract derivability conditions (`Arithmetic.loeb`, `Arithmetic.consistency_unprovable`), and the conclusion agrees with the library's independent proof (`Arithmetic.agrees_with_foundation`).

Theorem 30 (Consistency tower)**.** *For every sound stage S:*

(i) $S + \mathrm{Con}(S) \vdash \mathrm{Con}(S)$ *and* $S \nvdash \mathrm{Con}(S)$*;*

(ii) *S is strictly weaker than $S + \mathrm{Con}(S)$, and no sound stage contains its own zoom-out;*

(iii) *the finite tower $S, S + \mathrm{Con}(S), \ldots$ increases strictly, and the sound stages ordered by strict extension have no terminal stage;*

(iv) *every zoom-out adds a sentence true in $\mathbb{N}$, no sound stage proves every truth of $\mathbb{N}$, and every sound stage is strictly weaker than true arithmetic.*

The checked statements are `SoundStage.next_proves_consistency`, `SoundStage.not_proves_own_consistency`, `SoundStage.strictlyWeaker_next`, `SoundStage.not_next_le`, `SoundStage.tower_strictMono`, `noTerminalStage`, `SoundStage.next_new_truth`, `truth_not_reached` and `SoundStage.strictlyWeaker_trueArith`.

Unconditional at PA. The theorem is instantiated. PA and $\mathsf{I}\Sigma_1$ receive constructed Δ_1 presentations: a recognizer for the codes of universal closures of induction instances, correct over $\mathbb{N}$ and provably proper in $\mathsf{I}\Sigma_1$, joined with the finite axiom set of PA^- (`InductionSchemeDelta1.peanoDelta1`, `InductionSchemeDelta1.iSigmaOneDelta1`). The construction is the Foundation library's. The library version pinned here still postulates these presentations, so the proof is reproduced and the postulates are never imported. With the standard-model facts from the same library, PA and $\mathsf{I}\Sigma_1$ are sound stages (`peano`, `iSigmaOne`), so the type of sound stages is inhabited and the negative statements above are not vacuous. The tower PA, $\mathsf{PA} + \mathrm{Con}(\mathsf{PA})$, $\ldots$ has the following checked properties:

- it increases strictly and its levels are pairwise distinct (`peano_tower_strictMono`, `peano_tower_injective`);
- no level contains every level (`peano_tower_no_top`);
- each level's consistency is proved one level up and not at the level itself (`peano_tower_certified`, `peano_tower_not_self_certified`);

- each level misses a truth of $\mathbb{N}$ (`peano_tower_true_unprovable`).

The base cases are explicit: $\mathsf{PA} \nvdash \mathrm{Con}(\mathsf{PA})$ and $\mathsf{PA} + \mathrm{Con}(\mathsf{PA}) \vdash \mathrm{Con}(\mathsf{PA})$, and the same for $\mathsf{I}\Sigma_1$ (`peano_not_proves_consistency`, `peano_next_proves_consistency`, `iSigmaOne_not_proves_consistency`, `iSigmaOne_next_proves_consistency`). All of these use exactly `propext`, `Classical.choice` and `Quot.sound`; no postulated definability instance is involved.

Negative control. The Δ_1 condition is what excludes a top. True arithmetic extends $\mathsf{I}\Sigma_1$, is true in $\mathbb{N}$ and proves every truth of $\mathbb{N}$, so it satisfies every condition except one. It has no Δ_1 presentation (`trueArith_not_delta1`). Without that condition true arithmetic would be a terminal stage: every theory true in $\mathbb{N}$ is contained in it.

What is claimed, and what is not.

- $\mathrm{Con}(S)$ is intensional: it is built from the chosen Δ_1 presentation of S [120]. The tower is the tower of these presentations.
- Only the finite levels are built. Progressions along ordinal notations, with limit stages that collect the earlier ones [121, 122], need a uniform arithmetic presentation of the stages, and they are open here. So is the completeness of such progressions. Turing's progression reaches every true Π_1 sentence, and Feferman's uniform-reflection progression reaches every true arithmetic sentence, but only along a path through notations that is not effectively selected.
- The whole tower is proved in Lean, so it is also an instance of the fourth claim of the previous subsection, with Lean and the three axioms above as the ambient theory; Lean remains part of the metatheoretic account. Strict ascent avoids same-level self-certification; it does not remove the incompleteness theorems.
- The tower is one strand with genuine strength: each step adds a true Π_1 sentence that the previous stage does not prove. This is the kind of edge content that the linearization test above requires; the shape of the index diagram contributes nothing.

12.3 Conditional coupling of the atlas and qualification tower

The atlas and qualification tower should be connected by explicit maps, not poetic analogy:

$$\begin{array}{ccc} \text{route-rich world } R_{p,n} & \xrightarrow{q_{p,n}} & \text{observed world } E_{p,n} \\ \downarrow & & \downarrow \\ \text{later opposite model } M_{\mathsf{op}p,n+1} & \models & T_{p,n}. \end{array}$$

The upper row is coalgebraic observation; the lower row is semantic qualification. They become one theorem only after the model interpretation is proved natural with respect to $q_{p,n}$ and stage transport. Keeping the rows separate permits multiple host foundations and multiple observers to cross-check the same finite theorem without pretending that representation diversity is semantic diversity.

13 What is already clarified

The following architectural points are now stable enough to serve as selection criteria.

1. Operational occurrences and histories are proof-relevant latent capacity. Erasure requires an observer-factorization theorem.

2. One authentic typed rho communication specimen realizes that boundary: endpoint-indexed receipt families descend, while a revision-indexed continuation family does not and is not a constant family.

3. At each context, simple type objects embed as constant dependent families, while their display maps may still depend on that context; extensionality is a separate axis.

4. Route equality, identity proofs, and observational equality are related but not definitionally identical.

5. Prime's next foundation is planned around CBPV with qualified strategy choices. The current revision-keyed call-by-need nucleus remains the concrete sharing reference. Agreement with CBN or eager evaluation requires a declared fragment and observer; it is not automatic with effects or nondeterminism.

6. Dependent sequencing retains selected values and receipts by default.

7. Cost, evidence, rank, budget, ownership, and revision are independent coordinates unless a theorem relates them.

8. Reflection obeys beta at the executable boundary; eta is an observer-specific collapse license, not a universal quotation law.

9. Opaque cross-world reference is distinct from transparent containment.

10. Alternating qualifications are ordinary later-model qualifications; a countable directed helix linearizes and gains no strength from its shape.

11. Finitary directed unions preserve no-bottom, while undirected unions need not.

12. A family-enclosing Tarski-universe operator recovers the universe-above-data interface at the capability and semantic-model levels; same-level self-coding remains impossible in the predicative model.

13. Extensional HOTG selection and computational DTT enclosure are two faces joined by a total-forward, code-indexed-backward adequacy shape, not rival spellings of one construction. The abstract constructor algebra and the HOTG family-enclosure theorem are checked; their concrete contextual composition remains open.

14. The exact reversible intensional/extensional core is $C/{\sim_d} \simeq \mathrm{EssIm}(d)$. Semantic quotienting removes code multiplicity but does not represent extensional-only host objects.

15. At a fixed base, dependent total space, ordinary constant families, and dependent sections form the checked adjoint triple $\Sigma_\Gamma \dashv \Delta_\Gamma \dashv \Pi_\Gamma$. For a multi-point base, Δ_Γ is not full; the separately fully faithful contextual simple-to-dependent inclusion has richer, context-dependent source morphisms.

16. An open-ended PQI-inspired indexed total space can retain physical/extensional, qualitative/intensional, and indexical coordinates without collapse. It carries no exhaustiveness component.

17. Family enclosure, superuniverse closure, and Mahlo reflection are distinct capabilities. Their proof-theoretic analyses form a genuine escalation in pinned DTTs, but comparison with HOTG requires a common interpretability or equiconsistency argument rather than comparing sufficient model assumptions.

18. Every finite list of universe-stage requirements has a strict later host, while the natural-numbered tower has no final stage.

19. An explicit tail reduced power supplies relative infinity without ultrafilter selection; its eventual order is provably partial.

20. Principal ultrapowers collapse to their selected coordinate. A free ultrapower can add nonstandard points, but this is a model-theoretic axis and not a proof-strength or non-well-foundedness theorem.

21. Moral Paradigm Equivalence holds exactly at the checked judgment boundary in all six ordered directions, and at the checked model boundary under an inhabited semantic alignment. Encoding cost and learnability remain separate.

22. Rice's theorem constrains computable exact encodings of extensional program behavior; it does not refute semantic paradigm equivalence.

23. An empirical optimizer can be realized as a target-centered virtue, while FCA and WM–PLN supply observer-relative concept closure and positive/negative evidence without replacing the ethical ontology.

24. Local thinness of the selected comparison sector is a certificate at one boundary dimension: it is equivalent to invariance of every proposition-valued observer on the parallel-cell fibres there, it licenses erasure of those fibres to a point, and no finite thin prefix forces thinness one dimension higher.

25. Parallel routes or context cells are distinct exactly when some lawful dependent transport distinguishes them; uniqueness of parallel routes is equivalent to the absence of transport discriminators.

26. Mode-comparison thinness is independent of identity-route uniqueness, endpoint reflection, evidence-fibre size, finite observation horizon, and cell height.

27. Finite positive evidence is monotone and sound at every stage, eventual completeness is a separate property, and finite exhaustion is not refutation; a rule with countably many premises can be derivable in the union of finite stages while derivable at none.

28. Disjoint provenance does not entail stochastic independence, and revision and extraction soundness do not determine forgetting.

29. Capture-avoiding substitution over a typed higher-order packet is strictly weaker than $\beta\eta$-aware unification, and agreement on canonical generated examples does not qualify a native checker.

30. Equations are data of a language definition and act through the composite step $E; R; E$; the presentation's collection tags and declared algebras generate their own laws, a substitution interpretation is a separate declaration rather than a law, every OSLF formula and the admissible structural-modal observations descend to equation classes, a separate Hennessy–Milner

fragment is adequate over image-finite quotient systems, and the Lean rho canonical stepper establishes forward image-finiteness for closed binder-safe rho processes. The generated OSLF formula language is the two-direction Hennessy–Milner language over that step and remains sound in full; its box-free fragment is adequate for forward bisimilarity. A typed infinite family of inequivalent predecessors proves that the corresponding backward finiteness hypothesis fails for rho, so full bidirectional completeness is not claimed. Least-context adequacy is conditional on the redex-relative-pushout universal property, while a separate parallel-partner logic is adequate for its declared labeled system. Proving that the C runtime key realizes the Lean canonicalizer on all inputs remains open.

The finite colored-object formalization proves opaque-reference injectivity, foreign-member invisibility, local well-foundedness, symmetry, a global two-node cycle, observer nonfaithfulness, the finite-width obstruction, and an unbounded successor-reflection chain. It deliberately does *not* claim a final coalgebra or added foundational strength; it is principally a negative boundary specimen. The generic rule-system theory proves finite-stage capture, directed-union no-bottom, later-opposite model qualification, and their paired negative controls.

14 Formal ethics as a cognitive-architecture discriminator

An ethical language tests whether the architecture can retain distinct kinds of reason while comparing their consequences. The active formalization follows the Formal Ethics Ontology and the open Ethical Paradigm Equivalence conjecture [81, 82, 83]. Its core distinguishes ethical from metaethical theories; theories from their fields; native sentences from supporting background theories; arguments from semantic validity; and mere theory membership from justified truth in an explicit model. Target-centered virtues retain all four source coordinates: field, basis, agent-indexed mode of responsiveness, and target. Hitting a target once is not identified with possessing a stable disposition.

14.1 Exact equivalence and the cost of encoding

Let

$$X = \mathsf{Agent} \times \mathsf{Situation} \times \mathsf{Action}, \qquad V = \{\mathsf{good}, \mathsf{bad}, \mathsf{permissible}\}.$$

A fully evaluated ethical theory is a function $j : X \to V$. The checked theory families are guarded deontic rules, integer-valued utility theories with sign classification, and full target-centered virtue theories. Each family has an explicit constructor from an arbitrary j.

Theorem 31 (Evaluation-level Moral Paradigm Equivalence)**.** *Every determinate theory in each of the deontic, utility, and target-centered virtue families has an exactly judgment-equivalent theory in each of the other two families. Hence all six ordered translations preserve the complete agent/situation/action judgment function.*

The result does not identify a deontic rule with a utility or a virtue. An arbitrary source classifier may reappear inside a rule guard, score map, or virtue target. Exact expressibility is therefore compatible with unusual, large, noncomputable, or difficult-to-learn encodings. This is the precise form of the conjecture's steelman: differences commonly attributed to "paradigms" may instead be differences among especially concise or natural theories within equally expressive families.

At the set-of-sentences level, explicit maps also exist in all six directions. For semantics that commute with these maps, source and target theories have the same model class. This hypothesis is

inhabited rather than merely postulated: any integer score semantics induces aligned deontic, value, utility, and virtue-target readings. The current model theorem concerns these pinned sentence fragments; lifting it to arguments, justification, learning dynamics, and richer ontology clauses remains open.

14.2 Computability is a qualifier, not a counterexample

For program-coded actions, let C be a moral classifier and let the consequence of a code be its full partial-recursive function. Rice's theorem yields the following exact boundary.

Theorem 32 (Moral Rice boundary)**.** *No nontrivial decidable classifier is invariant under equality of full partial-recursive consequence functions. Moreover, every exact computable and faithful encoding of the responsiveness classifier into another theory transfers this undecidability.*

This theorem does not refute Moral Paradigm Equivalence. Every classifier still has an exact one-duty and one-target encoding. It says instead that an exact encoding of a nontrivial extensional property must retain its undecidable content somewhere. Finite theories whose component duties or active-target tests are all computable cannot make that content disappear. Semantic faithfulness, computability, size growth, approximation error, naturalness, and learnability must therefore be coordinates of a translation profile rather than one Boolean notion of equivalence.

14.3 Optimization, virtue learning, and uncertain concepts

An empirical optimizer is itself realizable as a target-centered virtue: its field selects the decision situations, its basis names the optimized value, its mode admits the relevant actions, and its target says that no alternative has a higher declared score. The checked bridge proves that support by this virtue is equivalent to score optimality and that its learned policy selects the same action as the optimizer. This does not identify every virtue with reward maximization; it proves that optimizer behavior is one legitimate virtue theory.

Approximation is controlled by a margin theorem. If a learned score differs uniformly from the target score by at most ε and the selected action has margin greater than 2ε, strict and weak maximizing decisions are preserved. A tie is the negative control: uniform approximation without a margin cannot force the same selected action. Generic function-approximation results become ethically useful only after being composed with such a decision stability theorem.

Formal Concept Analysis is connected to the actual virtue aspects rather than used as a replacement ontology: cases and field, basis, mode, and target facts form an incidence relation; concepts and implications are indexed by an explicit observer view. Enlarging a view can refute an implication that held over fewer cases. Proof-carrying support and refutation events are separately encoded as positive and negative WM–PLN evidence. Ethical concepts can therefore be learned and revised while their ontological roles remain typed.

14.4 Native plurality for a multi-agent reasoner

A reflective ethical agent should natively support at least three reason forms: rule compliance or explicitly authorized override; empirical optimization with an optimality receipt; and disposition update followed by virtue-guided action with target evidence. The checked decision type is a dependent sum whose evidence fibre depends on the selected reason form. Two modes may choose the same visible action while retaining different justifications.

For multi-agent reasoning, three roles must remain distinct: an observer has a view; a care-member's interests enter the value relation; and an agent has authority to act or commit a receipt.

Processes can be observers, affected parties, agents, or several of these, but none of these implications is automatic.

The checked allocation specimen makes this distinction operational. A coordinator chooses between a concentrated and a shared burden allocation. Both choices reach the same resolved state and have the same net impact, while their person-indexed impacts, affected-party consent claims, and work/span receipts differ. The protocol response is itself a dependent ethical decision, so rule-following, empirical optimization, virtue learning, and an authorized override retain their native evidence inside the event.

Theorem 33 (Multi-agent observer adequacy boundary)**.** *Endpoint and aggregate impact factor through the coarse audit observer. The person-indexed impact and work/span functions do not. The consent-claim family does not descend through that observer because the concentrated and shared events have respectively one- and two-element claim fibres. All three do descend through the finer allocation observer. Moreover, deontic, consequential, and virtue capabilities can choose the same shared allocation, but justification mode does not factor through allocation plus endpoint. Only an authority-bearing coordinator can construct a commit receipt; care membership alone cannot construct one. Event-indexed sequencing retains each consent claim beside its event and commutes with ordered-list-to-occurrence-bag observation.*

This is a falsifiable architecture rule rather than a slogan about ethical richness: any optimizer or scheduler that prunes the two events using only the coarse observer is valid for endpoint and aggregate consumers, but invalid for distribution-, consent-, reason-, or cost-sensitive consumers. Scheduler fairness, bargaining, and conflict among values remain separate extensions of the same event structure.

For a highly capable compassionate reasoner, the architectural consequence is plurality with explicit comparison: reason natively in each ethical family; translate exactly when possible; expose the computational and statistical cost of each translation; preserve the reason receipt beneath any extensional verdict; revise learned concepts with evidence; and widen the sphere of care only through an explicit, inspectable policy. No single score, rule list, or learned disposition is licensed as the whole moral ontology.

15 Modes, proof authorities, bootstrap, and the remaining braid

The architecture now has enough checked examples to state what the word "mode" may and may not mean. This is a necessary restraint. Revision, cost, evidence, observation, proof authority, evaluation order, and semantic strength all interact, but they are not one kind of mathematical structure. Making every distinction a mode would merely relocate the ambiguity into an oversized mode category.

15.1 A criterion for modes and their neighbours

Definition 34 (Mode criterion)**.** An axis earns the status of a *mode* when it changes which contexts, variables, or substitutions are available and is connected to another such regime by a modality with coherent context locking. Parallel transformations between modalities are 2-cells. Data that only measures, witnesses, observes, checks, or realizes a judgment belongs to a different layer unless an independent theorem shows that it also changes contextual availability.

The resulting classification is as follows.

Phenomenon	Primary structure	Discriminator
Contextual accessibility	Mode, modality, and lock	Does crossing the axis change which assumptions may be used, and is substitution coherent across that crossing?
Comparison of modalities	2-cell or modification	Is this a coherent transformation between two context translations rather than another context translation?
Revision availability	Guarded or temporal mode	A later context may use a controlled past context; the lock and Löb rule are structural.
Revision history	Displayed family of receipts	Two executions with the same endpoint can still have different event- or revision-indexed inhabitants.
Work, span, memory, risk	Monoidal grading or quantitative enrichment	The same modality may admit several lawful cost models; cost therefore is not determined by the mode category.
Extensional readout	Functor, quotient, or observer	It forgets distinctions and creates a descent question; it need not alter the source context grammar.
Value/computation polarity	CBPV adjunction and effect structure	It becomes an additional mode axis only if the selected calculus gives values and computations distinct context disciplines.
Proof authority	Typed NIK node and checked route	It decides which certificates establish which judgments; it is not a context modality.
Semantic host	Interpretation or model	It gives meanings and relative soundness, not an additional native inference rule.
Runtime	Realization with an observation theorem	It executes the calculus; equivalence requires a proved operational comparison, not a shared label.
Ethical paradigm	Native reason structure and learned policy	It can become a cognitive control mode for an agent, but it is not automatically an MTT mode of the foundational calculus.

One feature can legitimately occupy two rows. Revision is the clean example: availability of earlier assumptions is modal, whereas the concrete event trail is a displayed dependent object. Erasing the latter is not justified merely because the former has a later modality.

15.2 The mode-theory option ladder

The mode theory is a parameter, not a foregone conclusion. Four candidate levels should therefore be compared before any quotient or dependent syntax is hardened:

Level	Added structure	Discriminator and present status
M_0	Three modes and freely composed 1-cells only	Checked as the free path category. It retains finite modality routes, but generator count proves that it cannot express either $qF \Rightarrow Q$ or an adjunction unit.
M_1	Locally preordered 2-cells	The primitive authored generator fibre is now proved subsingleton at every fixed boundary, so pointwise thinning loses no primitive generator identity. A generic construction now closes base structural cells and authored generators under vertical composition and whiskering, then takes the pointwise thin reflection; this is an actual locally thin bicategory. Its three-mode instance contains all six selected modalities, all eight authored cells, the three adjunctions, and the factorization isomorphism. Raw constructor histories are not thin, while the checked categorical model identifies the tested factorization round-trip with reflexivity. A point-probe theorem now proves global semantic descent and constructs the resulting pseudofunctor into **Cat**. A generic mapping-out theorem also gives existence and fibrewise uniqueness once the object and one-cell action is fixed and locally thin on its image. The full bicategorical equivalence, including transformations and modifications, remains open. This is the natural boundary for a *mitten* experiment.

Level	Added structure	Discriminator and present status
M_2	Proof-relevant parallel 2-cells	The checked raw two-computad retains this level before coherence equations. It becomes necessary only if two parallel mode transformations survive the selected equations and act differently on a required dependent consumer. That criterion is now a checked theorem: parallel context cells are unequal exactly when some whiskered dependent family distinguishes them. Distinct observer names, terms, reason routes, or execution receipts elsewhere do not by themselves satisfy this discriminator. In the corrected pseudo-CwF bicategory even the displayed action is forced by the context-level natural transformation, so a dependent distinction must either change that base transformation or justify a genuinely weaker transformation notion. The rho and graph specimens now exhibit distinct actions of parallel transformations between point substitutions inside their context categories. They prove that the dependent floor is genuinely proof-relevant, but they are not parallel cells between O/I/E modalities and therefore do not yet force M_2.
M_3	Further guarded or value/computation mode axes	Admit an axis only when contexts genuinely obey a distinct lock discipline. Revision receipts and quantitative grades remain displayed families or enrichments unless a dependent interaction theorem forces mode status.

These levels are not a linear strength hierarchy in every sense. M_0 is the one-dimensional shadow of either M_1 or M_2; M_1 is a propositional collapse of parallel-cell identity; and M_3 varies the object and 1-cell signature independently of whether its homs are preorders or proof-relevant categories. The selection rule is therefore observational: choose the least level whose induced cell actions retain every distinction required by the rho, graph, revision, and ethical-reason specimens, and choose it per certified sector rather than once for the whole language.

The pointwise discriminator is now exact. For one fixed pair of parallel 1-cells, local thinning is the quotient of its cell fibre by the indiscrete equivalence relation. An observation of cells factors through this quotient if and only if it is constant on that fibre, and the quotient unit is injective if and only if the original fibre was already subsingleton. Thus a valid M_2 witness must supply two cells in the same fibre and an observation that separates them. This criterion prevents proof-relevant data at a different index from being silently promoted to proof-relevant mode structure.

15.3 Local truncation certificates instead of a global cell ceiling

The discriminator has since become a theorem rather than a criterion. For parallel routes in a groupoid of routes, two routes are distinct exactly when some lawful dependent transport family acts differently on them. The forward direction is constructive: the covariant representable family

based at the route source is the discriminator, and evaluating the transported identity route recovers the original route. Consequently uniqueness of parallel routes is equivalent to the absence of every transport discriminator, and a route structure with decidable route identity makes every lawful transport invariant. The same representable construction lifts to natural transformations between category-valued translations: two parallel context cells are unequal exactly when some whiskered dependent family distinguishes them. The two-element automorphism group is the positive canary, since identity and negation transport a Boolean to different outputs; ordinary lifted equality is the negative control. This is a small transport specimen, not a model of univalence and not an object-language identity eliminator.

Applied to the selected O/I/E sector, the semantic translation cells have no dependent-family discriminator, whereas the raw factor-cell round trip remains syntactically distinct from reflexivity, rho revision histories have a communication-history discriminator, and direct and detour graph proofs have a proof-family discriminator. Locally thin mode semantics and proof-relevant operational cargo therefore coexist; neither promotes nor collapses the other.

Thinness is moreover a local certificate, not a global bound on cell dimension. For a globular tower, local thinness at one boundary dimension is equivalent to invariance of every proposition-valued observer on every parallel-cell fibre at that dimension. A local truncation certificate licenses erasing an inhabited thin fibre to a point, and every result-valued observer factors through that erasure. Distinct parallel cells always supply a distinguishing observer. Conversely, for every finite horizon there is an inhabited tower that is thin at every lower dimension and genuinely thick at the horizon, so no finite thin prefix establishes a universal cutoff. The strict globular shadow of the actual operational-to-extensional hom sector carries exactly such a two-cell certificate, and consumers of the authored factor comparison factor through the one-point representation. The general tower is a globular set: it supplies cells and their boundaries, not weak higher composition or coherence, so the result settles the truncation policy without selecting a synthetic higher-category theory.

Three further non-implications keep the certificate from overreaching. The same locally thin comparison sector is placed over all four combinations of identity-route uniqueness or its failure with endpoint reflection or its failure, and both identity choices are paired with a constructively thick comparison fibre; so mode-comparison thinness neither selects nor excludes J-only, scoped-strict, or cubical identity. One concrete profile combines the thin comparison sector, two distinguishable typed higher-order substitutions over fixed endpoints, and an observation problem whose tail survives every finite prefix; so thin comparisons do not make evidence fibres subsingletons, do not impose a uniform finite observation horizon, and do not imply a global cell cutoff. Finally, globular cell height and finite observation horizon are independent, realized by profiles in all four Boolean combinations.

The resulting doctrine is symmetric. A higher cell earns semantic admission when a required consumer acts differently on parallel lower cells. An erasure earns a fast path when a truncation or invariance theorem proves every declared consumer constant on the erased fibre. Neither the mere availability of higher structure nor its present absence is a global design verdict. The ambient language may therefore remain open to higher cells while individual sectors are proved, one certificate at a time, to need none.

15.4 Four concrete mode-theory candidates

The ladder above measures two independent choices: the number of mode axes and the identity retained between parallel 2-cells. It is therefore not itself a list of rival final systems. The following four candidates are the smallest serious alternatives to carry through the same specimens. Each

card fixes what the candidate asserts, what it cannot assert, and the observation that decides whether its extra structure is necessary.

Core locally thin triangle $\mathcal{M}^{\leq}_{\mathrm{OIE}}$.

Its objects are $\mathsf{O}, \mathsf{I}, \mathsf{E}$; its generating modalities are $F, U, q, \Delta, \Gamma, Q$; and its generated parallel-cell fibres are propositions. It asserts the three adjunctions and $qF \cong Q$ while deliberately forgetting how a valid comparison cell was composed. The checked locally thin bicategory is the syntactic candidate, and the checked diagram in **Cat** is its intended nondegenerate model. It can express context locks, the two intensional–extensional reflections, and operational observation. It cannot distinguish two parallel transformations after they have the same boundary. Global semantic descent is now proved for the selected **Cat** model, as is the fixed-object-and-one-cell mapping-out law. The remaining deciding obligations are the full bicategorical universal property, a DRA-aligned modal-CwF account of the triangle (and any separately desired covariant pseudo-CwF comparisons), and an explicit connection from a parallel mode-cell fibre to a required dependent action. The checked rho and graph families do separate parallel context cells, but those cells are transformations between point substitutions rather than between O/I/E modalities. Once a mode-indexed account exists, the checked local-faithfulness theorem transfers base-image thinness to its corrected dependent transformations.

Strict guarded product $\mathcal{M}^{\leq}_{\mathrm{OIE}} \times \mathcal{G}$.

Here $\mathcal{G}$ is the checked guarded-time mode category and the cartesian product makes guarded locking commute strictly with every O/I/E modality. This is a strong, falsifiable baseline rather than a default law. It can express later contexts, finite horizons, and guarded recursion independently of the semantic triangle. It cannot express an interaction in which guarding before readout differs from readout before guarding. The deciding square is supplied by the revisioned rho receipt: compare guarded endpoint readout in both orders while retaining the theorem that advancing the guard does not retag cached evidence. If the square commutes only up to coherent isomorphism, replace the product equation by an invertible 2-cell; if its actions differ, retain the noncommutative interaction explicitly.

Internal versus modal CBPV.

The conservative alternative gives each selected mode its own value/computation adjunction and effect algebra without adding mode objects. The modal alternative takes a product with distinct value and computation modes and makes the value-to-computation boundary a context lock. The already checked $F \dashv U$ between O and I records evidence completion and forgetting; it must not be renamed a CBPV value/computation split without a comparison theorem. One dependent effectful term decides the fork: return a witness, let the continuation type depend on that witness, and substitute the term across the adjunction. If the internal organization validates the required substitution and handler laws, the additional mode axis is redundant. If dependence is legal only behind a lock, the modal alternative has earned its extra objects.

Selected proof-relevant quotient $\mathcal{M}^{2}_{\mathrm{OIE}}$.

This retains the raw generated cell syntax modulo the named adjunction, inverse, associativity, unitor, and selected interchange equations, rather than identifying every parallel cell. It can record genuinely distinct ways of transforming one context modality into another. Its additional identity is not justified merely by distinct terms, observers, rule receipts, or histories: those may live in dependent fibres below a single modal transformation. The decisive

witness must be a pair $\alpha, \beta : \mu \Rightarrow \nu$ together with a required type, term, or substitution whose actions under α and β differ. Until such a witness survives the selected equations, this candidate is an important comparison object rather than the chosen mode theory.

M_0 remains an ablation baseline: it checks which claims require genuine 2-dimensional structure, but the existing adjunction units already rule it out as the final theory. Identity discipline and universe strength are not additional cards here because neither changes contextual availability by itself; they cross every candidate as independent type-theoretic axes.

15.5 The checked modal core and the exact MTT gap

At the semantic level there is a checked nondegenerate diagram

$$F : \mathsf{O} \rightleftarrows \mathsf{I} : U, \qquad q, \Gamma : \mathsf{I} \longrightarrow \mathsf{E}, \qquad \Delta : \mathsf{E} \longrightarrow \mathsf{I},$$

with

$$F \dashv U, \qquad q \dashv \Delta \dashv \Gamma, \qquad qF \cong Q.$$

Here O contains raw dynamics, I contains reflexive route evidence, E is the extensional carrier category, and Q is direct operational observation. The factorization is accompanied by canaries showing that the adjunctions are not equivalences. Thus the semantic triangle commutes without collapsing.

The generating two-dimensional boundary is now checked as well. There are three mode objects and six primitive 1-cells $F, U, q, \Delta, \Gamma, Q$. Their finite paths form a free bicategory, and a separate two-computad freely adjoins the units and counits of $F \dashv U$, $q \dashv \Delta \dashv \Gamma$, together with both directions of $qF \cong Q$. A pseudofunctor sends the free paths to the concrete semantic functors in **Cat**, and the raw authored 2-cells have a unique structural interpretation in the corresponding natural transformations. Two negative controls make this construction substantive: generator count is invariant under every bare free-bicategory coherence cell, so bare coherence can produce neither the comparison $qF \Rightarrow Q$ nor even the unit $1_{\mathsf{O}} \Rightarrow UF$. Conversely, the authored comparison exists; its forward–backward composite remains a non-reflexive raw receipt while its semantic image is the identity. This separates retained derivation history from semantic coherence exactly where intended.

The local-thinness audit sharpens this boundary. Every primitive authored 2-generator fibre is subsingleton, so its pointwise thin-reflection map is injective. By contrast, raw cell syntax retains the explicit factor-forward/factor-backward route and a raw-shape observer separates it from reflexivity. The semantic interpretation nevertheless maps those two raw cells to the same natural transformation. Hence raw proof identity is not yet evidence that the final mode theory must be M_2: at least one raw distinction is administrative history already validated for erasure by the independent categorical model.

The coherent locally thin candidate is now checked rather than conjectural. For an arbitrary base bicategory and a family of authored parallel 2-generators, the construction includes both base structural cells and authored cells, closes them under vertical composition and left and right whiskering, and takes the indiscrete quotient separately in each parallel-cell fibre. The resulting hom categories are thin and carry a genuine bicategory structure whose associators and unitors are inherited from the base. A negative control over a locally discrete free-monoid base proves that the construction does not manufacture a cell between unrelated 1-cells; adding one authored generator creates the selected reachability needed for the positive control. Instantiating this construction at the three-mode two-computad gives three bicategorical adjunctions and the comparison isomorphism, with their triangle and inverse laws following lawfully from local thinness.

This construction also makes the remaining M_1 obligation exact. Every raw mixed cell has a canonical interpretation in **Cat**, obtained by interpreting base coherence through the free pseudofunctor and authored cells through their independently checked natural transformations. On each fixed parallel boundary, that interpretation descends through thinning if and only if it is constant on the raw cell fibre. The factorization round-trip passes this test, whereas a raw-shape observer correctly fails it.

Global constancy is now proved for the selected **Cat** semantics. Each source mode has a one-point probe, the images of all point maps remain jointly surjective under every generated modality path, and every target path sends a one-point probe to an object with subsingleton carrier. Naturality along those point maps therefore forces any two transformations between parallel path functors to agree on every carrier element. Quotient modalities require the substantive step: their representatives supply the required preimages. This proves every semantic cell fibre subsingleton, discharges every raw-fibre descent obligation at once, and yields an actual pseudofunctor from the entire locally thin bicategory to **Cat**. A negative control shows that the intensional semantic category still has distinct endomorphisms, so the result does not follow from an ambient thin category. A generic construction further proves that any pseudofunctor locally thin on its image, together with selected images of the authored generators, extends uniquely on every fixed parallel cell fibre. This is the fixed-object-and-one-cell mapping-out law. The remaining categorical obligation is the full bicategorical universal property relating pseudofunctors, transformations, and modifications, not semantic existence or cellwise uniqueness.

The revisioned rho communication specimen supplies the complementary result. Two typed COMM receipts at the same visible endpoint retain revisions zero and one, and revision observation cannot factor through a thin quotient of that receipt fibre. Its continuation family also cannot descend to endpoints. These are genuine dependent operational distinctions, but the receipts are terms in an endpoint-indexed family, not automatically transformations between modalities. A thin factor-comparison cell can therefore coexist with the non-collapsed receipt family. Moving from M_1 to M_2 still requires an explicit action theorem connecting parallel mode cells to such a consumer.

The graph-walk native type theory now supplies the same boundary independently. A direct walk and a longer detour inhabit one endpoint-indexed native reachability fibre, both carry generated certificates, and their exact lengths are distinct. Length observation therefore cannot descend through a thin reflection of the proof fibre, while the selected mode factor cell remains inhabited and thin. Thus neither rho receipt identity nor graph proof identity alone decides M_1 versus M_2; both must first be connected to the action of parallel mode transformations.

The proof-relevant dependent floor is now checked directly rather than inferred from these two examples. Its contexts are small categories, substitutions are functors, dependent families are covariant functors into **Type**, terms are natural sections, and comprehension is the Grothendieck category of elements. The discrete one-point category supplies the chosen terminal context. This gives a category with families in which route identity is retained by the context itself.

Natural transformations between substitutions now have their exact dependent action as well. Right whiskering acts on every reindexed family, sends natural sections coherently between the resulting families, and induces the associated functor between categories of elements. Identity and vertical composition laws hold for all three actions. Covariant representables are jointly faithful: two transformations between fixed substitutions are equal if and only if their actions agree on every dependent family. Thus the dependent semantics neither invents nor erases context-level 2-cell identity.

Applying this criterion separates the two categorical levels decisively. In the selected O/I/E image, every parallel semantic-cell fibre is subsingleton, so all induced family actions agree; in particular, the non-reflexive raw factorization round trip and reflexivity have identical dependent

actions after interpretation. By contrast, each rho revision path and each native graph proof induces a transformation between the corresponding point substitutions. The communication-history family distinguishes the revision-zero and revision-one actions, and the native-proof family distinguishes the direct and detour graph actions. These are authentic proof-relevant context cells, not counterexamples to local thinness of the presently selected mode semantics.

This also corrects a possible typing mistake in the architecture. The O/I/E semantic categories and their six functors are contexts and substitutions inside this one category-indexed CwF. They are not, merely by that fact, three separate CwF models and six pseudo-CwF morphisms. To lift the triangle one dimension higher, one must first define the mode-indexed dependent doctrines and then prove the required morphism, adjunction, and coherence laws. The context-level action theorem is reusable input to that construction, not a substitute for it.

There is also an exact erasure theorem. Replace each hom type $\mathcal{C}(x, y)$ by the proposition that it is inhabited. A dependent family $A : \mathcal{C} \to \mathbf{Type}$ descends to this proposition-valued support if and only if

$$A(f)(a) = A(g)(a) \qquad \text{for every } f, g : x \to y \text{ and } a : A(x).$$

The forward implication follows because all inhabitants of one support proposition are equal; the converse chooses one morphism from each inhabited hom and uses parallel invariance to prove that transport is independent of the choice. Constant families give the positive control. A one-object Boolean-xor category acting by identity and negation gives the negative control.

Both operational specimens now instantiate this same theorem. For rho, the one-step evidence is a semantic revision paired with an exact typed strict-core COMM occurrence; finite evidence paths form the context, and the family at a state consists of retained histories from the closed communication source. Revision-zero and revision-one paths use the same typed occurrence and the same endpoints but act differently on the identity history, so the family cannot descend. For graphs, native reachability proofs form the homs, walk append is composition, and the family at a vertex consists of native proofs from vertex zero. The direct and detour proofs act differently and again cannot descend. This proves that proposition-valued route existence is a licensed readout only for route-insensitive families; it does not narrow what the proof-relevant dependent layer can carry, and it still does not manufacture distinct mode transformations.

The corrected pseudo-CwF bicategory closes a further possible loophole. Its comprehension-coherence square uniquely determines every displayed family component from the natural transformation on contexts. The base projection is therefore locally faithful, and two transfer theorems are now checked: a subsingleton fibre of base natural transformations induces a subsingleton fibre of corrected pseudo-CwF transformations, and any observation separating two corrected transformations also separates their projected **Cat** 2-cells. In particular, proof relevance in dependent terms cannot hide an additional corrected modal transformation over one fixed base cell. A Boolean-negating displayed map supplies the negative control: it has the apparent fibrewise boundary of an identity cell but fails the required comprehension square. Consequently, a future M_2 witness must exhibit either two genuinely distinct context-level transformations in the selected semantic model, or a weaker or lax notion of modal transformation together with an independent semantic argument that its extra displayed action is lawful. The actual O/I/E account in corrected pseudo-CwFs therefore remains a conditional selection theorem rather than a completed dependent model of the triangle. Moreover, the six current O/I/E functors cannot simply be relabelled as pseudo-CwF morphisms: in the category-indexed model they are substitutions between contexts of one CwF. A genuine pseudo-CwF account requires separately specified dependent doctrines at the three modes.

There is now a sharper obstruction at the q edge. For any route type X, equip the quotient map $\mathrm{Quot.mk} : X \to X/\!\sim$ with a selected representative of each quotient class. The checked

theorem says that every dependent family over X descends up to fibrewise equivalence if and only if this quotient map is injective. Discrete route types pass: their relation is equality. The codiscrete Boolean route type fails at the family level: its quotient identifies false and true, while a family with fibres 1 and 2 above those points cannot descend. Constant families still do descend through the same quotient. Thus q is a lawful object readout but is not a lossless operation on arbitrary dependent types.

This rules out only the naive all-family descent route. The orientation of an MTT model is important here: the left context action may be $L = q$, while its dependent right action supplies the modal type former Δ. Such a model does not require every intensional family to be the pullback of an extensional one. Indeed, universal descent would say that the dependent image of Δ exhausts the intensional doctrine, contradicting the intended non-collapse.

The resulting options are therefore layered rather than mutually exclusive. The baseline MTT candidate uses q at the context-lock level and Δ as the right-adjoint type operation. Families carrying coherent descent data form the extensional-readable subdoctrine. An additional pushforward from arbitrary intensional families may be constructed only if a consumer requires it, and then its substitution, comprehension, and Beck–Chevalley laws must be proved without claiming lossless recovery. A covariant pseudo-CwF morphism in the q direction likewise needs one of these explicit family actions; it is not supplied merely by the object-level adjunction.

The dependent quotient edge is now an actual category with a dependent right adjoint in the sense of CwDRA semantics [41]. Intensional contexts are route types and substitutions are route-preserving maps. A type over a route context is a family equipped with functional transport along every declared route; a term is a section commuting with that transport. Context comprehension is the dependent total route type, whose routes pair a base route with the equation saying that transport reaches the target fibre value. These operations satisfy the CwF identity, composition, projection, variable, pairing, beta, and eta laws, and the one-point route type is terminal. The extensional mode carries the ordinary set-family CwF on bare carriers.

The context functor of the checked CwDRA is the actual quotient $q : X \mapsto X/{\sim}$. Given an extensional family B over qX, its dependent right action is

$$(\Delta_X B)(x) = B([x]),$$

with transport along $r : x \;\mathsf{Route}\; y$ given by equality transport along $[x] = [y]$. Extensional sections of B are equivalent to route-natural sections of $\Delta_X B$. The easy direction precomposes with quotient classes. The reverse direction is genuine dependent descent: a route-natural section defines a relation-invariant map into the dependent total space $\sum_{z:qX} B(z)$; quotient lifting produces a map over qX, whose second component is the descended section. Both inverse laws and substitution naturality are checked.

The right action is deliberately proper. Constant families inhabit it, but the family with fibres 1 and 2 at the two points of the codiscrete Boolean route context cannot even be fibrewise equivalent to any pulled-back extensional family: route-related fibres in every $\Delta_X B$ are equivalent. Thus lawful modal integration and intensional non-collapse coexist in one model. Independently, explicit family-factorization data remains stable under every commuting observation square; it characterizes a useful readable layer without replacing the stronger transport-aware CwF structure.

The comprehension comparison for this right action is now exact. For every $B : qX \to \mathcal{U}$, quotienting the pulled comprehension gives a canonical equivalence

$$q(X.\Delta_X B) \;\simeq\; \sum_{z:qX} B(z).$$

The forward map sends (x, b) to $([x], b)$. Its inverse is defined by heterogeneous quotient recursion, and coherence proves that changing the representative of $[x]$ changes neither the resulting quotient class nor its dependent value. Both inverse laws are checked, as are the corresponding inverse substitutions between the two extensional contexts. Thus the comparison does not depend on a hidden representative-selection convention.

The first type-former audit over this CwF is now checked. Covariant route transport is sufficient for dependent sums: fibrewise dependent pairs, introduction, both projections, both beta laws, and substitution stability are defined for arbitrary route families. It also supports substitution-stable fibrewise equality formation and reflexivity. This selected equality model is proof-irrelevant and endpoint-reflecting; it is therefore an extensional fibrewise identity structure, not yet the intended intensional identity with intrinsic J or a cubical path structure.

Dependent products expose a real variance boundary. Transporting a dependent function forward along $r : x \to y$ requires transporting its argument backward from the fibre over y to the fibre over x. A checked product construction therefore assumes that every transport map of the domain family is the forward map of a fibre equivalence. Under precisely this capability, pointwise dependent functions form a route family with lambda, application, beta, and substitution laws. Every quotient-pulled family $\Delta_X B$ satisfies the capability, since its transports are equality casts.

The restriction is not an artefact of that construction. On a directed two-point context $0 \to 1$, let the domain fibre be 1 at 0 and 2 at 1, with forward transport selecting one Boolean. A dependent codomain can be inhabited over the source and over the selected Boolean while empty over the other Boolean. The pointwise product is then inhabited at 0 and empty at 1. No covariant route family can transport the former into the latter, even if its fibres are only required to be equivalent to the pointwise products. Thus Σ is available throughout the covariant doctrine, whereas Π selects an invertible-transport subdoctrine that includes the extensional right action.

The exact comparison permits a stronger preservation audit than fibrewise agreement. Pull an extensional codomain over $\sum_{z:qX} B(z)$ back through the comprehension equivalence. Dependent-sum transport then agrees with the pullback of extensional Σ transport, not merely with its fibres. Dependent-product transport agrees in the same sense on the invertible-transport fragment, and fibrewise extensional identity agrees on all pulled families. These statements explicitly account for the casts induced by quotient equality. The associated term constructors now commute as well: dependent pairing and both projections, lambda and application on the admissible product fragment, and reflexivity for extensional identity. The second-projection theorem includes the induced equality between its two substituted codomain families rather than comparing only untyped values. The transport of this audit to the category-indexed proof-relevant CwF is no longer open. For arbitrary category-indexed families, dependent sums now have pairing, both projections, both beta laws, and functor-substitution stability; fibrewise extensional identity has stable formation, reflexivity, proof irrelevance, and endpoint reflection. Pointwise dependent products have lambda, application, beta, and substitution stability whenever the domain action is fibrewise invertible. Every family over a groupoid supplies that action from the groupoid inverse.

The same category-indexed model has its own sharp negative control. In the walking directed-edge category, take a domain whose source fibre is 1, whose target fibre is 2, and whose forward map selects one Boolean. Use the covariant representable of routes out of the source element as the dependent codomain. The source pointwise product contains the identity route, while the target product is empty because the unselected Boolean has no incoming total-space route. Consequently no category-indexed family has even fibrewise-equivalent pointwise-product fibres. The variance restriction is therefore intrinsic to covariant dependent transport, not an artefact of the proposition-valued route model. The remaining closure obligations are intrinsic J, universes, and exact comparisons connecting these operations to the quotient CwDRA and the eventual MTT cell

action.

Separately, the checked modal CwF spine supplies contexts, substitutions, dependent types and terms, comprehension, dependent products, a Tarski universe, Fitch-style locks, modal types, and their substitution and coherence laws over a small strict *category* of modes. Products of such mode categories give commuting independent axes, and cost gradings remain additional monoidal structure. A revision mode category supplies strict guards, later, Löb induction, finite horizons, and commutation with a second product axis. The contextual semantics also has a genuine bicategory of pseudo-CwF morphisms and corrected transformations.

The checked unityped–simple–dependent family chain is reusable input but not the missing O/I/E lift. It consists of a strict inclusion followed by the fully faithful constant-family pseudo-CwF morphism; it does not supply the six O/I/E modalities, their three adjunctions, or the factorization cell. Calling that chain the dependent O/I/E triangle would conflate dependency depth with operational, intensional, and extensional semantics. The actual lift must either construct the lawful covariant pseudo-CwF morphisms and corrected transformations, or connect the context locks directly to their dependent right adjoints in the MTT orientation. At the quotient edge any covariant action must additionally state which dependent families are admitted or which pushforward doctrine is used; universal lossless descent is now disproved by the codiscrete Boolean control.

These results are substantial, but their conjunction is not yet MTT in the sense of Gratzer, Kavvos, Nuyts, and Birkedal [40]. MTT is parametrized by a mode 2-category: objects are modes, 1-cells are modalities, and 2-cells are transformations between modalities. Its locks and substitutions must account for those 2-cells, and its canonicity theorem is obtained uniformly by gluing. The checked two-computad supplies the generators and their semantic images, and the checked locally thin construction supplies one actual bicategorical quotient in which triangle, inverse, interchange, and all parallel-cell equations hold by thinness. The point-probe theorem now makes the selected **Cat** interpretation an actual pseudofunctor out of this quotient. The fixed-object-and-one-cell mapping-out law is also checked. What remains is to upgrade that law to the full bicategorical universal property and compare it with a proof-relevant quotient imposing only the selected equations. The current one-categorical modal CwF is still exactly the free path-category shadow: it retains finite modality paths and forgets authored 2-cells. It therefore still lacks their action on locks, types, terms, and substitutions.

Layer	Checked boundary	Remaining theorem
Semantic triangle	Six primitive functors, three adjunctions, commuting observation, a free path bicategory, eight authored 2-generators, a pseudofunctor into **Cat**, an actual locally thin candidate, global point-probe descent, the fixed-action mapping-out law, and positive/negative non-collapse controls	Prove the full bicategorical universal property and form the proof-relevant quotient by precisely the adjunction, inverse, and selected interchange equations.
Modal CwF	Dependent contextual operations, locks, modal types, laws, coherence, and selected term quotation	Add explicit 2-cells, lock reindexing along them, modal introduction/elimination, and all coherence laws.

Layer	Checked boundary	Remaining theorem
Weak categorical semantics	Pseudo-CwF morphisms, corrected transformations, bicategory laws, and a locally faithful base pseudofunctor	Construct the pseudofunctor from the chosen mode 2-category and connect it to the modal CwF semantics.
Independent axes	Product modes commute; guarded time and cost do not collapse into one another	Decide which commutations are structural equations, which are invertible 2-cells, and which deliberately fail.
Syntax and checking	Several contextual and dependent fragments have executable decisions	Construct initial intrinsically typed MTT syntax, elaboration, conversion, and a checker for the selected mode theory.
Metatheory	Many component substitution, adequacy, and non-descent theorems	Prove subject reduction, normalization or another conversion argument, canonicity, consistency relative to a named model, and conservativity of extensions.

The generator layer of the natural candidate now exists: it is free on $F, U, q, \Delta, \Gamma, Q$, the units and counits of the three adjunctions, and an invertible comparison $qF \Rightarrow Q$. The next construction is not another signature. It is the quotient by exactly the adjunction triangle equations, the two inverse equations for the comparison, and whatever interchange laws are explicitly selected. Guarded revision can then be combined by a product only after those interchange cells have been stated. CBPV polarity should initially remain an internal adjunction; it should become another mode axis only if the dependent context experiment shows that this buys a real law rather than duplicate notation.

15.6 Would Agda make the final MTT tranche easier?

Agda is a valuable comparator, but switching to Agda does not by itself supply general MTT. Agda 2.9 has built-in systems for irrelevance, run-time erasure, flat/cohesive use, and polarity; these share an ordered and compositional modality mechanism but have feature-specific typing rules [45]. It also offers an outer strict universe hierarchy under `-two-level`, computational paths and univalence under `-cubical`, and ticks and locks for guarded recursion [46, 47, 48]. These make it unusually good for comparing the scoped-UIP, cubical, and guarded branches of the dependent spectrum.

They do not amount to an arbitrary mode 2-category. The current research programme for full MTT in Agda explicitly identifies categorical substitution as the central obstacle and pursues three separate routes: refactoring Agda's existing modal features, a new lock calculus for full MTT in Agda, and a generic modal proof assistant [49, 50]. This is evidence that the proposed integration is promising, not that it is already an off-the-shelf shortcut.

Two neighbouring systems sharpen the choice. *mitten* implements bidirectional MTT checking and normalization by evaluation for decidable *preordered* mode theories, reducing type checking to the mode-theory word problem [42]. It is an excellent experiment if all parallel modalities may have at most one relevant 2-cell. BiSikkel is a deep Agda embedding with modes, modalities, and 2-cells, presheaf semantics, substitution, normalization, and a logical framework, but its program

language is multimode *simple* type theory rather than full dependent MTT [43, 44].

The tool ruling is therefore:

1. Keep Lean as the canonical consolidation and proof ledger for the present tranche. It already contains the semantic triangle, dependent-family experiments, pseudo-CwF bicategory, HOTG comparison, GSLT semantics, and NIK nodes that must be connected.

2. Use Agda for a bounded discriminator, not a migration: carry one common dependent specimen through ordinary intensional identity, two-level strict identity, and cubical paths, then compare computation, universe bookkeeping, and extracted evidence.

3. Instantiate the proposed mode signature in *mitten* only if a locally preordered quotient preserves every transformation the architecture needs. A negative result here is useful: it proves that proof-relevant 2-cells are not optional.

4. Use BiSikkel for a small presheaf and mode-composition experiment, not as evidence that dependent substitution, Π, Σ, identity, or universe laws have been discharged.

5. Reconsider a full Agda implementation after the mode 2-category and the vertical specimens have fixed the required syntax. Before that point, a new host mainly duplicates unresolved design choices.

Thus Agda likely makes the *identity and guarded branches* easier to explore, but not the final architecture easier to settle. The best immediate gain is an independent semantic and ergonomic comparison; the central MTT work remains the lock/substitution/coherence problem in any host.

15.7 NIK is a family of native services, not another mode

A Native Inference Kernel is an indexed family of native services over independently stated semantic targets. A service has one of four faces: direct decision, native proof, native operation, external certificate boundary. Only the fourth is a certificate contract, and only it declares

$$(\mathsf{Judgment}, \mathsf{Certificate}, \mathsf{check}, \mathsf{scope}, \mathsf{meaning}, \mathsf{soundness}),$$

where scope records the admitted fragment and profile, and meaning states what accepted judgments assert; a direct decision declares its kernel, a native proof service its intrinsic proof objects and their exact judgment, a native operation its executable arrow with its preservation law. Services over targets are not a category: every face transports along an equivalence of targets, a native operation pushes forward along any admitted operation, and a decision kernel does not pull back along a meaning-preserving operation that fails to reflect meaning (checked in `SemanticServiceFamily`). A certificate-transport route between boundary services must separately state:

1. translation of judgments and certificates;

2. commutation with native checking;

3. preservation of scope and semantic meaning;

4. reflection, fullness, or the exact essential image when any reverse claim is made;

5. proof-fibre, occurrence, universe-shift, and resource behaviour.

The maximal-native-calculus principle follows: a domain theory is checked at the strongest appropriate native inference layer, not reduced to opaque replay when the kernel can validate its quantifiers, connectives, conversion, or domain-specific rules. Replay remains valuable at an export boundary, but it cannot substitute for a native derivation when the latter is constructible. The direct SUMO experiment supports feasibility for a guarded higher-order core with source-style quantification and domain constraints; it does not yet establish complete SUMO coverage. Likewise, the dependent, HOL, Metamath, Megalodon, and GSLT nodes retain distinct meanings rather than becoming tags on one universal checker.

NIK selection is itself an open semantics. Given a claim, the dispatcher must choose a node whose scope contains it, prefer a native authority over a lossy translation, retain all route receipts, and reject incomparable claims rather than silently coercing them. Selection may optimize trusted-code size, proof size, checking work, semantic strength, or portability, but those objectives must remain explicit coordinates.

15.8 A typed interchange core for higher-order problems

Higher-order problem interchange supplies a concrete test of the authority discipline. The TPTP typed higher-order form TH0 is the community standard for simply typed higher-order problems [84]; it is to HOL *problems* what a proof exchange format is to HOL *theorems*. The checked development treats it as a typed, algorithm-neutral waist rather than as a prover:

named TH0 source
↓
admitted intrinsically typed formula
↓
portable declaration-aware packet, observed as structured data by dialect runtimes
↓
proof-relevant substitution events in an operational GSLT
↓
extensional syntactic-unifiability meaning
↓
exact checker refinement toward a native realization

The typed object is the semantic waist. Runtimes that store, inspect, transform, or schedule the packet are representation and orchestration environments; none becomes the authority for binding, typing, $\beta\eta$-conversion, higher-order unification, or validity. A candidate from any search role acquires meaning only through independent replay, and a finite miss says only that the chosen stage found no accepted witness.

Two checked boundaries fix the scope. The positive canary compiles a named identity formula, obtains the same intrinsic formula through two dialect adapters, checks the identity substitution, and constructs the associated operational events. The negative boundary is that a correct capture-avoiding simultaneous substitution service accepts syntactic unifiers but rejects a one-step β-unifier; substitution is strictly weaker than $\beta\eta$-aware unification. Implementing higher-order substitution over the packet is therefore necessary for a higher-order calculus and not sufficient for one. Superposition with λ-terms additionally needs $\beta\eta$ conversion under a stated equality profile, pattern or general unification with its completeness boundary—general higher-order unification is undecidable [87, 88]—typed orderings and selection, explicit extensionality, choice, and Henkin profiles, substitutions and parents inside proof objects, and an independent checker [89]. These are separate

checked services sharing the waist; their combination is an algorithm profile, not a redefinition of the core.

Monomorphic TH0 pins only the extensional pole of the present envelope. The readout ladder toward the dependent core runs through rank-1 polymorphic TH1 [85] to dependently typed higher-order logic [86], whose typing lands on the constant-family slice of the dependent envelope described above. Parsing every official TPTP family is source preservation and selects no execution semantics; Horn execution in particular is an optional plugin licensed only by an exact fragment recognizer, a least-model theorem, and replay. Scoped elaboration into the intrinsic core, with an explicit residual for what it does not cover, is a design target. Exact all-input checker refinements are checked to compose across source, structured intermediate, and native layers, and the negative control is essential: a final checker that agrees with the source on every canonical encoded example yet accepts malformed inputs cannot inhabit the native contract. A generated native parser and checker are therefore qualified by that all-input refinement, never by agreement on canonical examples.

15.9 Bootstrap diversity and object-theory strength are independent

The bootstrap question has at least five coordinates: certificate language, checker implementation, object theory, qualification base, and semantic model. None determines the others. In particular, a low bootstrap index does not imply a weak object theory, and a strong semantic host does not make a checker small.

The current seed doctrine is a matrix rather than a single privileged root:

NIK seed x over qualification base y	PA/Tpr	HOL
OpenTheory	seed cell	seed cell
Pure or MM0	seed cell	seed cell

with an optional stronger-model lane kept separate. The labels in this matrix are not assumed to denote homogeneous roles: each cell must state whether its x is the native certificate syntax, the checking logic, or an interchange format, and exactly what "over y" means. Ambiguity here is a bootstrap bug, not harmless shorthand.

Qualification has three independent axes. A theorem must be valid under its complete stated dependencies; its instantiated term must elaborate as the independently generated expected proposition at the actual endpoints, observer and universe bounds; and its replay evidence must be current for those source artifacts. Resolving a theorem name or matching a content hash does not discharge the second axis. Conversely, a refuted instance of a conditional construction disqualifies that application, not the conditional theorem itself. Kernel theorems, external-checker evidence and runtime fixtures retain their distinct guarantees in the same manifest.

Every retained cell needs two independent checking realizations, a complete grammar and static semantics, a soundness theorem in its declared base, exact import/export round trips, negative corrupted-certificate tests, resource bounds, and a severance test showing that exported checking no longer depends on the generator. A semantic model may prove relative consistency of a lower theory; no cell may infer its own soundness from successful self-replay.

The bootstrap ablation order is consequently:

1. freeze the finite NIK metatheory and certificate contract;
2. realize at least two seed cells with independent implementations;
3. prove translation and conservative-fragment theorems between surviving cells;

4. replay the complete declared corpus, not a miniature sample;

5. remove a cell only when the remaining graph still supplies independent checking, model qualification, and portability.

15.10 OSLF over equation classes

The construction follows the operational-semantics-in-logical-form recipe of Meredith and Stay and the sharper presentation in Meredith's *Finding Mind*: a presentation (Σ, E, R) denotes terms modulo E, reduction must respect those classes, and least reactive labels require a redex-relative pushout rather than an arbitrary enabling context [23, 24]. The executable `mettail-rust` implementation is an independent engineering witness: it materializes contextual equation closure and lets authored rewrites match through the resulting equivalence relation [25]. It does not, however, generate collection laws from declarations or prove modal adequacy; those are obligations of the construction here.

A language definition carries its equations as data, and the operational step of the generated GSLT is the composite $E; R; E$: an equation move, one authored rewrite, an equation move. The generated equation theory is the least contextual equivalence containing two kinds of generator. Authored equations contribute their instances. The presentation itself contributes laws it cannot help meaning: a bag tag makes every closed bag permutation-invariant with multiplicity retained, a set tag adds idempotence, a vector carries no law, and a declared collection algebra on a bare collection constructor supplies flattening of nested occurrences, absorption of the declared unit, collapse of the singleton, and identification of the empty collection with the unit, each guarded by the sort of the declaring constructor so that two bag sorts never exchange laws. When a presentation authors no equation and declares no bag, set, or algebra, the generated theory is syntactic equality and the saturated step is the authored step. Binder names are metadata of the locally nameless carrier and generate no law. The generated setoid is a semantic relation and need not be decidable or admit a canonical representative. A normalizer is optional runtime structure: when provided, it must be proved sound and complete for the setoid. The rho normalizer below supplies such a theorem for that presentation; it is not part of the generic definition of a GSLT.

For the rho calculus this generic theory is exactly the established canonical structural congruence on closed processes: every canonicalization step is a chain of generated equations, every generator preserves the canonical form, and the reflective canonical-equality generator that had been admitted for completeness is conservative. The step relation does not transfer the same way. Both rule interpretations match the communication redex with the same bindings, but the five-field syntactic interpretation instantiates the contractum's substitution node structurally and leaves a receiver that drops the received name with the drop of a quotation, whereas the declared reflective interpretation exposes the payload. The two contracta have different canonical forms, so no authored or derived law relates them; a closed source witnesses the disagreement in both directions at the primitive step level. The step of a language-generated GSLT is fixed only once its substitution interpretation is declared, and for rho the declaration is the reflection profile. Identifying the inert drop of a quotation with its payload would change the calculus and is not a repair.

Every formula of the OSLF language over the saturated step with equation-invariant atoms denotes a predicate on equation classes, negation and implication included. The structural-modal spatial former, which reads the outermost constructor of the representative it is given, does not: in rho the singleton parallel wrapper and the quote/drop law both change the outermost shape without changing the process. The behavioural fragment is invariant over every language, and reading the spatial former modulo the equations, on some equivalent representative, makes the

whole structural-modal language invariant while agreeing with the direct reading on the behavioural fragment.

Hennessy–Milner adequacy is proved over the quotient for a GSLT with an explicit observation set and a labeled family of steps, both respecting the equations. Bisimilar terms satisfy the same formulas of the fragment with truth, atoms, conjunction, negation, and diamond. Image-finiteness must be read on classes, since the saturated step reaches every representative of a successor class; under finitely many successor classes per label, logically equivalent terms are bisimilar, the negation-free logical preorder is the simulation preorder, and both statements descend to the quotient. The equations themselves form a bisimulation, so every such notion is a property of classes.

The completeness converse has a separate constructive qualification. An actual infinitely running, finitely branching GSLT provides a checked lower bound: unrestricted image-finite modal reflection implies LLPO. Given a binary sequence with at most one true entry, two roots agree on every finite formula, while bisimilarity of the roots is exactly the disjunction that all even entries vanish or all odd entries vanish. Each formula needs only finite lookahead; a coherent infinite matching needs the stronger disjunction. The classical adequacy theorem remains available with its recorded dependency. Finite branching and agreement at every finite depth alone do not discharge that constructive obligation.

An independently checked finite-depth bridge uses authored successor lists and a complete finite vocabulary. Its observer reads every scaled formula of depth at most n; equality of that readout is exactly zero in the constructed depth-n bound. An actual material family has an exact decoder through this observer precisely when its carriers agree on that kernel. A selected result has its own, stronger compatibility criterion, with a separate exact decoder and an inverse comparison of complete sections. Forgetting depth constructs observation-class maps satisfying identity and composition; compatible families and selected values commute with them. These results do not turn a positive approximation error into dependent identity transport.

One transition spends one observation level. The constructed occurrence span therefore runs from depth $n+1$ to depth n. Positive discount derives outgoing endpoint lifting and exact successor-modal pullback at those depths. The event readout retains the entire authored occurrence, but a matched transition need not have that same occurrence witness. Incoming lifting and exact-occurrence lifting remain separate laws, with checked failures even when the outgoing law holds. A proved stabilization stage supplies another constructive route: its finite readout kernel is full graded bisimilarity, and outgoing lifting then holds at a fixed depth. An actual root advancing to a loop has a constructed depth-zero stabilization proof and identical futures, while only the loop has an incoming transition. Stabilization therefore does not discharge the predecessor obligation.

The graded readings have an explicit ordinary modal signature: an atom (o, v) tests whether reading o has value v, while actions retain the authored labels. Agreement on these atoms is exactly agreement on all reading values, and ordinary bisimulation is exactly graded bisimulation. Old atomic signatures enter only through their proved formula interpretations; such interpretations need not exist. At a positive-discount stabilization stage, the actual system of represented readout classes has derived outgoing lifting and preserves and reflects every full Hennessy–Milner formula. Its bisimilarity is equality, proved by pulling class bisimulations back to the source. The existing formula-generated native predicates consequently descend, while an arbitrary native predicate descends exactly when it is constant on the actual readout kernel. An authored cyclic control agrees under a coarse reading at every depth but separates an old state atom; adding depth cannot recover an omitted signature. This bridge neither derives predecessor lifting nor asserts coverage of all underlying execution steps by a separately authored labelled action relation.

An independently constructed two-sided observer includes authored finite predecessor lists as well as successor lists, with soundness and coverage modulo the equations in each direction. Its

actions retain both direction and label; the forward formula language embeds into it and induces a surjective map of represented readout classes. Positive discount and a stabilization certificate for this two-sided system derive both source- and target-endpoint lifting. Future diamonds and the actual universal predecessor box then preserve and reflect their predicates. A separate authored event space retains occurrence and provenance; endpoint lifting does not identify those event witnesses. An infinite phase-cycle control has both endpoint laws while refuting exact occurrence lifting. A past control has identical forward readings at every depth and is separated at depth one by the two-sided observer. Thus adding the past is an explicit observer refinement with proved consequences.

Reading the same bounded formulas after every admissible context constructs the largest context-closed relation contained in ordinary finite-depth agreement. The context class need not be finite. Identity and composition act on complete observation fibres, and kernel-preserving substitutions act without selecting representatives; commuting authored substitutions give the corresponding commuting class square. Predicate, family and term descent are characterized by this precise contextual kernel. A wrapping context exposes a flag hidden by the original terminal observer: its selected material result cannot descend through the ordinary observer but does descend, with its actual decoder, through the contextual one.

The runtime obligation closes the loop. For closed rho processes, every step from any representative is matched, up to canonical form, by a depth-one communication on the canonical representative, because substitution commutes with canonicalization on binder-safe bodies and the communicating pair survives flattening, unit absorption, and sorting. A canonical bag admits no congruence descent, so its successor frontier is the finite list of its communication contracta; hence a closed rho process has finitely many successor classes, so the forward, box-free Hennessy–Milner fragment is adequate for this quotient transition system under any chosen equation-invariant observation set. The Lean canonicalizer computes precisely this equation theory (flatten, absorb nil, sort under alpha-invariant keys, collapse wrappers, and orient the name law $@\{*x\} \to x$). The hand-written C runtime implements the same recipe, including unquoting only in a matched drop position and leaving a free drop inert, but no all-input refinement theorem yet identifies its keys and steps with the Lean functions. A generated rho profile therefore has three acceptance obligations: its key must equate exactly the presentation's laws, its frontier must be the depth-one contracta of the canonical representative for every respelling of a source, and its substitution reading must be the declared one.

The generated formula language is now connected to that adequacy result exactly. Over the saturated step with equation-invariant atoms, the OSLF language with its step-future diamond and step-past box is the labeled Hennessy–Milner language over the two directions of the step: every OSLF formula translates to a two-direction formula with the same satisfaction, and every two-direction formula is expressible in the OSLF language, so OSLF equivalence is two-direction bisimilarity under image-finiteness modulo the equations in both directions. The box-free syntactic fragment is complete for the forward language, so equivalence on that fragment is ordinary bisimilarity under forward image-finiteness alone; the admissible structural-modal formulas embed the same way, and under the same finiteness their positive fragment characterizes the simulation preorder. A finite successor-class enumeration, the runtime contract of a quotient endpoint, discharges the forward hypothesis for every labeled system whose labeled steps are steps of the theory from a label-determined term. For rho, on the established presentation and on the public interpreted one alike, box-free OSLF equivalence is therefore forward bisimilarity with no remaining hypothesis.

The distinction is essential for rho. For each natural number n, a closed communication can use as its channel the quotation of a process containing exactly n outputs while its receiver ignores the message. Every member of this family takes one semantic $E; R; E$ step to the inert

process, and structural congruence preserves the input/output count, so different indices determine different equation classes. Thus the inert process has infinitely many predecessor classes. The full step-future/step-past language is still well-defined and bisimulation-sound, but the standard finite-branching completeness converse is false as an unconditional rho theorem.

Context labels receive the same care. The context-decorated logic reads a diamond $\langle K \rangle \varphi$ as a step of $K[u]$, and its adequacy statement must compare that logic with bisimilarity of the context-labeled system, not with reduction bisimilarity of the GSLT: a four-state theory has two reduction-bisimilar terms that one context diamond separates. With the correct notion, context-decorated equivalence is context bisimilarity under image-finiteness per context, and the least-enabling-context systems inherit the same finiteness from an enumeration. The generic least-enabler interface requires both firing and the factorization universal property. Rho's evaluation-context grammar supplies candidate one-hole contexts, but it no longer manufactures a minimal-context instance: deriving the genuine labels requires the redex-relative-pushout theorem and the admissible context class that excludes holes under quotation, exactly as *Finding Mind* warns [24]. Separately, parallel partners form a useful declared diagnostic system. Plugging respects the canonical equations, the canonical stepper makes every partner system image-finite, partner-labeled equivalence is partner bisimilarity unconditionally, and every partner formula is a native type of the generated OSLF. This partner result is not presented as the least-context theorem: arbitrary enabling contexts may over-distinguish. It does strictly refine reduction bisimilarity—a receiver waiting on a channel and the inert process are reduction-bisimilar, while one output partner separates them.

Binder alpha-equivalence is already closed once, at admission to the locally nameless semantic carrier: erasing display metadata is idempotent, and alpha equivalence is literal equality on admitted open, closed, and reflective terms. Adding alpha again to E would quotient the same distinction twice. Still open are the rho redex-relative-pushout construction, the C refinement, emission of generated runtime artifacts, and the two-sided comparison under effects, cost, ownership, and scheduling observations.

15.11 Vertical specimens that decide the architecture

No component diagram can select the native theory by itself. Each candidate must carry the same dependent objects through syntax, semantics, observation, native checking, and realization.

The operational interlingua already fixes the first stages of that carriage. Every GSLT supplies a reduction span, from which the operational-semantics logical framework (OSLF) derives the diamond/box modal adjunction; the construction does not make a relation executable, and executable evidence for an edge remains a separate checker authority. The modal doctrine attaches to the operational equipment as contravariant structure over a subcategory of tight translations: exact diamond pullback needs outgoing coverage and exact box pullback additionally needs incoming coverage, so not every operational arrow supports exact modal reindexing. The positive native fragment—top, conjunction, disjunction, spatial heads, and diamond—has finite proof-relevant certificates, sound and complete on its declared fragment, whereas box is a universal predecessor obligation with no certificate constructor. GSLT–IL itself is the typed interlingua for indexed operational calculi, routes, elaboration relations, and execution catalogs. It has two command forms, computing inside one fibre or exposing a requested transport whose route kind is retained as data, so that forward translations, exact covered translations, theory extensions, world revisions, observer changes, and physical realizations are admitted by different catalog entries. It is not the ontology of all logical meaning. The Hennessy–Milner theory of the generated OSLF runs through the same interlingua. Every labeled system descends to the semantic theory of its GSLT and the quotient map is a cover, so formulas and bisimilarity agree on authored terms and on their classes; returning

a fibre state as a command is a cover, so the native types of a fibre are exactly the native types of its returned commands; a pending transport request is a translation but not a cover, because the transport step leaves the fibre; along a covered stage map every fibre formula is preserved and reflected and bisimilarity is exact, while along a merely forward stage map only the negation-free formulas are preserved. This is the formula-level face of the modal doctrine: exact transport of native types needs coverage, and forward maps transport them laxly. Prime is not routed through this theory here; it enters only through the returned-fibre exact image, whose five named premises remain open. At the Prime edge, the returned-fibre fragment is checked to be an exact image: one-step witnesses, relational chains, open derivations, and admitted execution agree, while commands under a pending route and the act of applying a route lie provably outside that image and no full command decode exists. Five named premises separate this from a full internal-language universal property: intrinsic command syntax, typed transport, transport substitution, cell coherence, and initial factorization.

The proof-relevant operational layer now also supplies the dependent-context bridge required to test whether dependency is genuinely useful. Proof-relevant GSLTs and their evidence-preserving translations form a strict category before occurrence evidence is erased. Sending each such system to the free category of its finite evidence paths is functorial; composing this functor with the operational projection of a biform theory gives a context-valued reading of biform translations. In the smallest discriminator, two biform endomorphisms have the same logical theory map and event meaning, but one retains a Boolean occurrence tag while the other flips it. Their logical images coincide and their induced context maps differ. The representable family of histories transports its reflexive inhabitant to different results along the two parallel occurrences and therefore cannot descend to proposition-truncated reachability; a constant Boolean family does descend. Thus dependent typing is justified precisely when later data depends on which operational history occurred. When all parallel histories act alike, thin descent permits the simpler extensional or simply typed reading. This is a checked CwF-level discriminator, not yet an intrinsic J computation rule or a decision between intensional and cubical identity [3, 7].

Specimen	What it already discriminates	Remaining decisive braid
GSLT–IL core	Operational rules, typed paths, occurrences, natural transformations, a strict category of proof-relevant translations, its functorial action on dependent path contexts, derived native judgments, the OSLF modal doctrine on covered translations, formula-level exactness of covered stage maps and laxness of forward ones, and the returned-fibre exact image at the Prime edge	Intrinsic dependent syntax, substitution through all rule constructors, effects, a general syntax–semantics adequacy theorem, and the five named premises of the full internal-language universal property.

Specimen	What it already discriminates	Remaining decisive braid
Typed rho cell	A genuine communication receipt; dependent continuation; endpoint descent and revision non-descent; occurrence-bag naturality; exact NIK replay; operational and generated Cost1 executions; selected Cost2 fibre	Dependent CBPV translation, guarded modal term, HOTG code/readout, plural NIK dispatcher admission, and runtime full-abstraction boundary.
Graph walk	A grammar-generated native walk judgment, canonical adequacy, proof identity/composition/reversal/map, exact linear work, and forged-edge rejection	Invert every successful raw reverse-scan execution to prove the arbitrary-term no-junk direction; then transport the proof algebra through modes, NIK, and generated execution.
Dependent spectrum	Constant families, MLTT families, contextual reflection, dependent effects, and guarded receipts can be compared on one object	Carry the same object through each identity and universe branch with stated substitution, computation, and observer laws.
Ethical allocation	Rule, optimizer, and learned-virtue reasons can share an action while retaining distinct dependent evidence; coarse observation loses consent, distribution, cost, and reason	Complete source ontology, uncertain concept revision, native NIK checking, multi-agent scheduling, and runtime realization.
Direct SUMO core	Source-shaped higher-order quantification and domain guards can support native derivations rather than translated theorem replay	Broaden connective, variable, class, and source-axiom coverage with complete positive and negative conformance before calling it a SUMO kernel.

The dependent spectrum has a genuine inclusion chain, crossed by independent axes. Treating effects or guarded revision as later rungs would incorrectly claim that every intensional dependent theory is less expressive than every effectful or temporal one. The dependency-depth chain is:

D_0 Constant families and simple types: the exact fully faithful fragment where extensional readout is sufficient.

D_1 Intensional MLTT with Π, Σ, identity and J, and a cumulative Tarski hierarchy.

D_2 Contextual or reflective types, quotation, and a strict outer reasoning layer; compare scoped UIP/2LTT against retaining intensional inner identity.

Five axes cut across all three levels:

1. identity: intensional J, scoped strict equality, or cubical paths;
2. computation: pure terms, internal dependent CBPV, or a locked value/computation mode pair;

3. time: unguarded substitution, a strictly commuting guarded product, or a guarded interaction governed by nontrivial 2-cells;

4. universes: a finite cumulative hierarchy, level polymorphism, family enclosure, superuniverse or extended predicative Mahlo closure, and a separately stated extensional HOTG model host;

5. modal-cell identity: locally discrete, locally preordered, or proof-relevant according to the exact parallel-cell discriminator.

Cost, span, memory, evidence strength, and risk are quantitative enrichments of points in this grid. Receipts, occurrences, ownership claims, and learned ethical reasons are dependent inhabitants over those points. Neither group is promoted to a mode axis without an independent context-lock theorem. Cubical identity therefore does not choose a universe service; HOTG hosting does not choose an evaluation strategy; and a guarded receipt does not by itself prove that modal 2-cells must be proof-relevant.

15.12 The decision order

Before a new native implementation fixes representation or conversion choices, the design must expose the following boundaries. These are requirements on the next implementation, not a claim that the current Prime presentation is ratified.

1. Pin the intended `lang:`, `set:`, and `type:` profiles and their judgment, identity, effect and universe interfaces. Their upper envelope is a family of compatible profiles with explicit translations, not a claimed greatest consistent theory. Stronger enclosure and reflection services remain available as qualified extensions rather than becoming unnoticed premises of ordinary arithmetic.

2. Prove old-judgment and conversion stability for reusable components, and composition laws for feature combinations. A symbol dependency list is insufficient when conversion rules can alter the old fragment.

3. Specify recovery on each claimed lossless image and coherence of interleaved translations. Preserve distinctions required by later dependent consumers; identify explicitly any equation, observation or proof quotient.

4. Separate the direct mathematical service interface from the programming frontend. Bind both clients to the same admitted meanings without requiring one client's syntax or evaluator to justify the other's mathematical claims.

5. Connect authored rules to the generated operational calculus, and its independently specified execution to the emitted artifact. Existing physical schema-instantiation agreement leaves actual C source refinement open; packet admission alone is not full derivation or execution adequacy.

6. Fix the root trust and versioned qualification rules before relying on generated kernels to verify later generators. Internal earlier-stage metatheorems require their own syntax representation and strength assumptions, as distinguished in the qualification tower above. Interface stability and checked migrations can preserve components across revisions; no theorem yet promises arbitrary future changes without rework.

The shortest honest route to a final choice is now:

1. The locally thin semantic gate and fixed-action mapping-out law have resolved positively for the selected **Cat** model. Prove the full bicategorical universal property, record the exact nondegeneracy and essential-image statements, define the three mode-indexed dependent doctrines, construct their lawful comparison morphisms, and compare the result with the proof-relevant quotient. Local faithfulness now proves that a later dependent action can rule out M_1 only by distinguishing context-level transformations (or by motivating and validating a different transformation notion); fibre witnesses alone cannot do so.

2. Extend the modal CwF to a genuine MTT syntax with 2-cell action, substitution, Π, Σ, identity, universes, and a decidable selected mode theory.

3. Define one comparison-profile type shared by mode translations, NIK routes, bootstrap exports, and runtime realizations. It must record logical, computational, observational, resource, and strength coordinates.

4. Carry the typed rho and graph-walk specimens through $D_0 \subset D_1 \subset D_2$ and across the identity, computation, time, universe, and modal-cell-identity axes. A candidate failing either specimen is not the general native theory.

5. Connect the surviving calculi to plural NIK nodes, then instantiate at least two independent bootstrap seed cells.

6. Prove two-sided runtime comparison at declared observations and measure translation quirk, code growth, conversion work, and proof-fibre retention.

7. Select a native calculus only after the same matrix has exposed which candidate is simplest *without* erasing a required distinction.

This order deliberately postpones syntax branding. It does not postpone the mathematics: every step produces a reusable theorem or a falsifying specimen.

16 Proof-theoretic and model-theoretic ledger

The big-picture theory remains incomplete until the following columns are filled for each concrete calculus and each hosting profile. Initial syntax, executable checking, and one semantic model answer different questions.

16.1 Proof theory

Obligation	Required evidence
Rule package	Formation, introduction, elimination, computation, congruence, structural, and universe rules with no hidden meta-level conversion.
Substitution	Weakening, exchange where intended, substitution, and stability of every judgment under context morphisms.

Obligation	Required evidence
Conversion	Decidability or a precisely scoped semidecision; soundness, reflection on typed inputs, congruence, and work bounds.
Metatheorems	Subject reduction, canonicity, normalization or a reduction-free decision argument, and consistency relative to a named model.
Cut/composition	Cut admissibility or an explicit computational interpretation of cut; normalization of certificates modulo bureaucratic permutations.
Conservativity	For each extension, a translation or elimination theorem showing what old-language judgments remain unchanged.
Interpretability	Judgment translation plus preservation, and reflection when claimed; this induces the strength preorder used by bootstrap ablation.
Universe strength	Separate constructor inclusion, ordinal analysis, interpretability, and classical model assumptions for enclosure, superuniverse, Mahlo, and HOTG profiles.
Reflection	Exact distinction among an external soundness proof, an internally represented model, an internal consistency theorem, and a reflection rule.
Complexity	Certificate size, checking work, conversion work, and structural bounds needed for weaker arithmetic hosts.

Initiality supplies the least structure generated by the rules. It does not imply consistency, normalization, minimal proof-theoretic strength, or minimal trusted code. Those are separate orderings and may have incomparable minima.

16.2 Model theory

Obligation	Required evidence
Model signature	Carriers, worlds, context interpretation, equality, and the precise universe/smallness profile.
Soundness	Every rule preserves satisfaction, including conversion and substitution; bottom is refuted at a named world when consistency is claimed.
Completeness	A term, Henkin, realizability, or other canonical model, with the exact class of judgments for which semantic consequence reflects to syntax.
Categorical semantics	CwF/comprehension, fibrational, modal, CBPV, or coalgebraic structure, with preservation laws explicit rather than strict by convenience.
Observer semantics	Descent, adequacy, full abstraction where possible, and non-collapse witnesses where information is intentionally retained.
Model comparison	Morphisms, elementary embeddings, logical relations, or interpretations connecting HOL, dependent, arithmetic, and coalgebraic faces.

Obligation	Required evidence
Large-universe calibration	Sufficient large-cardinal models, lower bounds, and equiconsistency claims recorded separately; a stronger model assumption alone is not a stronger-theory theorem.
Nonstandard profiles	Which statements transfer to ultrapowers, hypersets, or open-ended atlases, and an explicit divergence witness showing where the model is more than notation.
Qualification provenance	The host theory and assumptions used to construct the model, plus the level at which its soundness theorem is checked.

For the observer-indexed programme, the crucial model-theoretic targets are the small-based final-coalgebra instantiation, the route-rich/extensional comparison, and a HOTG-internal construction with separate membership. The tail, principal, and free reduced-power boundary now supplies checked nonstandard-model controls, but a transfer theorem for the intended object language remains open. For the native type theory, the crucial proof-theoretic targets are typed dependent conversion, an inductive–recursive realization of family enclosure, the DTT-to-HOTG adequacy square, effectful substitution, and interpretability among the candidate calculi. Neither list should be allowed to disappear behind the other.

17 Status ledger: proved, refuted, conditional, open

Candidate provenance. Prime remains a language-design target, not the name of a foundation selected by an import or an implementation. Named reusable presentations include the cumulative tower, staged-reflective calculus, and branch-local Need reference. `PrimeCandidates` collects candidate integrations; `PrimeCeTTaCurrent` separates inspected runtime artifacts and bounded V1 calibrations. Compatibility proofs, necessity relative to stated requirements, experimental preference, and implementation qualification are four different claims. A Lean emitter is not required to start a parallel implementation, although that implementation cannot claim a refinement theorem before the theorem is supplied.

`DesignDetermination` makes necessity nonvacuous by requiring an inhabited admissible class as well as implication for every candidate in that class. Its quotation countermodel proves that preserving unevaluated quoted syntax alone does not force non-strict evaluation of ordinary arguments. `PrimeCandidates.DecisionSurface` combines the existing shared/fresh provider workload with a new obstruction: equal returned-value supports do not suffice to recover the distinct occurrence/effect observations. Neither result selects a universal evaluation strategy. Similarly, the declaration `CumulativeTower.Id.eliminate` names J in that presentation; its formation and qualified preservation proofs do not adopt global K/UIP or determine Prime's identity theory.

Theory-driven specification development. `DesignStudy.Specification` indexes requirements by their actual semantic predicates. Evidence is indexed by one candidate, so a completed requirement list establishes compatibility for that candidate rather than combining unrelated existence witnesses. Extending the list narrows the admissible class; necessity is retained only for a consistent extension. The existing `Razor.Criterion` supplies independent comparison orders, including product orders, without selecting a globally best language. The finite evaluation study proves both positive narrowing by an added unused-effect requirement and a negative control: individually satisfiable requirements may have no joint candidate. Requirements on a component can now be pulled back along the projection from an assembly; conjunction retains that

same assembly index. Adding evidence does not change old predicates. These laws do not manufacture compatibility: the agreement of interfaces remains a further predicate on the assembly. `ScopedComputationSpecification` instantiates this distinction with one native handler and an independently authored primitive-world function. Exact realization and preservation of dependent results qualify admitted source programs under refined substitution. Concrete misindexed and unrealized implementations refute either implication between these two requirements. The resulting inhabited shared study is not yet the whole candidate contract.

`SharedJudgmentFragment` now supplies a raw assembly of actual declarations, scoped execution, receipt reconstruction, intrinsic HOL proof production and contextual readouts. Seven independent semantic predicates qualify those same data. Rules derive from the declarations; contextual resumption uses the assembly's own primitive worlds and is related to its handler by exact realization. Coverage excludes an always-declining matcher. The HOL producer must agree with independently checked submitted certificates and bind the complete claim. A different admitted reflexivity reconstruction also qualifies, so the contract does not force the reference J syntax. Altered handlers, indices, declarations and answer-only views provide independent negative controls.

`SharedJudgmentServices` now gives the actual matching and HOL producers request-indexed invocation rules. Successful replies bind their native payload in the submitted dependent continuation; declined replies remain observable but select no continuation. The HOL payload is formed proposition syntax with a separate source proof, not native proof inhabitation of that proposition. Altered indices, outputs, premises, claims and submitted certificates provide independent non-admission controls.

`SharedJudgmentServiceOSLF` connects this protocol to an independently authored backend, `ContextualEffectTreeLanguage`. Six rules implement the pure, choice, read, write and intent constructors. Recursive premises execute the subprograms; the relation environment only appends lists and prefixes intents, never invoking the source evaluator. At sufficient structural depth, execution returns exactly one completion containing the entire ordered world list. Arbitrary returned patterns are proved to equal that completion. Insufficient depth yields no partial prefix and cannot establish semantic absence. The native codec retains payload scope, branch traces, private states, intent order and repeated occurrences. Exact execution and native dependent admission remain independently necessary: a misindexed handler can execute exactly and still fail admission.

The protocol's generated native type applies three semantic diamonds to an equation-invariant occurrence observation. Satisfaction is proved equivalent to the actual submitted request, selected reply and continuation, and an occurrence in its exact completed worlds. This is existential membership; multiplicity is retained by the separate complete-list laws. Actual matching worlds have equal native answers but different returned states, refuting general state recovery from answers alone. Erasing deferred intents also fails the exact observation. No claim is made that those intents were performed externally.

`NativeHOLFragmentDenotation` now interprets the actual scoped native HOL constructors and their simple-type arguments. Induction on source syntax proves the representation square; native meaning is not defined by a decoder into source HOL. Renaming, actual capture-avoiding substitution and native beta computation preserve the interpretation. Every interpreted value is admissible under an admissible Henkin valuation, without a full-domain assumption. Componentwise interpretation therefore also makes the submitted native substitution an admissible semantic environment transformation. At a fixed HOL context and type, independently constructed native denotations are now pointwise equal. The proof uses a heterogeneous logical relation: erased lambdas can have different hidden application domains, so uniqueness of those domains cannot be assumed. This coherence requires neither normalization nor full Henkin function domains.

The actual service response retains that constructorwise formula meaning. An observation of

its payload through the generated native type inherits conditional model truth from the qualified source proof and the interpreted substitution. Any independent constructorwise meaning of the same observed payload agrees with that value; this does not itself establish truth. Operational success does not discharge the source assumptions. The same returned native map-length formula is true in the standard model and false in the junk model; the latter satisfies the displayed equations but not the induction premise of the accepted proof. This is a native HOL image interpretation; it does not alone interpret the surrounding mixed contexts, dependent products, identity, lists or arbitrary native universe levels.

The matching payload now has a separate constructorwise interpretation in `NativeMatchedTransportDenotation`. Wire constructors and independently Data-typed variables receive their structural meanings; the endpoint-dependent native J spine is interpreted through the existing full set-family J. Actual capture-avoiding substitution commutes with this interpretation. An open Data parameter over the common native context can be filled by the selected pattern: native typing, substitution, consumed source and semantic J then commute. Encoded receipt data has no identity interpretation, and a formed Data expression outside the structural Wire grammar is explicitly excluded. Thus native formation does not silently supply this partial meaning.

`NativeWireDataDenotation` supplies a total free algebra for the actual Data constructors, including nonsymbol application heads and improper cons tails. Its initial-algebra law determines all constructor meanings; canonical Wire values embed injectively and every semantic value has an admitted native representative. Actual refined substitution retains the constructor meaning for arbitrary Data parameters in the mixed context. The former partial interpretation agrees on its image. The algebra is total, but its native denotation relation remains a constructor fragment, not a fundamental interpretation of all native functions returning Data.

`SharedJudgmentMatchingMeaning` constrains the actual assembly's reconstruction operation by that semantic property, separately from native admission. The original J reconstruction and its reflexivity alternative both satisfy the combined requirements without identifying their code. Their actual lifted payloads and generated OSLF observations inherit the endpoint meaning. This is not a semantics of arbitrary admitted motives or unrelated fields of the mixed context, and it selects no native K/UIP rule.

`FormationSensitiveCompletedRelatorPreservation` first establishes subject reduction for the completed native roots and their contextual and parallel development, including the full J motive. Authored conversion of two independently admitted terms has a common reduct admitted at both original displayed types. Consequently, `FormationSensitiveCompletedConversion` identifies authored conversion at a fixed displayed type with the equivalence closure of completed development on admitted terms. Its intermediate states are formed, and the relation is stable under actual refined substitution. This reconstructs an admitted path, not the route identity of every supplied raw receipt. A valid reflexive conversion certificate for a missing declaration still cannot supply its admission. Neither normalization nor a full native model is established by this result.

`SharedJudgmentInterpretation` attaches raw context, type, annotated-term and substitution relations to the same native assembly. Their totality and structural, substitution and conversion laws are independent predicates. For the stated strict interpretation class, soundness of the actual native conversion rule separates into formed completed-development invariance and transport across convertible annotations. A nonempty raw relation preserves fixed-annotation conversion yet fails the latter condition. An actual mixed HOL-list/wire projection illustrates the positive transport at a distinct beta-expanded Data annotation. This theorem transports an established meaning; it does not manufacture a full model, infer constructor meanings from totality, or impose strict equality on all possible hosts.

`SharedJudgmentUniverseInterpretation` supplies raw code constructors on this same

semantic CwF, with totality, decoding and cumulative meaning stated independently. Under these laws and actual formed substitution, native cumulativity yields a semantic code square. A concrete weakening supplies a positive native control; the tagged native formation interpretation excludes strict code uniqueness. The operations and conditional square do not by themselves construct a full universe model.

`SharedJudgmentServiceInterpretation` likewise separates the four raw NIK faces from qualification at independently supplied target meanings. It retains the exact accepted artifact and derives its separately qualified native judgment and meaning, including transport under formed substitution. A real wire decision and a nonidentity Data-parameter substitution exercise that connection. Changed parameters or payloads fail the corresponding meaning claim even when the native payload remains formed. The unchanged intrinsic HOL checker also cannot acquire unconditional junk-model truth. Neither arbitrary external proof-byte checking, required service coverage, nor a full common interpretation follows from these fragment controls.

`SharedJudgmentServiceRegistry` now fixes required interfaces outside the proposed implementations: the guest proof system, operation input target, certificate carrier and exact native surface are part of the caller's contract. Independent face/target qualification cannot replace these bindings. A native operation with an empty input domain satisfies the weaker face laws but fails the required interface; a differently scoped native payload fails the surface contract despite independent formation and meaning. Four actual wire invocations qualify on one shared fragment interpretation. This does not extend that interpretation to the whole native mathematical host.

`ContextualTypeOperations` supplies independent raw Pi, Sigma and identity constructor data over one CwF. Beta, full identity-context boundary and substitution laws are separate predicates on those same data. Strict type/term representative comparisons are an explicitly named interpretation class. A nonconstant set-family instance satisfies the laws, including full J motive reindexing; indiscrete identity on that same full family structure cannot qualify dependent elimination. These are semantic interfaces and controls, not an interpretation of every native constructor or universe.

`SharedJudgmentTypeInterpretation` connects raw Pi, Sigma, identity and based-J operations to the same native interpretation data. Context extensions are compared by maps rather than assumed equal. Independent coverage concerns actual admitted native families and motives, not all ambient families. Constructor agreement and typed-substitution laws derive the principal-result meaning of any admitted based-J application and its reflexivity computation. A concrete dependent telescope and weakening past a new identity witness connect source admission, method admission, both meanings and the semantic reindex/reflexivity square. Empty implication-shaped agreement relations fail coverage at this source. The separate full set-family operation laws do not construct the native interpretation; converted annotations and universe-code coherence remain distinct obligations.

`SharedJudgmentUniverseTypeCoherence` now separates the actual native Pi/Sigma code-value agreement from the semantic type-formation clauses. Independent decoding, coverage and qualified comprehension comparisons derive display-map isomorphisms for both constructors on the same admitted open family. The comparison condition concerns the actual base projection, not just the newest variable: a checked invertible map retains that variable while changing the base. A nonidentity connector repairs the square without imposing strict agreement of the original presentations. A genuine mixed-level chart has fibres of sizes two and three; its dependent product has six elements and its dependent sum five, excluding the wrong constructor code. This local chart is not a model of all native levels. These laws imply neither code equality nor automatic transport of every annotated-term meaning through an arbitrary isomorphism.

`SharedJudgmentServiceTypeCoherence` connects successful use of the required registry to that same returned payload's independently formed native type. Code totality and decoding supply its

code meaning; formed substitution transports the original code and the original interpreted payload value. A concrete type automorphism exchanges two admitted, interpreted payload values without preserving the first native term's meaning. The existing four-face wire registry is qualified at its constructor interpretation but has no universe-code meaning for its returned Data type. Thus service qualification does not supply the missing mathematical host.

`SharedJudgmentSemanticRequirements` collects 31 native semantic predicates, four declaration predicates, the source predicates and the externally fixed service contract on one record of raw assembly data, interpretation, universe operations, type operations, registry and declaration meanings. Required declaration instances are supplied by the caller: actual installation and closed formation precede semantic coverage. Normative references resolve separately from proof evidence. The source-only evidence list leaves every semantic and declaration obligation unlisted as discharged; this is bookkeeping, not logical independence of all clauses. Empty relations and a nonempty annotation-sensitive relation fail this collection. Qualification interprets the actual conversion rule, supplies the accepted-type code connection and extends required closed declaration meanings into formed caller contexts. Presheaf comparison and the remaining public operation/consumer contracts remain to be incorporated. This collection is neither a completed six-family specification nor a jointly inhabited model.

`SharedJudgmentDeclarationInterpretation` retains the name, schema and actual universe-level instance, with the effective installed lookup. Independent closed type and head meanings, together with the existing substitution clauses, derive contextual use. Declared application uses the actual Pi-meaning clause with an explicitly sufficient strict representative comparison; it is not a universal demand for strict equality. The installed natural declarations have nontrivial Value meanings. Changing seven's meaning to eight retains native formation and semantic type coverage but fails head agreement. Raising the requested Data instance remains formed but does not change the installed signature. Neither a name nor formation supplies the missing instance. Arbitrary definition bodies and delta soundness are not proved by this opaque-declaration control.

`SharedJudgmentPublicContext` retains original source, native environment and claimed admission metadata separately from actual backend configuration and completion. Expanding the implicit environment preserves and reflects completion with identical outputs, including the full reply and world records, without asserting a bijection of paths or costs. Two admitted programs nevertheless have identical authentic outputs and distinct source syntax. Output-only use and source inspection therefore require different declared readouts. Full CBPV, live dependency binding and an exhaustive public consumer family remain separate obligations.

`SharedJudgmentProgramContexts` closes the complete-world observation under seven actual source-context operations: substitution, either position of ordinary sequencing, either position of dependent sequencing, and either choice branch. Quantification over the actual target environment, initial state and branch derives congruence and exact backend completion in every finite path through this inventory. A genuine administrative detour is invisible to these computational contexts but remains visible as code. Equal present answers do not support an admitted continuation whose effects inspect retained state. This is a particular contextual view contract, not a global forgetting policy or a claim about all reflection, sharing and resampling contexts.

`SharedJudgmentPolarizedServices` supplies actual matching and HOL callbacks to the existing native-indexed polarized machine. A HOL environment template is filled by the actual captured native argument; successful values retain dependent native admission from the independently formed request environment. Unreadable input and rejected requests produce distinct retryable residuals, not a false proposition or an empty successful answer. The dependent invocation sidecar retains the submitted request and response before payload observation. Active thunk/function equations hold under the existing continuation consumer and preserve complete

worlds; the same wrappers remain distinguishable as stored producer origins. A captured valid receipt succeeds despite a changed caller argument, and a genuinely function-valued HOL argument instantiates the represented induction claim. These fixed-target-scope laws do not establish full polarized primitive typing or every receipt/sharing consumer. Cross-target transport is now qualified separately in `SharedJudgmentServiceNaturality`. The actual HOL callback commutes with native substitution under its argument binder; matching commutes on encoded wire arguments. More precisely, the complete retained matching invocation commutes if and only if decoding the substituted argument gives the same wire observation as before. A formed Data variable is unreadable before substitution and becomes an accepted receipt afterward, under an independently admitted substitution. Formation alone therefore does not imply this inspection law. The changed answer resolves a retryable residual; it is not a reversal of a logical refutation. A function-valued HOL argument crosses a genuinely extended target context without variable capture, retaining its submitted request and dependent result admission. These callback laws do not yet transport arbitrary captured heaps, sharing or resampling consumers.

`FormationSensitiveContextualCategory` constructs the structural base using actual refined context formation and substitution, with natural comprehension and a faithful forgetful functor to the earlier declaration-aware contextual category. The earlier category uses the older typing relation; its existence cannot replace this refined-judgment comparison. Under the existing explicit universe-regularity condition, every common-package admitted judgment is represented on its formed telescope. This unquotiented structure is not yet the required conversion-invariant model: a real mixed projection and its result determine distinct raw sections despite being convertible. Identifying distinct encoded wires, on the other hand, prevents recovery by the specifically required wire decoder. The model must therefore support the declared semantic comparisons without claiming that all code-inspection consumers factor through them.

`FormationSensitiveContextualComparison` supplies one explicit comparison before any quotient: independently formed convertible binder annotations yield isomorphic extensions by actual refined identity substitutions. Projection, reindexing and roundtrip laws preserve the native type and term codes. The newest-variable comparison uses an explicit conversion to the independently formed target annotation. A dependent $\mathsf{Id}\ \mathsf{Data}\ x\ x$ family and its reflexivity term move to a context annotated by a distinct beta expansion of Data, while raw context inspection still distinguishes the two contexts. This needs no object-level K/UIP or equality reflection; it is not equality of the context objects or the missing full semantic interpretation.

`FormationSensitiveContextualQuotient` and `FormationSensitiveConversionFibres` supply a conversion-invariant structural route. Actual admitted substitutions are quotiented by pointwise native conversion; Mathlib's quotient category makes exactly those identifications. Its objects remain formed native contexts. Type classes contain independently formed annotations; total term classes retain both annotation conversion and term conversion. Actual substitution descends to type and term presheaves on the quotient category, with a natural term-to-type projection. The real dependent identity family and reflexivity term agree after the convertible mixed substitutions, although their raw codes differ. For opaque extensions, term-class equality also has an entirely formed completed-conversion path after annotation conversion.

`FormationSensitiveQuotientCwf` constructs all structural CwF laws and a terminal empty context from these carriers. Comprehension chooses an actual formed type representative; a term class can be represented at that annotation by the existing refined conversion rule. Pairing, projection, variable and uniqueness laws are proved, not supplied as qualification assumptions. Reindexing agrees with the descended presheaves. An explicit context isomorphism relates the chosen extension to every actual formed annotation in its class, preserving the base projection. This is a syntactic CwF route in the standard organization of dependent syntax [91], without an initiality claim or

literal equality of raw context objects. The very same constructed term fibre identifies the mixed computation with its result while keeping seven and eight distinct.

This route intentionally forgets a selected universe witness, not universe membership: one actual Data class has independently proved membership at two cumulative levels. Neither every original witness nor every original term code can be reconstructed from the respective class. Those inspections remain available on retained presentation data; the quotient is not a universal observer.

`FormationSensitiveQuotientUniverses` constructs a Tarski hierarchy on this same CwF, using actual native terms of the universe type as codes. Decoding returns their independently formed type class. At each supplied level, a class has such a code exactly when it has an actual formation witness at that level; every type of this syntactic CwF has a code at some level. Decoding is injective on these fixed-level code classes. Cumulative lifting preserves the decoded class, is injective, and commutes with the actual quotient substitution. A larger-level code descends along a supplied level order exactly when its decoded type has the required lower formation witness. This does not recover an original raw code or selected witness. Under the proved opacity condition, native universe conversion is exactly equality of level-expression evaluations; successive universes remain distinct. No consistency theorem follows just from their distinction.

`FormationSensitiveQuotientProducts` adds actual native dependent products and sums, not externally supplied operations. Lambda/application and pairing/projections satisfy beta and their full typed substitution laws. Lifted context maps are compared by the proved comprehension property. Formation at independently supplied domain and codomain levels survives transport across the chosen binder representative. Consequently, `FormationSensitiveQuotientUniverseProducts` constructs product and sum codes at the maximum of those levels, decoding to those very same operations. Each level is closed under same-level products and sums; identifying the level expressions $\max(\ell, \ell)$ and ℓ here is not universe resizing. A dependent family $\mathsf{Id}\ \mathsf{Data}\ x\ x$ returns the actual native reflexivity term at its supplied wire. That dependent function is itself passed through a higher-order lambda/application. Dependent pair projections retain the wire and its reflexivity term, while pairs carrying seven and eight remain distinct. The code constructors have proved decoding laws; their own code-level substitution API is separate from the already proved substitution laws of the underlying products and sums.

`FormationSensitiveQuotientIdentity` makes identity formation and reflexivity substitution-stable in the same model. The existing native J declaration has a more specific scope: its element level is parameter zero and its motive level is parameter one. Every actual motive body admitted at that profile, including dependence on both the endpoint and path, is represented by a typed double lambda. Native iota gives beta in the actual term fibre, and the two-binder substituted J commutes with reindexing. Explicit isomorphisms compare the native based context with the CwF's chosen one; converted formed motive bodies and methods give the same J class at a fixed endpoint context. An actual lower-bound theorem for universe typing handles all refined conversion and cumulativity tails. It shows that identity/reflexivity can exist at a domain excluded by the fixed J declaration's argument type. This boundary concerns the declaration's scope, not a selection between J, K or UIP.

`FormationSensitiveLevelInstantiation` and `NativeIdentityLevelInstantiation` now qualify the explicit level-instantiation route. Refined judgments, formed contexts and typed substitutions transport through a transformed native and extension signature. Every admitted motive body at the instantiated element and motive levels has native J, beta, and caller-substitution preservation. The transformation respects identity and composition of level substitutions; it is not asserted injective or invertible. A higher-universe motive computes in the transformed environment and rejects a crossed result or endpoint. The original fixed environment cannot supply that same admission. Appending a same-named declaration leaves the native lookup unchanged. Conse-

quently, these theorems give a family of indexed environments, not simultaneous level-polymorphic instances in the current assembly. Embedding that family requires an explicit rule/environment index or a qualified declaration-elaboration route.

`FormationSensitiveQuotientInterpretation` supplies interpretation graphs on the actual formed native contexts, covering and reflecting admitted judgments under the existing universe-regularity condition. The graphs preserve native substitution and compare independently formed convertible annotations. Actual native families have comprehension coverage through the proved context isomorphisms. An exact raw-context-preserving total section from every raw context into this formed-context model is impossible when an unformed context is present. This is a domain obstruction for that exact section, not a theorem against arbitrary semantic maps or exploratory unformed syntax. Conversion invariance likewise cannot make two distinct raw comprehension objects literally the same chosen object.

`FormationSensitiveContextSourceObstruction` makes the formation boundary stronger. In the actual common rules, consider the raw telescope

$$A : U_{\max(0,1)}, \qquad f : A, \qquad x : f\ \mathsf{wire}_0.$$

The prefix is formed, but its final annotation cannot treat the abstract $f : A$ as a function. Substituting $A := \Pi(_ : \mathsf{Data}).U_0$, $f := \lambda_.\mathsf{Data}$ and $x := \mathsf{wire}_0$ gives a refined-typed substitution into the formed empty context. The last annotation becomes a formed beta redex, yet this does not make the original telescope formed. Thus independently formed target and typed substitution do not supply source formation. The shared semantic interface now indexes formed native contexts explicitly. Raw service requests, terms and proof records remain representable; a retained unformed application request cannot acquire the qualified native-admission predicate.

`SharedJudgmentQuotientInterpretation` attaches these actual graphs to that corrected interface. The same data prove admission coverage and reflection, substitution totality and constructors, type/term substitution, conversion, annotation transport, comprehension and dependent-family coverage. `SharedJudgmentQuotientUniverses` adds sort meaning, code coverage, decoding, cumulative meaning and universe substitution on the same CwF. An actual open type code crosses a new Data binder and two cumulative lifts with a coherent decoded meaning. Changing its universe carrier to the successor is rejected; cumulative admission is not universe equality.

`FormationSensitiveQuotientDeclarations` and `SharedJudgmentQuotientDeclarations` attach installed declarations to the same interpretation. The table selects exactly independently installed and formed instances; their closed types and heads have meanings that extend to arbitrary formed callers and commute with actual substitution. A real HOL universal declaration retains its class through a filled mixed HOL/wire caller. A formed annotation does not install a missing name, and changing seven to eight does not preserve the actual selected meaning.

`SharedJudgmentQuotientComprehension` proves that the independently qualified projection and newest-variable meanings uniquely determine the comprehension comparison. Every qualified dependent family consequently has its actual canonical quotient representation; this does not identify distinct raw context objects. On that basis, `SharedJudgmentQuotientProducts` proves all eight shared Pi/Sigma clauses, including lambda, application, pairing, projections and beta. `SharedJudgmentQuotientProductCodes` identifies their actual native code values at the fixed result universe, even with mixed input levels and an arbitrary qualified comparison. The product clauses take only product operations; proving them no longer requires an unrelated total J field. `SharedJudgmentQuotientIdentity` similarly attaches native identity formation and reflexivity. A changed endpoint remains separately admitted but cannot receive the original computation's meaning.

SharedJudgmentQuotientServices attaches the original required four-face specimen to this same interpretation. Its targets, submitted inputs and native-surface contract are unchanged; actual caller substitution preserves the response meaning. Wrong-face, empty-source and separately meaningful but changed-payload services fail that independent contract. This is qualification of the stated constructor-classification specimen, not all future services or an assertion that callback reexecution commutes with every substitution.

FormationSensitiveQuotientBasedContextRepresentation and FormationSensitive QuotientIdentityGeometry prove the actual projection, endpoint, path and reflexivity-section comparisons for the two-binder native telescope. These maps retain different raw annotations; the two context presentations are compared by typed inverse maps rather than silently identified. SharedJudgmentQuotientMotiveCoverage uses them to construct a qualified frame for every original admitted native parameter tuple, in every formed source context. No constant-motive or closed-term restriction is introduced. Independent projection, endpoint, path, motive and method meanings uniquely determine that frame. The same construction gives the submitted native J its motive meaning and returns its actual method at reflexivity. The fully variable telescope and a path-dependent mixed HOL/list/wire input exercise coverage; endpoint and path remain distinct.

FormationSensitiveQuotientBasedJRepresentation separately proves representative independence when all four admitted parameters—type, left endpoint, motive and method—are converted. The resulting J terms agree in the compared dependent fibre, even when the submitted raw contexts differ. Changing an admitted method from seven to eight changes the J result, witnessed by its actual reflexivity computation. These are coherence laws for retained requests, not identification of all proof objects and not K/UIP.

The attachment now supplies component-level proofs corresponding to twenty-eight of the thirty-one native predicates and three of the four declaration predicates. Required-entry presence remains a demand from the caller, proved for actual installed instances rather than inferred from coverage. Native motive coverage and boundary laws require only frame operations. Three predicates use the full total-operation interface: J meaning, beta and substitution. The canonical family-only attachment to the exact source quotient is now refuted by the J-meaning clause, as described below; the beta and substitution clauses are not independently refuted. The twenty-eight component proofs remain valid, but the full shared semantic record has not been jointly qualified.

A motive family can forget a function that native J observes. FormationSensitiveQu otientIdentityInputCoherence constructs two independently admitted requests in the actual open parameter telescope:

$$(A, x, P, d), \qquad (A, x, \lambda y.\lambda q.\, P\, y\, q, d).$$

Their interpreted type, left endpoint, applied motive family and reflexivity method agree. Indeed, abstracting the applied body preserves that entire semantic input for every admitted request. The two open J outputs nevertheless differ in this presentation's exact conversion quotient: both are neutral under every parallel development, and their retained motive arguments have different constructors. The conversion relation has beta and the declared native contractions, not function eta. Both requests compute to the same method at reflexivity, so testing only that computation misses the distinction.

SharedJudgmentQuotientMotiveAbstraction connects this example to the unchanged constructor requirement. With the canonical frames, no function of the family-only input has the exact submitted native J meaning. Restricting the function's domain cannot repair this while both required requests remain admitted. The result refutes SEM-ID-004 for this particular attachment; it does not refute J, arbitrary CwFs or every interpretation of the source language.

There is a positive comparison on the same source: equality of the actual typed motive-function classes, together with the type, left and method classes, gives representative-independent dependent J transport. The mixed HOL/list/wire workload exercises this law with genuinely changed representatives. Retaining this function is sufficient for that comparison; no least-representation theorem is claimed. Separately, `FormationSensitiveQuotientIdentityReindexing` constructs the general quotient based-context map and proves its base, endpoint, path and reflexivity-square laws along every semantic substitution, without presupposing any J operation.

A constructed retained-input identity interface. `ContextualRetainedIdentityOperations` adds a typed motive function to the family/method input. Its domain predicate does not assert that elimination succeeds: it identifies independently admitted input tuples. `FormationSensitiveQuotientRetainedIdentityInterface` constructs the operation on the complete semantic image of those tuples. Equality of inputs implies equality of outputs for every representative, not just a chosen source witness. The operation computes to the submitted method at reflexivity. Its domain is closed under every semantic quotient substitution, and the output commutes with substitution for every method heterogeneously equal to the transported one.

`SharedJudgmentQuotientRetainedIdentity` relates this operation to every independently qualified native frame. Frame coverage, scope coverage, exact constructor meaning, beta and substitution hold together on the same interpretation. The old family-only requirements are not silently weakened: they remain a separate, failed attachment. The retained interface does not supply elimination on unrelated off-image semantic inputs, a normalization proof, an independent set/presheaf model, or a K/UIP policy.

The companion `SharedJudgmentRetainedSemanticRequirements` reuses exactly twenty-eight original predicate bodies and identifiers and adds five separately identified retained-input clauses. None of the three changed original identities is reused. In `SharedJudgmentQuotientRetainedSemantics`, one actual record satisfies all thirty-three semantic clauses, the seven source clauses, the original six service/binding clauses and four declaration clauses. Required entries include independently named wire and HOL declarations; the caller's requirements are not defined to mean whatever is admitted. A changed but independently admitted service payload passes the semantic clauses yet fails the fixed service contract. A required absent declaration fails despite having a formed type. These controls distinguish shared qualification from a collection of unrelated valid components.

The ergonomic fork is therefore about an interface's permitted forgetting: ordinary motive application may use the family view, while exact native inspection and reuse must retain distinctions they observe. Alternatives are a retained-function interface, a separately qualified coarser mathematical observation, or an explicit change to conversion or elaboration. Adding semantic eta would lose the present conversion-reflection contract; it need not lose admission reflection when native formation remains an independent premise. None of these alternatives selects K/UIP or a final language profile.

`ContextualBasedIdentityScope` isolates the extra requirement in its current total-operation field. A scoped raw operation extends, agreeing on its scope, exactly when every off-scope output fibre is inhabited. This is a statement about operation data, not a proof of off-scope beta or substitution. A nonempty substitution-stable scope of pulled-back motives on an indiscrete identity structure has an eliminator, yet an endpoint-sensitive motive with an empty off-reflexivity fibre prevents a boundary-correct total extension. The separating scope is not proposed as a restriction on native consumers: all admitted native parameter/motive inputs and their admitted semantic substitutions must still be covered. Thus the total field is a stronger candidate-class restriction, not

a derived desideratum. The motive-function collision above is a separate obstruction even within the required native scope; the off-scope extension criterion alone cannot repair it. Indexed environment instantiation and qualified implicit elaboration may be composed, but do not automatically qualify a compatible constructor interface or select K/UIP.

The retained native component therefore has joint constructor and required service qualification. The full six-family assembly still needs its operational-presheaf, independent host and public-consumer comparisons; normalization and K/UIP are not established. A Hofmann–Streicher universe comparison remains a separate classification and size-sensitive construction [92], not a consequence of forming this syntax quotient. TRACE-004 and ASM-005 remain open for the remaining shared bindings, public consumer predicates and joint qualification.

`HOL.ProofSyntax` retains all 29 rules of the existing extensional HOL calculus as indexed proof data, including hypothesis occurrences and typed binder arguments. Its erasure proves exactly the original judgment; the converse establishes existence of syntax, not recovery of an original proof. A two-node direct proof and a five-node detour have the same intrinsic admission but different structural observations. No decoder from that admission recovers both originals. `UniformListChartProofSyntax` and `SharedJudgmentProofEvidence` now provide the richer output for the actual submitted chart requests. Structural reconstruction yields a 50-node HOL proof or a 52-node detour; erasure agrees with the unchanged shared producer. Both support the same theorem application, while a structural strategy consumer distinguishes them. Restoring the original strategy requires its retained request and successful replay. Pairing that request with another valid proof fails producer faithfulness. This is a bounded structural consumer, not general proof learning, external proof-byte checking or live authorization.

`SharedJudgmentAssemblyServices` now drives the existing NIK interfaces with the actual assembly callbacks. Its receipt decision has an independent decoded occurrence/index meaning; a separate theorem retains the actual returned native proof and selected proposition. Two admitted callbacks agree as decisions but return different proof syntax. The HOL adapter passes the actual producer output to the unchanged native-proof service at the original claim. A deliberately certificate-ignoring callback returns a valid proof of the same claim, yet fails faithful replay of the submitted history. This separates a proof-of-goal service from a replay/strategy consumer; it does not require all native proof services to factor through certificates.

`EquationPremiseStability` separates open program rewriting from qualified mathematical equation use. Retaining equation text alone is insufficient: a validated example adds an actual rewrite $C \to D$ which enables the unchanged conditional equation $A = B$. Agreement of the retained equations' actual premise judgments, together with retained grammar, preserves the complete contextual equation equivalence, including presentation-derived laws. A nontrivial premise-free $A = B$ remains stable under the same rewrite addition. These are sufficient semantic conditions, not a requirement to protect a namespace from ordinary definitions or to promote every program rewrite into judgmental equality.

Propositions as inputs to live programs. `EvidenceConditionedBranching` fixes the meaning of a claim at an explicit scope. Accepted support for that exact claim selects the then body; accepted evidence of its semantic negation selects else. An independently supplied complete decision kernel has exact positive and negative specializations. A rejected submitted proof, malformed found tree, closed search traversal or exhausted budget does not supply negation. Failed search retains the requested claim, polarity, actual goal and original outcome. The existing authority outcomes express evidence status, not three-valued truth or a choice of the object language's proposition universe.

The contextual effect-program combinator runs exactly the selected body, retaining evidence, branch, state and recorded intents. Its residual case runs neither body and changes no state or intents. Actual SAT assignment and complete-refutation services exercise both branches; a rejected assignment and an empty candidate search can both concern a satisfiable proposition. Changing the assumption formula changes the checking scope. These laws do not turn an intentionally empty collection into an error, prescribe all search-resumption policies, or supply a textual/C implementation. An explicitly defeasible action remains possible as a separately specified policy, not as a relabeling of unfinished work as mathematical refutation.

An empty answer and a recoverable attempt are different views. `AuthorityOutcomeViews` characterizes when the partial Boolean answer determines the existing four-way public status: exactly when the boundary and incomplete payload types are not both inhabited. A specified consumer distinguishing authority refinement from budget refinement cannot factor through that answer if both cases are present. This is a carrier-relative result, not a requirement that every producer expose both cases. Public status itself still forgets distinctions between incomplete receipts.

The actual assignment sampler in `OutcomeViewControls` illustrates both sides. It has no outside-fragment constructor, so its status factors through the answer view. But an empty submission and a rejected singleton submission give the same answer and status, whereas their submitted-work counts are zero and one. Neither coarse view can implement even this inspection. Retaining the unresolved payload with its explicit theory/query context supports the inspection. The construction does not preserve positive proof histories, implement a resumption scheduler or prescribe a global receipt store. Whether syntax exposes an empty collection, a residual expression or an outcome value remains a presentation choice subject to the promised consumer operations; an outcome value is not an extra truth value.

`FiniteSetRepresentationService` connects ordinary list computation to this brancher without extending trusted conversion. An independently fixed relation identifies the abstract set represented by a list. Actual append implements union under its source-admission condition, returning the ordered list $[2, 1, 2, 3, 1]$ unchanged. Its finite membership decision selects then for 3 and else for 4, retaining exactly the selected body's worlds. A deliberately right-dropping implementation fails the same external union contract on admitted input. Reordered and deduplicated lists represent the same set but cannot all be recovered from that set meaning. Unadmitted input still computes, without acquiring the promised representation theorem. This is a native-operation/decision specimen, not a full set-theory host or a CeTTa realization. Representation/refinement is therefore a concrete alternative to making every executable body a trusted delta rule.

The universe boundary is now attached to the actual common judgment. `NativeUniverseModelBoundary` proves each native successor formation and specifies its minimal semantic data: the upper level codes both the lower code carrier and its decoded types. Embeddings compose. Under the separate predicative-rank condition, distinct finite successor stages require distinct semantic levels, excluding every finite level family and in particular the existing two-level set-family span. This is not a language ceiling. An explicit finite-rank family realizes every formation head, respecting level conversion and instantiation, but fails dependent-product closure at rank three. Unbounded formation alone is therefore insufficient.

`FamilyEnclosingUniverseTower` constructs a conditional alternative from an independently supplied small family-enclosing operator. Each next stage encloses the previous code carrier and decoding family. The same tower has successor embeddings, canonical cumulative paths and Pi/Sigma/identity closure. The native universe boundary interprets actual successor/max lev-

els, respects level substitution and head equality, and builds mixed-level Pi/Sigma codes at their native maximum. The cumulative paths satisfy identity and composition. This does not construct the small enclosing operator. In particular, the concrete ambient enclosing operator raises the code universe and cannot silently supply this same-level iteration.

This enclosing operator is one sufficient model-construction route, not a requirement forced by the desiderata. It encloses every ambient family and supplies full set-function domains, W-types, Nat and host-equality identity. INV-TT-006 instead requires agreement with a presheaf-universe interpretation, and INV-TT-009 rejects making the sealed pure fragment the ceiling. All actual native levels and admitted motives must still receive their required interpretation. A relative construction must expose its host assumptions; neither its conditional correctness nor a two-level control supplies an independently inhabited full model. Other interpretation routes need their own qualification rather than silent substitution.

`TarskiClosureRankObstruction` now derives predicative ranks from full dependent products and sums together with a Boolean code. If a level coded its own code carrier, dependent sum closure would represent its total decoded carrier, and dependent product closure would represent all Boolean functions on that carrier. Their injection back into the carrier contradicts Cantor's theorem. The actual enclosing envelopes contain Booleans as the sum of two units, so the conditional tower satisfies this rank condition without another assumption. Connected to the native successor formations, these closure operations alone exclude every finite semantic level family. Restricted Henkin function domains are outside this result's premises; neither operator existence nor the full native interpretation follows.

`TarskiDecodedFamilyCoherence` derives contextual substitution laws from independent code constructors and decoding equivalences. Products and sums agree with the existing set-family CwF operations after the explicit decoding transports. Arbitrary motives over full comprehension commute with reindexing and composition. A genuinely varying example changes a value from three to one under a nontrivial comparison; its dependent motive then has fibres of different finite cardinalities. Keeping only the code and omitting the chosen decoding transport would change the meaning. Semantic beta/eta here is not an addition to native conversion.

`TarskiCumulativeCodeCoherenceBoundary` exposes a further distinction. For arbitrary dependent inputs, forming a product or sum and then lifting its code has an equivalent decoded carrier to lifting the inputs and forming above. A tag adapter preserves the entire enclosing interface, including identity and W closure, but makes these actual codes different. In the same lower universe, a semantic motive selects the unit code at one tag and the empty code at the other. It cannot descend through any observation identifying the two routes, including decoded cardinality. The result is conditional on the supplied small operator and proves non-entailment of code/motive coherence from that interface. The tag test is not proved definable in the native language. It neither establishes native unsoundness nor requires a particular code quotient or observer; coherence for admitted motives remains to be qualified.

`NativeCumulativeFormationCoherenceBoundary` connects this semantic obstruction to the actual refined native formation rules. In the context $A : U_l, B : \Pi x : A.U_l$, the open type $\Pi x : A.B\,x$ is formed and then raised, or formed after raising both premises. Both constructions give the same admitted term at the same upper displayed universe, and this formation survives refined substitution. A strict interpretation satisfying the independently specified Pi and cumulativity equations must identify their two semantic codes. The tagged operator cannot satisfy both equations, even though their decoded dependent function spaces are equivalent. This identifies a compatibility obligation for the supplied interpretation operations; it does not require a native K rule, a definable tag test or a chosen quotient.

`ContextualIdentitySubstitution` supplies the set-family model's full dependent J substitu-

tion square and its reflexivity computation square. The motive may depend on both endpoints and the identity witness. This semantic equality model validates proof irrelevance without selecting native K or UIP. Separately, `IndexedPolynomialListRelSmall` reconstructs the existing list relator from a fixed-endpoint fibre in the relation's universe, retaining every witness. Its equivalence commutes with evidence maps and arbitrary endpoint reindexing. No order between the element and relation universes is needed. Interpreting these operations in the actual mixed native judgments remains a distinct obligation.

`HenkinLogicalCodeDescent` isolates another required interface. A carrier-preserving map of HOL types into semantic codes supports global admissibility and logical-equality policies exactly when each policy agrees at code collisions, after transporting its arguments. The resulting universal and equality operations have their Henkin meanings on the retained image; off-image logical axioms and native product/substitution compatibility are not inferred. No full-domain premise is needed for this criterion. One noninjective, carrier- and application-preserving Boolean/proposition coding supports both policies in the standard model but obstructs both in the stated monotone Henkin model. The latter fails the additional `FunctionsRespectEqv` condition and is not a countermodel in the narrower institution requiring it. Standardizing its domains changes a sentence's truth, so that repair is a model change, not a lossless presentation change. Conversely, the positive example excludes any general requirement that code names be injective.

The Pattern-language backend covers one typed service crossing and a finite native body, not raw external proof parsing or unrestricted OSLF formula checking. `SharedJudgmentServicePrograms` separately supplies arbitrary finite source compositions of actual matching/HOL requests, ordinary and dependent sequencing, and choice. Capture-avoiding substitution acts on submitted environments and bound continuation bodies. The callback-relative interpretation has exactly the independently defined ordered source worlds, retaining private state, intents, branch occurrences and chronological responses. The old one-crossing source embeds with unchanged worlds. `SharedJudgmentServiceProgramAdmission` gives an independent admission relation and proves native result preservation under formed contexts and refined substitutions. A nested Sigma workload retains both service values and an identity witness; an admitted changed-claim program instead stops without producing a value. Admission does not imply successful evidence.

`SharedJudgmentServiceProgramBackend` now gives these finite compositions an independently authored typed GSLT. Explicit request, invocation, response and continuation states retain the submitted environments. Native bodies execute through the effect-tree LanguageDef, not a rule whose premise runs the complete source evaluator. Every completion is exactly the full ordered source-world list, including private states, recorded intents, stopped outcomes and chronological request/reply histories. Capture-safe substitution and independent native result admission transfer to these runs. Reversing or deleting the nested matching/HOL history is rejected by the exact completion law. This typed backend is not a Pattern codec for whole programs, an external proof-byte checker or an unrestricted proof-search algorithm.

`SharedJudgmentServiceProgramOSLF` derives the complete-run view directly from finite paths of that typed backend, without a Pattern codec. Generic GSLT synthesis yields native types whose satisfaction is exactly the chosen predicate of the complete source-world list, under execution qualification. Actual substitution preserves that meaning. Existential output and finite-position predicates deliberately retain less: the real single and double matching/HOL programs both expose the same value, but complete-list predicates distinguish one occurrence from two, and a second-position predicate separates them. Stopped results remain observable without supplying successful native values. The generated right adjoint is the existing predecessor-universal box, not an unrestricted forward must-run decision procedure. Native term admission requires the independent source judgment in addition to behavioral observation.

The neutral `SemanticServiceResumption` composes the four existing service faces at one semantic target. Its qualification theorem preserves source admission and actual invocation for every chronological receipt, proves meaning for every successful result, and identifies a rejected last receipt for stopped execution. A four-face sequence constructs later requests from earlier returned values. Malformed proof and certificate submissions stop without refuting their meaningful claim; a false direct decision is a distinct response. Executing an operation on an inadmissible source does not produce qualification. These general callback-relative results do not yet provide a common native interpretation of all four service faces. The typed backend above implements repeated matching/HOL requests, not that further four-face integration.

The next semantic target is one fundamental interpretation of the actual refined native judgments, contexts, substitutions and conversion rules. The HOL and matching fragment meanings must be its agreement corollaries. The total Data algebra and decoded-family transport laws are now available, but must still enter that same interpretation of all admitted motives and conversion rules. The conditional enclosing tower supplies formation and type-former codes in one family; an inhabited unbounded universe-code model remains a separate requirement. A conditional fundamental theorem would not by itself construct that model.

The seven fragment predicates do not replace the six normative decision families. Intended-model and universe interpretation and common native four-face qualification remain open. The ledger records these connections as partial evidence, not a completed language.

`PrimeCandidates.Specification` now connects six normative questions to actual theorem terms, scoped alternatives and remaining investigations. Its prose questions are not themselves necessity theorems. The versioned `DEC-ID-20260912` record selects scoped identity and optional retained presentation references as a development direction, not as a fully qualified foundation. Its remaining proof obligations stay explicit; the other five families have no selection record. `PrimeComparisons` independently joins all six decision families to pinned source observations. The additional identity, mathematical-host and reasoning comparisons distinguish computed conversion from native identity elimination, deferred universe formation from a cumulative host, and rechecked inhabitants or imported unit-superposition certificates from a qualified HOL service. Certificate-internal conversion replay also remains distinct from binding a result to a caller's requested claim and premise revision. The separate experimental scoped-judgment draft review now adds five recorded runs with 74 literal outputs. Supported beta/eta and an anonymous-index dependent round trip succeed on both reviewed binaries; the experimental prefix handler suppresses otherwise ordinary definitions, and publication loses proof/dependency distinctions. Five corresponding probe topics retain these observations alongside the relevant candidate laws. These are historical observations of the first pinned draft, not claims about its later source. A separate second-pass source snapshot records exact operation-name ownership, retained proofs and dependency-sensitive reuse, alongside the remaining raw-record qualification and structural-conversion fallbacks. Its eight source observations include the authored live-proposition program. A separate isolated probe of the existing binary records five outputs: positive and mismatched-proof controls, accepted use of a directly inserted theorem record, inert explicit recheck and an empty stored proof. That path acts as an unchecked premise; its record label is not checked theorem provenance. This is no C rebuild or full-gate result. Preserving the earlier runs, later source inspection and isolated probe separately prevents any from silently replacing another's evidence. Earlier proposed experiments remain unexecuted, and the C-execution correspondences remain open. No runtime qualification follows from source hashes, recorded examples or the abstract checker's correctness. The common set-family capability theorem explicitly uses the same universe parameter throughout its statement.

The scoped-identity named-context connection now uses actual native proof presentations in the common HOL/List/wire declaration environment. Direct J congruence and conditional export

followed by discharge have the same typed claim, distinct syntax and a proved beta conversion. Original inspection resolves a retained proof reference; current application separately resolves a named binding and performs capture-avoiding substitution. The resulting native judgment follows from source typing and the independently typed substitution, not from a record label. An unavailable live hypothesis binding blocks this application without invalidating the original conditional proof. A retained old binding still permits application, and unrelated revision changes preserve it. A six-to-seven-variable control shifts the retained motive and selects the old witness rather than the fresh binder. These statements neither impose proof replay on reuse nor equate failed binding resolution with logical non-derivability.

For datatype-specific identity proofs, an additional declaration-extension bridge instantiates the native J schema while retaining the extension's genuine computation roots. The opaque-extension interface cannot supply this property: its roots are deliberately empty. The bridge proves J application typing and preserves the inherited reflexivity computation. It does not claim preservation or normalization for arbitrary added datatype equations.

The internal Hedberg core is now represented by native proof terms, not imported from the external route-family theorem. Let $c_y : \mathrm{Id}_A(x, y) \to \mathrm{Id}_A(x, y)$ be a based selector. Native J first constructs inverse, composition and cancellation, then proves that $c_x(\mathrm{refl}_x)^{-1} \cdot c_y(p)$ is propositionally equal to p. A supplied native proof $c_y(p) = c_y(q)$, transported through this normalization, therefore yields a native proof $p = q$. The seven arguments are abstracted into a closed Pi theorem, and substitution instantiates its checked schema in extensions with genuine datatype roots. The selector and its constancy proof are actual typed object-language inputs; they are neither a UIP axiom nor a host-language decision procedure. A separate internal adapter now constructs those inputs from a native equality decision for any $A : U_0$. A decision is a dependent pair containing a Boolean tag and either an identity witness or a function from that identity type to the small-type empty encoding $\Pi X : U_0.\, X$. Native Boolean elimination constructs a weakly constant selector; the Hedberg core then produces UIP. Abstracting A, its decision procedure, both endpoints and both paths gives a closed six-argument native theorem. Positive and negative decision constructors compute through the actual Boolean, pair and beta rules. A rejected candidate proof is not such a negative decision: the latter contains an actual refutation.

The Boolean datatype supplies genuine region evidence rather than a declared type name with an assumed decision procedure. Its four declarations have fresh names and formed types; its dependent eliminator retains both iota roots. Native J transports through a constructor-discriminating type family to obtain a refutation from an opposite-constructor path. Boolean elimination constructs the required selector and constancy proof, and a further native elimination over the left endpoint yields

$$\Pi x, y : \mathrm{Bool}.\, \Pi p, q : \mathrm{Id}_{\mathrm{Bool}}(x, y).\ \mathrm{Id}_{\mathrm{Id}_{\mathrm{Bool}}(x,y)}(p, q).$$

Both endpoints may be neutral variables; this is not a host-language check of two closed cases. The resulting derived proof is retained and applied through a named binding in this same Boolean/J rule instance. The four-to-five variable substitution keeps the original endpoints and paths, while missing live bindings and available retained bindings have distinct, checked outcomes. There is no uniqueness hypothesis in this context. The actual computational extension now has a proved conversion boundary and subject reduction for all seven installed roots: two Boolean, two List, native J, and two relator roots. Declaration-spine recovery starts from arbitrary admitted source typing, recovers the dependent arguments, and preserves the original displayed type, including its conversion and cumulativity adjustments. The proof-only parallel completion retains conversion-sensitive duplicated metadata and has a proved diamond; it is not a new runtime reduction relation. Pi/Sigma injectivity and constructor separation follow for the authored conversion. In particular,

the two Boolean constructors are not convertible, without a fuel bound.

The syntactic quotient CwF is instantiated with these same Boolean/J rules. It interprets the actual closed Hedberg term, derived Boolean region, retained native motive function, J beta, and substitution through every semantic arrow class. Named application is interpreted by the same substitution operation; an unavailable live binding does not erase the original judgment or its interpretation. A beta-expanded original supplies a positive example of equal semantic classes with distinct retained syntax. An inhabited propositional identity type does not identify its endpoints in the conversion quotient.

An explicitly assumed region is a local native binder, not a new computation rule. Export retains the region premise as a dependent Pi; discharge supplies a typed witness and computes by beta. The scope extension commutes with typed substitution. A genuinely derived Boolean region simultaneously witnesses path uniqueness and distinguishes the submitted path codes under raw inspection. More strongly, its semantic identity type is inhabited while the two submitted path values remain distinct in the actual conversion quotient. Thus path collapse is valid only for compatible observations. A guest whose transport realizes both identity and Boolean swap cannot also identify those paths; this is a conditional compatibility boundary, not a constructed univalent universe. No propositional path law is inserted into definitional conversion. Global normalization, new guest models and runtime conformance are separate claims, not consequences of this identity qualification.

A separately pinned later draft audit exercises these public boundaries constructively and adversarially. Native conditional congruence and explicit retention of an actual HOL proof succeed. Three expectations fail: a writable `signature` origin tag bypasses admission, a supposedly non-publishing `try` changes admission state, and an explicit-space type query loses its context routing under `try`. The same failures were reproduced on a second copied binary. The comparison ledger preserves independently authored expected and actual outputs rather than promoting the observed failures into the specification. None refutes the scoped mathematical construction. A later draft report records repairs and a structural-definition prototype; those runtime claims require a new pinned comparison and are not established by the present theoretical qualification.

The audited known-proof path does not rerun the proof checker, but it serializes and hashes the retained record and recursively visits dependencies. Small process measurements with independently checked inputs expose this work; they include parsing and output and are not isolated kernel timings or a speedup claim. The available runtime counters report zero even on a positive proof check and do not expose scoped counters, so they cannot establish zero allocations, traversals or executions. Qualifying an optimized current-use contract remains different from requiring proof replay or mandatory history.

Recursive-definition admission has a separate mathematical boundary. A true predicate induction principle does not by itself make its displayed cases free constructors. Boolean false and negation satisfy natural-number-shaped predicate induction, and negation is injective, but no function into the natural numbers can satisfy $h(\neg x) = h(x) + 1$: the two-cycle would force $h(\mathsf{false}) = h(\mathsf{false}) + 2$. The generic theorem characterizes this predicate induction precisely by finite successor reachability, not unique constructor decomposition. A constructed natural-number fold supplies the positive recursion and uniqueness control. Thus admitting recursive equations needs a justified recursion principle or suitable datatype construction, beyond extracting apparent cases from an induction statement. This boundary does not assert that a particular runtime accepts the invalid definition and does not reopen the qualified scoped-identity fragment.

The versioned `DevelopmentPlan`, version `2026-09-13.1`, retains 30 scoped obligations across six milestones. Every row records its acceptance condition, normative references and earlier dependencies. Proved rows carry theorem terms; partially supported rows remain open; mechanical

isolation rows require fresh source-gate results. The status readout separates closed/total, partial evidence, unmet predecessors and collection build coverage. A separate planning readout counts closed rows together with selected development directions: selecting a direction does not change `isClosed`, discharge the row's acceptance condition, or supply a theorem. The unchanged identity acceptance condition is now discharged by one fixed-fragment qualification combining scope laws, seven-root preservation and the actual retained-J interpretation. With the five source-isolation gates and ledger gate passed, the readout is 20 of 30 closed obligations, rather than 19 closed and one merely selected. This does not close the remaining public-observation, host, evaluation, admission, reasoning or joint-runtime comparisons. These counts concern this specification plan, not all future language research or the readiness of a runtime. In particular, a fully narrowed decision family is not the same thing as a family with several proved constraints.

Preserving answers for a single client is only one admissibility condition. Uniform representation, substitution, proof transport and learning over a family of tasks also matter: one induction principle with reusable instantiation is not assessed merely by comparison with a finite list of instances. Specialized first-order or propositional execution charts may retain a higher-order representation, subject to their own recognition, transport and reconstruction laws. This is a comparison programme, not a claim of a universal optimal host or an implemented general optimizer.

`HOL.UniformListInduction` now supplies an object-HOL specimen of that programme. One predicate-quantified induction principle is instantiated to prove that map preserves length uniformly in the function. Actual equational base and step derivations feed a proof-reconstruction theorem. The standard list model validates the theory; an explicit two-point junk model validates the displayed equations but rejects both induction and the map-length conclusion. Soundness therefore excludes an equation-only derivation. This negative control concerns those displayed equations, not an arbitrary first-order theory augmented with Peano or constructor axioms.

Named value contexts and actual native execution. `NamedValueContexts` separates an explicit versioned reference from an explicit live name and separates context storage from the live revision selector. Fresh publication checks that the target version is unoccupied; accepted publication preserves each existing versioned binding. Pure inspection returns the original code and a requested finite binding footprint. It has no evaluator argument. The independent `NamedContextDataflow` application proves that agreement on the actual instruction operands preserves the existing readiness evaluator, including missing operands. For one unchanged addition instruction a captured context continues to yield ten while its live counterpart yields twelve after a binding revision. An unrelated binding edit preserves readiness. Missing context and missing operand are distinct, and a separately exhibited firing step writes the output that inspection leaves untouched.

`SharedJudgmentNamedContext` supplies the candidate attachment to the existing service-program source and independently authored backend. Its table payload is the actual native substitution, not just a typing context. Resolved execution preserves complete backend outputs and the independently admitted result judgment. Fresh publication preserves both an occupied version's computation and its original admission; no checker rerun is used to derive that law. Substituting scoped source agrees with precomposing the named environments at the same final native scope. The actual typed wire program distinguishes stable and live binding changes. These are semantic interfaces and preservation laws, not a copying algorithm, universal history requirement, runtime speed claim, or default capture policy. A lexical snapshot does not freeze mutable objects reachable through a binding.

Evaluation scope, retained references, and effects. `ContextualEvaluationReferences` places two independently selected references over the existing contextual program monad: an origin reference O and the current evaluation reference E. This is a reader environment, not a mutable heap. Its scoped operation satisfies

$$(\mathsf{atSpace}\ B\ p \gg\!= k)(O, E) = p(O, B) \gg\!= (x \mapsto k(x)(O, E)).$$

An outside continuation resumes at E, whereas a continuation inside the scope remains at B. The scope law holds for the existing state, choice and intent syntax before choosing a handler. This separates the operations from their effect interpretation in the sense of algebraic handlers [16]. The two reference setters commute; arbitrary effects on the referenced spaces need not. Concrete executions distinguish origin and current reads, restore a caller's evaluation reference after nested evaluation without rolling back writes, and retain an earlier value while a fresh read observes an update. Neither coordinate alone recovers the paired-read consumer. The component does not choose the surface meaning of `&self`, make raw stored syntax a closure, prescribe context allocation, or establish asynchronous read consistency. A captured reference and a delayed symbolic lookup remain different promises, which a source-level binding interface must distinguish coherently. Resolving a reference value and immediately reading through it gives the same effect program as direct role use. Relocation after that resolution cannot redirect the saved reference. Executions distinguish a saved field reference read from a remote scope (23), delayed current-role lookup there (31), and the saved reference read after its field was updated (99). Capturing identity does not freeze contents or preserve a role's later meaning. This value law does not demand eager resolution by a surface binding form. Passing a delayed role computation and running it in another scope is a different, coherent operation; its changed result is not a violation of reference-value substitutability.

The same references in native named execution. `SharedJudgmentEvaluationReferences` selects the actual typed named substitution with the evaluation coordinate and retains the origin for pure source/environment inspection. It lowers the existing service interpreter to the existing effect program. Its decoded worlds agree exactly with the named runner and the independently authored backend completion relation. Relocation commutes with source substitution and environment precomposition at that effect-program level, before observing answers, replies, branch histories or intents. Two independently typed context morphisms preserve the dependent result judgment in their common final native context. Concrete controls use nonidentity parameter substitution and matching followed by HOL induction and native dependent-pair/reflexivity construction. Fresh publication preserves retained origin inspection while deliberately live evaluation changes its actual output. A missing evaluation reference is an explicit outer resolution failure, not a fallback to origin or an empty successful enumeration. This connects the components in one interpretation; it does not identify runtime space handles with logical contexts or prove naturality of arbitrary effect handlers between different native scopes. Named resolution here happens at program entry from the supplied table and version selector. The live native control compares executions before and after publication; within-execution mutation and rereading of that named table are not yet connected to the native interpreter. The separate generic state example does exercise fresh reads across an intervening heap write.

Handling a rejected binding without accepting it. `ContextualBindingConflict` connects explicit rejection handling to the existing ground refinement semantics. Acceptance is exactly the original refinement step. Rejection supplies the unchanged input store to a different continuation; projecting only successful updates recovers precisely the old partial operation. Both outcomes

preserve every prior binding. Moreover, no information-preserving extension can establish a rejected assignment: from $x = a$ and $a \neq b$, accepting $x = b$ remains impossible even after successful recovery work. Concrete controls retain $x = a$, reject $x = b$, and then admit $y = c$, while the conjunction $x = a \wedge x = b$ remains unsatisfiable. Thus the inconsistency of a proposal need not destroy its consistent surrounding context. These are ground-store and dispatch laws, not non-ground unification, automatic rollback of arbitrary effects, or a mandate to allocate diagnostics on ordinary execution paths.

Ground binding as a categorical common extension. `GroundBindingCategory` gives the preceding refinement algorithm an actual categorical universal property. Objects are partial ground stores; an arrow adds information without changing any known binding. If the actual `refineStep` accepts proposal (x, a) in store Γ, its output is the coproduct of Γ and the singleton assignment $[x \mapsto a]$ in this category. The constructed colimit witness factors every compatible extension, not just the extensions produced by the algorithm. For a finite run, the exact law is

$$\Gamma' \leq \Delta \quad \Longleftrightarrow \quad \Gamma \leq \Delta \ \wedge \ \Delta \models \mathit{proposals}, \qquad \mathsf{refineRun}(\Gamma, \mathit{proposals}) = \mathsf{some}(\Gamma').$$

Existence of a compatible extension is equivalent to success of that computation. Accordingly, joining compatible information intersects its sets of possible future extensions. Completion is a computable functor on the compatible-input subcategory, with a natural map from each input to its completion. The restriction matters: adding $x = a$ can make a previously acceptable proposal $x = b$ incompatible. When $a \neq b$, the two singleton assignments have no cocone at all, not merely no result within some fuel bound. Positive controls add a genuine fresh binding and have no arrow back to the less informative input.

These results concern the same logical name across ground constraints, not imperative replacement or distinct lexical binders with the same spelling. They do not establish the general substitution-category property of a non-ground most-general unifier. Nor does the absence of a common extension determine what a program reports: two handlers for the identical rejected step return, respectively, no report and a conflict report, with precisely the same absent successful-update projection. Retaining the old store is also possible, but does not make the rejected assignment true.

Absence, ordinary data, and failure observation. The existing finite-occurrence model in `EmptyDemandCoherence` distinguishes every singleton datum, including an atom named `Empty` or printed as `()`, from zero occurrences. `EmptinessTaxonomy` proves that ordinary success sequencing cannot produce a report from an empty input: there is no answer on which to invoke its continuation. An explicit zero observer can instead choose its empty branch. This distinguishes continuation algebra from reporting policy; it neither mandates a quiet interface nor requires a new diagnostic namespace. The separate finite-outcome model in `CollapseSuperposeRoundTrip` retains values, coverage and faults. Its value-only collection cannot recover discarded coverage or faults; re-expansion recovers the complete outcome exactly when that outcome was complete and fault-free. These are scoped observer laws, not a mandatory runtime record or a claim that a currently empty, unfinished search has been exhausted. A known ground conflict is different again: the no-cocone theorem rules out future solutions without revision of the existing constraints.

Retained originals and reconstructed presentations. `RetainedPresentationViews` separates a local observation from an explicit versioned reference. Capturing an existing presentation computes the supplied observation once; recovering the original is a table lookup with no evaluator,

observer or proof-checker input. Coarsening the visible value retains that lookup, and accepted fresh publication preserves it within the same store lineage. A missing reference returns no presentation. In contrast, an observation quotient can reconstruct a chosen representative only under a supplied section law. Exact reconstruction of every original would force the observation to be injective. The actual indexed-conversion instance supplies both a nonconstant observation and a collision: a retained five-constructor path is recovered unchanged, whereas representative recovery chooses a one-constructor path with the same nonzero result. Another live version may also select that shorter path without changing the captured one. Same-key overwrite and unrelated-store controls show why a key and a matching mathematical observation do not authenticate an original. The preservation law assumes the declared publication discipline; it does not establish proof authority, garbage-collection safety or mandatory history retention.

Constructed native proofs for conditional identity regions. The `SharedJudgmentIdentityRegions` module constructs congruence using the actual native J declaration and its submitted motive. From independently formed types B, C, a typed $f : B \to C$, and $r : \mathrm{Id}_B(p, q)$ it derives an admitted term of $\mathrm{Id}_C(f(p), f(q))$. Taking $B = \mathrm{Id}_A(a, b)$ gives the pointwise region rule without adding K or a new conversion equation. The construction exports an assumed r as a conditional Pi theorem; applying a separately typed witness discharges that same condition and beta-reduces to the explicit native body. Substitution preserves the body and retained motive exactly, and reflexivity computes by the native iota rule. Under the stated opacity condition a variable witness has no root J computation. Formed five- and six-entry native contexts exercise conditional and assumed use while retaining distinct endpoint syntax. These are native proof terms, not a transport-preservation axiom. The construction does not itself produce uniqueness from decidability, implement every named context transition, or qualify a complete univalent guest.

Uniform higher-order fusion. `HOL.UniformListMapFusion` constructs an inspectable object-HOL proof of

$$\forall f\, g\, xs. \quad \mathrm{map}\ f\ (\mathrm{map}\ g\ xs) = \mathrm{map}\ (\lambda x.\, f(g(x)))\ xs$$

in the unchanged list theory. The proof uses the single predicate-quantified induction sentence, explicit base and cons proofs, and object-level beta and congruence steps. Its universal result is actually specialized to arbitrary symbolic functions and lists, is stable under capture-avoiding substitution, and may be used beneath a typed higher-order application. Soundness gives validity in the existing standard list model. Reversing the composition order yields a false universal statement, even though it has object proofs for every empty-list sample and every diagonal function pair on arbitrary lists. The model's noncommuting-function counterexample excludes a proof of that statement from the full original theory. This establishes a uniform mathematical discriminator, not completeness of a learning or search algorithm or coverage of a new imported-library host.

Observer-relative replacement of equality evidence. `MonoidIndexedTransportCoherence` extends independently interpreted primitive conversion steps by an actual structural fold on accepted paths. For paths $p, q : A \to B$ with transport equivalences T_p, T_q, replacement is harmless for an observer o precisely when

$$\forall b \in B. \quad o(T_q(T_p^{-1}(b))) = o(b).$$

This is an observer-invariance condition on the induced loop, not uniqueness of equality proofs. For the existing monoid-indexed family, the fold agrees with the independently specified canonical-word

transport; hence parallel paths act identically on its nonconstant fibres. Two accepted paths still have different inspectable syntax. In particular their sizes are one and five, whereas their action on a nonzero two-element-fibre cursor agrees. Actual index substitution preserves the transport square, including an expansion from two to three values and an erasure from two to one. The latter map is provably noninjective, so its naturality cannot be justified by assuming an equivalence of the pre- and post-substitution fibres. This is a qualified conversion-family result, not a model of unrestricted native J, global UIP or permission to erase evidence inspected by another consumer.

`GroundUnaryEquationalChart` supplies a reusable specialized scope: registered HOL atoms and unary-application spines at one declared type. An independently stated grammar is recognized completely at sufficient depth, including duplicate registrations and atom-priority shortcuts. Successful reification has an exact syntactic round trip. An accepted equational certificate reconstructs an actual HOL derivation bound to the source assumptions and goal. The list-induction workload uses three-node and nine-node certificates for its base and step, then rebuilds the original predicate-quantified proof. These are replay-tree counts, not runtime or whole-proof cost comparisons. Changed premises, goals, malformed trees and insufficient depth are rejected by the chart; non-admission does not refute the HOL formula. The outer induction principle remains outside the chart and is still necessary in the source theory. General first-order/propositional search and candidate-native proof admission are not established.

Joint matching and source-proof boundaries. `MatchedIndexDependentTransport` connects actual completed selected-match replies to an executable request validator and index lookup. The resulting native J term has an endpoint-dependent identity family; typed substitution establishes its source judgment and the checked J step preserves admission of its result. The receipt remains encoded Data, not an identity proof. Changed indices, outputs and occurrence bindings are rejected. The construction is parametric in the input list and valid index, but makes no claim to implement a native Fin type.

`UniformListChartService` reconstructs the retained HOL induction proof from independently supplied equational certificates and premise lists. Its intrinsic NIK proof service binds the resulting proof to the complete source claim. The intrinsic binding checker is not a checker for arbitrary external HOL proof bytes. Missing induction excludes an equations-only proof; native proposition formation cannot repair that missing assumption.

`HOLNativeRelatorCompatibility` transfers the same proposition and assumption representations into a fresh common HOL/wire/List/J/relator environment. A newly admitted mixed payload contains an abstract HOL sequence inside a native List, paired with wire Data; checked projection preserves its judgment. The two sequence/list carriers are not thereby identified.

The reusable `OpaqueRelatorExtension` theorem qualifies every extended-admitted source, including terms not admitted in the original environment. Absence of new computation and definition unfolding preserves the exact conversion-certificate checker and full step preservation. A defined universe alias refutes extending that claim to arbitrary declarations. A separate declaration-collision control shows that opacity alone does not establish freshness or component embedding. These are scoped compatibility results, not consistency, normalization or host-optimality theorems.

One connected cognitive workload. `UniformListCognitiveWorkload` enumerates a bounded expression grammar and filters its hypotheses using actual samples. The empty-list sample underdetermines the answer; a nonempty sample and the standard list model reject the overfitted zero-length hypothesis. The retained map-length claim is reconstructed from independently supplied equational certificates and the same source induction assumption. General pipeline laws bind the

output to that generated formula, exact theory and consulted premise dependencies. A consulted revision changes reuse from current to historical and invalidates the original certificate against the new live premises; an unrelated revision preserves currentness. Historical derivability remains intact. Generation and recognition bounds, malformed proof data and semantic countermodels have different meanings. This is a connected bounded workload, not unrestricted higher-order learning, model finding or external proof-byte parsing.

Operation-stable contextual views. `PolicyFamilyOperationClosure` adds a sufficient constructive closure condition: each requested target observation after an operation is computed from a named source-policy coordinate and a dependent result translation. This derives congruence for policy equivalence, an operation on observational classes, and an executable canonical-vector action. Identity, composition and closed-restriction squares are proved. The condition is sufficient, not a necessity theorem for every congruent operation: some operations combine several coordinates.

The refresh control retains a count and its cached emptiness answer while forgetting an unrelated tag. That compact readout supports refresh exactly. The cached Boolean alone answers the current question but cannot determine the refreshed answer when it hides distinct counts. For a general sufficient readout, the theorem reconstructs the requested post-operation answers, not automatically the whole next readout. The canonical vector, quotient and explicit compact specimen have their stated stronger commuting laws. Carrier-level dependent descent and object-language substitution remain distinct requirements on any proposed integration.

`SyntacticPolicyDependentDescent` now connects them for a declared code-and-formation-level family over actual syntactic contexts. Its code-only restriction admits both raw term fibres and independently refined values. Actual capture-avoiding substitution acts on the descended quotient fibres without recovering a representative. The refined J computation commutes with the same family identification and substitution; an arity-changing weakening provides a concrete instance. A consumer requiring the retained formation level cannot descend through the code-only view, even after substitution. Thus the scoped substitution/descent obligation closes without adopting K/UIP or claiming descent for arbitrary dependent consumers.

`ScopedComputationObservation` supplies a second operational boundary. The ordered answer list retains native result admission for the shared implementation, but cannot reconstruct state-sensitive continuation intents. Bounded prefixes retain genuine positive answers; a missing prefix answer does not establish absence from the complete finite result list. The admission readout here is proposition-valued, not an executable type checker.

`ScopedComputationContextViews` now closes the declared consumers under one actual native resumption. Ordered answer/state-bit lists are exact and informationally least at the producer stage; dependent answers and new-intent suffixes are exact at the terminal stage. The compact resumption agrees with the existing interpreted `sequenceSigma` semantics. Admitted source programs and dependent result judgments witness both the positive protocol and the failure of answer-only observation. Branch histories and old intents may be hidden only while they are not public consumers: adding either invalidates this readout. The theorem does not quantify over arbitrary later continuations or select a global evaluation strategy.

`OpaqueRelatorScopedComputation` qualifies the unchanged handler and world semantics under arbitrary extended native signatures. Its admission theorem covers newly admitted programs, contexts and refined substitutions, not merely the image of old typing derivations. In the common environment, actual mixed HOL-list/wire payloads pass through substituted arguments, the original evaluator and the same contextual operation square. Independent wrong-handler and wrong-dependent-index controls remain invalid. The result connects these scoped computations to

their actual shared rules; it neither interprets arbitrary added declarations nor proves general Need equations.

The staged-reflective observation tables are authored requirement policies, not derivations forcing that architecture. `PresentationEvidence` separates model reports, operational adequacy and artifact binding without an authority ranking. Generic reduction-view laws and presentation evidence are separated from candidate instances. An import and qualified-name gate checks these declared boundaries and full collection coverage across separate library and exporter targets. The staged-reflective presentation, revisioned occurrence proof flow, returned-fibre internal language, first-class route interface, gluing-morphism laws, operational space/activation interfaces and PLN scheduler bridge are now separated from their selected instances. The shared rule-schema projector is independent of both candidates and current calibrations; an explicit cache profile preserves the existing calibration wire output. Reviewed concrete applications and build entries may import candidates, but no such entry becomes a foundation by doing so. The lexical check complements imports and does not replace compilation. The gate checks the declared source classifications, not an arbitrary mathematical claim made in a module's prose.

This section is the one place where the state of the programme is recorded without narrative. Every entry names the checked module or theorem that carries it; an entry without a name is a claim, not a result. The layer vocabulary is fixed here as well, because several earlier drafts conflated levels: a *framework* (simple type theory, dependent type theory, a logical framework, an institution) is not an *object theory* (HOTG, ZF, a Henkin theory, a closed theory of a node) expressed in it, which is not a *model* of that theory, which is not a *proof kernel* that decides or checks it, which is not an *operational presentation* (a language definition and the GSLT it generates) or a *realization* (a runtime), which is not a NIK *semantic target* (a carrier with its meaning predicate), which is not a NIK *service*, of which an external *certificate boundary* is one face among four. An earlier linear picture, frameworks at one end, operational semantics in the middle, and NIK certificates at the other end, is retracted; the checked shape is the diagram below.

17.1 The typed diagram

$$\begin{array}{ccccc}
\mathbf{Inst}_{\mathrm{mod}} & \xrightarrow{\text{consequence}} & \mathbf{Inst}_{\pi} & & \\
\uparrow{\scriptstyle \text{Grothendieck}} & & \uparrow{\scriptstyle \text{Grothendieck}} & & \\
\mathbf{Node} & \xrightarrow{\text{pre}} & \mathbf{Th} & \xleftarrow{\text{logical}} & \mathbf{Biform} \\
\uparrow{\scriptstyle \text{reducts}} & & \downarrow{\scriptstyle \text{target}} & & \downarrow{\scriptstyle \text{operational}} \\
\int^{c}\mathbf{Mod} & & \mathbf{Target} & & \mathbf{GSLT}
\end{array}$$

Here $\mathbf{Th} = \int \text{closedTheories}$ and $\mathbf{Node} = \int(\text{closedTheories} \circ \text{consequence})$. Services form a family over semantic targets, with four service forms; operational presentations have cover maps and an OSLF readout.

Each leg has a stated kind. Forgetting models is a functor and is lossy (`InstitutionConsequence.consequenceFunctor`, `consequenceFunctor_forgets_model_map`). The theory graph and its model-valued refinement are Grothendieck constructions (`TheoryGraph.closedTheories`, `TheoryGraphModels.Node`). Satisfying models form the contravariant Grothendieck construction of a functor $\mathbf{Node}^{op} \to \mathbf{Cat}$, and the projection to nodes is a Grothendieck fibration whose cartesian lifts are reducts (`TheoryGraphModels.modelsOf`, `toNode_isFibered`); model expansion in the opposite direction requires additional hypotheses. A closed theory induces a semantic target (`TheoryGraph.semanticTargetFunctor`); services over targets are an indexed family with an action of target equivalences (`SemanticServiceFamily`). Biform theories project to closed theories and to proof-relevant GSLTs, and their arrows are exactly the compatibility locus (`BiformTheory`). Passes

between operational presentations are semantic covers, and OSLF is adequate over equation classes (`IRPass`, `HennessyMilnerDirections`). No leg identifies a framework with a theory, a theory with a model, or a target with a service.

17.2 Proved by checked construction

1. **Execution-contract components have checked composition laws.** Resource reification and reflected actions transport through a frame-lifting resource map and compose through retained middle relation witnesses. Actual pure occurrence-query reads compile to complete ordered identifier/payload dependencies and preserve adaptive observations under validated reuse, bind and performed-prefix resumption. Dependent families of committed observers share the existing administration and erasure laws. Constructed controls distinguish missing frame lifts, reordered effects, stale child/absence reads, ambiguous identifiers and discarded duties. These results have the scopes stated in Section 3.13; they do not supply a complete native service, compensator or architectural minimality theorem.

2. **A checkable rational distinction-calculus fragment.** `DistinctionCalculus.derives_iff_checked` establishes agreement of raw certificates with independent derivations; `checkRawCompletion_sound` proves the least-extension property of an admitted finite completion. Apartness, goal descent, weighted balance and migration bounds have positive and negative controls. `PolarizedNeedDistinctionObserver.measured_loss_of_execution_distinctions` and `admission_does_not_factor_through_request` connect the quantitative observer to an actual reference-machine and native-checker workload. Section 3.4 states the scope.

3. **Raw mathematical replies cross an executed admission boundary.** Existing wire data has an exact, independently typed native representation. A fixed selected calculus checks raw requests and articles; replay at the exact expected request reconstructs the submitted derivation rather than trusting its reply tag. A reified source program binds that actual reply through allocation, owned force and cache before admission. Every completed answer has the exact checker result and cached world, and supplied finite articles complete from a slot-bounded heap. The exact final world retains that bound for subsequent calls without resetting its heap or history. Accepted MIL articles retain their raw erasure, relational evidence and independently qualified native judgments. An executed two-premise specimen exercises these contracts; wrong intermediate evidence, forged acceptance and stale requests fail. Data typing does not certify a theorem. A reified native-argument function and value-level caller now preserve this admission connection through actual higher-order passage, with exact lexical origins and final worlds. A separate executable materializer reconstructs arbitrary accepted trees of the existing mother/father/chain calculus and produces exactly their independently qualified native artifacts. It does not supply arbitrary learned-rule interpretations, an installed machine primitive or a C realization.

4. **Sharing a first-class inference provider preserves per-call inputs.** Actual source functions pass their bound arguments to the fixed MIL checker. Sharing a nondeterministic selection yields two correlated provider histories; selecting afresh yields four independent histories. The literal reply pair retains one accepted and one rejected article for the same mathematical goal, so caching the selected function does not collapse its inputs or cache all applications. The checked finite frontiers complete, and generic independent source typing excludes administrative domain faults at every bound. These are explicit source effects

and raw replies, not a cost model, interchangeable programs or two different proof-calculus implementations.

5. **First-class Need execution preserves independently admitted types.** Source-backed runtime values, captured environments, function bodies and cached answers transport through fresh cell allocation. Actual local interception and the complete allocation/commit protocol preserve the final computation type. Every bounded frontier and observed result is qualified from independent initial typing and primitive contracts; intermediate invariants are derived, not assumed. Native results regain formation-sensitive admission under a formed target context. Returned closures retain their actual qualified final heap. Administrative polarity faults are excluded for all bounds. An independently typed shared dependent function returns identity evidence at two supplied arguments; its setup effect occurs once and its call effect twice. The actual cached body remains typed in the retained final heap. Allowed retry shapes do not establish event provenance, and this preservation theorem does not supply independent natural adequacy, termination or a cached-origin denotation theorem.

6. **First-class source closures preserve the old owned Need execution.** A native-indexed polarized source has actual thunk/function constructors, index/closure packets and simultaneous three-scope substitution. Lexical execution distinguishes returned values from computation functions and ordinary forcing from shared demand. A conservative local-step interface enters captured code without adding effects or cells. The complete old machine embeds with exact ordered step/frontier/answer correspondence, retaining worlds, faults, occurrences and work. Higher-order, mixed-capture, indexed-closure and construction-versus-execution sharing controls execute the new source. Independent source typing preserves old derivations; simultaneous typed substitution and four principal source openings preserve the displayed result type. A general admitted producer's selected index remains packaged with its identity evidence. A first-class dependent function template executes in seven actual transitions for every loaded native argument; independent argument admission licenses its returned identity proof. Reverse typing conservativity and independent natural adequacy for the richer constructors are not consequences of these results.

7. **Mixed-scope binding and dependent computation-body application.** Simultaneous sorted-ABT substitution weakens every captured axis under the complete binder list. Its composition, renaming, opening, scope and signature laws are checked. A wrong cross-sort opening passes the same structural gates yet selects a different cached answer on the actual Need machine. Raw native-domain computation functions have independently qualified dependent application, substitution and captured-opening laws; actual evaluation feeds the resulting native judgment. Two forces share one effectful reflexivity result in the correct argument fibre. These components alone do not supply the richer source calculus's runtime typing or its full value/computation equational theory.

8. **Computation algebras and actual higher-order proof services.** The existing contextual Kleisli category embeds fully faithfully into Mathlib's algebras for its induced monad. Genuine computation products, including the empty product, and value-indexed powers have their contextual computation/stack universal properties. Sequencing, resume and reindexing commute through these interfaces. The free singleton returner remains nonterminal, and argument-dependent effects need not come from a computation returning a pure function. A product of actual native conversion reply algebras connects application to the checker and formed proof use. Provider selection once versus per call preserves acceptance support but

changes accepted artifact pairs, multiplicity and intents. These results qualify a semantic candidate, not a reified dependent calculus or a runtime migration.

9. **Contextual polarity and delayed dependent services.** The genuine Kleisli adjunction recovers existing return, map and bind; it separates pure values from effectful computations without claiming all CBPV type formers. The singleton-state/intent Kleisli instance has no terminal computation object, while the value category retains its terminal singleton: a precise empty-product obstruction to treating free returners alone as the full computation universe. First-class selected program values retain their dependent index through forcing, and qualified reuse/discard laws hold through arbitrary effectful selection and consumers. Independent native result-family qualification, typed answer substitution and exact evidence erasure connect raw delayed packets to admitted native Sigma results. State-change and wrong-index controls separate captured dependency from captured mutable state and execution from admission.

10. **Immediate need demand agrees with allocated eager sequencing.** Arbitrary producer derivations and completed actual runs correspond in both directions at the same exact outcome/world between immediate lazy demand and sequencing through an identity consumer. The comparison includes failure and allocation collision. Erasing a suspension or adding a forwarding cell changes protocol worlds even when native dependent answers agree; no unqualified strategy rewrite follows.

11. **A strategy rewrite is qualified by its effect fragment.** `ContextualStrategyQualification.repeat_eq_reuse` proves that one evaluation plus reuse replaces two consecutive evaluations of a structurally read-only program before every effectful continuation, under the existing isolated-world observation. An unused such computation may also be discarded. Choice, occurrence multiplicity, state and intent controls refute unrestricted versions. No cost law or complete CBPV realization follows from this observation-level result.

12. **Dependent results connect to effects and actual need consumption.** `ContextualDependentSequencing` proves the exact handler law for sequencing, including selected state, branch history and ordered intents. `FormationSensitiveDependentComputation` constructs actual native dependent pairs, commutes with native value substitution, and rejects a wrong selected identity index. `NeedDependentService` proves first and cached consumer runs of the existing owned machine, with full request-index validation and exact execution-answer erasure. `NativeConversionNeedConsumer` connects the actual finite checker to those runs, independently formed logical admission and effectful dependent use. These are qualified components, not a completed object-language CBPV calculus or C realization; the scoped source's whole-control invariant is established separately below.

13. **Reified computation bodies preserve dependent admission.** The `ScopedComputation` family independently constructs raw scoped syntax, formation-sensitive typing and direct ordered-world semantics. Typed substitution preserves every source constructor; interpretation commutes with source substitution and, under the primitive realization contract, exactly realizes the source worlds. Primitive result qualification implies native result typing for every admitted source program. Actual state/reflexivity operations discharge that condition for arbitrary admitted compositions. A changed-index primitive fails qualification despite realizing its own altered semantics. The finite non-aborting fragment has a result, but neither full CBPV progress nor normalization of its raw mathematical payloads follows.

14. **Scoped suspensions execute on the actual owned need machine.** `ScopedNeedComputation` adds independently scoped native and need-reference binders, structural laws, independent typing and an injective embedding of the earlier sequencing syntax. `ScopedNeedMachine` supplies captured-code actions to the unchanged Need protocol; exact local transition laws cover allocation, demand, native continuation opening, Sigma completion, effects and choice. A separately typed heap preserves typing under successful fresh allocation and qualified cache updates; a formed ambient context gives native admission of cached values. The separate control typing now retains source conversion evidence, distinguishes demand from allocation resumptions and aligns commit frames with exact cell types. Whole-control preservation derives the logical heap invariant from initial source typing and the independently qualified primitive implementation, throughout every actual successor and bounded frontier. Every returned native value is admitted in a formed ambient context. The primitive obligation is discharged for retention and dependent reflexivity over arbitrary admitted compositions; a changed-index implementation returns operationally but fails qualification. Independently, the generic allocation-slot invariant is preserved by every actual step, so fresh allocation succeeds from reachable empty-heap states without a caller-supplied success premise. Actual executions separate shared from independent choices, no-force allocation from eager work, stable from retryable failure, and unfinished fuel from a negative logical answer. Retry retains earlier effect receipts. These are scoped source execution and whole-control typing laws, not full CBPV qualification or a C implementation.

15. **Finite scoped Need evaluation has exact-world correspondence.** Independent proof-relevant source evaluation/force rules correspond in both directions to completed actual-machine executions, retaining outcome, heap, branch path, counters and receipts. Every bounded answer has a source derivation. General whole-control preservation upgrades native values to their declared judgments under independent qualification. Controls distinguish duplicate choice occurrences, cached reuse, retry effects, allocation collisions, wrong native indices and unfinished budgets. A saved-stack first-return law separately prevents reasoning from a path that has already consumed its intended return boundary. These existential finite-run results do not establish a derivation bijection, ordered answer-list equivalence, cost equality or termination.

16. **A formation-sensitive cumulative candidate has structural laws.** `FormationSensitiveTyping` retains formation at constant use, lambda and pair introduction, and conversion. Renaming and simultaneous substitution preserve those premises; regularity gives a formed result type in every formed context under the proved tower universe interface. `FormationSensitiveMIL` strengthens every existing primitive/chain program without changing its quotation or independent relational evidence. `FormationSensitiveNativeList` adds the corresponding formation and evidence law for every finite native list-relator constructor spine. The actual MIL eliminator now has a formed declaration and motive/branch types; its primitive/chain iota endpoints have refined judgments in formed canonical telescopes and through typed substitutions. A receipt-dependent consumer separates two displayed types at identical endpoint codes and rejects a crossed reduct. General primitive/chain typing preservation uses the package's independently proved Pi-conversion boundary; the actual contextual law is recorded below without remaining qualification premises. These results do not qualify arbitrary declaration computation or install a Prime kernel. The native List, identity and proof-relevant relator eliminators likewise have independently formed motives, declarations and branch endpoints under typed substitution. Declaration-rooted recovery

yields general primitive preservation for all five roots; the actual conversion boundaries are discharged as recorded below. Redirected recursive evidence and changed identity endpoints fail the original root relation.

17. **A formed judgment has a constructed finite restriction.** `FormationSensitive.Dependencies.judgment_supported` accounts for the telescope, intermediate declaration formation and conversion paths. `FormationSensitive.Dependencies.exists_restriction_with_reflection` constructs the retained declaration/root-equation package, preserves the judgment and reflects all restricted judgments. Structural closure of root equations is proved, not assumed of an arbitrary smaller table. The example removes a genuine polymorphic declaration and computation, while deleting a hidden formation dependency invalidates refined admission despite raw acceptance. Universe policy and dependent structural syntax are unchanged.

18. **The actual HOL interface reaches refined syntax.** `FormationSensitiveHOLInterface` proves successful representation typing and exact renaming/substitution laws for the supported HOL constructors under independently typed symbol interpretations. `FormationSensitiveHOLInvariant` uses seven actual declarations, including independently formed type-indexed universal and equality operators, to represent the call-guard invariant, function-valued specialization and predicate quantification. Independent source Henkin validity does not follow from formula formation. Equality at every represented simple type, including function types, is an explicitly declared HOL proposition-former, not native identity or a semantic extensionality axiom. `FormationSensitiveHOLUniformList` represents the actual list equations, predicate-quantified induction and map-length conclusion using thirteen opaque declarations with four distinct carriers. Formation and function-valued substitution are checked. The source HOL derivation remains distinct from formation of its represented native proposition. The specialized chart reconstruction reaches that same represented conclusion; it does not turn proposition formation into a native proof inhabitant. No proof-as-inhabitant translation or final Prime logic is selected.

19. **Actual Tower conversion has a proved Pi boundary.** `PureConversion` constructs complete development and Church–Rosser for the full root-empty cumulative package. Semantic universe heads are retained, not erased. The resulting `Tower.piConversionBoundary` discharges refined root beta preservation through `FormationSensitive.Typing.betaPi_tower`. Nonempty packages satisfying the separately constructed definitional expansion qualification inherit the boundary as well; arbitrary iota extensions and normalization remain separate.

20. **Native hypothesis conversion and contextual preservation are qualified.** `MILConversionCompletion.conversion_iff` preserves and reflects the exact authored conversion theory while adding only proof-side roots guarded by existing conversion evidence. Parallel development is sound for that theory and stable under binding and simultaneous substitution. The actual primitive and chain branches, followed by structural term induction, prove `MILConversionParallel.diamond` for all raw open terms. Joined auxiliary paths yield the native Pi/Sigma component and head boundaries. `FormationSensitiveMILQualification.steps_preserve` consequently preserves every admitted judgment through finite contextual runs of the unchanged native rules, with no assumed conversion boundary or primitive preservation. Controls reject incompatible metadata, a changed primitive symbol and a duplicated chain induction result. The completion does not equate runtime steps or proof paths, and does not establish normalization or a total conversion decision procedure.

21. **The combined List/identity/mapRel package is qualified.** `NativeRelatorConversionCompletion.conversion_iff` compares the exact original conversion with its conservative proof-only completion. Retained metadata coherence, binding laws and cofinal development give `NativeRelatorConversionParallel.diamond` and the actual Pi/Sigma boundaries. `FormationSensitiveNativeRelatorQualification.steps_preserve` preserves independently admitted judgments through all finite contextual runs of the original five-root package. Its checked-step interface needs no additional target-typing premise. Conversion cannot identify Pi or Sigma with a universe head; every purported finite code across either boundary fails. Neither original execution confluence nor normalization is claimed. The positive mixed-metadata completion control has no exact authored root, so the proof presentation is not a replacement runtime.

22. **Native conversion has executable finite evidence checking.** `StructuralConversionCode.Code.conversion_iff_checked` covers every original structural conversion when its root decoder is qualified. Actual finite root decoders instantiate it separately for MIL, List/identity and List/identity/mapRel, including arbitrary open terms, congruence under binders, symmetry and exact path composition. The executable checker uses the existing substitution and semantic level decision, not proof-side completed roots. `NativeConversionChecking.checked_mil_step_preserves` retains each admitted source judgment through an accepted forward step. Checked conversion changes a displayed type only with independent target formation; a dependent identity example exercises both endpoints. Corruption controls separate rejected evidence, nonconversion and missing formation. Existence of accepted evidence is not a total conversion search, raw normalization, or qualification of a union of native packages. `StructuralConversionCodeDependencies.Code.check_eq_of_agreeOnCheck` preserves both positive and negative verdicts from an executable comparison of the finite retained root requests. Used and unused branch-change controls distinguish proof-local reuse from whole-decoder agreement.

23. **Transparent definitions have a checked conversion interface.** `ConstantExpansion.Qualification.conversion_iff` derives full contextual conversion preservation and reflection from constant and root certificates. The actual signature specimen includes transitive aliases, an opaque value and a polymorphic identity. Refined beta preservation is instantiated for the nonempty package; `FormationSensitive.Judgment.delta` independently preserves a constant's displayed type using its checked body. A three-step dependent specialization retains refined typing throughout.

24. **Full contextual preservation has actual package instances.** `FormationSensitive.Judgment.steps_preserve_tower` preserves the formed context and displayed type through arbitrary finite runs of the actual root-empty Tower. Qualified transparent definitions have a general constructor, instantiated without remaining primitive obligations by `PreservationExamples.actual_runs_preserve`. Dependent pair components, a computational projection type and binder-domain reduction consume the theorem. Additional iota rules and normalization are not consequences of these instances.

25. **Declared applications recover their dependent parameter typings.** `DeclarationSpine.recoverTelescope` derives the complete typed substitution for an arbitrary finite declaration telescope under universe regularity and Pi qualification, retaining replay of the exact displayed result. The nonempty alias/identity signature instantiates the qualification without remaining conversion hypotheses: an observed ground result recovers a named argument

type, and an invalid lambda argument is excluded.

26. **Native proof soundness now feeds an operational theorem.** `HenkinProofSoundness` interprets successful checking for its recursive monomorphic proof fragment under independently meaningful hypotheses, named facts and definition equations, retaining the actual normalization pipeline. `HenkinTransitionInvariantImport` applies it to a checked native proof and the existing compiler model to obtain reference-result preservation through finite runs. No separate intrinsic preservation derivation is used. Altered definitions, insufficient fuel and unscoped hypotheses have different checked failure controls. Prefix proof polymorphism and foreign-checker implementation equivalence are not covered.

27. **The existing simple translation reaches refined typing.** `FormationSensitiveSimpleFragment` proves type/context formation, term typing, and substitution admission using the unchanged intrinsic atom/arrow translation. Its two substitution routes agree on raw syntax; existing conversion exactness on the image has refined endpoints. An open higher-order specialization supplies a capture-sensitive control. Exact recovery of internal type annotations remains refuted for this raw erasure, and full HOL consequence or inhabitation reflection is not inferred.

28. **A fixed proof has a finite rule-local reuse contract.** `InferenceRuleSupport.checkRaw_eq_of_agreement` preserves the exact checker verdict when complete lookups agree on all rule identifiers in the proof tree. Accepted derivations transfer with the same raw tree. An unused rule can be removed, while root agreement alone misses a required child rule. This does not prove logical-feature minimality or sufficiency of a simple-type kernel.

29. **Rewriting modulo equations is the sole operational step.** A language definition generates the GSLT whose step is $E; R; E$, the class rewriting relation of Peterson–Stickel; the equation-free case is the theorem `stepModuloEquations_iff_step_of_no_generators`, not a second implementation, and the core is named for it (`StepModuloEquations`, `gsltModuloEquations`, `ProtocolModuloEquations`). Presentation-derived laws (bag and set permutation, flattening, units) are generators of E, with their substitution and renaming stability discharged for the generic case and for rho; in the Cost lane a derived occurrence carries its declaring carrier or algebra rule as origin and no static colour, and for rho every such occurrence is a bag permutation absorbed by the base canonicalizer (`rho_costDerivedInstance_canonicalize_eq`).

30. **OSLF over equation classes.** Every formula of the observational logic and every admissible structural-modal observation descends to the quotient by E; the Hennessy–Milner fragment is adequate over image-finite quotients (`HennessyMilnerDirections`, `FormulaAdequacy`, `EnumeratedAdequacy`).

31. **Passes are covers.** A representation is a validated language definition with its relation environment; a pass is a semantic cover, the bounded morphism of modal logic taken up to target equations, and representations with passes form a category with a functor to covered operational theories (`IRPass`, `SemanticImplementation`). Covers preserve and reflect every Hennessy–Milner formula and bisimilarity; a term map with one escaping target step is the map of no pass (`no_pass_of_escape`).

32. **Machines as languages.** The binding-decision evaluator and the ordered-match machine of the compiled-rule lowering are language definitions, proved exact against typed reference

machines and adequate to the fused evaluator (`BindingDecisionAdequacy`, `OrderedMatchMachineAdequacy`). One complete run of the machine is one step of the cold call-guard language, and the identity on terms is a semantic cover into the machine's run view (`MainlineCallGuardMatchMachinePass.coldToMachinePass`); the run view itself is the big-step relation of a representation, closed under its equations (`IRRunView`).

33. **Institutions and the theory graph.** Model-valued institutions with comorphisms form a category; forgetting models to consequence is a functor to the category of π-institutions (`InstitutionConsequence.consequenceFunctor`). Closed theories over varying institutions form the theory graph, constructed as Mathlib's Grothendieck construction of the direct-image functor of closed theories (`TheoryGraph.closedTheories`); the closed theories of one institution embed faithfully as its fibre, and every arrow has a semantic target reading (`TheoryGraph.semanticTargetFunctor`).

34. **Models are fibred over theories, with the correct variance.** Over the theory graph of model-valued institutions, the satisfying models of a node form a full subcategory of the institution's models at the theory's signature; a theory-graph arrow reduces target models to source models through the comorphism's model view and the signature morphism, and the two satisfaction conditions carry satisfaction back (`TheoryGraphModels.satisfies_reductHom`). Reducts compose contravariantly, so satisfying models are a functor $\mathbf{Node}^{op} \to \mathbf{Cat}$ (`modelsOf`), and their total category is Mathlib's contravariant Grothendieck construction, whose projection to nodes is a Grothendieck fibration with reducts as cartesian lifts (`SatisfyingModel`, `toNode_isFibered`, `transport`). A real global point of the dependent-families institution transports along the constant-family route to the simple global point it encodes (`TheoryGraphModelsCanary.reduct_falseModel`).

35. **NIK services form an indexed family over targets.** The four faces are a classifier, pairwise disjoint; certificate boundaries are the subtype of one face and no native service lies in it (`SemanticServiceFamily.Service.face`, `ne_of_face_ne_certificateBoundary`). Along an equivalence of targets every face transports and the face is preserved (`Service.transport`, `face_transport`). Transport preserves composition and is inverted by the inverse target equivalence, with the native proof system and certificate carrier retained (`transport_trans`, `transport_symm_transport`, `transportEquiv`). A native operation pushes forward along any admitted operation (`pushforward`), and request-local selection returns a native operation whose runs preserve the target's declared meaning (`ofStronger_preserves`). A native operation or native proof exposes no external certificate (`isEmpty_externalCertificate_of_face_ne`).

36. **Type-derived interpretation transports HOL proofs.** A base sort may be mapped to an arbitrary simple type, with constants retyped accordingly. The intrinsic translation preserves context positions, commutes with weakening and term instantiation, and transports every rule of the extensional HOL derivation calculus (`ExtDerivation.mapTypes`). A quantified beta law over a base sort thereby becomes a theorem quantified over functions. This realizes a concrete part of the little-theories account of metalevel polymorphic reasoning in a monomorphic object logic [10]. Megalodon's plain type interpretation is exactly specialization of a schematic type in the existing HOL syntax (`HenkinTypeFragment.schematicType_specialize`); its native type-application operation commutes with that interpretation, with capture-avoiding shifts accounted for. These are not yet a translation of arbitrary Megalodon terms or proof objects.

37. **Native monomorphic terms have an exact intrinsic HOL interpretation.** The supported native constructors are variables, named and primitive symbols, application, lambda abstraction, implication, and universal quantification. Native contexts are reconstructed from plain context formation. The interpretation preserves the actual symbol names, primitive indices, lookup evidence, binder types, and term erasure (`HenkinTermInterpretation.interpret_native`). Only symbols used by the term require plain lookup types; unrelated polymorphic declarations remain allowed. Under the stated annotation and lookup conditions, successful native type inference is equivalent to the existence of an intrinsic term of the corresponding type with exactly that native erasure. Higher-order terms at nonzero type depth and a mixed environment supply positive controls; a type-mismatched application has no interpretation. The partial erasure has no image for native type abstraction, type application, or prefix proposition quantification. This is a boundary of this constructorwise interpretation, not an impossibility of representing polymorphism by a richer interpretation.

38. **The native binding operations agree with intrinsic HOL.** Erasure commutes with typed renaming implementing native index shifts, with weakening, and with capture-avoiding term instantiation (`HenkinTermInterpretation.erase_instantiate`). Consequently a native beta redex and its native contractum are exact erasures of intrinsic terms with equal denotations in every Henkin model, at every valuation (`HenkinTermInterpretation.native_beta_interpretation`). Neither full domains nor extensional argument congruence is needed for this local beta law. A nested-binder control distinguishes the correctly shifted contractum from variable capture. General native normalization and arbitrary proof acceptance are not consequences of this local theorem. Checked delta interpretation is established separately below.

39. **Successful native beta/eta and delta/beta/eta normalization preserve intrinsic meaning.** Successful erasure determines its intrinsic typed input uniquely. Native unused-variable removal agrees with typed strengthening, including beneath binders, and exposes eta's necessary no-use condition. The actual `normalizeOne` pass has a same-typed intrinsic output with equal literal denotation in every Henkin model (`normalizeOne_interpretation`). Induction on the actual finite-fuel evaluator extends this to `Tm.normalize` (`betaEtaNormalize_interpretation`). The full native delta-then-beta/eta pipeline has exact output erasure and preserves meaning under independently checked definition typing and model equations (`nativeNormalize_interpretation`). A beta/eta contraction, a two-definition expansion and an altered-definition model supply controls. These theorems concern successful runs on the supported plain term interpretation. They do not prove success at some fuel for every input, normalization completeness, raw annotation formation in general, or soundness of arbitrary native proof acceptance.

40. **Evidence-dependent judgments compose without collapsing their selected witnesses.** `Sequential.evidenceRunEquiv` gives an exact proof-relevant run view for sequential composition. The actual matcher followed by a bounded-vector-index service instantiates its acceptance theorem, with a general duplicate-count law and out-of-range and non-vector negatives. Guard support is lossless exactly for subsingleton evidence fibres; two receipts refute arbitrary recovery. Intrinsic HOL invariant transport is instantiated on the actual call-guard compiler, with a paired accepted Megalodon theorem and a rejected altered interpretation. See Section 3.7 for the contracts and the distinctions from a general dependent checker, source importer or full C realization.

41. **Closed-term constant interpretations act on genuine Henkin models.** The existing `substConst` induces a model reduct with unchanged carriers and admissible domains. Term closure is proved through substitution; it is not assumed as an extra field. Denotation and satisfaction commute with this interpretation, and its identity and composition laws hold as equalities of complete model records (`HenkinModel.constantSubstitutionReduct_comp`). Fullness and the existing extensional-congruence condition are each preserved and reflected, rather than required. In the non-full Boolean example, interpreting a higher-order constant by $\lambda f\, x.\, f(fx)$ validates the tested double-negation sentence; the equally typed $\lambda f\, x.\, fx$ refutes it. For an independently chosen model, full-term denotation preservation is equivalent to agreement on the individual constant denotations. A typed replacement alone does not establish that agreement.

42. **Native checked definition unfolding has an exact partial interpretation.** Eligible definition bodies are reconstructed from native inference, annotation formation and local lookup checks in the final environment. Partial substitution in the existing intrinsic HOL syntax then follows the actual evaluator's fuel convention: only traversing a named definition consumes fuel. Its success, erased result and failure agree exactly with native `deltaNormalize`; additional fuel preserves an already successful result (`deltaNormalize_eq_some_iff`, `deltaInterpretation_eq_none_iff`, `deltaInterpretation_mono`). Output annotation formation is proved independently from the raw evaluator, and native type synthesis is preserved (`native_infer_deltaNormalize`). A mixed library containing a real prefix-polymorphic definition supplies the two-definition-depth example under a binder at nonzero type depth; one depth of fuel fails. Delta expansion correctly retains the beta redex rather than silently performing another normalization phase.

    In a chosen Henkin model, independently stated declaration equations imply literal denotation and satisfaction preservation for successful expansion (`denote_deltaInterpretation`). At fixed fuel the exact criterion concerns the successful constant replacements, not equations inferred from failed evaluation. Neither full function domains nor termination is assumed. Definitions whose bodies use unsupported prefix constructors still need the richer interpretation; unrelated prefix declarations are not prohibited.

    The model condition is instantiated independently on this same native environment in `NativeDefinitionModel`. The declared identity function denotes identity, while the meanings of p and $q := \mathsf{id}\, p$ are independently chosen propositions. Exact erasure determines the checked bodies. The declaration equations hold exactly when the chosen propositions are equivalent; they imply meaning preservation for the actual two-level unfolding beneath the schematic binder. Assigning p true and q false preserves the native typing checks but refutes both those equations and unfolding preservation. This specimen uses a standard full model; the generic preservation theorem above also applies to non-full Henkin models.

43. **Definition typing, stable names and termination are distinct.** A native tail-checked list can declare $p : \mathsf{Prop}$, then $q : \mathsf{Prop} := p$, then shadow p by a function-typed parameter. In the resulting raw environment, q still infers as a proposition but unfolds to the function-typed p. A fresh-name extension preserves the comparison. Checking closed bodies in the final environment avoids this particular typing failure, but permits a self-cycle whose unfolding fails at every finite fuel. These are checked obstructions for raw environments (`DefinitionEnvironmentBoundary`); they are not counterexamples to the source document checker, whose content-addressed admission, normal-form and external-association checks require separate correspondence.

44. **Native proof checking has a finite, dependency-local frame theorem.** `NativeSupport` tracks term names, primitive indices and known proposition names through the actual native syntax. Shifting and type instantiation preserve support; term substitution, beta/eta reduction and successful normalization introduce no unaccounted dependencies. In `EnvironmentDependency`, local agreement compares complete lookup results, including absence and definition transparency. Dependency closure includes every name exposed by a selected definition body or known proposition. These independently stated conditions imply exact equality of native type synthesis and normalization.

    `NativeProofEnvironment.checkProof_frame` extends the result to the complete native proof syntax, including prefix polymorphism, term instantiation and known propositions. For a fixed checker and fixed inputs, the frame law has the form

    $$\mathsf{Supported}(\Gamma, \pi, \varphi, S) \wedge \mathsf{Closed}(\mathcal{E}, S) \wedge \mathsf{Agree}(\mathcal{E}, \mathcal{E}', S) \implies \mathsf{check}_{\mathcal{E}'}(\pi, \varphi) = \mathsf{check}_{\mathcal{E}}(\pi, \varphi).$$

    The theorem retains the same fuel bound and covers rejection and normalization fuel failure as well as acceptance. It requires neither a monomorphic interpretation nor logical soundness of the supplied declarations. Such soundness is not a consequence of the frame law. Equal returned results do not imply equal running time, allocation or lookup traces: even a fresh prepended declaration may add work to a list-based lookup. The laws concern the existing Lean `MathdataKernel` operations; refinement of an OCaml executable or emitted C artifact remains a separate obligation.

    The executable checks in `EnvironmentDependencyCheck` decide precisely the stated agreement and closure conditions, and the native proof-request check also verifies support of the assumptions, proof and goal. Their acceptance implies the frame law without an unverified dependency claim. A stored verdict still requires provenance linking it to the old check; manifest acceptance neither authenticates an arbitrary cache entry nor proves the submitted proposition. Every finite native environment admits a computably supplied closed manifest covering any requested finite support; the conservative construction includes dependencies from unused and shadowed bodies, and is explicitly not a least closure. The frame conditions are sufficient, not necessary or minimal. Prepending declarations with keys outside the protected support preserves agreement, even when some protected lookup was absent. Merely choosing names absent from the old declaration list does not protect previously missing references.

    A concrete chain $a := b$, $b := p$, $p := r$ retains its native proof verdict under an unrelated declaration. Changing the hidden body of p preserves the directly mentioned lookups and their inferred types but changes the verdict; making p opaque does so as well. A known proposition can hide the same dependency chain. A genuine prefix-polymorphic proof survives an unrelated extension, while filling a protected missing known lookup changes rejection to acceptance. These are native evaluator controls, not abstract replacement evaluators or invented proof authorities.

    Source inspection also distinguishes two real signature operations: `import_signatures` appends imported declarations, whereas `DocParam` may prepend a same-type opaque declaration and `def_of_tmh_r` stops at the first matching name. Thus content-addressed naming alone is not an exact operational reuse theorem. The alias-chain control is a raw-environment specimen, not an asserted accepted source document: `DocDef` skips bare aliases. No failure of document-checker soundness follows from these observations.

45. **Native type specialization agrees with intrinsic retyping.** At arbitrary substitution depth, retyping an intrinsic term erases to the native capture-avoiding type substitution (`HenkinTermInterpretation.erase_mapTypes_typeInstantiateAt`). Native binder annotations remain formed, and specialization preserves a raw native typing judgment through the reconstructed intrinsic term (`native_infer_typeInstantiateAt`). The same-library instance keeps names and primitive indices unchanged. Closed prefix-polymorphic lookup types suffice for every representable constant; unrelated prefix declarations need not be removed. A free variable in an unchecked global declaration refutes a name-preserving specialization map in that unchanged library. Nonzero-depth controls distinguish capture avoidance from mere well-scopedness: an incorrectly captured result can also pass formation.

46. **A genuine polymorphic equality declaration has checked instances.** The native preamble definition is represented literally as $\mathsf{eq}_A\, x\, y := \forall Q : A \to A \to \mathsf{Prop},\, Q\, x\, y \Rightarrow Q\, y\, x$. The generic definition checks at its declared prefix type; unfolding the actual named declaration and native type-beta reduction yield exactly the monomorphic body, which is also the erasure of its intrinsic specialization. One actual generic reflexivity proof checks against the source-level statement $\forall x : A,\, \mathsf{eq}_A\, x\, x$ for every closed plain type, with proposition formation checked separately (`NativeEqualitySpecialization.reflexivity_statement_checked`). A proof specialized at the wrong type and a prefix type placed in a plain argument slot are rejected. The native definition and reflexivity proof use neither set nor choice axioms and require no dependent object types; this does not identify their object logic with the Lean metatheory.

    In the existing term-closed Henkin semantic interface, the defined equality implies intrinsic extensional equality: lambda closure admits a separating predicate constructed with intrinsic equality. Agreement at all types and admitted values holds exactly when admitted functions respect that extensional equality (`HenkinEqualityInterpretation.functionsRespectEqv_iff_equality_agreement`). Full domains are unnecessary. The existing non-full monotone Boolean model gives a counterexample without the congruence condition. This is a comparison in that semantic interface, not a soundness theorem for all native proof rules or a semantics of arbitrary prefix-polymorphic sentences.

47. **Ordinary hypothesis formation is separate from prefix proposition formation.** Megalodon's source `PLam` rule requires ordinary type `Prop`; its prefix-permitting proposition check is used at a different boundary. The raw Lean checker now uses the ordinary guard. Definition-conversion and theory-admission implication rules carry the same ordinary typing premise, while theorem admission independently requires prefix-aware formation. Ordinary and genuinely polymorphic identity proofs remain accepted. A prefix-bound hypothesis is rejected, and a genuine prefix theorem has an accepted admission article. A deliberately unformed initial known inventory illustrates why raw proof lookup does not supply theorem-formation evidence. These controls repair a specific native-source fidelity discrepancy; they do not assert complete source equivalence or HOTG model-validity soundness of the checker. In particular, the source document check also requires normal-form input; the reflexive formation witness here does not impose that separate policy.

48. **Type-derived signatures compose, and their theories enter the existing theory graph.** The objects pair a base-type alphabet with the existing typed constant signature. Morphisms interpret base symbols by simple types and constants by symbols at their translated types, in the standard type-derived sense [11]. Identity and composition

are proved on all intrinsic terms, including open terms under binders; heterogeneous equality accounts for the equal translated indices, rather than identifying distinct source types. Closed formulas consequently form a sentence functor (`TypeDerivedSignature.sentence`). The fixed-base constant-only signature category embeds faithfully, and its original sentence map is recovered. The existing finite extensional derivation closure gives a π-institution (`HOLTypeDerivedConsequence.institution`), with its closed theories included through the existing theory-graph fibre. A concrete two-arrow interpretation changes both type expressions and numeric constant symbols and carries a beta theorem through this fibre. The incorrectly ordered symbol map is distinct, while sort collapse still fails to reflect consequence. No model-completeness premise is hidden in this proof-theoretic construction.

49. **Full-domain models form a type-derived institution.** Given a target model with full admissible domains, interpreting source base carriers by the denotations of their target types constructs a standard-domain source model. Recursive carrier equivalences preserve application, lambda, logical equality, and quantifiers. Consequently a source sentence holds in that reduct exactly when its translation holds in the target (`HenkinModel.models_mapTypes`). Structural recursion proves equality of the interpreted carrier types, and the canonical equivalence is the cast along that equality (`Ty.denoteSubstituteEquiv_eq_cast`). Consequently iterated and composite standard-domain reducts are equal as complete model records, including their constants (`HenkinModel.standardTypeReduct_comp`). This is not univalence or an identification of arbitrary isomorphic models. Composition does not require fullness of the original target, because both routes standardize it. The identity reduct is exactly standardization; equality with the original model holds exactly when its domains are full (`HenkinModel.standard_eq_iff_fullDomains`). This last criterion concerns model equality, not elementary equivalence. Full models therefore give a strict contravariant model functor and an institution (`FullDomainInstitution.institution`). Its model categories are discrete, as in the fixed-base construction; general model homomorphisms have not been added. The fixed-base extensional Henkin institution has a comorphism into it, with unchanged sentences and models forgotten in the reverse direction. The model-forgetting map has exactly the full models as its image, not an asserted coverage of all Henkin models.

50. **Full-domain semantics has a substantive theory-graph consumer.** Extensional HOL derivability maps soundly into the semantic-consequence institution (`HOLFullDomainTheory.soundness`); the converse fails for the current excluded-middle-free calculus. Full set-based models validate an excluded-middle instance whose nonderivability has an independent Kripke countermodel (`HOLFullDomainTheory.soundness_not_reflecting`). This distinguishes the stated proof calculus from this semantic profile, not every possible classical HOL calculus. Semantic theories enter the existing satisfying-model fibration. One theory requires two distinct elements of a base sort; a type-derived arrow interprets that sort by its endomorphism type. Two constant Boolean functions satisfy the translated axiom, and their model reduces along an actual arrow of the total model category (`HOLFullDomainTheory.FunctionSort.transported`). A singleton model refutes the source axiom, proving that this object-theory restriction is not merely an unqualified HOL theorem. Thus framework, chosen axioms, satisfying model, and proof calculus remain separate in the consumer.

51. **Biform theories.** Farmer's biform theories, a closed theory with a proof-relevant algorithm whose events carry meaning formulas, form a category whose arrows are exactly the compatibility locus of a logical and an operational translation (`BiformTheory`); the same holds over the theory graph (`BiformTheoryGraph`). A certificate boundary for a bi-

form theory is an independent checker proved fibrewise equivalent to the native calculus (`BiformCertificateBoundary`); certified biform theories over the theory graph form their own category (`CertifiedBiformTheoryGraph`).

52. **One real biform theory.** The cold call-guard compiler machine, run to completion, is the algorithm; the specification theory of the reference compiler is the logical node; the meaning of a run is the specification sentence it decided, and every run's sentence is a theorem because a denotation invariant proves whole-run correctness (`MainlineCallGuardBiformTheory.run_halts_with_specification`, `compilerBiform`). Semantic validity of a compiled family is, by its definition, exact production by that specification at the current authority. The cold presentation language runs the same computations on encoded states; encoding is a proof-relevant translation of run algorithms and gives a biform route whose meaning square commutes, hence a point of the compatibility locus (`encodeRoute`, `encodeRoute_compatible`).

53. **Operational models of theories.** A GSLT whose states satisfy a closed theory, with native consequence sound pointwise, is an operational model; modelled theories form a category with separate projections to closed theories and to covered operational theories, and a state-indexed model of a model-valued institution induces one (`OperationalModel`, `InstitutionConsequence.StateIndexedModel`).

54. **One concrete logical node of simple type theory.** The property-explicit Henkin simple type theory with excluded middle, schematic choice, and Dedekind infinity is a closed theory of its consequence institution (`HOLHenkinTheoryNode`).

55. **A vertical specimen on the set-operation fragment of the HOTG preamble.** Four objects kept apart: the membership institution of the fragment (framework), the closed theory generated by its five preamble principles as a node of the theory graph (object theory), Mathlib's ZFC sets as an object of the fibre of satisfying models (model), and Megalodon's Mathdata kernel at the environment declaring the principles as a native-proof service (kernel and service) (`PreambleFragmentVertical`). A derived theorem, subset reflexivity through powerset introduction, is accepted by the native kernel with a genuine Megalodon proof object computed through the kernel's own evaluator, is a theorem of the node, and holds in the ZFC model (`subsetPower_native_accepted`, `subsetPower_mem_fragmentNode`).

56. **A native proof survives a genuinely weaker axiom scope.** `PreambleDependencyRestriction` retains only powerset introduction and its two symbols, preserving the actual proof's checker verdict at every fuel. Generic native soundness establishes its meaning under the retained axiom. A Boolean-carrier Henkin model satisfies that axiom but refutes powerset elimination; the omitted axiom's native proof fails as well. Missing or altered used axioms fail both dependency checking and checking of the actual proof. The old set-operation model class already assumes both powerset directions, so proof-local environment restriction inside that class alone could not demonstrate semantic weakening.

57. **The simple fragment inside the dependent one, and the descent criterion.** Inclusion of constant families defines a comorphism from the simple-families institution to the dependent one, exact on its image (`CwfSimpleDependentInstitution.entails_iff`); over an inhabited context, a family descends to a simple type exactly when it is constant (`DependencyDescentCriterion.descends_iff_constant`); a simple theorem transports (`identity_transports`); transport along a path in the varying family refutes the

path false = true, and transport in the family of finite types has its computation rule at reflexivity while the family is not constant (`no_path_false_true`, `finTransport_refl`, `finFamily_not_constant`). The cumulative tower's own syntax is a node with a satisfying model, the empty telescope witnessing $U_0 : U_1$ (`TowerTheoryNode`).

58. **Simple syntax maps structurally into the tower.** The intrinsic simple function fragment has a syntactic CwF: typed simultaneous substitutions compose associatively, the empty context is terminal, and list extension satisfies the comprehension laws (`SimpleFragmentCwf`). Erasure extends to a strict CwF morphism preserving contexts, types, terms, substitution, projection, and the generic variable (`SimpleFragmentCwfMorphism.strictMorphism`). Satisfaction in the global-inhabitation readout is preserved at translated global points (`SimpleFragmentInstitutionBoundary.satisfaction_preserved`). This is a structural and forward satisfaction theorem; it does not establish reflection of inhabitation or syntactic conservativity.

59. **Conversion is conservative on the simple term image.** Contextual beta reduction and its equivalence closure are defined on the existing intrinsic simple syntax. Renaming and simultaneous substitution preserve this equality, including substitution by pointwise-convertible terms (`SimpleFragmentConversion`). Strong normalization follows by a type-preserving interpretation into the existing regular two-sort kernel; no injectivity of raw erasure is used. An executable, intrinsically typed normalizer returns a finite reduction and an irreducible result (`SimpleFragmentNormalization.strong_normalization`). Normal forms at a fixed context and type are determined by their erasure (`IntrinsicSTT.Normal.eq_of_erase_eq`). Tower skeleton confluence then proves, for arbitrary intrinsic simple terms $s, t : A$ in Γ,

$$\mathrm{erase}(s) =_{\mathrm{tower}} \mathrm{erase}(t) \quad \Longleftrightarrow \quad s =_\beta t.$$

    The tower conversion path may leave the simple image; the tower is the sealed presentation without additional declaration-specific computation (`SimpleFragmentConversionDecision.towerConv_iff_betaConv`). Thus erasure induces an injective map of conversion classes, and simple beta classes are equivalent to their intrinsic normal-form representatives. This is equational conservativity on an image, not reflection of inhabitation or fullness of a model reduct. The normalization method uses the standard distinction between strong normalization and uniqueness of normal forms [90].

60. **The simple conversion quotient is contextual.** A generic congruence on a simply typed CwF quotients substitutions and terms together, retaining contexts and types. Compatibility with composition, substitution, and pairing supplies all quotient CwF laws; the projection preserves terminal context and comprehension strictly (`ScwfQuotient`). This uses the scwf structural interface [91], without assuming eta or dependent type equality reflection. Intrinsic beta conversion instantiates this construction (`SimpleFragmentQuotientCwf`). Lambda and application descend, satisfy beta, and commute with substitution. Substitution classes have unique pointwise normal representatives; normalizing components before composition agrees with normalizing the composite. Tower conversion reflects these components. The raw projection is surjective on arrows but not faithful: it forgets genuine beta-redex distinctions, while distinct variables and the eta pair remain distinct. This is the simple quotient, not a constructed quotient of all dependent tower types and contexts.

61. **Native normalization and direct conversion decision share a semantic target.** Comparing the erased syntax of computed normal forms decides exactly the independently defined

intrinsic beta relation (`SimpleFragmentConversionDecision.decideConversion_correct`). Normalizing both endpoints is a native operation that preserves and reflects that relation, and changes raw syntax on the discarded-identity specimen. The decision and operation are distinct NIK service forms; neither consumes an external certificate. Normalization is idempotent and compatible with substitution modulo beta conversion.

62. **Identity notions and their licensed maps.** Conversion determines the equations, the equations determine behavioural equivalence under every observer and equality in every model, fine behavioural equivalence determines coarse, occurrence identity erases to endpoints, and lineage determines reachability only (`IdentityNotions`). A family descends through an observer exactly when it is invariant under that observer's behavioural equivalence (`descendsThrough_iff`).

63. **Observers.** Refining an observer adds distinctions; the induced map of behavioural classes is functorial, and coarsening is reversible exactly when no atoms or labels are added (`ObserverRefinement`).

64. **Circular propositions form a constructed final coalgebra with partial truth.** The strong Kleene propositions with a truth node are an M-type; every system of self-referential judgments unfolds into it by exactly one map (`System.unfold_unique`), which preserves and reflects grounded verdicts (`System.unfold_mem_least`, `System.Hom.mem_least_iff`). Grounded truth is consistent and agrees with stages and well-founded evidence; over encodable base facts the universe and its truth and falsity sets read faithfully as hypersets (`hyperset Universe`). Section 8.4 states the scope; the universe is propositional.

65. **The consistency tower is unconditional at PA.** Every sound Δ_1 extension of $\mathsf{I}\Sigma_1$ has its consistency proved one stage up and not at itself, and the stages increase strictly with no terminal stage. PA and $\mathsf{I}\Sigma_1$ are constructed sound stages, with constructed Δ_1 presentations and no postulated instance (`peano`, `peano_tower_strictMono`, `peano_tower_certified`, `peano_tower_not_self_certified`). Only finite levels are built (Section 12.2).

66. **The layer atlas.** Forty-one cells of combinations of the layers of a rewriting and knowledge-base language are checked theorems in `PrimeOptions.LayerAtlas`, each assembled from its owning module: CALM in both halves, frontier-stratified absence, non-confluence of consumption and of snapshot absence, the impossibility of a distributive law of distributions over sets, the per-answer tensor of choice, weights and cost, priority and epochs, late-bound forgery and its repairs, grounded knowledge, evidence derivation, and the cells listed in the next items.

67. **Evidence is derived from observation contracts.** Additive revision is exact exactly for disjoint ledgers (`EvidenceContracts.additive_revision_exact_iff_disjoint`); revision under unknown overlap lies between the maximum and the sum (`unknown_overlap_bounds`); same-world conjunction counts are exactly the Fréchet interval (`conjunction_counts_iff`); PLN's tensor is conditional-independence fusion of reports (`toPaired_tensor`, `strength_fuse_uniform`); evidential negation is monotone and intuitionistic negation is not (`against_mono`, `hnot_not_monotone`); PLN deduction is exact under screening off (`deduction_exact_of_screening`); the paired count is overdispersed (`PairedSampleDispersion.paired_variance`). Conclusion confidence is derived through credal intervals, with the propagation map fixed by the contract (`deduction_confidence_route`, `widthConfidence_eq_odds`, `idm_intervalWeight`). Section 3.5 states the scope.

68. **Possibility composes exactly over positive semirings** (`AnnotationCapabilities.support_bind_iff_positive`), and the speculative answer readout is a monad morphism from ordered runs (`SpecRunObserver.observe_bind`).

69. **Frontier absence is intuitionistic negation, certified by ownership.** `FrontierNegation.knot_iff_absent_frontier`, `refinement_sound`, `ownership_certifies_frontier` and `foreign_extension_certified` (Section 3.14).

70. **Schedulers, arbitration and fair service.** Probability envelopes are attained by deterministic schedulers (`SchedulerResolution.exists_deterministic_le`); arbitration by commit order or exclusive authority restores a single outcome (`Arbitration.first_committer_wins`, `gated_confluent`); bounded overtaking and deadlines bound waiting (`Run.fires_within_overtaking`, `deadline_fires_within`).

71. **Ranked knowledge-based programs are decided.** `RankedDecision.least_decides` and `ranked_implementation_unique` give a sufficient criterion for a decided least fixed point in the one-round case.

17.3 Refuted or proved impossible

Raw formation, supported indices and global closure do not by themselves establish cross-sort capture avoidance. The checked Mathdata-port example passes these gates while term substitution captures an outer type variable. The scoped-ABT Need example likewise passes both structural gates while selecting a different cached cell. The corresponding semantic theorems need their independent binding and fragment qualifications.

Each item is a checked negative result; the corresponding goal is closed, not postponed.

1. **Principal typing coherence does not cover arbitrary raw lambdas.** `DeclarationSpineExamples.raw_lambda_has_incomparable_function_types` gives one raw identity lambda with the formed types $G \to G$ and $(G \to G) \to (G \to G)$ in the actual Tower. Pi separation proves these types are not convertible. Declaration-rooted recovery cannot silently become unrestricted uniqueness of typing.

2. **Formed, substitution-stable equations need not normalize.** `FormationSensitiveExamples.ConversionCollapse` gives an actual rule package with $G \equiv G \to G$, both endpoints formed in the same universe, and a closed admitted beta-loop at G. Thus formation, renaming and substitution stability alone cannot qualify a normalizing conversion interface. The counterexample fails the Pi-conversion boundary.

3. **Conversion qualification does not guarantee termination.** `ConstantExpansion.Examples.CyclicDefinition` qualifies an actual one-declaration signature $c : G := c$ and checks its refined judgment. Its delta self-loop refutes accessibility. This separates a faithful conversion expansion from a terminating unfolding strategy even when Pi separation holds; the cyclic package is a negative control, not an installed profile.

4. **Conversion qualification does not check a definition's body.** `PreservationExamples.WrongBody` qualifies the actual signature $c : G := \lambda x.x$ for conversion and admits the constant at its formed declared type. Pi/head separation proves that the delta reduct cannot have that type. The resulting `no_root_preservation` excludes admission based only on the declaration type and conversion qualification.

5. **Reduction bisimilarity is not the observational equivalence.** The transcription of the context Hennessy–Milner theorem with reduction bisimilarity is refuted by an inert two-state system (`ReductionBisimilarityCanary.not_adequacy_sound`), and reduction bisimilarity is not partner-adequate for rho (`reduction_bisimilarity_not_partner_adequate`). Observational equivalence must be indexed by an observer.

6. **Rho's backward modality is not image-finite.** For every n there is a closed communication whose channel quotes a process with exactly n outputs, so predecessor classes are infinite and the box modality has no finite enumeration (`BackwardBranchingBoundary.directionalSystem_not_imageFiniteModulo`). Backward adequacy through enumeration is closed as a route.

7. **Literal covers miss runs modulo equations.** With $A = B$, a literal run $B \to C$, and $C = D$, the raw run relation cannot connect A to D (`IRRunViewEquationCanary.not_raw_a_to_d`); only the relation closed under equations does.

8. **A semantic cover cannot be built where a literal one is impossible only by choosing a different encoding** when a target step escapes every image class (`SemanticImplementation.no_semantic_cover_with_some_encoding`), and one invented transition from a stuck state breaks the call-guard cover (`MainlineCallGuardMatchMachinePass.negative_canary`).

9. **Forgetting models is lossy, and model expansion can fail.** Distinct model transformations induce the same consequence translation (`InstitutionConsequenceCanary.consequenceFunctor_forgets_model_map`); a comorphism that preserves consequence need not cover models or reflect entailment (`InstitutionCanary`); two distinct satisfying models have the same theory node (`TheoryGraphModelsCanary.forgetting_models_loses_information`). Along the arrow that adds the varying family as an axiom, the point selecting `true` satisfies the source theory but is the reduct of no model of the target theory (`no_expansion`). Thus reducts do not supply expansions of every source model. This counterexample does not exclude expansions for other arrows or under additional hypotheses.

10. **A theory is not one of its models.** A model satisfies a sentence its theory does not contain (`TheoryGraphModelsCanary.model_exceeds_theory`).

11. **Decision services do not automatically pull back along admitted operations.** A decision kernel cannot be pulled back along a meaning-preserving operation that does not reflect meaning (`ReindexingCanary.no_pullback_decision`). This rules out a total pullback with that meaning contract; it does not rule out every categorical organization of services.

12. **The current Henkin type translation omits prefix polymorphism.** Megalodon's types carry type variables and a prefix type quantifier; the implemented partial instantiation returns no image for the quantifier. Plain types of depth zero instantiate independently of the environment, while no fixed translation of a free type variable agrees with every environment assignment (`HenkinTypeFragment.instantiate_all`, `instantiate_plain_environment_independent`, `no_uniform_translation`). These are boundaries of the stated translation, not a theorem excluding every encoding of polymorphic logic. Plain types now instantiate at every finite context depth; only in-scope variable assignments matter, and shifting and native instantiation obey semantic reindexing laws. Full prefix-polymorphic sentence and kernel comparison remains open.

13. **Arbitrary type interpretation does not reflect theoremhood.** Take two source base sorts, with singleton and Boolean carriers, and the sentence saying that subsingletonness of the first implies subsingletonness of the second. The standard source model refutes it. Identifying the two sorts translates it to a provable implication of a sentence to itself (`TypeSubstitutionExample.collapseTypes_does_not_reflect_theorems`). No constants are involved. Thus proof preservation and satisfaction against a constructed reduct must not be confused with conservativity. The separating source model moreover has no target Henkin model agreeing on all translated sentences, without any fullness restriction on that target (`TypeSubstitutionExample.no_satisfaction_expansion`). The same obstruction is realized by the full-domain institution's model functor: a singleton/Boolean source model is outside the image of the reduct along the sort-collapse arrow (`FullDomainInstitution.BaseCollapse.no_expansion`).

14. **Standardization can change truth in a term-closed non-full model.** A logical relation with Boolean implication at the base type, the universal relation at propositions, and the usual function clause generates domains closed under every term, including lambda abstraction (`HenkinModel.ofLogicalRelation`). Its admissible Boolean endomaps are monotone, so each has a fixed point. Standardization admits negation, which has none. Thus the sentence $\forall f : b \to b.\, \exists x : b.\, f(x) = x$ changes truth even under the identity standard-domain reduct (`MonotoneBooleanModel.identity_reduct_changes_truth`). This witnesses failure for the term-closed `HenkinModel` structure; it must not be used as an example in the stronger fixed-base institution, whose models additionally require extensional-argument congruence. The missing congruence is explicitly refuted (`MonotoneBooleanModel.not_functionsRespectEqv`): the observations $h \mapsto h(\mathsf{false})$ and $h \mapsto h(\mathsf{false}) \wedge h(\mathsf{true})$ agree on admitted monotone endomaps, yet an admitted higher-order predicate distinguishes their behavior on excluded negation. Thus lambda closure and extensional congruence are genuinely different obligations. Neither the example nor the literal fullness criterion says that every non-full model is separated from its standardization by a sentence.

15. **Native scope is not model validity.** In the set-operation fragment the covered native scope is sound for validity and properly smaller than it (`PreambleFragmentVertical.native_scope_proper`); a theorem of the node is accepted by no certificate of the guarded checker (`theorem_not_accepted_by_guarded_checker`); universal membership is refuted by a model, outside the node, and rejected by the guarded checker with every proof (`universalMembership_separated`). The service's target, intrinsic theoremhood, and the node's target, semantic consequence, are distinct objects (`nativeProofService_target_meaning`).

16. **Dependent variation is not simple.** A varying Boolean family is inhabited at one global point and not at the other, while every translated simple type is point-independent; the constant-family comorphism is not surjective on sentences (`CwfSimpleDependentInstitution.varying_observation_not_in_simple_image`, `DependencyDescentCriterion.varying_outside_image`).

17. **The global-inhabitation institution is a readout, not a semantics of terms.** Point-satisfaction reads inhabitation only: distinct constant families with inhabited fibres are satisfied by the same points (`DependencyDescentCriterion.readout_identifies_inhabited_types`), and the tower's satisfaction observer forgets which term witnessed inhabitation (`CumulativeTower.CwfInhabitationInstitution.inhabitation_observer_is_not_witness_complete`).

18. **Raw simple-term erasure has no natural global-point reduct.** Two distinct closed terms of type $o \to o$ erase to the same tower term:

$$t_1 = \lambda x : o.\, (\lambda z : (o \to o).\, x)(\lambda y : o.\, y),$$
$$t_2 = \lambda x : o.\, (\lambda z : ((o \to o) \to (o \to o)).\, x)(\lambda y : (o \to o).\, y).$$

Both are well typed and have equal extensional meaning. Their intrinsic application types distinguish them, while unannotated erasure forgets those types (`SimpleFragmentErasureBoundary`). For a context functor F preserving the terminal context 1, a natural reduct $\beta : D(1, F(-)) \Rightarrow C(1, -)$ would satisfy $\beta_\Gamma(F\sigma) = \sigma$ for every closed substitution $\sigma : 1 \to \Gamma$, by naturality at σ and uniqueness of $1 \to 1$. Therefore such a reduct forces injectivity on global points (`GlobalPointReduct.pointTransport_injective`). The displayed collision contradicts that requirement. Consequently no institution comorphism has this raw signature functor, regardless of its sentence map (`SimpleFragmentInstitutionBoundary.no_comorphism_over_erasure`). The distinction between a CwF morphism and an institution comorphism concerns the extra natural model-reduct data [14]. This raw obstruction remains valid. The term-class map modulo conversion is now proved injective, as recorded above; that result does not supply a model reduct. The stronger inhabitation mismatch below survives even after passing to simple beta classes.

19. **The permissive tower does not reflect simple inhabitation.** The current `Presentation.HasType.lamIntro` rule does not request formation of its domain or codomain, and `HasType.conv` does not request formation of its target type. These omissions have a concrete consequence. With o the translated ground type, set

$$D = \lambda x.\, \Pi(y : x\, x).\, o, \qquad A = D\, D, \qquad \delta = \lambda x.\, x\, x, \qquad \Omega = \delta\, \delta.$$

One ordinary beta step gives $A =_\beta \Pi(y : A).\, o$. A variable $x : A$ can therefore be used as both function and argument; raw lambda introduction and conversion give $\delta : \Pi(y : A).\, o$ and $\delta : A$, hence $\Omega : o$ in the empty context (`SimpleFragmentInhabitationBoundary.omega_typed_ground`). Moreover $\Omega \to_\beta \Omega$. Its Pure skeleton is neither strongly normalizing nor regularly typable. In contrast the intrinsic simple ground type has no closed term, by its interpretation in the empty set. It has no beta class either. Thus unrestricted inhabitation reflection is false, and there is not even a function from this contextual tower ground fibre to the corresponding simple raw or quotient fibre (`no_contextual_ground_class_retraction`). The empty telescope and final ground type are both formed, so adding endpoint formation alone does not repair the missing internal premises. The existing `BoundaryJudgments.TowerMeaning` predicates for both checking and synthesis also hold for this example (`omega_tower_check_meaning`, `omega_tower_synthesis_meaning`); this is a semantic assertion, not acceptance by an executable checker. This is a boundary of the implemented permissive relation, not of Martin-Löf type theory, the regular two-sort kernel, or Lean's kernel. It is compatible with conversion conservativity between two already translated simple terms. The separate four-rule declaration-aware structural checker does not include the lambda, application, or conversion rules used by this counterexample; no acceptance claim about that checker follows here.

20. **Extensional eta equality is not the selected beta equality.** In context $f : o \to o$, the intrinsic terms f and $\lambda x : o.\, f\, x$ have the same denotation in every function interpretation, but are distinct beta-normal forms. They are not beta-convertible, and the native conversion decider rejects them (`SimpleFragmentConversionDecision.eta_not_betaConvertible`, `rejects_eta`). No eta law has silently been added to the tower.

21. **A token certificate is not an exact boundary.** A checker whose certificate type is a point preserves theorem existence but cannot be fibrewise equivalent to a calculus with two distinct proofs of one theorem (`BiformCertificateBoundary.no_unit_certificate_equivalence_for_tagged_theorem`).

22. **Coarsening loses information, and cannot mint identity.** The class map of a genuine observer coarsening is not injective (`ObserverRefinementCanary.refinement_not_injective`). In the temporal specimen a coarse observer identifies two moments that a fine observer separates and a lineage relates, while their identity is not derivable; coarse observation mints neither fine behavioural identity nor the equations, and the marker family descends through the fine observer only (`IdentityNotions.Temporal.lineage_without_identity`, `coarse_cannot_mint_fine`, `coarse_cannot_mint_equations`, `marker_not_descends_coarse`).

23. **An altered meaning is not a biform theory.** Declaring every compiler run declined is refuted by the run on an empty snapshot, without executing anything else (`MainlineCallGuardBiformTheory.declinedMeaning_not_sound`).

24. **Private operational state is invisible to the logical reading.** Two modelled theories with one identical semantic target differ by a hidden bit that no native sentence observes (`ModelledTheory.SemanticPredicateCanary.same_nik_object_but_private_operational_state`).

25. **Unrestricted pointwise dependent products do not descend** along the route-family right action; a directed countermodel rules them out even up to fibrewise equivalence (Section 18, item on the q edge).

26. **A circular universe has no two-valued truth and no classical deduction.** No Boolean valuation of circular propositions commutes with negation (`no_boolean_valuation`), and every Hilbert system over them with modus ponens, K and S is trivial (`hilbert_trivial`). No strong Kleene formula defines "undetermined" (`System.undetermined_not_definable`), and the least fixed point does not hold that the liar is untrue (`System.liar_revenge`).

27. **No sound arithmetic stage certifies itself or reaches the standard model.** A sound stage does not prove its own consistency (`SoundStage.not_proves_own_consistency`), no sound stage contains its zoom-out (`SoundStage.not_next_le`), and none proves every truth of $\mathbb{N}$ (`truth_not_reached`). The Δ_1 condition is necessary: true arithmetic satisfies every other condition and has no Δ_1 presentation (`trueArith_not_delta1`).

28. **Additive revision of overlapping evidence is unsound.** It counts the overlap twice (`EvidenceContracts.revision_counts_overlap_twice`, `Controls.overlap_double_counted`); source-keyed maximum under-counts without per-source nesting (`Controls.sourceMerge_undercounts`).

29. **The total of a paired product is not a sample size.** Its variance equals that of independent joint samples only for single samples (`PairedSampleDispersion.paired_variance_eq_joint_iff`).

30. **The posited deduction total is not a derived confidence.** The sum of the premise totals is above the interval-derived evidence weight in one case and below it in another, with equality in a third (`EvidenceContracts.Controls.deduction_posited_overstates`, `Controls.deduction_posited_understates`, `Controls.deduction_posited_agrees`).

31. **PLN's tensor is neither conjunction nor the deduction direct path**, and fusing posteriors counts a prior twice (`EvidenceContracts.Controls.fuse_not_direct_path`, `Controls.prior_counted_twice`). PLN deduction without screening off fails (`Controls.deduction_fails_without_screening`).

32. **No ring annotation keeps possibility** (`AnnotationCapabilities.not_positive_of_ring`), and intuitionistic negation is not monotone in any nontrivial Heyting algebra (`EvidenceContracts.hnot_not_monotone`).

33. **A frontier is not complete for refinements, and the owner can revoke it** (`FrontierNegation.Controls.refinement_incomplete`, `Controls.owner_revokes`).

17.4 Conditional: proved under a stated hypothesis

1. **Zero average distortion reflects every pair only with support.** `Distribution.distortion_zero_iff` requires positive mass at every state. Without it, the checked point-mass example has zero average distortion while a hidden pair loses its distinction. Coherence and nonnegative observation-loss scores do not certify a beneficial change: `same_distortion_opposite_value_changes` separates these claims.

2. **Primitive qualifications extend to contextual preservation.** `FormationSensitive.Typing.step_preserves` lifts primitive head and declared-root preservation, universe regularity and Pi/Sigma conversion boundaries to every contextual step. It uses the refined beta and projection theorems and retains original conversion/cumulativity tails. The pure, concrete transparent and native hypothesis packages discharge the premises, as recorded above. Arbitrary declaration-extended packages and their iota rules must supply their primitive obligations; independently typed canonical instances do not discharge the general root law.

3. **Kernel acceptance implies validity on the covered scope.** In the set-operation fragment, every covered formula is valid (`SetOperationSemanticAuthority.covered_valid`); outside the covered scope the raw Mathdata kernel's soundness for the membership semantics is not established.

4. **Semantic validity of a compiled family** is exact specification production together with two authority hypotheses, a well-formed snapshot and currency at the owning space (`CompiledGuardFamily.ValidFor`); the biform theory certifies the production, the hypotheses are the caller's.

5. **The cartesian lift is available for every arrow**; that its fibre is nonempty at a given node is a property of the node's theory, not of the fibration (the empty-theory node has models, a contradictory theory has none).

6. **Exactness of the simple-to-dependent route holds on the constant-family image only** (`CwfSimpleDependentInstitution.entails_iff`); outside that image the descent criterion says what fails. This compares the two set-family institutions; it is not a conservativity theorem for intrinsic simple syntax translated into the cumulative tower.

7. **PLN deduction holds under screening off** (`EvidenceContracts.deduction_exact_of_screening`); source-keyed revision is ledger union under per-source nesting (`ledgerEvidence_union_eq_sourceMerge`).

17.5 Open, with the smallest step that would decide each

1. **Distinction-calculus generation and broader semantics.** The finite completion checker is proved sound for least completion, but no general certificate-producing shortest-path algorithm, generator completeness theorem or C correspondence is supplied yet. The two papers' real-valued analysis, statistical allocation, partial correspondences, hive/gluing and metacognitive results remain outside this rational fragment. Extend them according to a required consumer; they are not all prerequisites for the first usable Prime draft.

2. **The full HOTG vertical remains unconstructed.** The set-operation fragment of the preamble is a node with a model and a native-proof service. The complete HOTG axiom theory and its intended universe model still require an explicit accounting of metatheoretic strength. This is not an expressibility obstruction from monomorphic HOL: Brown, Kaliszyk, and Pąk present HOTG using base types, propositions, and arrows [58]. Megalodon's prefix polymorphism additionally packages definitions and theorems uniformly across types; its preamble defines equality at a type parameter and states functional extensionality at two type parameters. Comparing those native schemes requires more than a constructorwise monomorphic translation, but is a separate obligation from presenting the HOTG axiom theory itself. The limited polymorphism has a documented predecessor: Egal uses it for choice, equality, extensionality, and reusable definitions and theorems, while deliberately restricting type variables in the hope of eliminating them [57]. A type-indexed family of monomorphic instances and a native polymorphic declaration therefore require an explicit instantiation comparison, not an informal identification. The plain-type instantiation laws, intrinsic HOL retyping, extensional proof transport, and full-domain satisfaction condition are now proved. A separate type-derived signature category now has identity, composition, a sentence functor, derivability transport, and a consequence-only theory-graph fibre; it faithfully includes the old constant-only category without altering the fixed-base Henkin institution. Full-domain reducts now compose as model records and give a model-valued institution, a soundness comorphism from proof consequence, and satisfying-model transport for a non-logical object-theory axiom. Native monomorphic terms now have an exact intrinsic interpretation, with native substitution and local beta-denotation compatibility. Capture-avoiding native type specialization now agrees with intrinsic retyping and preserves native typing on the supported fragment. The named generic equality declaration, its unfolding, and a generic reflexivity proof have checked instances, with an exact model-side equality criterion. Checked monomorphic definition unfolding now has exact finite-fuel success/failure, native typing preservation, and same-model denotation preservation under independent declaration equations. Successful plain-term normalization now has a typed, meaning-preserving interpretation. Native proof soundness covers the recursively specified monomorphic proof constructors and directly supplies an operational invariant theorem. Decisive next steps are prefix-polymorphic declaration and proof interpretation, the required completeness/termination scope, any claimed comparison with a foreign checker, ordered definition admission with stable dependencies and model expansion, and an explicit statement of the universe strength required by the intended HOTG model. Non-discrete model morphisms and general non-full Henkin reducts remain separate semantic extensions. The checked mixed-substitution boundary makes the fragment restriction essential: raw inference and scope alone cannot justify extending the monomorphic interpretation through arbitrary type abstractions. The new all-sort ABT operation preserves the tested binding, but does not change the existing port or prove a larger native checker sound. For non-full Henkin models, replacing a base carrier by an ambient function space need not pre-

serve admissibility; replacing it by an admissible-value subtype still requires a comparison between source base equality and target extensional function equality. Neither obligation is discharged by the full-domain theorem.

3. **The logical tower target still needs qualified conversion and consumer coverage before adoption.** The strict syntactic CwF morphism is constructed; a comorphism between its raw global-inhabitation readouts is refuted for that signature functor. A separate constant-family comorphism compares set-family models, whose contexts are sets rather than native telescopes. Conversion between translated simple terms is now preserved and reflected, and the simple quotient now includes substitutions and comprehension. However, inhabitation reflection from the current permissive tower is refuted, not merely open; quotienting source beta conversion cannot repair it. The separately named formation-sensitive candidate now has raw erasure, renaming, substitution, regularity, root beta preservation for the actual root-empty Tower package and qualified nonempty transparent definitions, all primitive/chain quotation admission, and finite native list-relator constructor admission. The MIL eliminator's declaration and canonical primitive/chain iota schemas now have formed judgments and typed substitution instances retaining their receipts. Declaration-spine recovery derives its general primitive preservation. A conservative proof-only completion and its proved parallel diamond now discharge the actual Pi/Sigma conversion boundaries, and full contextual preservation follows without those premises. Authored execution confluence and normalization are not asserted by this route. The unchanged five-root List/identity/mapRel package now also has independent refined declaration and schema formation, typed-substitution instances, proved Pi/Sigma boundaries and full displayed-type contextual preservation. This is a separate qualification, not inferred from the hypothesis package. Full contextual preservation is now proved for the actual root-empty Tower, both native packages separately and qualified transparent packages with independently checked bodies. The generic theorem reduces extensions to primitive preservation and conversion-boundary obligations. Finite conversion checking is now executable and exact for the original MIL, List/identity and List/identity/mapRel packages separately. Any advertised normalization or total conversion-search capability still needs its own justified scope; combining the packages also requires interaction qualification. Checked wrong-body and cyclic-definition controls separate preservation and termination from conversion safety. The existing intrinsic atom/arrow translation and its substitutions now land in refined typing as well, with exact conversion comparison on its image but without intrinsic annotation recovery. A separate declared interface now represents the supported actual HOL formulas, including the operational invariant, without identifying formation with proof. Finite dependency-closed declaration/computation restriction is constructed within this grammar; restriction to a simpler grammar or universe profile remains different work. A valid restriction can retain proofs and reflect all its judgments while failing preservation of a retained unfolding; the missing reduct declaration and a repair with full contextual preservation are now checked. The remaining decision is which native packages must share a computation context, and which required consumers can instead use explicit qualified service targets. A combined MIL/List/identity/mapRel context requires interaction qualification; the separate preservation results do not supply it. An executing restriction similarly requires its own schema/declaration-closed qualification, whereas proof-local restriction is already established. Stronger grammar readback or model interpretation is required only for a route the draft actually advertises. Normalization and canonical forms must be established wherever the selected value core's advertised checking contract requires them; finite positive evidence checking does not assert a total conversion decision. No canonical typing rule was changed by

the candidate tranche. A replacement or restriction of the canonical logical target requires an explicit design decision and a consumer migration; it is not implied by renaming a node.

4. **The planned CBPV foundation needs a qualified strategy interface.** The current need protocol provides revision-keyed sharing laws. Exact contextual sequencing now constructs admitted native dependent pairs, with native value substitution and effect-sensitive negative controls. Actual first/cached need consumption now validates dependent packets and feeds native conversion replies into formed admission. These connections now include reified scoped return/sequence/choice/call syntax, independent computation typing, typed substitution and native result preservation through exact handler realization. Separately scoped nonrecursive need binding and force now execute captured source code on the actual owned machine, with independent typing, local heap-extension and continuation laws, and sharing/failure controls. Whole-control preservation now derives the logical heap invariant through complete control/commit stacks from independent initial typing and primitive result qualification. Bounded frontiers retain that invariant even when unfinished. Independent natural semantics now corresponds exactly to finite completed runs, including fault status and full worlds; every observed bounded answer has a source derivation. Unfinished frontiers remain operational residuals, not negative judgments. A concrete Kleisli polarity interface now recovers the existing computation monad; selected semantic thunks support qualified reuse and native dependent admission, while immediate demand corresponds exactly to allocated eager sequencing. A fully faithful algebra embedding now supplies genuine computation products and value-indexed powers, with contextual sequencing, stack and reindexing laws. Actual native proof services exercise application and distinguish provider selection from per-call effects. Simultaneous mixed-sort ABT binding now has scope, substitution and opening coherence, with actual cached-cell observations detecting capture that structural checks miss. Native-domain computation bodies separately have dependent application and captured-environment admission through the existing Need realization. A native-indexed polarized candidate now adds actual first-class source thunks, computation functions, dependent index/closure packets and all-scope substitution. Independent typing, typed substitution and principal opening preservation retain the displayed native dependency; no reverse typing conservativity is implied. Lexical execution uses a conservative silent-step extension; the old machine's complete ordered frontiers embed exactly, with all world data and work retained. Its richer runtime now has general typing preservation through captured environments, cached functions, local control and complete commit stacks. Fresh allocation preserves existing declarations and admits new captured references; observed closures remain coupled to their qualified final heap. Administrative polarity faults are excluded for all bounded runs. Independent source evaluation now corresponds exactly to completed finite actual-machine runs, retaining the final heap and complete receipts for rich answers and raw faults. Its admission transfer preserves that same final heap; bounded noncompletion is not source refutation. Principal ordinary-force, lexical application and packet-opening equations now preserve exact completed answers and worlds. Their equivalence survives the same active higher-order consumer, with admission transferred from an independently typed complete source consumer. Immediate shared demand and allocated returned-value sequencing have an exact-world raw projection; independent returner typing makes their outcomes equal. These are not unrestricted source congruences or equal-cost rewrites. Actual unused producer capture refutes full-world congruence, and typed thunk-returning applications distinguish lexical opening from syntactic substitution at the literal closure observer; their native force probes agree. Required remaining obligations are any further chosen value/computation equations, their qualified observa-

tions and workload-required type formers. A raw mathematical-service data/admission route now executes checking and exact-request revalidation, with all-completed-answer reflection and a proof-relevant MIL consumer. Higher-order passage of the reified checker now retains that connection, and a separate executable exact-tree materializer supplies its native artifact. Actual input-sensitive provider selection also distinguishes shared and independent choice while retaining both raw articles and verdicts. These do not by themselves complete arbitrary provider composition or the remaining integrated workloads. Requested divergence, fairness or cost observations beyond finite correspondence also need their own contracts. Native mathematical payloads are not yet classified or normalized CBPV values. No implicit mutable-state handler follows from effect receipts. The occurrence-sensitive Kleisli instance's checked absence of a terminal computation object rules out using its free returners alone for all finite computation products. The constructed algebraic alternative answers that obstruction, but its connection to the reified polarized source and its operational equations remains to be established. The allocated-sequencing unit counterexample rules out unqualified equality at the full work/receipt observer. Neither construction is an adopted full dependent model. An explicit need binder and an implementation by qualified state are candidates, not interchangeable without proof. No default eager policy, cost optimum or unrestricted dependent sequencing is already established. The existing contextual-effect evaluator now has a general read-only reuse/discard law and checked effect-sensitive counterexamples; those laws cannot be generalized to divergence or other effects without qualification.

5. **StructuredC passes do not yet establish arbitrary C hosting.** `MainlineCallGuardStructuredCPass.coldToStructuredCPass` covers the canonically admitted cold language by generated StructuredC invocation runs. The hot source contains a separate step-level cover (`MainlineCallGuardExecuteStructuredCPass.executeToStructuredCPass`); the relevant umbrella now rebuilds against the current Lean executor. Its older mode-dispatch cover is only a mode projection. These constructions retain canonical-entry admission, explicit handlers and selected execution strategies. The remaining C-facing qualification and the strongest justified extension beyond admitted states must not be inferred from a generated program or from the mode-dispatch table. They belong to the lowering lane, not to selection of Prime's logical foundation.

6. **Modal syntax.** Locks and modal types have no explicit 2-cell action, and the M_1 versus M_2 gate is undecided (Section 18). Decisive step: exhibit or refute a transport discriminator inside one mode-cell fibre.

7. **Prime is not a fixed representation.** Its returned-fibre premises are undischarged, so it is routed neither through OSLF nor through the theory graph. Decisive step: the five premises, one by one.

8. **Bootstrap seeds.** No "NIK/x over y" cell exists in the tree; the seed matrix of the bootstrap section is entirely open.

9. **Transfinite consistency and reflection progressions.** Only the finite levels of the consistency tower are built. The smallest deciding step is a uniform Δ_1 presentation of the finite tower, from which the ω-th stage is a sound stage; then the strict climb through $\omega + 1$. Completeness of progressions along notations is a further step.

10. **A structural definition of typed self-interpretation.** The published criterion for a typed self-interpreter admits a value-carrying interpreter for System T, and the type of codes

cannot be simpler than the type of the program [131]. A satisfactory definition should present the interpreter as a structure-preserving homomorphism between calculi, which commuting with substitution alone does not capture. The library's code/behaviour retraction theorem (`exists_fixedPoint_of_staticBeta`) already gives the single-code-type obstruction. The smallest deciding step is a candidate definition, checked to hold for a type-indexed quotation of a small total calculus and to fail for the value-carrying representation.

11. **Deduction intervals with interval marginals.** Conclusion confidence is derived through credal intervals (Section 3.5); the corner theorem covers the deduction map with point marginals p_B, p_C, where it is bilinear. With interval-valued marginals the map divides by $1 - p_B$ and the exact image of the box is not given by its corners in general. The smallest decisive step is the exact envelope of the deduction map over the full box, or a corner theorem under a stated monotonicity condition.

12. **Decided knowledge-based programs over reachable states.** The ranked criterion covers one round; the multi-round case with must/can bounds remains open.

13. **A native absence source with ownership certificates.** The frontier and ownership theorems are proved over an over-approximation of firing; a frontier check over MM2 executables with owned heads, and its agreement with the kernel model, is the decisive step.

18 Open obligations and decision gates

18.1 Modes and multimodal dependent syntax

1. The locally preordered candidate reflection of the checked three-mode two-computad is now an actual bicategory with the three adjunctions and the factorization isomorphism, the full **Cat** semantics descends by the point-probe theorem, and the fixed-action mapping-out law is checked. Prove its full bicategorical universal property and exact nondegeneracy and essential-image statements.

2. Construct the corresponding proof-relevant candidate and either exhibit two parallel cells that survive those equations and have distinct actions on a required dependent consumer, or prove every currently required action fibre-invariant. The rho and graph point-substitution cells already provide positive context-level discriminators, but do not inhabit a mode-cell fibre. The gate itself is now exact: parallel cells differ if and only if a whiskered dependent family distinguishes them, so the obligation is to exhibit or refute such a witness inside a mode-cell fibre. This remains the decisive M_1 versus M_2 gate.

3. Embed the existing one-categorical mode theories as locally discrete 2-categories and prove that this conservatively recovers their modal CwF operations and product-axis laws.

4. Connect the descended semantic pseudofunctor to the modal CwF action, proving that selected unit, counit, interchange, and factorization cells act coherently on locks, types, terms, and substitutions. Reuse the checked jointly faithful action of context transformations on families, sections, and comprehension, while keeping substitutions inside one CwF distinct from morphisms between mode-indexed CwF doctrines.

5. Extend the checked dependent-right-adjoint structure at the q edge. The route-family and extensional-family CwFs, context functor q, right type action Δ, natural term equivalence, and

proper-image theorem are now checked. Dependent sums and extensional fibrewise identity are now checked on all route families; dependent products are checked on the invertible-transport fragment, every Δ-pulled family lies in that fragment, and a directed countermodel rules out unrestricted pointwise products even up to fibrewise equivalence. The exact comprehension equivalence for the right action, its inverse substitution laws, and full transport preservation for Σ, restricted Π, and extensional identity are checked. Pairing, both projections, lambda, application, and extensional reflexivity commute through the same comparison. The category-indexed proof-relevant CwF now independently supports the same Σ/extensional-identity/restricted-Π boundary, including its groupoid positive case and directed representable counterexample. Prove intrinsic J and universe closure; connect both doctrines and their operations to the MTT lock syntax; and carry the rho and graph specimens through them. Add a pushforward only if a named consumer needs one; arbitrary-family lossless descent is already ruled out.

6. Extend locks and modal types with explicit action of 2-cells; prove modal substitution, weakening, lock composition, and coherence without treating propositional equalities as definitional by convenience.

7. Add Σ, identity and J, cumulative universes, and the selected quotation operations to the MTT syntax; state the strict, intensional, and cubical identity branches separately.

8. Decide whether CBPV value/computation polarity is an internal adjunction at each mode or a further mode axis by carrying a genuinely dependent effectful term through both alternatives.

9. Combine guarded revision with the triangle, specifying which axes commute strictly, which commute by invertible 2-cell, and which interactions are intentionally noncommutative.

10. Construct intrinsically typed initial syntax, elaboration, conversion, and a checker for the selected decidable mode theory.

11. Prove subject reduction, normalization or an equally strong conversion argument, canonicity, gluing soundness, and consistency relative to a named nondegenerate model.

12. Run bounded Agda, *mitten*, and BiSikkel discriminators against the same mode signature and specimen, recording exactly which dependent and 2-cell obligations each system can express.

18.2 Native proof authority and NIK composition

1. Define the common service contract: an independently stated semantic target served by one of four faces; the judgment, certificate, checking, scope, meaning, and soundness contract is the certificate-boundary face only, and resource accounting is a coordinate of selection.

2. Give each DTT, GSLT, SUMO, HOL/OpenTheory, Metamath, and Megalodon node its actual native inference strength; do not replace available native rules with opaque certificate replay.

3. Define checked routes with judgment and certificate maps, checker commutation, semantic preservation, essential image, proof-fibre behaviour, universe shifts, and cost.

4. Extend the indexed family of services over targets, with transport along target equivalences and face-dependent partial reindexing. State the laws required for any further categorical

organization without manufacturing a common judgment meaning that none of the nodes has.

5. Define and verify the dispatcher: scope containment, native-node preference, ambiguity rejection, route provenance, and explicit optimization objective.

6. Distinguish theoremhood in a selected environment, external semantic validity, model satisfaction, and executable reduction at every node.

7. Complete the direct SUMO higher-order core across the needed connectives, quantifiers, variable discipline, class terms, domain restrictions, and source-axiom fragment, with source-derived positive and negative tests.

8. Prove at least one conservative round trip and one deliberately nonfaithful route between each adjacent authority family in the selected NIK graph.

9. Carry dependent rho, graph-walk, and ethical reason certificates through node selection and back, retaining their observer and resource profiles.

18.3 Bootstrap and independent qualification

1. Give a typed meaning to a seed cell "NIK/x over y", separating certificate syntax, checker logic, object theory, export format, and qualification base.

2. Instantiate at least two cells from the OpenTheory versus Pure-or-MM0 and PA/Tpr versus HOL matrix before selecting a preferred bootstrap path.

3. Supply two independently implemented checkers for every retained seed contract and compare their accepted certificate languages extensionally.

4. Prove each checker's soundness in its declared qualification base and state every induction, arithmetic, choice, extensionality, and universe assumption.

5. Prove exact import/export round trips, certificate canonicalization where required, and conservative-fragment theorems between seed cells.

6. Replay the complete declared NIK metatheory and target corpus; include malformed, mutated, wrong-scope, and wrong-theory negative certificates.

7. Perform generator-severance and dependency audits so that a seed checker does not silently depend on the system it is meant to qualify.

8. Maintain separate bootstrap-index, trusted-code-size, object-strength, checker-work, and semantic-strength orderings; prove comparisons rather than inferring one from another.

9. Add strong-model qualification only as a separate relative-model lane, never as self-soundness or as a replacement for syntactic checking.

18.4 Vertical interconnections and decisive specimens

1. Carry the typed rho communication cell through $D_0 \subset D_1 \subset D_2$ and across all identity, computation, time, universe, and modal-cell-identity branches, then through NIK and a runtime observation theorem.

2. Prove raw-term no-junk for the graph native-walk judgment by inverting successful guarded reverse scans into edge and Mathlib walk witnesses.

3. Generalize the graph result into a reusable theorem identifying when a LanguageDef plus operational GSLT generates a sound and complete native indexed judgment rather than merely a structural grammar.

4. Transport graph proof identity, composition, reversal, homomorphic map, and exact-work certificates through the chosen modes and prove operational naturality of the transported executions.

5. Instantiate the DTT-to-HOTG adequacy square on the same rho and graph objects, recording coded essential image, nonfaithful readout, and every level shift.

6. Define one comparison-profile algebra and composition law used by mode maps, theory interpretations, NIK routes, bootstrap exports, observers, and runtime realizations.

7. Prove a two-sided GSLT–runtime comparison at each declared observation, including effects, occurrences, revision, ownership, cost, and scheduling assumptions.

8. Use the ethical allocation as a high-level consumer: verify that the selected metatheory supports rule reasoning, empirical optimization, learnable virtues, multi-agent views, consent and authority receipts, and uncertain concept revision without identifying them.

18.5 Coalgebra and set theory

1. Instantiate the indexed final-coalgebra theorem for a small observer graph and the exact functor H_V.

2. Construct the route-rich functor F_V, its final coalgebra, and the induced erasure morphisms q_o.

3. Prove at least one non-collapse witness and characterize exactly which families descend along each q_o.

4. Interpret the construction inside a pinned HOTG theory while keeping host membership distinct from coalgebraic membership.

5. Build a two-stage Grothendieck–Indra fragment and prove coherence before claiming a class-sized atlas.

18.6 Type theory and computation

1. Realize the compositional contracts of Section 3.13 on complete retained MeTTa worlds: native matching, ordered occurrences, absences, take/update, consulted spaces and rules, binding lineage, shared producers and force authority. Compose their actual decoders and actions with the independently selected observation, oracle, progress and cost comparisons.

Construct an authorized obligation-disposition service and irreversible/uncertain controls; preserving a readout alone does not authenticate settlement or guarantee eventual discharge.

2. Construct a native inductive–recursive family-enclosing universe and prove its substitution and decoding laws.

3. Construct native superuniverse and extended-predicative Mahlo candidates with their exact elimination rules; prove the capability implications and run consumer-guided conservativity ablations against plain enclosure.

4. Establish a common-base interpretability or equiconsistency comparison before claiming that a Mahlo-DTT profile is above, below, or equivalent to HOTG.

5. Instantiate the checked abstract adequacy algebra with the checked HOTG family host, then prove contextual substitution naturality, constructor commutation, and explicit level shifts for one concrete native syntax.

6. Define the category of host objects equipped with native code witnesses and determine whether the chosen syntax-to-HOTG interpretation is full, faithful, or only essentially surjective after quotienting by a declared semantic equality.

7. Construct the relevant elementary quotient completion for the selected dependent doctrine, prove its coherence with substitution and comprehension, and establish conservativity or Morita equivalence. Do not substitute the checked set-level kernel-pair quotient for this obligation.

8. Run at least one category-of-models or presheaf consumer through the enclosure service before considering a stronger kernel rule.

9. Compare concrete intensional rule packages on typed Π conversion, J, scoped UIP/path structure, universes, quotation, and effects.

10. Prove CBN-to-CBPV and CBV-to-CBPV translations for the selected dependent fragments, including substitution and observation preservation. For the planned Prime basis, identify exactly which strategy transformations preserve the required demand/effect observations and under which hypotheses.

11. Prove the selected call-by-need realization with ownership, revision and sharing-sensitive choice/effect observations. Compare with CBN only where the fragment and observer justify that comparison.

12. Establish the two-sided realization square between the operational interlingua and a real runtime at a declared observation.

13. Make the planned CBPV foundation implementation-actionable: decide its surface elaboration and defaults, qualify sharing, and prove the selected connection to the current need nucleus. CBPV is no longer only a comparison target; the particular syntax, effects and realization still need selection.

18.7 Ultrafilter and nonstandard semantics

1. Extend the explicit tail construction to a constructive-real or rational profile with moduli or apartness, stating exactly which order and field laws survive.

2. Prove a language-level Loś theorem for the selected first-order or many-sorted fragment, rather than relying only on algebraic equation transfer.

3. Characterize the dependent families and observers that descend through the principal and free quotient maps.

4. Connect one free-ultrafilter perspective to a small indexed coalgebra and exhibit independent witnesses for nonstandard magnitude and non-well-founded depth.

5. State every use of choice, transfer, saturation, and universe enlargement in the model profile.

18.8 Formal ethics and multi-agent cognition

1. Complete the source-ontology hierarchy and axiom-dependency trace, especially action, intention, social relation, value conflict, and metaethical correspondence.

2. Lift the six MPE translations from evaluated judgments and pinned model fragments to arguments, entailment, justification, learning dynamics, and theory revision.

3. Define translation profiles recording computability, code-size growth, evaluation cost, approximation error, learnability, and observer boundary; prove at least one genuine succinctness or complexity separation.

4. Extend the FCA/WM–PLN bridge to uncertain concept revision, source scope, and conflicting evidence while retaining the typed ontology.

5. Extend the checked allocation core from care membership, consent claims, authority, aggregation, and cost to bargaining, reparative duties, scheduler fairness, and conflicts among values and virtues.

6. Carry the checked allocation specimen beyond proof-relevant GSLT events, dependent reason receipts, and extensional observation into NIK replay and a live runtime realization.

18.9 World-model evidence and probability

1. State every evidence-combination step with its observation contract: identified ledgers for revision, the same-world interval for conjunction, sample pairing for independent products, conditional independence for fusion of reports, and screening off for deduction (Section 3.5). Under unknown overlap, combine by the coordinatewise maximum, which is sound for every overlap. Derive every conclusion confidence through the credal-interval route, declaring the prior strength and the selector; replace posited confidence totals, such as the sum of premise totals in deduction and the weight product in conjunction.

2. Make authorities for probability assumptions explicit: factorization, conditional independence, Bayesian-network separation, credal bounds, and exchangeability. A compiled evidence-combination plan should discharge its side condition by a checked semantic decision. The finite d-separation decision and its occurrence-preserving discharge into a closed

derivation are checked; a correlated-packet model shows that disjoint provenance coexists with stochastic dependence, so source non-reuse cannot discharge the obligation.

3. Specify and compare lawful forgetting profiles for world models, including provenance retention, causal deletion, and observer-relative readout. The current revision and extraction soundness laws are checked not to determine forgetting. Two further laws are now checked as explicit fields of a forgetting profile: forgetting never increases extracted evidence, so observation counts never rise; and, with support tracking, retained support shrinks and avoids the forgotten footprint while nonzero evidence is always supported, so every conclusion supported only inside the footprint is deleted and every survivor keeps a source outside it. A reset-to-prior layer satisfies the earlier interface and violates the first law. The remaining obligation is to compare provenance-retaining, causal-deletion, and observer-relative profiles against these laws on the live world model.

4. Lift countable disjunction and conjunction to the monotone-certificate and obligation-tower interfaces inside the object language, proving eventual completeness from justification coverage rather than assuming it from fair scheduling.

18.10 Proof search and generated code

1. Prove lazy frontier-store refinement and unique inference ownership for the given-clause loop.

2. Connect fair schedules to refutational completeness and exact replay.

3. Express every compiler stage as an actual semantics-preserving transformation; generated C should be a compact residual program, not a table dump of the source relation.

4. Measure specialization, decision trees, partial evaluation, and staged interpreters against correctness-preserving baselines before retaining tables.

5. Parse every official TPTP record family in a generated native parser with a source-faithful carrier and physical include and provenance resolution; parsing selects no execution semantics.

6. Elaborate TH0 in scope to the intrinsic typed core with an explicit residual, and qualify the elaborator on positive and adversarial corpora.

7. Provide substitution, $\beta\eta$ conversion, higher-order unification, superposition, inductive learning, and proof planning as separate checked services over the same typed waist, each with its stated completeness boundary and independent replay.

8. Admit Horn execution only as a plugin with an exact fragment recognizer, a least-model or immediate-consequence theorem, and replay.

9. Instantiate the checked exact all-input refinement composition with the concrete generated parser and checker; agreement on canonical examples is proved insufficient.

10. Attach strategy profiles to execution spaces, tag every knob with the guarantee class it may affect, and prove the meet law for cross-space queries together with fairness of the combined schedule.

19 Conclusion

The proposed construction has a sound core, but only conditionally. A small-based observer-indexed functor in a suitable category of classes can have a final coalgebra; a bare set equation with unrestricted powerset cannot. HOTG can host the construction precisely when the non-well-founded membership is an internal interpreted relation and universe shifts are explicit. A coherent open-ended atlas is one possible contract for infinite reflection: every finite constellation is eventually reflected and no observer is final. Until its coalgebras and coherence maps are constructed, however, it is not an ultrainfinitist model or a source of foundational strength.

Dependent type theory can recover much of the elegance of Tarski–Grothendieck universe selection without pretending that types literally are Grothendieck sets: enclose the family at hand in a closed Tarski universe, price the predicative level shift, and interpret the result into a named extensional host. The practical purpose of the tower is controlled reflection: lower theories, models, checkers, observer worlds, and categories become data at a later level without licensing same-level self-soundness. The remaining gap is a native inductive–recursive calculus with satisfactory conversion and a proved end-to-end HOTG adequacy square. Superuniverses and Mahlo reflection are meaningful stronger candidates because they internalize selected and arbitrary family operators, respectively; whether either is stronger than HOTG in a common foundational ordering remains a theorem to prove, not a consequence of their different model constructions. The exact reverse bridge is the semantic-code quotient equivalent to the coded essential image, leaving extensional-only host sets honestly outside native computation. An explicit tail quotient supplies choice-free relative infinity at the price of partial order; a principal ultrafilter is a collapse control; and a free ultrafilter supplies a classical nonstandard point. None substitutes for the separate coalgebraic construction of non-well-founded depth.

Alternating qualification towers likewise add no strength through their shape: a countable directed helix linearizes. Their valid content is later-model soundness, finite certificate support, directed-union no-bottom, and whatever genuine interpretations or reflection principles their stages carry.

The operational–intensional–extensional diagram is therefore a checked semantic guide, not yet a completed multimodal dependent type theory. The next foundational step is to generate its mode 2-category, connect 2-cell action to locks and substitution, and prove a nondegenerate dependent model. Its comparison sector now carries a local truncation certificate: the thin fast path is licensed there by theorem, while higher cells remain available everywhere and are admitted exactly where a dependent consumer distinguishes them. The typed interchange core for higher-order problems shows the same discipline at the authority boundary: substitution, conversion, unification, search, and replay are separate services over one waist. NIK then sits beside this calculus as a family of native services over its semantic targets, while the bootstrap matrix independently qualifies selected authorities. Agda, its two-level and guarded fragments, *mitten*, and BiSikkel are valuable bounded comparators; none removes the need to prove the dependent substitution and coherence laws of the selected mode theory.

The same non-collapse discipline organizes dependent typing, CBPV, lazy proof search, quotation, cost, ethical reasons, and theory qualification. The typed rho cell shows why revision-dependent evidence must remain beneath endpoint readout, while the generated graph-walk judgment tests whether operational rules really induce an indexed native proof theory rather than a renamed host predicate. Exact Moral Paradigm Equivalence shows the complementary lesson: different native theory structures may encode the same judgments while paying different computational and learning costs. Preserve the proof-relevant total space; derive extensional faces through observers; require actual programs for transformations and actual models for qualification; and use

decisive experiments before selecting a native calculus. This gives the emerging theory a coherent north star without pinning a living language specification too early.

References

[1] T. Dinsdale-Young, L. Birkedal, P. Gardner, M. Parkinson, and H. Yang. Views: Compositional reasoning for concurrent programs. In *POPL*, 2013. https://cs.au.dk/~birke/papers/views.pdf.

[2] C. B. Jones. Tentative steps toward a development method for interfering programs. *ACM Transactions on Programming Languages and Systems*, 5(4):596–619, 1983.

[3] P. Martin-Löf. *Intuitionistic Type Theory: Notes by Giovanni Sambin of a Series of Lectures Given in Padua, June 1980*. Bibliopolis, 1984.

[4] P. B. Andrews. *An Introduction to Mathematical Logic and Type Theory: To Truth Through Proof*, second edition. Kluwer Academic Publishers, 2002.

[5] P. B. Andrews, D. A. Miller, E. L. Cohen, and F. Pfenning. Automating higher-order logic. In *Contemporary Mathematics*, vol. 29, 1984. https://www.lix.polytechnique.fr/~dale/papers/AutomatingHOL84.pdf.

[6] M. Hofmann. *Extensional Concepts in Intensional Type Theory*. PhD thesis, University of Edinburgh, 1995. http://hdl.handle.net/1842/399.

[7] J. Carette, W. M. Farmer, and Y. Sharoda. Biform theories: Project description. In *Intelligent Computer Mathematics*, LNCS 11006, pp. 76–86, 2018. https://arxiv.org/abs/1805.02709.

[8] W. M. Farmer and M. von Mohrenschildt. Transformers for symbolic computation and formal deduction. In *Workshop on the Role of Automated Deduction in Mathematics*, pp. 36–45, 2000. https://imps.mcmaster.ca/doc/transformers.pdf.

[9] W. M. Farmer, J. D. Guttman, and F. J. Thayer. Little theories. In *Automated Deduction—CADE-11*, LNCS 607, pp. 567–581, 1992. https://imps.mcmaster.ca/doc/little-theories.pdf.

[10] W. M. Farmer. Little theories: A method for organizing mathematical knowledge. NatFoM 2023 and LFPNML 2023, 7 September 2023, slides 17, 30, and 56. https://europroofnet.github.io/_pages/WG4/Cambridge23/farmer.pdf.

[11] M. Codescu. Higher Order Logic with Henkin Semantics. Manuscript, Definition 4 and Remark 1. https://iks.cs.ovgu.de/~codescu/henkin.pdf.

[12] P. Koepke, A. De Lon, and A. Lorenzen. ForTheL for type theory. In *Fifth Conference on Artificial Intelligence and Theorem Proving*, 2020. https://aitp-conference.org/2020/aitp20-proceedings.pdf.

[13] J. A. Goguen and R. M. Burstall. Institutions: Abstract model theory for specification and programming. *Journal of the ACM*, 39(1):95–146, 1992. https://publish.lfcs.inf.ed.ac.uk/reports/90/ECS-LFCS-90-106/.

[14] J. Goguen and G. Roşu. Institution morphisms. *Formal Aspects of Computing*, 13(3–5):274–307, 2002. https://doi.org/10.1007/s001650200013.

[15] D. Patterson and A. Ahmed. Linking types for multi-language software: Have your cake and eat it too. In *2nd Summit on Advances in Programming Languages*, LIPIcs 71, Article 12, 2017. https://doi.org/10.4230/LIPIcs.SNAPL.2017.12.

[16] G. D. Plotkin and M. Pretnar. Handling algebraic effects. *Logical Methods in Computer Science* 9(4:23):1–36, 2013. https://arxiv.org/abs/1312.1399v2.

[17] D. Turi and G. D. Plotkin. Towards a mathematical operational semantics. In *12th Annual IEEE Symposium on Logic in Computer Science*, pp. 280–291, 1997. https://doi.org/10.1109/LICS.1997.614955.

[18] J. E. Ghalayini and N. Krishnaswami. The denotational semantics of SSA. arXiv:2411.09347, 2024. https://arxiv.org/abs/2411.09347.

[19] MLIR Project. MLIR rationale: A multi-level intermediate representation. https://mlir.llvm.org/docs/Rationale/Rationale/.

[20] J. Merckx, A. Lopoukhine, S. Coward, J. Cheng, B. De Sutter, and T. Grosser. `eqsat`: An equality saturation dialect for non-destructive rewriting. arXiv:2505.09363, 2025. https://arxiv.org/abs/2505.09363.

[21] T. Altenkirch and C. McBride. Towards observational type theory. Manuscript, 2006. https://personal.cis.strath.ac.uk/conor.mcbride/ott.pdf.

[22] T. Altenkirch, P. Capriotti, and N. Kraus. Extending homotopy type theory with strict equality. In *25th EACSL Annual Conference on Computer Science Logic*, LIPIcs 62, Article 21, 2016. https://doi.org/10.4230/LIPIcs.CSL.2016.21.

[23] L. G. Meredith and M. Stay. Operational semantics in logical form. Technical report, 2020.

[24] L. G. Meredith. *Finding Mind: The Mortal Scientist and the Ecology of Thought.* Working draft 23, August 2026. https://github.com/F1R3FLY-io/publications/tree/main/FindingMind/draft23.

[25] The MeTTaIL contributors. `mettail-rust`: executable language definitions with contextual equation closure. Source repository, 2026. https://github.com/zariuq/mettail-rust.

[26] B. Ahrens. *Initiality for Typed Syntax and Semantics.* Doctoral thesis, 2012. https://arxiv.org/abs/1206.4556.

[27] J. Koenig and Z. Shao. CompCertO: Compiling Certified Open C Components. In *PLDI*, 2021. https://doi.org/10.1145/3453483.3454097.

[28] L. Zhang, Y. Wang, J. Wu, J. Koenig, and Z. Shao. Fully Composable and Adequate Verified Compilation with Direct Refinements between Open Modules. *Proceedings of the ACM on Programming Languages* 8(POPL), Article 72, 2024. https://doi.org/10.1145/3632914.

[29] Y. Zhang, J. Koenig, Z. Shao, and Y. Wang. Unifying Compositional Verification and Certified Compilation with a Three-Dimensional Refinement Algebra. *Proceedings of the ACM on Programming Languages* 9(POPL), Article 64, 2025. https://doi.org/10.1145/3704900.

[30] D. Patterson and A. Ahmed. Linking Types for Multi-Language Software: Have Your Cake and Eat It Too. In *SNAPL*, LIPIcs 71, Article 12, 2017. https://doi.org/10.4230/LIPIcs.SNAPL.2017.12.

[31] D. Patterson, N. Mushtak, A. Wagner, and A. Ahmed. Semantic Soundness for Language Interoperability. arXiv:2202.13158, 2022. https://arxiv.org/abs/2202.13158.

[32] J. Launchbury. A Natural Semantics for Lazy Evaluation. In *Principles of Programming Languages (POPL)*, 1993. https://doi.org/10.1145/158511.158618.

[33] K. Nakata and M. Hasegawa. Small-step and big-step semantics for call-by-need. arXiv:0907.4640, 2009. https://arxiv.org/abs/0907.4640.

[34] P. B. Levy. Call-by-push-value: A subsuming paradigm. In *Typed Lambda Calculi and Applications*, LNCS 1581, pp. 228–243, 1999. https://doi.org/10.1007/3-540-48959-2_17.

[35] P. B. Levy. Adjunction Semantics For Call-By-Push-Value. Theory seminar, 12 March 2004. https://pblevy.github.io/papers/adjuncttalk.pdf.

[36] S. Antoy and M. Hanus. *Curry: A Tutorial Introduction*, Section 3.12, Lazy Evaluation. https://www.curry-language.org/docs/tutorial/html/curry-tutorial.Ch3.S12.html.

[37] D. McDermott and A. Mycroft. Extended Call-by-Push-Value: Reasoning About Effectful Programs and Evaluation Order. In *Programming Languages and Systems (ESOP)*, LNCS 11423, pp. 235–262, 2019. https://doi.org/10.1007/978-3-030-17184-1_9.

[38] C. Torczon, E. Suárez Acevedo, S. Agrawal, J. Velez-Ginorio, and S. Weirich. Effects and Coeffects in Call-By-Push-Value (Extended Version). arXiv:2311.11795, 2023. https://arxiv.org/abs/2311.11795.

[39] M. Vákár. A framework for dependent types and effects. arXiv:1512.08009, 2015. https://arxiv.org/abs/1512.08009.

[40] D. Gratzer, G. A. Kavvos, A. Nuyts, and L. Birkedal. Multimodal dependent type theory. *Logical Methods in Computer Science* 17(3):11, 2021. https://doi.org/10.46298/lmcs-17(3:11)2021.

[41] L. Birkedal, R. Clouston, B. Mannaa, R. E. Møgelberg, A. M. Pitts, and B. Spitters. Modal dependent type theory and dependent right adjoints. *Mathematical Structures in Computer Science* 30(2):118–138, 2020. https://doi.org/10.1017/S0960129519000197.

[42] P. Stassen, D. Gratzer, and L. Birkedal. *mitten*: A flexible multimodal proof assistant. In *TYPES 2022*, LIPIcs 269, pp. 6:1–6:23, 2023. https://doi.org/10.4230/LIPIcs.TYPES.2022.6.

[43] J. Ceulemans, A. Nuyts, and D. Devriese. Sikkel: Multimode simple type theory as an Agda library. *Electronic Proceedings in Theoretical Computer Science* 360:93–112, 2022. https://doi.org/10.4204/EPTCS.360.5.

[44] J. Ceulemans, A. Nuyts, and D. Devriese. BiSikkel: A multimode logical framework in Agda. *Proceedings of the ACM on Programming Languages* 9(POPL), pp. 210–240, 2025. https://doi.org/10.1145/3704844.

[45] The Agda Team. Modalities, Agda 2.9 language reference. https://agda.readthedocs.io/en/latest/language/modalities.html.

[46] The Agda Team. Two-level type theory, Agda language reference. https://agda.readthedocs.io/en/latest/language/two-level.html.

[47] The Agda Team. Cubical, Agda 2.9 language reference. https://agda.readthedocs.io/en/latest/language/cubical.html.

[48] The Agda Team. Guarded type theory, Agda language reference. https://agda.readthedocs.io/en/latest/language/guarded.html.

[49] KU Leuven DistriNet. Syntax and implementation principles for multimodal dependent type theory, research project 3E240493, 2024–2028. https://research.kuleuven.be/portal/en/project/3E240493.

[50] KU Leuven DistriNet. Implementation of multimodal type theory for Agda, research project 3E250760, 2025–2029. https://research.kuleuven.be/portal/en/project/3E250760.

[51] J. C. Blanchette, Q. Qiu, and S. Tourret. Given Clause Loops. Archive of Formal Proofs, 2023. https://isa-afp.org/entries/Given_Clause_Loops.html.

[52] B. van den Berg and F. De Marchi. Models of non-well-founded sets via an indexed final coalgebra theorem. *Journal of Symbolic Logic* 72(3), 2007; arXiv:math/0508531. https://arxiv.org/abs/math/0508531.

[53] B. van den Berg and F. de Marchi. Non-well-founded trees in categories. arXiv:math/0409158, 2004. https://arxiv.org/abs/math/0409158.

[54] P. Aczel. *Non-Well-Founded Sets*. CSLI Lecture Notes 14, 1988.

[55] S. Abramsky. A Cook's Tour of the Finitary Non-Well-Founded Sets. arXiv:1111.7148, 2011. https://arxiv.org/abs/1111.7148.

[56] J. Worrell. On the final sequence of a finitary set functor. *Theoretical Computer Science* 338(1–3):184–199, 2005.

[57] C. E. Brown and K. Pąk. A Tale of Two Set Theories. In *Interactive Theorem Proving*, LIPIcs 141, 2019. https://arxiv.org/abs/1907.08368.

[58] C. E. Brown, C. Kaliszyk, and K. Pąk. Higher-order Tarski–Grothendieck as a foundation for formal proof. In *Interactive Theorem Proving*, LIPIcs 141, 2019. https://doi.org/10.4230/LIPIcs.ITP.2019.9.

[59] D. J. Chalmers. *Constructing the World*. Oxford University Press, 2012.

[60] P. Dybjer. Internal type theory. In *Types for Proofs and Programs*, LNCS 1158, pp. 120–134, 1996.

[61] M. Bezem, T. Coquand, P. Dybjer, and M. Escardó. Type theory with explicit universe polymorphism. arXiv:2212.03284, revised 2024. https://arxiv.org/abs/2212.03284.

[62] M. Bezem, T. Coquand, P. Dybjer, and M. Escardó. A generalized algebraic theory for type theory with explicit universe polymorphism. arXiv:2603.04010, 2026. https://arxiv.org/abs/2603.04010.

[63] Daniel Gratzer, Michael Shulman, and Jonathan Sterling. Strict universes for Grothendieck topoi. arXiv:2202.12012, revised 2024. https://arxiv.org/abs/2202.12012.

[64] P. Aczel. The type theoretic interpretation of constructive set theory. In *Logic Colloquium '77*, pp. 55–66, 1978.

[65] Gyesik Lee and Benjamin Werner. Proof-irrelevant model of CC with predicative induction and judgmental equality. *Logical Methods in Computer Science* 7(4:5), 2011. https://doi.org/10.2168/LMCS-7(4:5)2011.

[66] N. Gambino and P. Aczel. The generalised type-theoretic interpretation of constructive set theory. *Journal of Symbolic Logic* 71(1):67–103, 2006.

[67] M. E. Maietti and G. Rosolini. Elementary quotient completion. arXiv:1206.0162, 2012. https://arxiv.org/abs/1206.0162.

[68] R. Bocquet. Coherence of strict equalities in dependent type theories. arXiv:2010.14166, revised 2022. https://arxiv.org/abs/2010.14166.

[69] C. Kapulkin and Y. Li. Extensional concepts in intensional type theory revisited. arXiv:2310.05706, revised 2025. https://arxiv.org/abs/2310.05706.

[70] M. A. Shulman. Set theory for category theory. arXiv:0810.1279, 2008. https://arxiv.org/abs/0810.1279.

[71] Z. L. Low. Universes for category theory. arXiv:1304.5227, 2014. https://arxiv.org/abs/1304.5227.

[72] E. Palmgren. On universes in type theory. In *Twenty Five Years of Constructive Type Theory*, Oxford Logic Guides 36, 1998. https://www2.math.uu.se/~palmgren/universe.pdf.

[73] M. Rathjen, E. R. Griffor, and E. Palmgren. Inaccessibility in constructive set theory and type theory. *Annals of Pure and Applied Logic* 94(1–3):181–200, 1998. https://doi.org/10.1016/S0168-0072(97)00072-9.

[74] A. Setzer. Extending Martin-Löf type theory by one Mahlo-universe. *Archive for Mathematical Logic* 39:155–181, 2000. https://csetzer.github.io/articles/mahlo.pdf.

[75] A. Setzer. Proof theory and Martin-Löf type theory. Manuscript, 2008. https://csetzer.github.io/articles/setzerProofTheoryAndMartinLoefTypeTheory.pdf.

[76] R. Kahle and A. Setzer. An extended predicative definition of the Mahlo universe. In *Ways of Proof Theory*, pp. 315–340, 2010. https://doi.org/10.1515/9783110324907.315.

[77] P. Dybjer and A. Setzer. The extended predicative Mahlo universe in Martin-Löf type theory. *Journal of Logic and Computation* 34(6):1032–1063, 2024. https://doi.org/10.1093/logcom/exad022.

[78] P. Dybjer and A. Setzer. Predicativity of the Mahlo universe in type theory. Manuscript, 2025. https://csetzer.github.io/articles/dybjerSetzerWeylVolume2025/dybjerSetzerWeylVolume2025Final.pdf.

[79] Y. Takahashi. Interpretation of inaccessible sets in Martin-Löf type theory with one Mahlo universe. *Logical Methods in Computer Science* 21(4):16:1–16:34, 2025. https://doi.org/10.46298/lmcs-21(4:16)2025.

[80] U. Waldmann, S. Tourret, S. Robillard, and J. C. Blanchette. A comprehensive framework for saturation theorem proving. *Journal of Automated Reasoning* 66(4):499–539, 2022.

[81] Z. Goertzel. Formal Ethics Ontology. Public ontology repository, 2026 snapshot. https://github.com/zariuq/Formal-Ethics-Ontology.

[82] Z. Goertzel. Moral Paradigm Equivalence. *Garden of Minds*, 2025. https://gardenofminds.art/esowiki/main/ethical-conjectures/moral-paradigm-equivalence/.

[83] Z. Goertzel. Target-Centered Virtue Ethics. *Garden of Minds*, 2024. https://gardenofminds.art/esowiki/ethics/virtue-ethics/target-centered-virtue-ethics/.

[84] G. Sutcliffe and C. Benzmüller. Automated reasoning in higher-order logic using the TPTP THF infrastructure. *Journal of Formalized Reasoning* 3(1):1–27, 2010.

[85] C. Kaliszyk, G. Sutcliffe, and F. Rabe. TH1: The TPTP typed higher-order form with rank-1 polymorphism. In *Practical Aspects of Automated Reasoning (PAAR 2016)*, CEUR Workshop Proceedings 1635, pp. 41–55, 2016.

[86] C. Rothgang, F. Rabe, and C. Benzmüller. Theorem proving in dependently-typed higher-order logic. In *Automated Deduction – CADE 29*, LNCS 14132, 2023. https://arxiv.org/abs/2305.15382.

[87] G. P. Huet. A unification algorithm for typed λ-calculus. *Theoretical Computer Science* 1(1):27–57, 1975.

[88] W. D. Goldfarb. The undecidability of the second-order unification problem. *Theoretical Computer Science* 13(2):225–230, 1981.

[89] A. Bentkamp, J. C. Blanchette, S. Tourret, P. Vukmirović, and U. Waldmann. Superposition with lambdas. *Journal of Automated Reasoning* 65(7):893–940, 2021.

[90] P. Selinger. Lecture notes on the lambda calculus. arXiv:0804.3434, revised 2013, Sections 4, 6, and 7. https://arxiv.org/abs/0804.3434.

[91] S. Castellan, P. Clairambault, and P. Dybjer. Categories with families: unityped, simply typed, and dependently typed. arXiv:1904.00827, 2019. https://arxiv.org/abs/1904.00827.

[92] S. Awodey. On Hofmann–Streicher universes. *Mathematical Structures in Computer Science*, 2024. https://www.cambridge.org/core/journals/mathematical-structures-in-computer-science/article/on-hofmannstreicher-universes/97FD8FFB986B0336F9E1072427284B13.

[93] F. Blanqui, G. Dowek, E. Grienenberger, G. Hondet, and F. Thiré. A modular construction of type theories. *Logical Methods in Computer Science* 19(1):12:1–12:28, 2023. https://doi.org/10.46298/LMCS-19(1:12)2023.

[94] C. E. Brown, C. Kaliszyk, M. Suda, and J. Urban. Hammering higher order set theory. Manuscript, 2025. https://ckaliszyk.github.io/d/25/cbckmsju-cicm25.pdf.

[95] A. Bauer and A. Petković Komel. An extensible equality checking algorithm for dependent type theories. *Logical Methods in Computer Science* 18(1):17:1–17:42, 2022. https://arxiv.org/abs/2103.07397.

[96] MORK contributors. Minimal MeTTa 2 (MM2). Project specification wiki, accessed September 2026. https://github.com/trueagi-io/MORK/wiki/Minimal-MeTTa-2-(MM2).

[97] MORK contributors. Sources and sinks. Project specification wiki, accessed September 2026. https://github.com/trueagi-io/MORK/wiki/Sources-and-Sinks.

[98] B. Goertzel. The distinction calculus: d-forms for hives of reflexively self-modifying systems. Manuscript, corrected September 2026.

[99] B. Goertzel. The distinction calculus of metacognitive self-modification. Manuscript, revised September 2026.

[100] C. Fields and M. Levin. Competency in navigating arbitrary spaces as an invariant for analyzing cognition in diverse embodiments. *Entropy* 24(6):819, 2022. https://doi.org/10.3390/e24060819.

[101] Y. Futamura. Partial evaluation of computation process, revisited. *Higher-Order and Symbolic Computation* 12:377–380, 1999.

[102] A. Assaf and G. Burel. Translating HOL to Dedukti. *EPTCS* 186:74–88, 2015. https://arxiv.org/abs/1507.08720.

[103] A. Assaf and R. Cauderlier. Mixing HOL and Coq in Dedukti (extended abstract). *EPTCS* 186:89–96, 2015. https://arxiv.org/abs/1507.08721.

[104] A. Assaf. Conservativity of embeddings in the lambda Pi calculus modulo rewriting. Long version of the TLCA 2015 paper. https://arxiv.org/abs/1504.05038.

[105] V. F. Turchin. The concept of a supercompiler. *ACM Transactions on Programming Languages and Systems* 8(3):292–325, 1986. https://doi.org/10.1145/5956.5957.

[106] M. T. Bennett. The optimal choice of hypothesis is the weakest, not the shortest. In *Artificial General Intelligence (AGI 2023)*, LNCS 13921, 2023. arXiv:2301.12987.

[107] D. Otten and M. Spadetto. The biequivalence of path categories and axiomatic Martin-Löf type theories. arXiv:2503.15431, 2025.

[108] F. Kamareddine and R. Nederpelt. A refinement of de Bruijn's formal language of mathematics. *Journal of Logic, Language and Information*, 13(3):287–340, 2004.

[109] F. Pfenning. Intensionality, extensionality, and proof irrelevance in modal type theory. In *Proceedings of the 16th IEEE Symposium on Logic in Computer Science (LICS 2001)*, pages 221–230, 2001.

[110] V. Blot, G. Dowek, T. Traversié, and T. Winterhalter. From rewrite rules to axioms in the $\lambda\Pi$-calculus modulo theory. In *Foundations of Software Science and Computation Structures (FoSSaCS 2024)*, LNCS 14575, pages 3–23, 2024.

[111] T. Coquand. Π-injectivity and subject reduction for $U : U$, by domain theory. Agda formalisation, 2026. https://github.com/coquand/domain-semantics.

[112] M. Carneiro. Lean4Lean: Verifying a typechecker for Lean, in Lean. arXiv:2403.14064, 2024.

[113] R. Diaconescu. Grothendieck institutions. *Applied Categorical Structures*, 10(4):383–402, 2002.

[114] M. Cerioli and J. Meseguer. May I borrow your logic? (Transporting logical structures along maps). *Theoretical Computer Science*, 173(2):311–347, 1997.

[115] R. Garcia, A. M. Clark, and É. Tanter. Abstracting gradual typing. In *Proceedings of the 43rd ACM SIGPLAN Symposium on Principles of Programming Languages (POPL 2016)*, pages 429–442, 2016.

[116] S. Kripke. Outline of a theory of truth. *The Journal of Philosophy*, 72(19):690–716, 1975.

[117] M. Fitting. A Kripke–Kleene semantics for logic programs. *The Journal of Logic Programming*, 2(4):295–312, 1985. https://doi.org/10.1016/0743-1066(85)80005-4.

[118] F. W. Lawvere. Diagonal arguments and cartesian closed categories. In *Category Theory, Homology Theory and their Applications II*, Lecture Notes in Mathematics 92, pages 134–145. Springer, 1969.

[119] H. B. Curry. The inconsistency of certain formal logics. *The Journal of Symbolic Logic*, 7(3):115–117, 1942.

[120] S. Feferman. Arithmetization of metamathematics in a general setting. *Fundamenta Mathematicae*, 49:35–92, 1960.

[121] A. M. Turing. Systems of logic based on ordinals. *Proceedings of the London Mathematical Society*, s2-45:161–228, 1939.

[122] S. Feferman. Transfinite recursive progressions of axiomatic theories. *The Journal of Symbolic Logic*, 27(3):259–316, 1962.

[123] H. Nakano. A modality for recursion. In *Proceedings of the 15th Annual IEEE Symposium on Logic in Computer Science (LICS 2000)*, pages 255–266, 2000.

[124] L. Birkedal, R. E. Møgelberg, J. Schwinghammer, and K. Støvring. First steps in synthetic guarded domain theory: step-indexing in the topos of trees. *Logical Methods in Computer Science*, 8(4:1), 2012. https://doi.org/10.2168/LMCS-8(4:1)2012.

[125] P. Severi. A light modality for recursion. *Logical Methods in Computer Science*, 15(1:8), 2019. https://doi.org/10.23638/LMCS-15(1:8)2019.

[126] B. C. Smith. Reflection and semantics in LISP. In *Proceedings of the 11th ACM SIGACT-SIGPLAN Symposium on Principles of Programming Languages (POPL 1984)*, 1984. https://doi.org/10.1145/800017.800513.

[127] M. Wand and D. P. Friedman. The mystery of the tower revealed: a non-reflective description of the reflective tower. In *Proceedings of the 1986 ACM Conference on LISP and Functional Programming*, 1986. https://doi.org/10.1145/319838.319871.

[128] O. Danvy and K. Malmkjær. Intensions and extensions in a reflective tower. In *Proceedings of the 1988 ACM Conference on LISP and Functional Programming*, 1988. https://doi.org/10.1145/62678.62725.

[129] N. Amin and T. Rompf. Collapsing towers of interpreters. *Proceedings of the ACM on Programming Languages*, 2(POPL):52, 2018. https://doi.org/10.1145/3158140.

[130] M. Brown and J. Palsberg. Breaking through the normalization barrier: a self-interpreter for F-omega. In *Proceedings of the 43rd ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages (POPL 2016)*, 2016. https://doi.org/10.1145/2837614.2837623.

[131] A. Bauer. On self-interpreters for System T and other typed λ-calculi. Note, 2016.

[132] S. Ding and Q. Zhang. The normalization barrier revisited. In *Workshop dedicated to Jens Palsberg on the occasion of his 60th birthday (JENSFEST 2024)*, 2024.

[133] T. J. Green, G. Karvounarakis and V. Tannen. Provenance semirings. In *Proceedings of the 26th ACM Symposium on Principles of Database Systems (PODS 2007)*, pages 31–40, 2007.

[134] E. Grädel and V. Tannen. Semiring provenance for first-order model checking. arXiv:1712.01980, 2017.

[135] J. M. Hellerstein and P. Alvaro. Keeping CALM: when distributed consistency is easy. *Communications of the ACM*, 63(9):72–81, 2020.

[136] T. J. Ameloot, F. Neven and J. Van den Bussche. Relational transducers for declarative networking. *Journal of the ACM*, 60(2):15, 2013.

[137] P. Wang. *Non-Axiomatic Logic: A Model of Intelligent Reasoning*. World Scientific, 2013.

[138] A. Jøsang. Belief calculus. arXiv:cs/0606029, 2006.

[139] A. Jøsang. Fission of opinions in subjective logic. In *Proceedings of the 12th International Conference on Information Fusion*, 2009.

[140] A. P. Dempster. Upper and lower probabilities induced by a multivalued mapping. *Annals of Mathematical Statistics*, 38(2):325–339, 1967.

[141] G. Shafer. *A Mathematical Theory of Evidence*. Princeton University Press, 1976.

[142] P. Walley. Inferences from multinomial data: learning about a bag of marbles. *Journal of the Royal Statistical Society, Series B*, 58(1):3–57, 1996.

[143] G. Shafer and V. Vovk. *Game-Theoretic Foundations for Probability and Finance*. Wiley, 2019.

[144] V. Vovk and R. Wang. E-values: calibration, combination and applications. *Annals of Statistics*, 49(3):1736–1754, 2021.

[145] C. Genest and J. V. Zidek. Combining probability distributions: a critique and an annotated bibliography. *Statistical Science*, 1(1):114–135, 1986.

[146] M. Fréchet. Sur les tableaux de corrélation dont les marges sont données. *Annales de l'Université de Lyon, Section A*, 14:53–77, 1951.

[147] S. J. Julier and J. K. Uhlmann. A non-divergent estimation algorithm in the presence of unknown correlations. In *Proceedings of the American Control Conference*, 1997.

[148] N. D. Belnap. A useful four-valued logic. In J. M. Dunn and G. Epstein, editors, *Modern Uses of Multiple-Valued Logic*, pages 5–37. Reidel, 1977.

[149] D. Nelson. Constructible falsity. *Journal of Symbolic Logic*, 14(1):16–26, 1949.

[150] W. Carnielli and A. Rodrigues. An epistemic approach to paraconsistency: a logic of evidence and truth. *Synthese*, 196:3789–3813, 2019.

[151] D. Varacca and G. Winskel. Distributing probability over non-determinism. *Mathematical Structures in Computer Science*, 16(1):87–113, 2006.

[152] M. Zwart and D. Marsden. No-go theorems for distributive laws. *Logical Methods in Computer Science*, 18(1), 2022.

[153] K. Keimel and G. D. Plotkin. Mixed powerdomains for probability and nondeterminism. *Logical Methods in Computer Science*, 13(1), 2017.

[154] A. Knapp, H. Mühlberger and B. Reus. Interpreting knowledge-based programs. In *Programming Languages and Systems (ESOP 2023)*, 2023. Extended version arXiv:2301.10807.